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[Intervention Review]

Medical methods for first trimester abortion

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ABSTRACT

Background

Medical abortion became an alternative method of pregnancy termination following the development of prostaglandins and antiprogesterone in the 1970s and 1980s. Recently, synthesis inhibitors of oestrogen (such as letrozole) have also been used to enhance efficacy. The most widely researched drugs are prostaglandins (such as misoprostol, which has a strong uterotonic effect), mifepristone, mifepristone with prostaglandins, and letrozole with prostaglandins. More evidence is needed to identify the best dosage, regimen, and route of administration to optimise patient outcomes. This is an update of a review last published in 2011.

Objectives

To compare the effectiveness and side effects of different medical methods for first trimester abortion.

Search methods

We searched CENTRAL, MEDLINE, Embase, Global Health, and LILACs on 28 February 2021. We also searched Clinicaltrials.gov and the World Health Organization's (WHO) International Clinical Trials Registry Platform, and reference lists of retrieved papers.

Selection criteria

We considered randomised controlled trials (RCTs) that compared different medical methods for abortion before the 12th week of gestation. The primary outcome is failure to achieve complete abortion. Secondary outcomes are mortality, surgical evacuation, ongoing pregnancy at follow-up, time until passing of conceptus, blood transfusion, side effects and women's dissatisfaction with the method.

Data collection and analysis

Two review authors independently selected and evaluated studies for inclusion, and assessed the risk of bias. We processed data using Review Manager 5 software. We assessed the certainty of the evidence using the GRADE approach.

Main results

We included 99 studies in the review (58 from the original review and 41 new studies).

1. Combined regimen mifepristone/prostaglandin

Mifepristone dose: high-dose (600 mg) compared to low-dose (200 mg) mifepristone probably has similar effectiveness in achieving complete abortion (RR 1.07, 95% CI 0.87 to 1.33; $I^2 = 0\%$; 4 RCTs, 3494 women; moderate-certainty evidence).

Prostaglandin dose: 800 µg misoprostol probably reduces abortion failure compared to 400 µg (RR 0.63, 95% CI 0.51 to 0.78; I^2= 0%; 3 RCTs, 4424 women; moderate-certainty evidence).

Prostaglandin timing: misoprostol administered on day one probably achieves more success on complete abortion than on day three (RR 1.94, 95% CI 1.05 to 3.58; 1489 women; 1 RCT; moderate-certainty evidence).

Administration strategy: there may be no difference in failure of complete abortion with self-administration at home compared with hospital administration (RR 1.63, 95% CI 0.68 to 3.94; I^2 = 84%; 2263 women; 4 RCTs; low-certainty evidence), but failure may be higher when administered by nurses in hospital compared to by doctors in hospital (RR 2.69, 95% CI 1.39 to 5.22; I^2 = 66%; 3 RCTs, 3056 women; low-certainty evidence).

Administration route: oral misoprostol probably leads to more failures than the vaginal route (RR 2.38, 95% CI 1.46 to 3.87; I^2 = 39%; 3 RCTs, 1704 women; moderate-certainty evidence) and may be associated with more frequent side effects such as nausea (RR 1.14, 95% CI 1.03 to 1.26; I^2 = 0%; 2 RCTs, 1380 women; low-certainty evidence) and diarrhoea (RR 1.80 95% CI 1.49 to 2.17; I^2 = 0%; 2 RCTs, 1379 women). Compared with the vaginal route, complete abortion failure is probably lower with sublingual (RR 0.68, 95% CI 0.22 to 2.11; I^2 = 59%; 2 RCTs, 3229 women; moderate-certainty evidence) and may be lower with buccal administration (RR 0.71, 95% CI 0.34 to 1.46; I^2 = 0%; 2 RCTs, 479 women; low-certainty evidence), but sublingual or buccal routes may lead to more side effects. Women may experience more vomiting with sublingual compared to buccal administration (RR 1.33, 95% CI 1.01 to 1.77; low-certainty evidence).

2. Mifepristone alone versus combined regimen

The efficacy of mifepristone alone in achieving complete abortion compared to combined mifepristone/prostaglandin up to 12 weeks is unclear (RR of failure 3.25, 95% CI 0.81 to 13.09; I^2 = 83%; 3 RCTs, 273 women; very low-certainty evidence).

3. Prostaglandin alone versus combined regimen

Nineteen studies compared prostaglandin alone to a combined regimen (prostaglandin combined with mifepristone, letrozole, estradiol valerate, tamoxifen, or methotrexate). Compared to any of the combination regimens, misoprostol alone may increase the risk for failure to achieve complete abortion (RR of failure 2.39, 95% CI 1.89 to 3.02; I^2 = 64%; 18 RCTs, 3471 women; low-certainty evidence), and with more diarrhoea.

4. Prostaglandin alone (route of administration)

Oral misoprostol alone may lead to more failures in complete abortion than the vaginal route (RR 3.68, 95% CI 1.56 to 8.71, 2 RCTs, 216 women; low-certainty evidence). Failure to achieve complete abortion may be slightly reduced with sublingual compared with vaginal (RR 0.69, 95% CI 0.37 to 1.28; I^2 = 87%; 5 RCTs, 2705 women; low-certainty evidence) and oral administration (RR 0.58, 95% CI 0.11 to 2.99; I^2 = 66%; 2 RCTs, 173 women). Failure to achieve complete abortion may be similar or slightly higher with sublingual administration compared to buccal administration (RR 1.11, 95% CI 0.71 to 1.74; 1 study, 401 women).

Authors' conclusions

Safe and effective medical abortion methods are available. Combined regimens (prostaglandin combined with mifepristone, letrozole, estradiol valerate, tamoxifen, or methotrexate) may be more effective than single agents (prostaglandin alone or mifepristone alone). In the combined regimen, the dose of mifepristone can probably be lowered to 200 mg without significantly decreasing effectiveness. Vaginal misoprostol is probably more effective than oral administration, and may have fewer side effects than sublingual or buccal. Some results are limited by the small numbers of participants on which they are based. Almost all studies were conducted in settings with good access to emergency services, which may limit the generalisability of these results.

PLAIN LANGUAGE SUMMARY

Are medical methods for early termination of pregnancy effective and do they cause unwanted effects?

Key messages

- Medical abortion is a safe and effective way to terminate pregnancy in the first three months.

- Mifepristone combined with misoprostol is more effective than using these medications on their own.

- Misoprostol is more effective when placed in the vagina than when swallowed, and is less uncomfortable than placing it under the tongue or in the cheek.

What is medical abortion?

Medical abortion uses one or more medicines alone or in combination to end a pregnancy. The most common medicines are the hormones prostaglandin and mifepristone. Other medicines include methotrexate (a kind of chemotherapy), and letrozole, which slows the production of the hormone oestrogen. These medicines work by softening the cervix (neck of the womb) and causing the uterus (womb) to contract. They may be swallowed (taken orally), put under the tongue or in the cheek, or put in the vagina. They can be given by a nurse or a doctor in hospital, or taken by women at home.

Medical abortion methods may cause unwanted effects such as heavy bleeding, pain, nausea, vomiting and diarrhoea, Failed abortion is an infrequent but important complication of medical abortion. Medical methods for early abortion are already widely available in some countries, and new medicines are being developed.

What did we want to find out?

We wanted to find out which medicines were most successful in achieving complete abortion in the first three months of pregnancy, and if the dose or way they were given made a difference. We also wanted to know if there were any unwanted effects

What did we do?

We searched for studies that investigated different medicines, doses and ways of giving the medicines to women having medical abortion in the first three months of pregnancy.

Study characteristics

We included 99 studies that investigated 24 different medicine combinations, doses, and ways of giving the medicine.

Main results

Misoprostol put into the vagina is probably more effective to achieve abortion than taken orally, and may be associated with less stomach discomfort than if put under the tongue or between the tongue and cheek. Misoprostol alone and mifepristone alone may result in more failed abortions than misoprostol and mifepristone taken together. There may be little or no difference in the success rate of abortions based on whether the medicines are given at home or in hospital, the dosage of mifepristone, or single versus repeated doses of prostaglandin. However, abortions may be more successful if the medicines are given by a doctor in hospital rather than a nurse in hospital.

What are the limitations of the evidence?

Overall, our confidence in the evidence is limited or very limited for several reasons. Most studies included enough participants and used adequate methods to select them and allocate them to a particular treatment. However, it was difficult to ensure that they and the doctors treating them didn't know what treatment they had received. Several studies did not publish their aims before they started, so it is hard to evaluate whether they measured and reported all their points of interest. Almost all the studies took place in high-income countries, where women can return for a check-up. We don't know if results would have been different in low-income countries.

How up to date is the evidence?
This is a update of a review last published in 2011. The evidence is up to date to February 2021.

Summary of findings 1. Mifepristone (600 mg) plus prostaglandin versus mifepristone (200 mg) plus prostaglandin for first trimester abortion

Mifepristone (600 mg) combined with prostaglandin compared to mifepristone (200 mg) combined with prostaglandin for first trimester abortion

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: mifepristone (600 mg) combined with prostaglandin
Comparison: mifepristone (200 mg) combined with prostaglandin

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with mifepristone (200 mg) combined with prostaglandin	Risk with mifepristone (600 mg) combined with prostaglandin			
Failure to achieve complete abortion	Study population		RR 1.07 (0.87 to 1.33)	3494 (4 RCTs)	⊕⊕⊕⊝ Moderate[a]
	86 per 1000	92 per 1000 (75 to 114)			
Nausea	Study population		RR 1.02 (0.95 to 1.09)	2432 (2 RCTs)	⊕⊕⊕⊝ Moderate[a]
	450 per 1000	459 per 1000 (428 to 491)			
Women's dissatisfaction with the procedure	Not reported				
Blood transfusion	Not reported				

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to inconsistency of intervention.

Summary of findings 2. Misoprostol 800 µg plus mifepristone 200 mg versus misoprostol 400 µg plus mifepristone 200 mg for first trimester abortion

Misoprostol (800 µg) compared to misoprostol (400 µg) for first trimester abortion as combined with mifepristone

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: misoprostol 800 µg combined with mifepristone 200 mg
Comparison: misoprostol 400 µg combined with mifepristone 200 mg

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with misoprostol 400 µg combined with mifepristone 200 mg	Risk with misoprostol 800 µg combined with mifepristone 200 mg			
Failure to achieve complete abortion	Study population		RR 0.63 (0.51 to 0.78)	4424 (3 RCTs)	⊕⊕⊕⊝ Moderate[a]
	94 per 1000	59 per 1000 (48 to 73)			
Nausea	Study population		RR 0.99 (0.94 to 1.05)	4424 (3 RCTs)	⊕⊕⊕⊝ Moderate[a]
	479 per 1000	474 per 1000 (450 to 503)			
Women's dissatisfaction with the procedure	Study population		RR 0.75 (0.60 to 0.93)	4420 (3 RCTs)	⊕⊕⊕⊝ Moderate[a]
	82 per 1000	61 per 1000 (49 to 76)			
Blood transfusion	Not reported				

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
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Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to inconsistency of intervention.

Summary of findings 3. Medical abortion self-administered at home compared to administered in hospital for first trimester abortion

Medical abortion self-administered at home compared to administered in hospital for first trimester abortion

Patient or population: women undergoing first trimester abortion
Setting: at home or hospital outpatient
Intervention: self-administered at home
Comparison: administered in hospital

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with hospital-administered	Risk with self-administered at home			
Failure to achieve complete abortion	Study population		RR 1.63 (0.68 to 3.94)	2263 (4 RCTs)	⊕⊕⊝⊝ Low[a,b]
	50 per 1000	82 per 1000 (34 to 199)			
Nausea	Study population		RR 1.09 (0.74 to 1.61)	1532 (3 RCTs)	⊕⊕⊝⊝ Low[a,b]
	236 per 1000	258 per 1000 (175 to 380)			
Women's dissatisfaction with the procedure	Study population		RR 1.63 (0.95 to 2.80)	2155 (4 RCTs)	⊕⊕⊝⊝ Low[a,b]
	50 per 1000	82 per 1000 (48 to 140)			
Blood transfusion	Study population		RR 0.33 (0.01 to 8.18)	731 (1 RCT)	⊕⊝⊝⊝ Very low[c,d]
	3 per 1000	1 per 1000 (0 to 22)			

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to risk of bias, as no allocation concealment, no blinding.
[b]Downgraded one level due to inconsistency of intervention.
[c]Downgraded two levels due to risk of bias, as unclear randomisation method, no allocation concealment, no blinding.
[d]Downgraded one level due to imprecision, as the 95% confidence intervals is wide and overlaps no effect.

Summary of findings 4. Medical abortion administered by nurses compared to by doctors for first trimester abortion

Medical abortion administrated by nurses compared to by doctors for first trimester abortion

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: administrated by nurses
Comparison: administrated by doctors

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with administrated by doctors	Risk with administrated by nurses			
Failure to achieve complete abortion	Study population		RR 2.69 (1.39 to 5.22)	3056 (3 RCTs)	⊕⊕⊝⊝ Low[a,b]
	25 per 1000	68 per 1000 (35 to 133)			
Nausea	Not reported				
Women's dissatisfaction with the procedure	Study population		RR 2.70 (0.16 to 44.49)	1988 (2 RCTs)	⊕⊝⊝⊝ Very low[a,b,c]
	6 per 1000	16 per 1000 (1 to 266)			
Blood transfusion	Study population		not estimable	1068 (1 RCT)	⊕⊕⊝⊝ Low[d]
	0 per 1000	0 per 1000 (0 to 0)			

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
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Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to risk of bias, as no blinding.
[b]Downgraded one level due to inconsistency of intervention.
[c]Downgraded one level due to imprecision, as the 95% confidence interval is wide and overlaps no effect.
[d]Downgraded two levels due to risk of bias, as no blinding and attrition.

Summary of findings 5. Oral misoprostol compared to vaginal misoprostol for first trimester abortion in combined regimen mifepristone/prostaglandin

Oral misoprostol compared to vaginal misoprostol for first trimester abortion in combined regimen mifepristone/prostaglandin

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: oral mifepristone and oral misoprostol
Comparison: oral mifepristone and vaginal misoprostol

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with vaginal misoprostol	Risk with oral misoprostol			
Failure to achieve complete abortion	Study population		RR 2.38 (1.46 to 3.87)	1704 (3 RCTs)	⊕⊕⊕⊝ Moderate[a,b]
	43 per 1000	103 per 1000 (63 to 168)			
Nausea	Study population		RR 1.14 (1.03 to 1.26)	1380 (2 RCTs)	⊕⊕⊝⊝ Low[a,b]
	482 per 1000	549 per 1000 (496 to 607)			

Women's dissatisfaction with the procedure	Not reported
Blood transfusion	Not reported

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
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Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a] Downgraded one level due to risk of bias, as no allocation concealment, no blinding.
[b] Downgraded one level due to inconsistency of intervention.

Summary of findings 6. Sublingual misoprostol compared to vaginal misoprostol for first trimester abortion in combined regimen mifepristone/prostaglandin

Sublingual misoprostol compared to vaginal misoprostol for first trimester abortion in combined regimen mifepristone/prostaglandin

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: oral mifepristone and sublingual misoprostol
Comparison: oral mifepristone and vaginal misoprostol

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with vaginal misoprostol	Risk with sublingual misoprostol			
Failure to achieve complete abortion	Study population		RR 0.68 (0.22 to 2.11)	3229 (2 RCTs)	⊕⊕⊕⊝ Moderate[a]
	86 per 1000	58 per 1000 (19 to 180)			
Nausea	Study population		RR 1.11 (0.93 to 1.33)	3543 (3 RCTs)	⊕⊕⊕⊝ Moderate[a]

	536 per 1000	595 per 1000 (499 to 713)			
Women's dissatisfaction with the procedure	Study population		RR 1.67 (0.80 to 3.50)	3303 (2 RCTs)	⊕⊕⊕⊝ Moderate[a]
	66 per 1000	110 per 1000 (52 to 230)			
Blood transfusion	Not reported				

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to inconsistency of intervention.

Summary of findings 7. Mifepristone alone compared to mifepristone/prostaglandin for first trimester abortion

Mifepristone alone compared to mifepristone/prostaglandin for first trimester abortion

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: mifepristone alone
Comparison: mifepristone/prostaglandin

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with mifepristone/prostaglandin	Risk with mifepristone alone			
Failure to achieve complete abortion	Study population		RR 3.25 (0.81 to 13.09)	273 (3 RCTs)	⊕⊝⊝⊝ Very low[a,b,c]
	139 per 1000	451 per 1000 (113 to 1000)			

Nausea	Not reported
Women's dissatisfaction with the procedure	Not reported
Blood transfusion	Not reported

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to risk of bias, as no allocation concealment, no blinding.
[b]Downgraded one level due to inconsistency of intervention.
[c]Downgraded one level due to imprecision, as the 95% confidence interval is wide and overlaps no effect.

Summary of findings 8. Prostaglandin alone compared to combined regimen for first trimester abortion

Prostaglandin alone compared to combined regimen for first trimester abortion

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: prostaglandin alone
Comparison: combined regimen

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with combined regimen	Risk with prostaglandin alone			
Failure to achieve complete abortion	Study population		RR 2.39 (1.89 to 3.02)	3471 (18 RCTs)	⊕⊕⊝⊝ Low[a,b]
	135 per 1000	323 per 1000 (255 to 408)			

Nausea	Study population		RR 0.90 (0.74 to 1.10)	2722 (12 RCTs)	⊕⊕⊝⊝ Low[a,b]
	412 per 1000	371 per 1000 (305 to 453)			
Women's dissatisfaction with the procedure	Study population		RR 1.65 (0.75 to 3.64)	1485 (5 RCTs)	⊕⊕⊝⊝ Low[a,b]
	127 per 1000	209 per 1000 (95 to 461)			
Blood transfusion	Study population		RR 0.33 (0.03 to 3.13)	300 (1 RCT)	⊕⊕⊝⊝ Low[a,c]
	20 per 1000	7 per 1000 (1 to 63)			

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
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[a]Downgraded one level due to risk of bias, as no allocation concealment, no blinding.
[b]Downgraded one level due to inconsistency of intervention.
[c]Downgraded one level due to imprecision and small sample size, as the 95% confidence interval is wide and overlaps no effect.

Summary of findings 9. Oral misoprostol alone compared to vaginal misoprostol alone for first trimester abortion

Oral misoprostol alone compared to vaginal misoprostol alone for first trimester abortion

Patient or population: women undergoing first trimester abortion
Setting: hospital outpatient
Intervention: oral misoprostol alone
Comparison: vaginal misoprostol alone

Outcomes	Anticipated absolute effects* (95% CI)	Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)

	Risk with vaginal misoprostol alone	Risk with oral misoprostol alone			
Failure to achieve complete abortion	Study population		RR 3.68 (1.56 to 8.71)	216 (2 RCTs)	⊕⊕⊝⊝ Low[a,b]
	53 per 1000	195 per 1000 (83 to 462)			
Nausea	Study population		RR 1.21 (0.92 to 1.61)	216 (2 RCTs)	⊕⊕⊝⊝ Low[a,b]
	283 per 1000	343 per 1000 (261 to 456)			
Women's dissatisfaction with the procedure	Study population		RR 1.25 (0.65 to 2.39)	100 (1 RCT)	⊕⊕⊝⊝ Low[a,b]
	240 per 1000	300 per 1000 (156 to 574)			
Blood transfusion	Not reported				

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to risk of bias, as no allocation concealment, no blinding.
[b]Downgraded one level due to imprecision and small sample size.

Summary of findings 10. Sublingual misoprostol alone compared to vaginal misoprostol alone for first trimester abortion

Sublingual misoprostol alone compared to vaginal misoprostol alone for first trimester abortion

Patient or population: first trimester abortion
Setting: hospital outpatient
Intervention: sublingual misoprostol alone

Comparison: vaginal misoprostol alone

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)
	Risk with vaginal misoprostol alone	Risk with sublingual misoprostol alone			
Failure to achieve complete abortion	Study population		RR 0.69 (0.37 to 1.28)	2705 (5 RCTs)	⊕⊕⊝⊝ Low[a,b]
	199 per 1000	137 per 1000 (74 to 254)			
Nausea	Study population		RR 1.61 (0.78 to 3.33)	2505 (4 RCTs)	⊕⊕⊝⊝ Low[a,b]
	226 per 1000	364 per 1000 (176 to 753)			
Women's dissatisfaction with the procedure	Study population		RR 0.14 (0.07 to 0.29)	220 (1 RCT)	⊕⊕⊝⊝ Low[a, c]
	464 per 1000	65 per 1000 (32 to 134)			
Blood transfusion	Study population		RR 5.00 (0.24 to 102.96)	220 (1 RCT)	⊕⊝⊝⊝ Very low [a, d]
	0 per 1000	0 per 1000 (0 to 0)			

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **RCT:** randomised controlled trial; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

[a]Downgraded one level due to risk of bias, as no allocation concealment, no blinding.
[b]Downgraded one level due to inconsistency of intervention.
[c]Downgraded one level due to imprecision and small sample size.

[d] Downgraded two levels due to imprecision and small sample size, as the 95% confidence interval is wide and overlaps no effect.

BACKGROUND

Description of the condition

It is estimated that around 40 million abortions occur every year, either legal or illegal, resulting in an abortion rate of 35 per 1000 women (Sedgh 2016). Medical abortion has the potential to expand abortion services, where surgical services are limited, and to expand women's choice of abortion method and experience.

Description of the intervention

Medical abortion became an alternative method of pregnancy termination with the availability of prostaglandins in the early 1970s followed by the development of an antiprogesterone in the 1980s. Large, uncontrolled studies suggested that early medical abortion with mifepristone and a prostaglandin would be an effective method for pregnancy termination (Urquhart 1997).

Various drugs have been used for first trimester medical abortion. The most widely researched are prostaglandins alone, mifepristone alone, methotrexate alone, mifepristone with prostaglandins and methotrexate with prostaglandins. Prostaglandins soften the cervix, cause uterine contractions and are used orally (swallowed; put on sublingual mucosa or buccal mucosa) or vaginally (put on vaginal mucosa) for ripening the cervix before surgical or medical abortion. The most commonly used prostaglandins are misoprostol administered either orally or vaginally.

How the intervention might work

Misoprostol is a prostaglandin analogue registered for use in nonsteroidal anti-inflammatory drug (NSAID)-induced gastric ulcer prevention and treatment. It has a strong uterotonic effect and is used alone to induce pregnancy terminations, illegally in some parts of the world (Blanchard 1999; Costa 1998), as well as legally, in areas where mifepristone is not available. The reported complete abortion rate for misoprostol alone varies between 61% for single use and 93% for repeat doses (Bugalho 1996; Carbonell 1997a). Gemeprost used alone appears to be less effective in inducing complete abortion than when used in combination with mifepristone (Norman 1992).

Mifepristone, an antiprogestogen, blocks the receptors for progesterone and glucocorticosteroid and increases the sensitivity of the uterus to prostaglandins (Bygdeman 1985). This blockage results in the breakdown of maternal capillaries in the decidua, the synthesis of prostaglandins by the epithelium of decidual glands and inhibition of prostaglandin dehydrogenase (WHO 1997).

Mifepristone has been licensed in France and China since 1988, in the UK since 1991 and, in the USA and India since 2000 and 2002, respectively. Mifepristone given alone has been shown to result in abortion only in 60% to 80% of cases, depending on the gestational age and the dose given (WHO 1997). However, in combination with a prostaglandin at up to 63 days of amenorrhoea, it leads to complete abortion in about 95% of pregnancies or more. The effect of mifepristone develops over a time period of 24 to 48 hours; therefore, prostaglandins have usually been administered after 36 to 48 hours. Currently, different regimens are in use. The recommended regimen by the manufacturer is mifepristone 600 mg followed by misoprostol (between 400 µg to 800 µg) or gemeprost (0.5 mg to 1 mg vaginally) and is registered for abortion in pregnancies up to 49 days in France and up to 63 days of amenorrhoea in the UK. However, a reduced dose of mifepristone combined with a prostaglandin has similar effectiveness and has the advantage of being much less expensive (WHO 1997).

Methotrexate has been used successfully for the treatment of unruptured tubal pregnancy. It is a folic acid antagonist that inhibits purine and pyrimidine synthesis and is cytotoxic to the trophoblast. The use of methotrexate with misoprostol for first trimester abortion was first introduced in 1993 (Creinin 1993; Grimes 1997). This combination was more effective when misoprostol was administered seven days after methotrexate compared to three days, leading to a complete abortion rate of 98% (Creinin 1995).

Letrozole is one of the aromatase inhibitors, which is used to block the synthesis of oestrogen, and stimulate ovulation. It is suggested that prescription of aromatase inhibitors prior to misoprostol or mifepristone for inducing medical abortion, might increase efficiency of the treatment regimen and decrease the need for surgical interventions (Yeung 2012).

Why it is important to do this review

Side-effects of medical methods are heavy bleeding, pain, nausea, vomiting and diarrhoea, varying in severity according to the protocols and gestational age (Baiju 2019). In two studies included in the Cochrane Review of the topic, compared to surgical procedures, medical methods are associated with a longer duration of bleeding (Say 2010).

Failed abortion is an infrequent but important complication of medical abortion. Both methotrexate and misoprostol may lead to fetal anomalies if the pregnancy persists (Grimes 1997). However, other reports state that none of the malformations reported could be conclusively related to medications used for medical abortion (Wiebe 2006).

Some women prefer medical to surgical abortion. Reasons some women opted for a medical abortion included 'more natural', 'being easier', 'more private', and 'can be done earlier in pregnancy' (Creinin 1996). Characteristics such as the method being newer, less invasive and the possibility of verifying the expulsion were reported by others (Bachelot 1992).

Medical methods for first trimester abortion are already widely available in some countries and increasingly available throughout the world. New medication has been identified for use in medical abortion, it is therefore important to identify the best available agents and regimen for use. This review is an update of a previous review, which was first published in 2004, and updated twice, in 2004 and 2011. Comparison of medical methods with surgical evacuation in the first trimester is the subject of another Cochrane Review, Say 2010.

OBJECTIVES

To compare the effectiveness and side effects of different medical methods for first trimester abortion.

METHODS

Criteria for considering studies for this review

Types of studies

We considered randomised controlled trials (RCTs) that compared different medical methods (e.g. single drug or combined, ways of application, or different dose regimens for medical abortion.

Types of participants

Women, pregnant in the first trimester, undergoing medical abortion.

Types of interventions

Different medical methods used for first trimester abortion, compared with each other.

Types of outcome measures

Primary outcomes

The main outcome measure was failure to achieve complete abortion two weeks after medical abortion.

Sonography was performed to verify abortion completion. If the sonographer reported that the uterus was empty, with no products of conception identified, complete success and efficacy of the drug were recorded.

Secondary outcomes

We also assessed other unwanted outcomes, such as side effects:

- surgical evacuation (as emergency procedure, non-emergency procedure, or undefined);
- ongoing pregnancy at follow-up, time until passing of conceptus (greater than three to six hours);
- blood transfusion;
- blood loss (measured or clinically relevant drop in haemoglobin);
- days of bleeding;
- pain resulting from the procedure (reported by the women or measured by use of analgesics, especially for abdominal pain);
- additional uterotonics used (reported as repeated medical abortion in some studies);
- women's dissatisfaction with the procedure;
- Nausea;
- Vomiting;
- Diarrhoea.

Although mortality is considered an important outcome, we did not anticipate analysing abortion-related mortality within the context of these studies. To date, no studies have reported mortality.

Search methods for identification of studies

The Cochrane Fertility Regulation Information Specialist conducted a search for all published, unpublished, and ongoing studies, without restrictions on language or publication status. The search strategies for each database were modelled on the updated search strategy designed for MEDLINE ALL (all update search strategies are available in Appendix 1, previous search strategies are available in Appendix 2). We contacted authors of included studies for data clarification and further information. We considered adverse effects described in included studies only. Searches were date limited from 2010 to February 2021 to extend from and include literature published since the last review. POPLINE database ceased publication in the interim and thus was not searched for this update. Searches were conducted in 2003 and 2011 for the original reviews, with update searches in February 2021.

Electronic searches

We searched the following databases:

- Cochrane Central Register of Controlled Trials (CENTRAL) via EBM Reviews (Ovid, January 2010 to February 2021);
- MEDLINE ALL (Ovid) (January 2010 to February 2021);
- Embase.com (January 2010 to February 2021);
- Global Health (Ovid) (January 2010 to February 2021);
- LILACs lilacs.bvsalud.org/en/ (January 2010 to February 2021).

We searched the following trials registries:

- ClinicalTrials.gov www.clinicaltrials.gov (January 2010 to February 2021);
- World Health Organization International Clinical Trials Registry Platform www.who.int/trialsearch (January 2010 to February 2021).

Searching other resources

We handsearched reference lists from articles identified by the search, as well as key review articles, to identify additional articles.

Data collection and analysis

See: Characteristics of included studies; Characteristics of excluded studies; Additional tables

Selection of studies

Two review authors (ZJ, ZK) independently selected studies for inclusion in the review using Covidence, after employing the search strategy described previously. The review authors evaluated studies under consideration for appropriateness for inclusion and methodological quality without consideration of their results. We included only RCTs.

After the initial screen of titles and abstracts retrieved by the search conducted by ZJ and ZK, we retrieved the full texts of the potentially eligible studies for further classification. Two review authors (ZJ, ZK) independently examined the full text of the studies claimed to be randomised. Only the studies with random allocation were included in this updated review. We resolved disagreements by discussion or by consultation with a third review author (SD). We documented the selection process with a PRISMA flow chart (Page 2021; Figure 1).

Figure 1. Study flow diagram of updated process

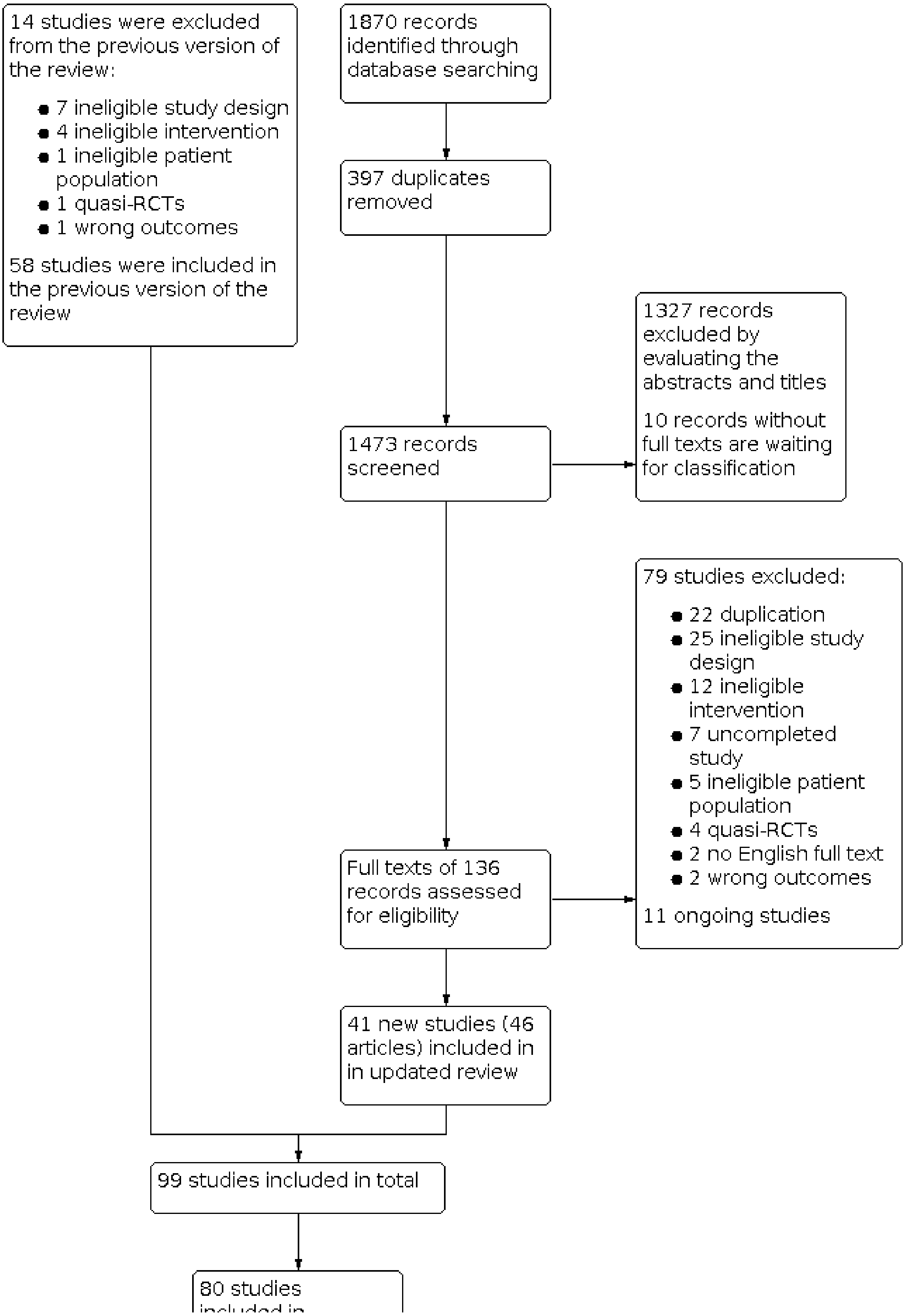
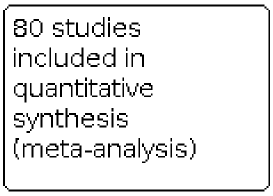


Figure 1. (Continued)



Data extraction and management

Two review authors (ZJ, ZK), independently extracted data using Covidence. If there were any discrepancies between review authors in either the inclusion or exclusion of studies, or in data extraction, we resolved them by consensus. We attempted to obtain additional information from study authors if required (ZJ, LX). We examined whether the primary study carried out an intention-to-treat analysis.

Two review authors (ZJ, ZK) independently extracted the study characteristics and outcome data of the included studies using a data extraction form designed according to Cochrane guidelines (Li 2021; Characteristics of included studies). Discrepancies were described and resolved by discussion (ZJ, ZK, SD).

Failure to achieve complete abortion is defined as an abortion that is not completed by the described intended method within two weeks. Other outcomes are failure of expulsion after four to six hours, expulsion time, side effects (nausea, vomiting, diarrhoea, abdominal pain), and mean duration of days of bleeding, bleeding volume, and blood transfusion (see Types of outcome measures). We further divided studies into early (49 days or fewer of amenorrhoea) and late (more than 49 days) gestational age at the time of abortion for subgroup analysis. Complications are defined as any serious complication described by the study authors and that was not a failure or side effect.

Assessment of risk of bias in included studies

Two review authors (ZJ, ZK) independently assessed the risk of bias for the update included studies using Cochrane tools and criteria (Higgins 2017). These were the six domains of: selection (random sequence generation and allocation concealment); performance (blinding of participants and personnel); detection (blinding of outcome assessors); attrition (incomplete outcome data); reporting (selective reporting); and other bias (such as funding). We assigned each study a judgement of low risk of bias, high risk of bias, or unclear risk of bias. We stated and resolved disagreements by consensus (ZJ, ZK, SD). We described all judgements fully and presented our conclusions in the risk of bias table (Characteristics of included studies). We described the risk of bias data in detail in the Results and facilitated the interpretation of the reliability of the results by quoting the original text or listing the reasons for judgements in the risk of bias table (see: Risk of bias in included studies and Characteristics of included studies). We took care to search for within-study selective reporting, which is studies failing to report obvious outcomes or reporting them in insufficient detail. We searched for published protocols of studies online and compared the outcomes specified in the protocol versus those reported in the final published study (LX).

Measures of treatment effect

For dichotomous data, we used the numbers of events in the control and experimental groups of each study to calculate Mantel-Haenszel risk ratios (RRs). For continuous data, if all studies reported exactly the same outcomes, we planned to calculate mean difference (MD) between treatment groups. If they reported similar outcomes on different scales, we planned to calculate the standardised mean difference (SMD). We reported confidence intervals (95% CI) for all outcomes. We conducted intention-to-treat (ITT) analysis to measure the treatment effect. We converted continuous data reported as medians and quartiles in studies to means and standard deviations (SD).

Unit of analysis issues

The unit of analysis was each woman randomised. Three update studies had multiple treatment groups; we included only one study with four arms in the meta-analysis (von Hertzen 2010). We synthesised the four arms into two arms based on the identical application route or dosage of misoprostol. We could not pool the other two studies, but have narratively synthesised them (Chen 2015a; Li 2015).

Dealing with missing data

We described for each included study, and for each outcome, the completeness of data including attrition and exclusions from the analysis. We stated whether studies reported attrition and exclusions and the numbers included in the analysis at each stage (compared with the total randomised participants), reasons for attrition or exclusion where reported, and whether missing data were balanced across groups or were related to outcomes. Where studies reported, or study authors could supply sufficient information, we planned to re-include missing data in the analyses that we undertook. We analysed the data on an ITT basis for the studies with participant depletion. We imputed individual values for attrition of dichotomous data, assuming events existing in the control group, but absent in the experimental group, such as the outcome of failure to achieve complete abortion.

Assessment of heterogeneity

We evaluated clinical heterogeneity by looking at the variability in participant factors (such as baseline characteristics and diagnostic criteria), interventions (such as therapeutic strategies, co-intervention, and time of treatment), and the outcomes studied (such as definition, format and time points of measurement). We assessed methodological heterogeneity from differences in the design factors of studies (such as randomisation methods).

The clinical heterogeneity documented in Characteristics of included studies was high among these studies, especially for the different interventions, gestational week and follow-up duration.

We performed subgroup analysis based on different gestational week for resolving clinical heterogeneity.

Clinical and methodological diversity can contribute to statistical heterogeneity. We used two methods to evaluate statistical heterogeneity: the Chi^2 test (P value) and I^2 statistic (Higgins 2003). A low P value (< 0.10) qualitatively indicated heterogeneity of intervention effects. The I^2 statistic quantitatively estimated the heterogeneity across studies: an I^2 statistic greater than 50% was taken to indicate substantial heterogeneity (Deeks 2021).

Assessment of reporting biases

In view of the difficulty of detecting and correcting for publication bias and other reporting biases, review authors aimed to minimise their potential impact by ensuring a comprehensive search for eligible studies and by staying alert for duplication of data. We evaluated selective reporting by comparing the protocols of RCTs with the published full-texts if the protocols were available.

We assessed potential publication bias using a funnel plot for the outcome of failure to achieve complete abortion, if there were 10 or more included studies (Analysis 15.1). Possible sources of asymmetry of funnel plots include publication bias and other biases, so the visual inspection of the funnel plot may be misleading, and we will not place excessive emphasis on it.

Data synthesis

We processed data using Review Manager 5 software (Review Manager 2020). The studies in this field use various combinations of agents, doses, intervals between the antiprogesterone and prostaglandin, and route of administration for prostaglandin. Since all of these variables may affect the outcomes, we did not consider it appropriate to combine similar studies into meta-analysis in many cases. However, it was possible to identify an experimental intervention and a constant (fixed) intervention, which enabled us to group the studies as follows, then use the random-effects model for significant clinical heterogeneity.

Combined regimen mifepristone/prostaglandin

- Comparison 1: different doses of mifepristone
- Comparison 2: different doses of prostaglandin
- Comparison 3: type of prostaglandin (gemeprost versus misoprostol; PGF2alpha versus misoprostol)
- Comparison 4: timing of prostaglandin
- Comparison 5: administration strategy of prostaglandin
- Comparison 6: route of administration of misoprostol - oral versus vaginal
- Comparison 7: route of administration of misoprostol - buccal versus vaginal
- Comparison 8: route of administration of misoprostol - buccal versus oral
- Comparison 9: route of administration of misoprostol - sublingual versus vaginal
- Comparison 10: route of administration of misoprostol - sublingual versus oral
- Comparison 11: route of administration of misoprostol - sublingual versus buccal
- Comparison 12: single versus split dose of prostaglandin
- Comparison 13: single versus continuous misoprostol
- Comparison 14: mifepristone alone versus combined regimen mifepristone/prostaglandin
- Comparison 15: prostaglandin alone versus a combined regimen (all)

Single regimen

- Comparison 16: prostaglandin alone - route of administration
- Comparison 17: mifepristone alone - high versus low dose

Combined regimen methotrexate/prostaglandin

- Comparison 18: timing of prostaglandin
- Comparison 19: route of methotrexate - intramuscular versus oral
- Comparison 20: dose of methotrexate
- Comparison 21: route of prostaglandin

Tamoxifen versus methotrexate (combined with prostaglandin)

- Comparison 22: low-dose tamoxifen (40 mg)
- Comparison 23: high-dose tamoxifen (160 mg)

Combined regimen mifepristone/prostaglandin versus mifepristone/prostaglandin plus tamoxifen

- Comparison 24: combined mifepristone/prostaglandin versus mifepristone/prostaglandin plus tamoxifen

Subgroup analysis and investigation of heterogeneity

Where possible we performed subgroup analyses for early and late first trimester abortions as the performance of some methods may differ with gestational age.

- Abortion up to 49 days
- Abortion over 49 days of amenorrhoea, such as in the comparison group of 1, 2, 4, 8, 10, 11, 13, 15 and 16

Sensitivity analysis

We conducted sensitivity analyses (if we identified at least three studies) for the outcome of failure to achieve complete abortion to determine whether conclusions were robust to different decisions made regarding eligibility and analysis (Analysis 15.1). These analyses included consideration of whether review conclusions would have differed if we had restricted eligibility to studies at low risk of bias (studies with low risk of bias in the domains of random sequence generation and allocation concealment) or if we had restricted analyses to studies without imputed data.

Summary of findings and assessment of the certainty of the evidence

We used the GRADE approach to assess the certainty of the evidence relating to the main outcomes (failure to achieve complete abortion, nausea, women's dissatisfaction and blood transfusion) for 10 comparisons relating to administration dose, strategy, route and regimen as below.

- Mifepristone (600 mg) combined with prostaglandin versus mifepristone (200 mg) combined with prostaglandin
- Misoprostol 800 µg combined with mifepristone 200 mg versus misoprostol 400 µg combined with mifepristone 200 mg

- Medical abortion self-administered at home versus hospital-administered
- Medical abortion administered by nurses versus administered by doctors
- Mifepristone/prostaglandin: misoprostol oral versus vaginal
- Mifepristone/prostaglandin: misoprostol sublingual versus vaginal
- Mifepristone alone versus mifepristone/prostaglandin
- Prostaglandin alone versus combined regimen
- Prostaglandin alone: misoprostol oral versus vaginal
- Prostaglandin alone: misoprostol sublingual versus vaginal

We used GRADEpro GDT to import data from Review Manager 5 (Review Manager 2020), in order to create summary of findings tables. We produced a summary of the intervention effect and a measure of certainty for the main outcomes using the GRADE approach. The GRADE approach uses five considerations (risk of bias, inconsistency, imprecision, indirectness and publication bias) to assess the certainty of the body of evidence for the outcomes. The evidence can be downgraded from 'high certainty' by one level for serious (or by two levels for very serious) limitations, depending on assessments for risk of bias, indirectness of evidence, serious inconsistency, imprecision of effect estimates or potential publication bias. Two review authors (ZJ, SD), working independently, made judgements about evidence certainty (high, moderate, low or very low), with disagreements resolved by discussion (ZJ, ZK, SD). We justified and documented our judgements, and incorporated them into reporting of results for the outcomes.

RESULTS

Description of studies

The search for the previous latest version of the review was conducted in 2011 and there were 58 included studies. The updated search, conducted in February 2021, identified 1870 records.

Results of the search

During this updated search, we screened 1473 records after excluding 397 duplicates. Of these 1473 records, we excluded 1327 records for the following reasons: non-RCT, surgical abortion, second trimester abortion or duplications after screening abstracts. We tried to retrieve 136 full texts for further evaluation; we excluded 79 for different reasons (Characteristics of excluded studies). We could not assess the full texts for 10 records; we contacted study authors for more information (Characteristics of studies awaiting classification). Eleven ongoing studies are waiting for completion and publication (Characteristics of ongoing studies). We added and updated 41 new studies (including 46 articles), and included 99 studies in total in this updated review (Characteristics of included studies). Of the 41 included studies, five studies presented data in two different publications respectively, including 10 published articles (Klingberg Allvin 2015; Iyengar 2015; Kopp Kallner 2015; Warriner 2011; von Hertzen 2003). The PRISMA study flow of the updated review was illustrated in Figure 1 (Page 2021).

Included studies

See Characteristics of included studies.

Of the 41 newly added studies, four were multiple-arm RCTs (Chen 2015a; Li 2015; Souizi 2020; von Hertzen 2010); the others were two-arm RCTs. The participants include women with missed abortion (an abortion in which the fetus dies but is retained within the uterus, without bleeding or abdominal pain), incomplete abortion (a miscarriage in which some fetal or placental tissue remains in the uterus, with bleeding or abdominal pain) and normal first trimester pregnancy. The follow-up duration of the included studies ranged from one day to 30 days. We added different strategies of administering medical abortions in this updated review, with seven included studies (Analysis 15.1). Administration was by a nurse or a doctor, at home or in hospital. Several studies divided women by gestational age at the time of abortion into subgroups: early (49 days or fewer of amenorrhoea) and late (more than 49 days of amenorrhoea) (Blum 2012; Chai 2013; Chong 2012; Lee 2011; Ngoc 2011; Raghavan 2010; Sheldon 2019; Song 2018). One study explored the efficacy of traditional Chinese medicine combined with mifepristone and misoprostol (Chen 2015a). Letrozole has recently been used to treat missed abortion, so we added studies that compared misoprostol and misoprostol combined with letrozole to the present review. However, tamoxifen and methotrexate have been used less frequently in recent years, so few studies referred to these abortion methods.

Our main outcome was failure to achieve complete abortion with the method intended. Three studies reported time until passing of conceptus as mean and SD (Javadi 2015; Sinha 2018; Torky 2018). Few studies reported blood loss measured in haemoglobin (Abdelshafy 2019; Chai 2013; Ng 2015; Qian 2015), and blood transfusion. Some studies chose the outcome bleeding more than expected (Iyengar 2015; Klingberg Allvin 2015; Kopp Kallner 2015; Ngoc 2011; Raghavan 2010; Schreiber 2018; Sonsanoh 2014; Tanha 2010; Torky 2018; Warriner 2011). Hospitalisation and unscheduled visits were the primary outcomes in studies performing medical abortion at home (Iyengar 2015; Li 2017; Song 2018; Shrestha 2014). Acceptability and completion were further reported in studies administrating medical abortion by nurse (Kopp Kallner 2015). Time spent in hospital, cost, and menstruation were evaluated (Li 2017; Kopp Kallner 2015). Pelvic infection was reported by only one study (Schreiber 2018). Five studies defined additional uterotonics used as repeated medical abortion (Iyengar 2015; Ngoc 2011; Sheldon 2019; Sinha 2018; Teimoori 2019). As well as nausea, vomiting, diarrhoea and abdominal pain, several studies also reported other side effects, such as fever, chills, dizziness and headache (Iyengar 2015; Klingberg Allvin 2015; Kopp Kallner 2015; Ngoc 2011; Raghavan 2010; Sheldon 2019; Shrestha 2014; Song 2018; Sonsanoh 2014; Souizi 2020; Tanha 2010; Torky 2018; Verma 2017; von Hertzen 2010).

Of the 58 previously included studies, one study presented data in two different publications (von Hertzen 2003). One study used two different comparisons (Wiebe 1999a, Wiebe 1999b). Five studies used different regimens, doses or timing of the medicines that could not be combined with any of the other regimens in the comparisons and were therefore listed separately in Table 1 (Liao 2004; Wang 2000; WHO 1989; WHO 1991; Wiebe 2006). Instead of failure to achieve complete abortion, 14 studies used either other definitions (i.e. surgical intervention) or administered additional prostaglandins (Carbonell 1997b; Creinin 1994; Creinin 1995; Creinin 1996a; Creinin 1997; Creinin 2001a; Creinin 2001b; Hamoda 2005; Jain 1999; Koopersmith 1996; Ozeren 1999; Schaff 2000; Wiebe 1999a; Wiebe 1999b).

Excluded studies

In total, we excluded 93 studies from the review, for different reasons (Characteristics of excluded studies). Of the 79 newly excluded studies, 22 were duplicate abstracts, 25 were ineligible study designs, 12 were ineligible interventions, seven were incomplete studies, five were ineligible patient population, four were quasi-RCTs, two were ineligible outcomes, and two were non-English full texts without translation. Of the 14 previously excluded studies, seven were ineligible study designs, four had ineligible interventions, one an ineligible patient population, one had ineligible outcomes, and one was a quasi-RCT.

Risk of bias in included studies

For a summary of each risk of bias item across all included studies, see Figure 2 and Figure 3. Further information including the reasons for judgements may be found in the risk of bias sections of the Characteristics of included studies.

Figure 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

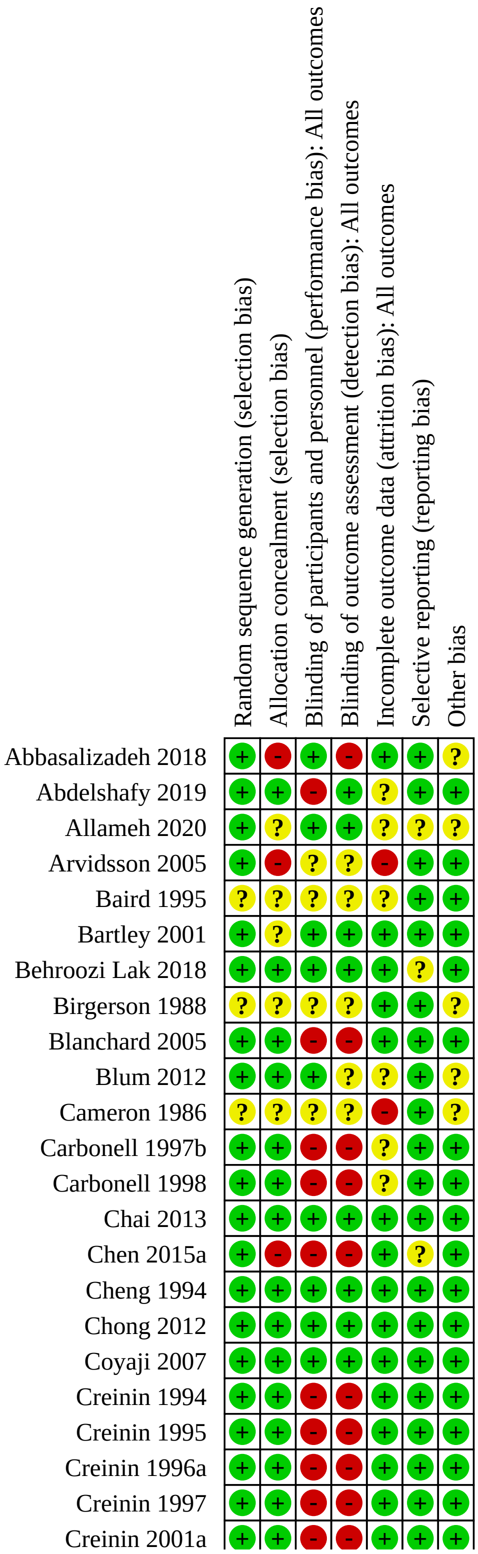


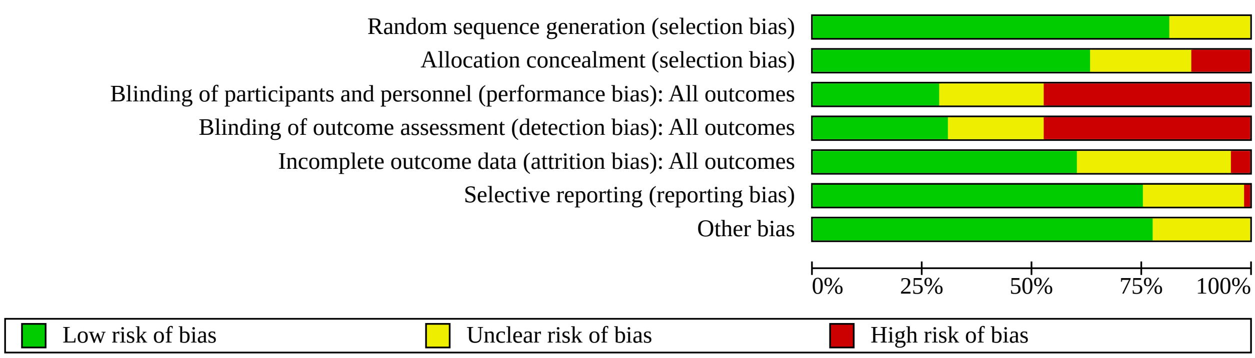
Figure 2. (Continued)

Creinin 1997	+	+	-	-	+	+	+
Creinin 2001a	+	+	-	-	+	+	+
Creinin 2001b	+	+	-	-	+	+	+
Creinin 2004	+	+	-	-	+	+	+
Creinin 2007	+	+	-	-	+	+	?
Dahiya 2011	+	-	-	-	?	?	+
El-Refaey 1994	+	+	?	?	?	+	+
El-Refaey 1995	+	+	?	?	?	+	+
Fekih 2010	+	+	-	-	+	+	+
Garg 2015	+	-	-	-	+	?	+
Goel 2011	+	+	-	-	+	?	+
Guest 2007	+	+	-	-	?	+	+
Hamoda 2005	+	+	-	-	?	+	+
Iyengar 2015	+	+	-	-	+	+	+
Jain 1999	+	+	+	+	+	+	+
Jain 2002	+	+	+	+	?	+	+
Javadi 2015	?	-	+	+	+	?	+
Klingberg Allvin 2015	+	+	-	-	?	+	+
Koopersmith 1996	?	?	?	?	+	+	+
Kopp Kallner 2015	+	+	-	-	-	+	+
Lee 2011	+	+	+	+	+	+	+
Li 2015	+	+	+	+	+	?	+
Li 2017	?	-	-	-	?	?	+
Liao 2004	+	+	+	+	+	+	?
Marwah 2016	+	-	-	-	+	?	+
McKinley 1993	?	?	?	+	+	+	+
Middleton 2005	+	+	-	-	+	+	+
Mizrachi 2017	+	+	-	-	+	+	+
Ng 2015	+	-	-	-	?	?	+
Ngoc 2011	+	?	+	+	?	+	?
Olavarrieta 2015	+	+	-	-	+	?	+
Ozeren 1999	+	+	?	?	+	+	+
Paritakul 2010	+	+	-	-	?	?	+
Qian 2015	+	?	-	-	?	?	?
Raghavan 2009	+	+	?	?	?	+	?
Raghavan 2010	+	+	-	-	?	?	+
Rodger 1989	?	?	+	+	+	+	?
Sandstrom 1999	?	?	?	?	+	+	?
Sang 1994	+	?	?	?	+	+	+
Sang 1999	+	+	?	?	+	+	?
Schaff 2000	+	+	-	-	?	+	?
Schaff 2001	+	?	-	-	?	+	+
Schreiber 2018	+	+	-	+	+	+	+
Shannon 2006	+	+	-	-	+	+	?
Sheldon 2019	+	+	?	+	?	+	+
Shrestha 2014	+	-	-	-	+	?	+
Sinha 2018	+	+	+	+	+	+	+

Figure 2. (Continued)

Shrestha 2014	+	-	-	-	+	?	+
Sinha 2018	+	+	+	+	+	+	+
Song 2018	?	-	-	-	-	?	+
Sonsanoh 2014	+	+	-	-	+	?	+
Souizi 2020	+	+	-	-	?	+	?
Swahn 1989	?	?	?	+	?	+	?
Tang 2002	+	?	?	?	?	+	+
Tang 2003	+	?	+	+	+	+	+
Tanha 2010	+	+	-	-	+	?	+
Teimoori 2019	?	-	-	-	+	-	+
Torky 2018	+	+	+	+	+	+	+
Vahid Roudsari 2010	?	?	-	-	+	?	+
Verma 2011	+	-	-	-	?	+	+
Verma 2017	+	?	-	-	+	?	?
von Hertzen 2003	+	+	+	+	+	+	+
von Hertzen 2007	+	+	+	+	?	+	+
von Hertzen 2009	+	+	+	+	?	+	?
von Hertzen 2010	+	+	+	+	+	+	+
Wang 2000	?	?	?	?	?	?	+
Warriner 2011	+	+	-	-	?	?	+
WHO 1989	?	?	?	?	+	+	?
WHO 1991	+	+	?	?	?	+	+
WHO 1993	+	+	+	+	+	+	+
WHO 2000	+	+	+	+	+	+	+
WHO 2001a	+	+	+	+	+	+	+
WHO 2001b	+	+	+	+	+	+	?
Wiebe 1999a	+	+	?	?	?	+	+
Wiebe 1999b	+	+	?	?	?	+	+
Wiebe 2004	+	+	?	?	+	+	+
Wiebe 2006	?	?	-	-	+	+	+
Winikoff 2008	+	+	-	-	+	+	+
Wu 1993	?	?	?	?	?	+	+
Zheng 1989	?	?	?	?	+	+	?

Figure 3. Review authors' judgements about each risk of bias item presented as percentages across all studies



Allocation

The random sequence generation methods were adequate in 81 studies (see Figure 3). Most used computer or random number tables, but 18 were unclear because of the absence of detailed randomisation methods (Baird 1995; Birgerson 1988; Cameron 1986; Javadi 2015; Koopersmith 1996; Li 2017; McKinley 1993; Rodger 1989; Sandstrom 1999; Song 2018; Swahn 1989; Teimoori 2019; Vahid Roudsari 2010; Wang 2000; WHO 1989; Wiebe 2006; Wu 1993; Zheng 1989; see Figure 2).

The risk of selection bias in allocation concealment was high in 13 studies without concealment (Abbasalizadeh 2018; Arvidsson 2005; Chen 2015a; Dahiya 2011; Garg 2015; Javadi 2015; Li 2017; Marwah 2016; Ng 2015; Shrestha 2014; Song 2018; Teimoori 2019; Verma 2011), unclear in 23 studies without enough information about concealment (Allameh 2020; Baird 1995; Bartley 2001; Birgerson 1988; Cameron 1986; Koopersmith 1996; McKinley 1993; Ngoc 2011; Qian 2015; Rodger 1989; Sandstrom 1999; Sang 1994; Schaff 2001; Swahn 1989; Tang 2002; Tang 2003; Vahid Roudsari 2010; Verma 2017; Wang 2000; WHO 1989; Wiebe 2006; Wu 1993; Zheng 1989; see Figure 2), and low in the other 63 studies with adequate allocation concealment. Some participants would receive additional misoprostol if the gestational sac was present at the first follow-up visit. It is not clear how these participants were distributed by treatment group (Schaff 2000).

Blinding

Blinding of participants and outcome assessment was difficult for most studies. About 30% of included studies achieved adequate blinding by using placebo (Abbasalizadeh 2018; Abdelshafy 2019; Allameh 2020; Bartley 2001; Behroozi Lak 2018; Blum 2012; Chai 2013; Cheng 1994; Chong 2012; Coyaji 2007; von Hertzen 2003; Jain 1999; Jain 2002; Javadi 2015; Lee 2011; Li 2015; Liao 2004; McKinley 1993; Ngoc 2011; Rodger 1989; Schreiber 2018; Sheldon 2019; Sinha 2018; Swahn 1989; Tang 2003; Torky 2018; von Hertzen 2003; von Hertzen 2009; von Hertzen 2010; WHO 1993; WHO 2000; WHO 2001a; WHO 2001b; Figure 3). Some studies did not report details about blinding methods and we evaluated them as unclear risk of bias (Arvidsson 2005; Baird 1995; Birgerson 1988; Cameron 1986; El-Refaey 1994; El-Refaey 1995; Koopersmith 1996; Ozeren 1999; Raghavan 2009; Sandstrom 1999; Sang 1994; Sang 1999; Tang 2002; Wang 2000; WHO 1989; WHO 1991; Wiebe 1999a; Wiebe 1999b; Wiebe 2004; Wu 1993; Zheng 1989; Figure 2). If the studies were open label, then we evaluated them as high risk of bias (Blanchard 2005; Carbonell 1997b; Carbonell 1998; Chen 2015a; Creinin 1994; Creinin 1995; Creinin 1996a; Creinin 1997; Creinin 2001a; Creinin 2001b; Creinin 2004; Creinin 2007; Dahiya 2011; Fekih 2010; Garg 2015; Goel 2011; Guest 2007; Hamoda 2005; Iyengar 2015; Klingberg Allvin 2015; Kopp Kallner 2015; Li 2017; Marwah 2016; Middleton 2005; Mizrachi 2017; Ng 2015; Olavarrieta 2015; Paritakul 2010; Qian 2015; Raghavan 2010; Schaff 2000; Schaff 2001; Shannon 2006; Shrestha 2014; Song 2018; Sonsanoh 2014; Souizi 2020; Tanha 2010; Teimoori 2019; Vahid Roudsari 2010; Verma 2011; Verma 2017; Warriner 2011; Wiebe 2006; Winikoff 2008; Figure 2).

Incomplete outcome data

Attrition bias was high in four studies with unbalanced and large percentage of attrition but without ITT analysis (Arvidsson 2005; Cameron 1986; Kopp Kallner 2015; Song 2018), unclear in 36 studies with unbalanced and small percentage of attrition but without ITT analysis (Abdelshafy 2019; Allameh 2020; Baird 1995; Blum 2012; Carbonell 1997b; Carbonell 1998; Dahiya 2011; El-Refaey 1994; El-Refaey 1995; Guest 2007; Hamoda 2005; Jain 2002; Klingberg Allvin 2015; Li 2017; Ng 2015; Ngoc 2011; Paritakul 2010; Qian 2015; Raghavan 2009; Raghavan 2010; Schaff 2000; Schaff 2001; Sheldon 2019; Souizi 2020; Swahn 1989; Tang 2002; Verma 2011; Verma 2017; von Hertzen 2007; von Hertzen 2009; Wang 2000; Warriner 2011; WHO 1991; Wiebe 1999a; Wiebe 1999b; Wu 1993; see Figure 2), and low in the other 59 studies with balanced attrition or with ITT analysis performed by study authors or with no attrition.

Selective reporting

One study did not report side effects as stated in their published protocol, and we judged the risk of bias for reporting as high (Teimoori 2019). We judged risk of bias as unclear in 23 studies without enough information from protocols (Allameh 2020; Behroozi Lak 2018; Chen 2015a; Dahiya 2011; Garg 2015; Goel 2011; Javadi 2015; Li 2015; Li 2017; Marwah 2016; Ng 2015; Olavarrieta 2015; Paritakul 2010; Qian 2015; Raghavan 2010; Shrestha 2014; Song 2018; Sonsanoh 2014; Tanha 2010; Vahid Roudsari 2010; Verma 2017; Wang 2000; Warriner 2011; Figure 2), and low in the other 75 studies with efficacy and side effects outcomes as per their protocols.

Other potential sources of bias

Risk of bias for other potential sources of bias was unclear in 22 studies (Abbasalizadeh 2018; Allameh 2020; Birgerson 1988; Blum 2012; Cameron 1986; Creinin 2007; Liao 2004; Ngoc 2011; Qian 2015; Raghavan 2009; Rodger 1989; Sandstrom 1999; Sang 1999; Schaff 2000; Shannon 2006; Souizi 2020; Swahn 1989; Verma 2017; von Hertzen 2009; WHO 1989; WHO 2001b; Zheng 1989), for pharmaceutical company support; funding from an anonymous donor; unexpectedly high number of ongoing pregnancies; cessation of study; and participants with eight to 20 weeks' gestation were included, and not stratified by gestational weeks. The other 77 studies without these factors, we judged as low risk (see Figure 2; Characteristics of included studies).

Effects of interventions

See: **Summary of findings 1** Mifepristone (600 mg) plus prostaglandin versus mifepristone (200 mg) plus prostaglandin for first trimester abortion; **Summary of findings 2** Misoprostol 800 µg plus mifepristone 200 mg versus misoprostol 400 µg plus mifepristone 200 mg for first trimester abortion; **Summary of findings 3** Medical abortion self-administered at home compared to administered in hospital for first trimester abortion; **Summary of findings 4** Medical abortion administered by nurses compared to by doctors for first trimester abortion; **Summary of findings 5** Oral misoprostol compared to vaginal misoprostol for first trimester abortion in combined regimen mifepristone/prostaglandin; **Summary of findings 6** Sublingual misoprostol compared to vaginal misoprostol for first trimester abortion in combined regimen mifepristone/prostaglandin; **Summary of findings 7** Mifepristone alone compared to mifepristone/prostaglandin for first trimester abortion; **Summary of findings 8** Prostaglandin alone compared to combined regimen for first trimester abortion; **Summary of findings 9** Oral misoprostol alone compared to vaginal misoprostol alone for first trimester abortion; **Summary of findings 10** Sublingual misoprostol alone compared to vaginal misoprostol alone for first trimester abortion

Due to the many interventions, we grouped included studies into comparisons, as listed below. The main outcome for which we performed meta-analyses was failure to achieve complete abortion with the method intended. Data on side effects could be combined for some comparisons. Subgroup analysis was performed based on different gestational ages where possible (less than or equal to 49 days, and greater than 49 days). One study used two different comparisons, and was therefore listed as two different studies (Wiebe 1999a; Wiebe 1999b). Three, four and five arms were included in four studies respectively (Chen 2015a; Li 2015; Souizi 2020; von Hertzen 2010). Seven studies in total used different regimens/doses/timing of the drugs that could not be combined with any of the other regimens in the comparisons and were therefore listed separately in additional Table 1 (Chen 2015a; Li 2015; Liao 2004; Wang 2000; WHO 1989; WHO 1991; Wiebe 2006).

1. Combined regimen mifepristone/prostaglandin

Comparison 1: different doses of mifepristone

We included 10 studies in this comparison; six were included in the meta-analysis of different comparison groups. The comparisons were 600 mg versus 200 mg, 200 mg versus 100 mg, and 200 mg versus 50 mg of mifepristone (McKinley 1993; von Hertzen 2009; WHO 1993; WHO 2000; WHO 2001a; WHO 2001b). We present details about split doses of mifepristone (i.e. administered as several doses or one dose; Liao 2004; WHO 1989; WHO 1991), and five different doses of mifepristone (150 mg, 125 mg, 100 mg, 75 mg, 50 mg; Li 2015) in Table 1.

Mifepristone 600 mg versus 200 mg

Six studies reported this comparison (McKinley 1993; WHO 1989; WHO 1991; WHO 1993; WHO 2000; WHO 2001a), of which we included data from four studies with 3494 participants in the meta-analysis (McKinley 1993; WHO 1993; WHO 2000; WHO 2001a). McKinley 1993 used misoprostol 600 µg orally, WHO 1993 and WHO 2001a used gemeprost 1 mg vaginally, WHO 2000 used misoprostol 400 µg orally. Compared to 200 mg of mifepristone, 600 mg of mifepristone probably has a similar effect on failure to achieve complete abortion (RR 1.07, 95% CI 0.87 to 1.33; I^2 = 0%; 4 RCTs, 3494 women; moderate-certainty evidence; Analysis 1.1; Summary of findings 1). The pooled analysis of the two studies that used the same dose and type of prostaglandin (gemeprost 1 mg) after 600 mg or 200 mg of mifepristone, also showed that 600 mg of mifepristone might have a similar effect on complete abortion as 200 mg (RR 1.02, 95% CI 0.72 to 1.45; 2 RCTs; 1685 women; Analysis 1.1). Nausea was similar between the two groups (RR 1.02, 95% CI 0.95 to 1.09; I^2 = 83%; 2 RCTs, 2432 women; moderate-certainty evidence; Analysis 1.2; Summary of findings 1). Time until passing of conceptus longer than three to six hours was similar for the two groups in the two studies that reported it (1116 women; Analysis 1.3). The five studies that reported ongoing pregnancy at follow-up (McKinley 1993; von Hertzen 2009; WHO 1993; WHO 2000; WHO 2001b), showed similar efficacy between the two groups (Analysis 1.4). The dosage or route of misoprostol were different among these four included RCTs, so pooled analysis was not performed.

Mifepristone 200 mg versus 100 mg

One study was included in this comparison (von Hertzen 2009). This was a four-arm study, comparing 200 mg versus 100 mg of mifepristone followed by 800 µg misoprostol vaginally after 24 or 48 hours. Compared to 100 mg of mifepristone, 200 mg of mifepristone probably made little difference in failure to achieve complete abortion (RR 0.89 95%; CI 0.61 to 1.29; 1182 women; Analysis 1.1).

Mifepristone 200 mg versus 50 mg

WHO 2001b used 200 mg or 50 mg followed by 0.5 mg or 1 mg of gemeprost vaginally. The group receiving mifepristone 50 mg and gemeprost 0.5 mg was discontinued after 249 women were enrolled because the complete abortion rate was below the pre-determined cut-off.

Mifepristone 150 mg, 125 mg, 100 mg, 75 mg versus 50 mg

Li 2015 was a five-arm study that compared five different dosage of mifepristone followed by 200 µg misoprostol orally after 24 hours. Failure to achieve abortion was similar between the groups (one study, 2500 women; Table 1). However, the side effects of nausea, vomiting, diarrhoea and abdominal pain might be higher when the dosage of mifepristone was 100 mg or higher (Table 1).

Comparison 2: different doses of prostaglandin

We included eight studies in the review for this comparison; we were able to include data from seven of them in the meta-analysis of different comparison groups as below (Arvidsson 2005; Chong 2012; Coyaji 2007; Rodger 1989; Shannon 2006; von Hertzen 2010; WHO 2001b).

Combined dosage comparison: gemeprost 1 mg versus 0.5 mg with different dosage of mifepristone

Two studies compared gemeprost 1 mg versus gemeprost 0.5 mg in 1034 women (Rodger 1989; WHO 2001b). The largest study in this comparison used a factorial design (mifepristone 50/200 mg and gemeprost 1/0.5 mg; WHO 2001b). Looking at the group with mifepristone 200 mg only, the failure to achieve complete abortion between the two doses of gemeprost was similar. The arm with the smallest dose (mifepristone 50 mg and gemeprost 0.5 mg) was stopped prematurely after 249 women were enrolled, as the effectiveness was below the predetermined cut-off point. Rodger 1989 included 120 women in the study using mifepristone of 600 mg, the first 60 women were not randomised; therefore, only data for the second 60 women were reported in this review (Rodger 1989).

Misoprostol 800 µg versus 400 µg

Five studies compared different doses of misoprostol after 200 mg of mifepristone. von Hertzen 2010 used four groups, comparing misoprostol 800 µg administered sublingually, 400 µg administered sublingually, 800 µg administered vaginally, and 400 µg administered vaginally. We pooled the outcome data of 800 µg sublingually and 800 µg vaginally; 400 µg sublingually and 400 µg vaginally, respectively, for exploring the efficacy of different doses with similar routes. The route of misoprostol between groups was similar in three (Chong 2012; Coyaji 2007; von Hertzen 2010), but different in two studies (Arvidsson 2005; Shannon 2006). Coyaji 2007 compared misoprostol 400 µg to 800 µg (given orally; 800 µg was administered as a repeat dose of 400 µg after three hours). Chong 2012 compared misoprostol 400 µg to 800 µg (given buccally; 400 µg was administered using two pills of misoprostol and two pills of placebo). Shannon 2006 used three groups, comparing misoprostol 400 µg, 600 µg and 800 µg. We included data from the 400 µg and 800 µg groups in the subgroup of misoprostol 400 µg orally versus 800 µg vaginally with Arvidsson 2005.

In total, 800 µg misoprostol probably showed a lower failure to achieve complete abortion than 400 µg (RR 0.63, 95% CI 0.51 to 0.78; I^2= 0%; 3 RCTs, 4424 women; moderate-certainty evidence; Analysis 2.1; Summary of findings 2) when the misoprostol was prescribed using a similar route between groups. Results from the largest study (von Hertzen 2010), using misoprostol sublingually or vaginally, carried most weight (RR 0.61, 95% CI 0.48 to 0.77; 3005 women; Analysis 2.1). However, the failure to achieve complete abortion was probably similar between groups when using misoprostol orally or buccally, no matter whether gestation was less than 49 days or not. The difference between 800 µg vaginally and 400 µg orally may also be similar.

Misoprostol 800 µg probably resulted in fewer ongoing pregnancies (RR 0.41, 95% CI 0.26 to 0.65; I^2 = 5%; 4 RCTs, 5060 women; Analysis 2.3), lower dissatisfaction with the procedure (RR 0.75, 95% CI 0.60 to 0.93; I^2= 0; 3 RCTs, 4420 women; moderate-certainty evidence; Analysis 2.8; Summary of findings 2), but similar nausea (RR 0.99, 95% CI 0.94 to 1.05; I^2 = 0%; 3 RCTs, 4424 women; moderate-certainty evidence; Analysis 2.4; Summary of findings 2) and more severe diarrhoea (RR 1.70, 95% CI 1.43 to 2.03; I^2 = 0%; 2 RCTs, 3302 women; Analysis 2.6) when compared to the 400 µg group. Surgical evacuation (Analysis 2.2), vomiting (Analysis 2.5), and abdominal pain (Analysis 2.7), were similar between the groups.

Comparison 3: type of prostaglandin

Gemeprost versus misoprostol

We included two studies that used different doses of misoprostol and different routes of administration (Baird 1995; Bartley 2001). Therefore, we did not combine the results in a meta-analysis. However, according to data from a single study (Bartley 2001), misoprostol used at a higher dose (800 µg) and administered vaginally, might be more effective than gemeprost 0.5 mg (failure to achieve complete abortion: RR 2.86, 95% CI 1.14 to 7.18; 910 women; Analysis 3.1). Misoprostol might result more vomiting (RR 1.49, 95% CI 1.06 to 2.10; 910 women; Analysis 3.2) and diarrhoea (RR 2.66, 95% CI 1.35 to 5.26; 910 women; Analysis 3.2) compared to gemeprost, but might have a similar effect on ongoing pregnancy (Analysis 3.3) and time until passing of conceptus longer than three to six hours (Analysis 3.4).

PGF2α versus misoprostol

Misoprostol (600 µg, orally) probably leads to a similar effect on complete abortion compared to PGF2α (failure to achieve complete abortion: RR 0.84, 95% CI 0.44 to 1.58; I^2 = 47%; 2 RCTs, 17,974 women; Analysis 3.1 (Sang 1994; Sang 1999)).

Comparison 4: timing of prostaglandin

We included nine studies for this comparison. Three studies used different dose regimens as well as time intervals (Creinin 2001b; Guest 2007; Schaff 2000); therefore, we have presented the results for each study separately.

Day 3 versus day 1

Misoprostol administered on day one was probably more effective than on day three following mifepristone (failure to achieve complete abortion: RR 1.94, 95% CI 1.05 to 3.58; 1489 women; Analysis 4.1) in the one study that reported this comparison (Schaff 2001). The follow-up for women was on day eight after mifepristone. Misoprostol administered on day three might slightly increase need for surgical evacuation (RR 1.49, 95% CI 0.78 to 2.83; 1489 women; Analysis 4.3), increase ongoing pregnancy (RR 1.56, 95% CI 0.51 to 4.73; 1489 women; Analysis 4.4) or increase women' dissatisfaction with the method (RR 0.99, 95% CI 0.80 to 1.23; 2 RCTs, 1429 women; Analysis 4.5).

Day 3 versus day 2

Misoprostol administration on day three might lead to slightly higher failure to achieve complete abortion compared with day two (RR 1.69, 95% CI 0.95 to 3.01; 1521 women; Analysis 4.1) based on evidence from one study (Schaff 2000).

Day 2 versus day 1

Misoprostol administration on day two might lead to similar efficacy on failure to achieve complete abortion compared with day one when combining results for gestational ages up to 63 days based on four studies (RR 0.99, 95% CI 0.61 to 1.61; I^2 = 48%; 4 RCTs, 3887 women; Analysis 4.1). However, in the subgroup analysis based on gestational weeks, failure might be higher on day two compared to day one in women at more than 49 days' gestation (RR 1.57, 95% CI 1.09 to 2.27; I^2 = 0%; 2 RCTs, 1207 women; Analysis 4.1), but lower in women at 49 days or fewer of gestation (RR 0.90, 95% CI 0.58 to 1.38; I^2 = 0%; 2 RCTs, 1116 women; Analysis 4.1).

Day 2 versus day 0

Three studies compared misoprostol on day two versus day zero for failure to achieve a complete abortion (Creinin 2001b; Guest 2007; Verma 2017). Creinin 2001b compared mifepristone 600 mg followed by misoprostol 400 µg; Guest 2007 compared mifepristone 200 mg followed by misoprostol 800 µg, and Verma 2017 compared mifepristone 200 mg followed by misoprostol 400 µg. Misoprostol administered at 36 to 48 hours might lead to lower failure to achieve complete abortion (RR 0.53, 95% CI 0.25 to 1.09; I^2 = 26%; 711 women; Analysis 4.1) and less need for surgical evacuation (RR 0.40, 95% CI 0.19 to 0.86; I^2 = 0%; 2 RCTs, 511 women; Analysis 4.3), when compared to six hours after mifepristone.

Women using misoprostol on day two might experience less nausea than those using mifepristone and misoprostol concurrently (RR 0.75, 95% CI 0.58 to 0.98; I^2 = 0%; 3 RCTs, 644 women; Analysis 4.2). There was no difference in the occurrence of side effects (vomiting, diarrhoea, abdominal pain, ongoing pregnancy, and women's dissatisfaction with the procedure) among the different time interval groups of misoprostol.

Day 1 versus day 0

Three studies used the same route, but different dosages of misoprostol (Creinin 2004; Creinin 2007; Goel 2011). Mifepristone 200 mg followed by misoprostol 800 µg vaginally or 400 µg vaginally (Goel 2011), administered on day one might be more effective than administration six hours or less after mifepristone (failure to achieve complete abortion: RR 0.65, 95% CI 0.46 to 0.91; I^2 = 0%; 3 RCTs, 2236 women; Analysis 4.1), no matter whether the gestational week was more than 49 days or not.

Passing time of conceptus, bleeding time and additional usage of uterotonics were reported by only one study (Goel 2011). There was no difference between day one versus day zero (Analysis 4.6; Analysis 4.7; Analysis 4.8).

Comparison 5: administration strategy of prostaglandin

We included seven studies with a total of 5319 women in the review and meta-analysis. Four studies compared home- (self-administered) and hospital-administered medical abortion (Iyengar 2015; Li 2017; Shrestha 2014; Song 2018). Three studies compared misoprostol provided by a nurse in the hospital versus a doctor in the hospital (Kopp Kallner 2015; Olavarrieta 2015; Warriner 2011).

The administration groups used different doses and routes of misoprostol. Iyengar 2015 used 800 µg misoprostol (oral, vaginal or sublingual) administered by women themselves two days after 200 mg oral mifepristone. The abortion outcomes were assessed by women at home (followed up by phone) or by nurses and or doctors in hospitals. Li 2017 used 75 mg oral mifepristone, followed by 400 µg oral misoprostol 24 hours later by home administration or hospital administration. Shrestha 2014 compared home administered versus hospital-administered vaginal misoprostol (800 µg) 24 hours after 200 mg oral mifepristone. Song 2018 applied 100 mg oral mifepristone in hospital, followed by 200 µg of sublingual misoprostol 24 hours later at home versus in hospital.

Home administration might result in a similar effect on complete abortion compared to hospital administration when combining results for gestational ages up to 63 days, based on four studies (RR 1.63, 95% CI 0.68 to 3.94; I^2 = 84%; 4 RCTs, 2263 women; low-certainty evidence; Analysis 5.1; Summary of findings 3). However, hospital administration showed lower failure in abortion for women at less than 49 days' gestation (Li 2017; Song 2018).

Oral administration of 200 mg mifepristone followed by 800 µg misoprostol (vaginally or buccally) one to two days later was provided by nurses or doctors (Kopp Kallner 2015; Olavarrieta 2015; Warriner 2011). Misoprostol provided by doctors might lead to lower failure in complete abortion than that provided by nurses (RR 2.69, 95% CI 1.39 to 5.22; I^2 = 66%; 3 RCTs, 3056 women; low-certainty evidence; Analysis 5.1; Summary of findings 4). However, among women at less than 10 weeks' gestational age, failure to achieve complete abortion might be similar when the medication was administered by nurses or doctors (RR 1.33, 95% CI 0.50 to 3.55, 884 women; Analysis 5.1; Olavarrieta 2015).

Mifepristone and misoprostol administered at home (self-administered) versus administered in the hospital were similar in terms of requiring subsequent surgical evacuation (RR 1.71, 95% CI 0.37 to 7.87; I^2 = 91%; 2 RCTs, 1331 women; Analysis 5.2). However, administration by doctors, compared with nurses showed lower incomplete abortion requiring surgical evacuation (RR 3.36, 95% CI 1.67 to 6.78; 2 RCTs, 2172 women; Analysis 5.2).

It is uncertain whether misoprostol administered by a nurse or at home has a similar effect on ongoing pregnancy or not, when compared to misoprostol administered by doctors or in hospital respectively. The clinical heterogeneity among populations is too significant to draw any conclusion. The side effects (nausea, vomiting, diarrhoea, abdominal pain, women's dissatisfaction with the procedure, bleeding days and blood transfusion) might be similar between the different administration groups of misoprostol.

Comparison 6: route of administration of misoprostol - oral versus vaginal

We included seven studies in the review for this comparison (Arvidsson 2005; Creinin 2001a; El-Refaey 1995; Qian 2015; Schaff 2001; Shannon 2006; Tang 2002), and we included three RCTs with a total of 1704 women in the meta-analysis (El-Refaey 1995; Qian 2015; Schaff 2001). El-Refaey 1995 used mifepristone 600 mg and Schaff 2001 used mifepristone 200 mg. Both used misoprostol 800 µg orally or vaginally after 48 hours (El-Refaey 1995), and at least 24 hours (Schaff 2001), after mifepristone. Qian 2015 used 200 mg oral mifepristone followed by oral (every three hours) or vaginal (every six hours) misoprostol 400 µg (no more than four doses).

Misoprostol administered orally probably led to more failure in complete abortion than the vaginal route (RR 2.38, 95% CI 1.46 to 3.87; I^2 = 39%; 3 RCTs, 1704 women; moderate-certainty evidence; Analysis 6.1; Summary of findings 5), and might lead to more nausea (RR 1.14, 95% CI 1.03 to 1.26; I^2 = 0%; 2 RCTs, 1380 women; low-certainty evidence; Analysis 6.2; Summary of findings 5) and more diarrhoea (RR 1.80 95% CI 1.49 to 2.17; I^2 = 0%, 2 RCTs, 1379 women; Analysis 6.2). Unexpectedly, vomiting occurred more often in the vaginal group in one study, and reporting error cannot be excluded (Schaff 2001). Only Qian 2015 reported bleeding volume and conceptus passing time but without differences between groups.

Three studies used different doses orally and vaginally and we could not combine them in meta-analysis (Arvidsson 2005; Creinin 2001a; Shannon 2006). We meta-analysed the comparison of 400 µg oral misoprostol versus 800 µg of vaginal misoprostol in comparison 2: dose of prostaglandin (Analysis 2.1; Arvidsson 2005; Shannon 2006), so we did not meta-analyse this route comparison again. One study (Tang 2002), used a combined regimen oral/vaginal in one group and repeated oral misoprostol doses in another group, and these data were therefore not included in the meta-analysis.

Comparison 7: route of administration of misoprostol - buccal versus vaginal

We included two studies with 479 women using 200 mg of oral mifepristone followed by 800 µg of misoprostol given buccally or vaginally for this comparison (Garg 2015; Middleton 2005). Misoprostol administered buccally might lead to lower failure in abortion than vaginally administered misoprostol (RR 0.71, 95% CI 0.34 to1.46; I^2 = 0%; 2 RCTs, 479 women; Analysis 7.1).

There were more women with diarrhoea in the buccal compared to the vaginal group (RR 1.51, 95% CI 1.12 to 2.03; 2 RCTs, 479 women; Analysis 7.2).

From evidence from one study (Garg 2015), it was uncertain whether misoprostol administered buccally was better than misoprostol administered vaginally on need for surgical evacuation (RR 0.33, 95% CI 0.01 to 7.81; 1 study, 50 women; Analysis 7.3) and women's dissatisfaction with the procedure (RR 0.20, 95% CI 0.01 to 3.97; 1 RCT, 50 women; Analysis 7.4).

Comparison 8: route of administration of misoprostol - buccal versus oral

We included one study with 847 women in this comparison (Winikoff 2008). Misoprostol administered buccally probably leads to lower failure in complete abortion than the oral route (RR 0.62, 95% CI 0.40 to 0.96; 847 women; Analysis 8.1) for all gestational ages and for women at more than 49 days' gestation (RR 0.37, 95% CI 0.18 to 0.73, 429 women; Analysis 8.1); and might reduce failure to achieve complete abortion slightly for women at 49 days' gestation or less (RR 0.72, 95% CI 0.25 to 2.04, 418 women; Analysis 8.1).

Misoprostol administered buccally might slightly reduce overall ongoing pregnancy (RR 0.31, 95% CI 0.09 to 1.08; 847 women; Analysis 8.2) and for women at 49 days' gestation or less (RR 0.64, 95% CI 0.11 to 3.80; 418 women; Analysis 8.2), while it might reduce ongoing pregnancy for women with more than 49 days' gestation (RR 0.18, 95% CI 0.04 to 0.78, 429 women; Analysis 8.2) compared to oral misoprostol.

Misoprostol administered buccally probably leads to more nausea (RR 1.10, 95% CI 1.01 to 1.19, 830 women; Analysis 8.3), while it might result in similar dissatisfaction with the procedure (RR 1.21, 95% CI 0.76 to 1.91, 835 women; Analysis 8.4) compared to oral misoprostol.

Comparison 9: route of administration of misoprostol - sublingual versus vaginal

We included three studies with 3543 women in this comparison (Hamoda 2005; Tang 2003; von Hertzen 2010). Misoprostol administered sublingually probably slightly reduced failure to achieve complete abortion (RR 0.68, 95% CI 0.22 to 2.11; I^2 = 59%; 2 RCTs, 3229 women; moderate-certainty evidence; Analysis 9.1; Summary of findings 6), number of needed surgical evacuations (RR 0.80, 95% CI 0.18 to 3.53; 1 study, 327 women; Analysis 9.2), ongoing pregnancy (RR 0.67, 95% CI 0.13 to 3.47; I^2 = 43%; 2 RCTs, 3229 women; Analysis 9.3), but might slightly increase dissatisfaction with the procedure (RR 1.67, 95% CI 0.80 to 3.50; I^2 = 64%; 2 RCTs, 3303 women; moderate-certainty evidence; Analysis 9.4; Summary of findings 6) compared to misoprostol administered vaginally. In one study women received additional doses of misoprostol if abortion was incomplete at follow-up and the results were not presented for the intended method used and were therefore not totaled (Hamoda 2005).

Misoprostol administered sublingually probably slightly increased nausea (RR 1.11, 95% CI 0.93 to 1.33; I^2 = 78%; 3 RCTs, 3543 women; moderate-certainty evidence; Analysis 9.5; Summary of findings 6), vomiting (RR 1.21, 95% CI 0.88 to 1.66; I^2 = 87%; 3 RCTs, 3543 women; Analysis 9.5), and diarrhoea (RR 1.83, 95% CI 1.33 to 2.50; I^2 = 79%; 3 RCTs, 3543 women; Analysis 9.5) than the vaginal route. Furthermore, Tang 2003 reported that more women in the sublingual group experienced side effects: nausea (RR 1.67, 95% CI 1.21 to 2.29) and vomiting (RR 2.93, 95% CI 1.69 to 5.06). Hamoda 2005 did not use an ITT analysis; loss to follow-up was identical in both groups (n = 13).

Comparison 10: route of administration of misoprostol - sublingual versus oral

We included two studies with 564 women for this comparison, that used 200 mg of oral mifepristone followed by 400 µg of misoprostol (Dahiya 2011; Raghavan 2009).

Misoprostol administered sublingually probably reduced failure to achieve complete abortion compared to the oral group (RR 0.26, 95% CI 0.10 to 0.68; I^2 = 0%; 2 RCTs, 564 women; Analysis 10.1), especially for women at 49 days' gestation or less (RR 0.28, 95% CI 0.08 to 0.99; 1 study, 422 women; Analysis 10.1).

It was uncertain whether misoprostol administered sublingually could reduce ongoing pregnancy (RR 0.36, 95% CI 0.01 to 8.50, 1 study, 93 women; Analysis 10.2) or need for surgical evacuation (RR 0.36, 95% CI 0.08 to 1.67, 1 study, 93 women; Analysis 10.3) compared to the oral group.

Misoprostol administered sublingually might slightly increase women's dissatisfaction with the procedure compared to misoprostol administered orally (RR 1.96, 95% CI 0.94 to 4.09, 1 study, 471 women; Analysis 10.4).

It was uncertain whether misoprostol administered sublingually was different from oral misoprostol on side effects including nausea (RR 0.62, 95% CI 0.27 to 1.41; I^2 = 78%; 2 RCTs, 564 women; Analysis 10.5), vomiting, and diarrhoea.

Comparison 11: route of administration of misoprostol - sublingual versus buccal

We included two studies with 640 women in this comparison (Chai 2013; Raghavan 2010). Both studies compared oral mifepristone 200 mg, Chai 2013 compared misoprostol 800 µg sublingually or buccally 48 hours later and Raghavan 2010 used misoprostol 400 µg sublingually or buccally 24 hours later.

Misoprostol administered sublingually may slightly increase failure to achieve complete abortion (RR 1.43, 95% CI 0.64 to 3.23; I^2 = 0; 2 studies, 640 women; Analysis 11.1), need for surgical evacuation (RR 1.72, 95% CI 0.75 to 3.96; I^2 = 0%; 2 RCTs, 640 women; Analysis 11.2), and ongoing pregnancy (RR 2.59, 95% CI 0.88 to 7.61; I^2 = 0%; 2 RCTs, 640 women; Analysis 11.3), when compared with buccal administration.

It was uncertain whether misoprostol administered sublingually reduced vomiting (RR 1.33, 95% CI 1.01 to 1.77; I^2 = 0%; 2 RCTs, 640 women; Analysis 11.4), nausea (RR 1.06, 95% CI 0.88 to 1.27; I^2 = 19%; 2 RCTs, 640 women; Analysis 11.4), diarrhoea (RR 2.19, 95% CI 0.56 to 8.51; I^2 = 80%; 2 RCTs, 640 women; Analysis 11.4), and abdominal pain (RR 1.10, 95% CI 0.79 to 1.52; I^2 = 92%; 2 RCTs, 640 women; Analysis 11.4) compared to buccal administration.

Misoprostol administered buccally might reduce women's dissatisfaction with the procedure when compared with sublingual administration as reported by Raghavan 2010 (RR 2.54, 95% CI 1.14 to 5.66; 550 women; Analysis 11.5).

The haemoglobin level (MD 0.00, 95% CI −0.40 to 0.40; 1 study, 90 women; Analysis 11.6) were similar between groups, but passing time of conceptus might be less in the sublingual group (MD −0.88, 95% CI −2.44 to 0.68; 1 study, 90 women; Analysis 11.7), as reported by Chai 2013.

Comparison 12: single versus split dose of prostaglandin

We included one study with 154 women in this comparison (El-Refaey 1994). Administration of 800 µg of misoprostol as a single dose might slightly reduce failure in complete abortion compared to two doses of 400 µg, two hours apart (RR 0.70, 95% CI 0.21 to 2.39; 1 study, 154 women; Analysis 12.1). It was uncertain whether side effects following misoprostol administered as single dose were different from the split-dose group.

Comparison 13: single versus continuous misoprostol

We included two studies for this comparison (Tang 2002; von Hertzen 2003). Honkanen reported on the same study as von Hertzen 2003, but on different outcomes. Tang 2002 and von Hertzen 2003 compared oral misoprostol 400 µg twice daily continued for seven days after either an initial oral (group A) or vaginal 800 µg (group B) and single vaginal dose (group C) among 150 women. All women had received mifepristone 200 mg 48 hours prior to misoprostol.

Misoprostol all administered orally (group A) probably increased failure to achieve complete abortion compared to the vaginal and continuous oral misoprostol group (RR 1.48; 95% CI 1.01 to 2.16; I^2 = 0%; 2 RCTs, 1581 women; Analysis 13.1) and might increase ongoing pregnancy (Analysis 13.5); probably induced more diarrhoea compared to the vaginal and continuous oral group (RR 1.83, 95% CI 1.11 to 3.01, 1 study, 1481 women; group B; Analysis 13.2) and single vaginal group (RR 2.09, 95% CI 1.24 to 3.53, group C; Analysis 13.2). The all-oral group (A) probably experienced less nausea (Analysis 13.3), and less vomiting (Analysis 13.4).

Comparison 14: mifepristone alone versus combined regimen mifepristone/prostaglandin

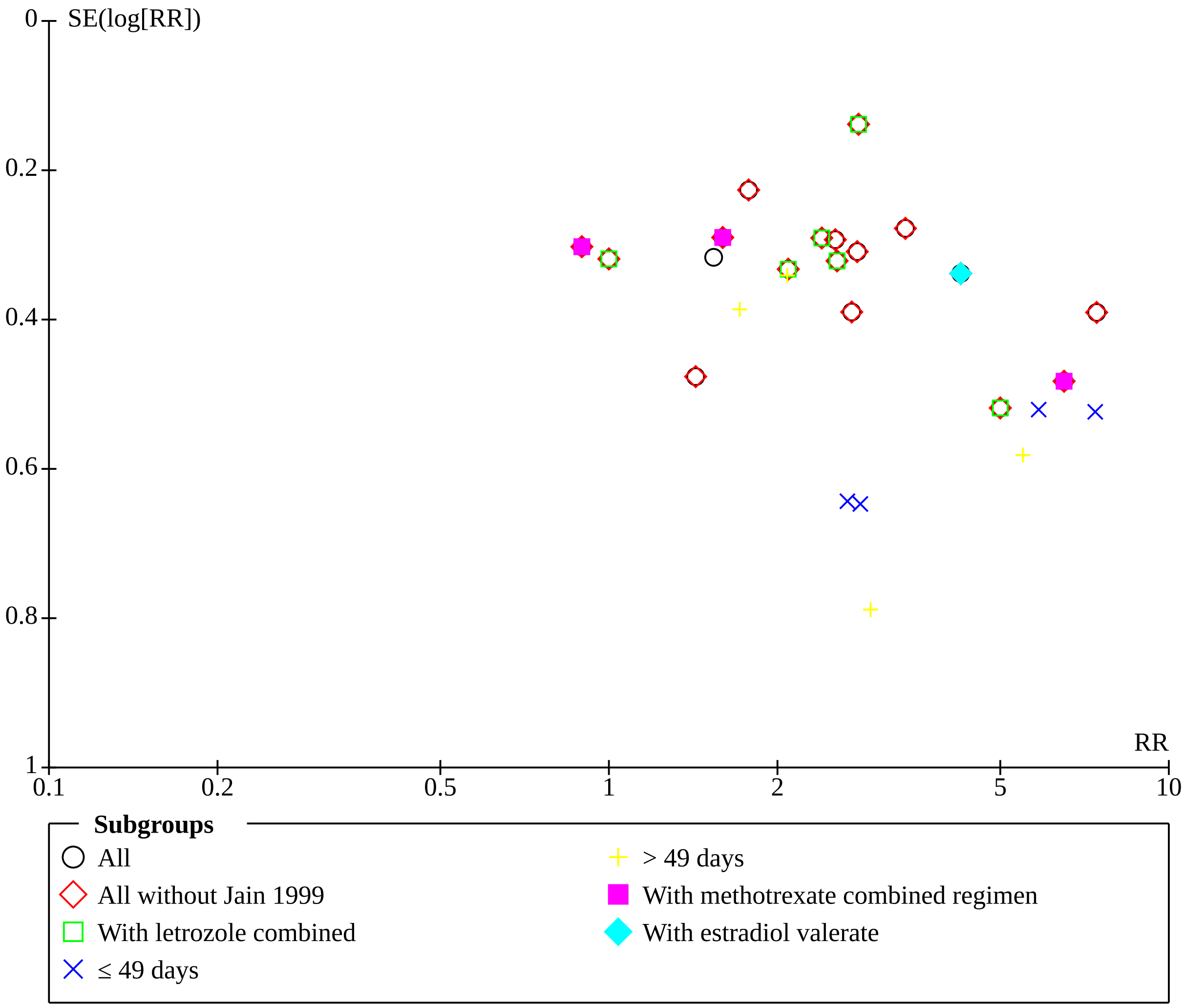
We included three studies for this comparison (Cameron 1986; Swahn 1989; Zheng 1989).

It was uncertain whether the combination regimen could reduce the failure to achieve complete abortion compared to mifepristone alone (RR of failure 3.25, 95% CI 0.81 to 13.09; I^2 = 83%; 3 RCTs, 273 women; Analysis 14.1; very low-certainty evidence; Summary of findings 7).

Comparison 15: prostaglandin alone versus a combined regimen (all)

We included 19 studies in this comparison and 18 (3471 women) of these in meta-analyses of different comparison subgroups. We evaluated publication bias as low by examining funnel plots Figure 4. Misoprostol combined with letrozole (Abbasalizadeh 2018; Allameh 2020; Behroozi Lak 2018; Javadi 2015; Lee 2011; Torky 2018), or estradiol valerate (Teimoori 2019), were added in the current review. Wiebe 2006, compared methotrexate combined with 400 µg misoprostol vaginal with misoprostol 400 µg sublingual; we did not include 400 µg vaginal in the meta-analysis, but presented data in Additional Table 1. One study used additional doses of prostaglandin and did not specify which women received them (Jain 1999).

Figure 4. Funnel plot of comparison 15 prostaglandin alone vs combined regimen (all), outcome 15.1 failure to achieve complete abortion



The studies consistently demonstrated that compared to a combination regimen, misoprostol alone might be less effective in achieving complete abortion (RR 2.39, 95% CI 1.89 to 3.02; I^2 = 64%, 18 RCTs, 3471 women; low-certainty evidence; Analysis 15.1; Summary of findings 8). Sensitivity analysis, excluding Jain 1999, also showed similar results (RR 2.45, 95% CI 1.92 to 3.14; I^2 = 64%; 17 RCTs, 3321 women; Analysis 15.1). Subgroup analysis based on gestational week also showed higher failure of complete abortion in the misoprostol-alone group. Misoprostol combined with letrozole might reduce failure to achieve complete abortion (RR 2.29, 95% CI 1.62 to 3.24; I^2 = 54%; 6 RCTs, 1008 women; Analysis 15.1).

Time to pass conceptus was shorter in the combination group (SMD 0.43, 95% CI 0.05 to 0.80; I^2 = 86%; 6 RCTs, 988 women; Analysis 15.2), fewer women needing surgical evacuation (RR 2.90, 95% CI 2.26 to 3.73; I^2 = 0%; 7 RCTs, 1307 women; Analysis 15.3), less use of additional uterotonics (RR 2.05, 95% CI 1.26 to 3.34; I^2 = 26%; 3 RCTs, 592 women; Analysis 15.4), and fewer women with ongoing pregnancy (RR 3.63, 95% CI 1.18 to 11.16; I^2 = 75%; 5 RCTs, 1353 women; Analysis 15.5). However, women receiving misoprostol alone might experience more diarrhoea compared to the combined regimen (RR 1.26, 95% CI 1.11 to 1.43; I^2 = 26%; 11 RCTs, 2342 women; Analysis 15.6), which might be caused by the higher dosage used in the misoprostol-alone group; but might experience slightly less nausea (RR 0.90, 95% CI 0.74 to 1.10; I^2 = 77%; 12 RCTs, 2722 women; low-certainty evidence; Analysis 15.6; Summary of findings 8). No difference was found for vomiting, pelvic infection and abdominal pain.

Prostaglandin administered alone might slightly increase women's dissatisfaction with the procedure (RR 1.65, 95% CI 0.75 to 3.64; I^2 = 90%; 5 RCTs, 1485 women; low-certainty evidence; Analysis 15.7; Summary of findings 8), but might slightly reduce the need for blood transfusion (RR 0.33, 95% CI 0.03 to 3.13; 1 study, 300 women; low-certainty evidence; Analysis 15.8; Summary of findings 8). Prostaglandin alone might shorten the number of bleeding days (Analysis 15.9) and improve haemoglobin level (Analysis 15.10) compared to the combination regimens.

Single regimen

Comparison 16: prostaglandin alone: route of administration

We included 11 studies in this comparison group, with 4667 women (Abdelshafy 2019; Klingberg Allvin 2015; Marwah 2016; Mizrachi 2017; Ng 2015; Paritakul 2010; Sheldon 2019; Sonsanoh 2014; Souizi 2020; Tanha 2010; von Hertzen 2007), five studies compared sublingual versus vaginal administration, one sublingual versus buccal, two sublingual versus oral, two oral versus vaginal, one repeat versus single dose, and two studies had different administration models (home versus hospital; nurse versus doctor). One study had three arms that compared misoprostol sublingually, orally or vaginally (Souizi 2020). The dosage of misoprostol among the studies ranged from 400 µg to 2400 µg. Except for two studies (Sheldon 2019; von Hertzen 2007), the others included women with missed or incomplete abortion. The follow-up duration ranged from two days to 30 days.

Misoprostol alone provided vaginally may reduce the failure to achieve complete abortion compared to misoprostol provided orally (RR 3.68, 95% CI 1.56 to 8.71; I^2 =0%; 2 RCTs, 216 women; low-certainty evidence; Analysis 16.1; Summary of findings 9). Misoprostol alone provided sublingually may reduce failure to achieve complete abortion compared with the vaginal route (RR 0.69, 95% CI 0.37 to 1.28; I^2 =87%; 5 RCTs, 2705 women; low-certainty evidence; Analysis 16.1; Summary of findings 10), may also reduce failure compared to the oral route (RR 0.58, 95% CI 0.11 to 2.99; I^2 =66%; 2 RCTs, 173 women; Analysis 16.1); but may slightly increase failure compared to the buccal route (RR 1.11, 95% CI 0.71 to 1.74; 1 study, 401 women; Analysis 16.1). A repeat dose of misoprostol may result in more failure to achieve complete abortion compared to a single dose (RR 1.30, 95% CI 0.78 to 2.15; 1 study, 180 women; Analysis 16.1; Mizrachi 2017).

We included two studies with different administration models of misoprostol: nurse-administered versus doctor-administered (Klingberg Allvin 2015), and at home versus in hospital (Ng 2015). Misoprostol alone administered by a doctor might reduce failure to achieve complete abortion compared to nurse administration (RR 3.84, 95% CI 2.16 to 6.83; 1 study, 1010 women; Analysis 16.1). However, home versus hospital administration might have similar effectiveness in achieving complete abortion (RR 1.00, 95% CI 0.37 to 2.72; 1 study, 154 women; Analysis 16.1).

It was uncertain whether misoprostol alone administered sublingually was different from misoprostol alone administered vaginally on blood transfusion (RR 5.00, 95% CI 0.24 to 102.96; 1 RCT, 220 women; very low-certainty evidence; Analysis 16.2; Summary of findings 10), ongoing pregnancy (RR 0.57, 95% CI 0.21 to 1.51; 2 RCTs, 340 women; Analysis 16.3; Sonsanoh 2014; Tanha 2010). It was also uncertain whether misoprostol administered sublingually was different from misoprostol administered buccally on ongoing pregnancy (RR 1.11, 95% CI 0.50 to 2.45, 1 RCT, 401 women; Analysis 16.3) and additional uterotonics (RR 1.89, 95% CI 0.77 to 4.63; 1 RCT, 401 women; Analysis 16.4; Sheldon 2019).

Surgical evacuation was probably lower (sublingual versus vaginal: RR 0.47, 95% CI 0.26 to 0.85; I^2 = 68%; 3 RCTs, 540 women; nurse versus doctor: RR 3.84, 95% CI 2.16 to 6.83; 1 study, 1010 women; Analysis 16.5), and women's dissatisfaction with the procedure was probably lower (sublingual versus vaginal: RR 0.14, 95% CI 0.07 to 0.29; 1 study, 220 women; low-certainty evidence; Summary of findings 10; nurse versus doctor: RR 2.18, 95% CI 1.40 to 3.41; 1 study, 1010 women; Analysis 16.10) in the sublingual- or doctor-administered groups, compared with the vaginal- or nurse-administered groups, respectively.

Nausea (RR 1.16, 95% CI 1.02 to 1.32; Analysis 16.6) and vomiting (RR 1.35, 95% CI 1.08 to 1.68; 1 study, 1010 women; Analysis 16.7) might occur much more frequently in the nurse-administered group than when the doctor administered the therapy (Klingberg Allvin 2015).

More women experienced diarrhoea (RR 1.73, 95% CI 1.41 to 2.12; I^2 = 25%; 5 RCTs, 2725 women; Analysis 16.8) and abdominal pain (RR 1.38, 95% CI 0.49 to 3.86; I^2 = 98%; 3 RCTs, 459 women; Analysis 16.9) in the sublingual group, compared with the vaginal route. However, the heterogeneity was high for the abdominal pain among studied, which might be induced by Souizi 2020 reporting higher ratio of abdominal pain, and the other two studies just counted severe pain (Sonsanoh 2014; Tanha 2010), When misoprostol was administered vaginally, compared to oral administration, less time was needed to expel the products of conception (SMD 0.71, 95% CI 0.31 to 1.12; 1 study, 100 women; Analysis 16.12; Marwah 2016). However, sublingual administration was faster to achieve complete abortion than the vaginal route (SMD −1.10, 95% CI −1.34 to −0.87; I^2 = 0%; 2 RCTs, 320 women; Analysis 16.12; Abdelshafy 2019; Sonsanoh 2014). Bleeding time was similar between sublingual and vaginal groups (SMD −0.22, 95% CI −0.50 to 0.06; 1 study, 200 women; Analysis 16.11), and between different misoprostol administration models (Analysis 16.11). The reduction of haemoglobin (Ng 2015), and the actual level of haemoglobin (Abdelshafy 2019), were each reported by one study (Analysis 16.13). Less reduction was observed in the sublingual or home group, compared with vaginal or hospital group.

Comparison 17: mifepristone single dose, high dose versus low dose

We included one study with 101 women in this comparison (Birgerson 1988). Low-dose (140 mg) mifepristone may have similar effectiveness to high-dose (700 mg) mifepristone for failure to achieve complete abortion (RR 1.32, 95% CI 0.74 to 2.38; Analysis 17.1). However, the low-dose group might experience much more nausea, but similar vomiting, compared to the high-dose group.

Combined regimen: methotrexate/prostaglandin

Comparison 18: timing of prostaglandin

We included three studies in the review for this comparison (Carbonell 1997b; Carbonell 1998; Creinin 1995), and data from two studies were included in the meta-analysis (Carbonell 1997b; Carbonell 1998). It was uncertain whether misoprostol given on day seven after methotrexate might have a similar effect to misoprostol given on day three in failure to achieve complete abortion (RR 0.14, 95% CI 0.02 to 1.10; 1 study, 86 women; Analysis 18.1), but misoprostol might slightly reduce failure to achieve complete abortion on day five compared to day three (RR 0.72, 95% CI 0.36 to 1.45; I^2 = 0%; 2 RCTs, 387 women; Analysis 18.1), on day five compare to day four (RR 0.75, 95% CI 0.38 to 1.49; I^2 = 0%; 2 RCTs, 394 women; Analysis 18.1), and on day four compared to day three (RR 0.97, 95% CI 0.52 to 1.80; I^2 = 0%; 2 RCTs, 393 women).

Comparison 19: route of methotrexate - intramuscular versus oral

One study (100 women) compared intramuscular versus oral administration of methotrexate (Wiebe 1999b). It was uncertain whether methotrexate given intramuscularly was different from methotrexate given orally on failure to achieve complete abortion (RR 2.04, 95% CI 0.51 to 8.07; Analysis 19.1) or on side effects (nausea: RR 0.52, 95% CI 0.22 to 1.25; vomiting: RR 4.89, 95% CI 0.57 to 42.21; diarrhoea: RR 1.22, 95% CI 0.18 to 8.34; Analysis 19.2).

Comparison 20: dose of methotrexate

Two studies (40 women) were eligible to be included in the review (Creinin 1996a; Creinin 1997). Both studies had a very small sample size (10 women in each group); they used differently dosed regimens and we therefore presented them separately. It was uncertain whether the different dosage of methotrexate were similarly effective in failure to achieve complete abortion (Analysis 20.1).

Comparison 21: route of prostaglandin (misoprostol)

One study with 309 women (Wiebe 2004), compared buccal versus vaginal administration of misoprostol three to six days after methotrexate. Women received additional misoprostol but it was unclear how many or in which treatment group. The vaginal route might be more effective in achieving complete abortion than the buccal route (RR of failure 1.43, 95% CI 1.08 to 1.90; Analysis 21.1). Women administered misoprostol vaginally after methotrexate might experience similar nausea, vomiting and diarrhoea as via the buccal route (Analysis 21.2).

Tamoxifen versus methotrexate (combined with prostaglandin)

One study compared methotrexate to tamoxifen, both followed by misoprostol. The study was conducted in two phases: phase one used low-dose tamoxifen (40 mg) and phase two used high-dose tamoxifen (160 mg). We have therefore referred to this study as Wiebe 1999a (low dose) and Wiebe 1999b (high dose).

Comparison 22: low-dose tamoxifen (40 mg)

Women administered low-dose tamoxifen (40 mg) might experience slightly increased failure to achieve complete abortion (RR 2.04, 95% CI 0.86 to 4.84; 198 women; Analysis 22.1), reduced nausea (RR 0.56, 95% CI 0.33 to 0.97; 198 women; Analysis 22.2); similar vomiting (RR 1.70, 95% CI 0.42 to 6.92; 198 women; Analysis 22.2) and diarrhoea (RR 1.53, 95% CI 0.26 to 8.96; 198 women; Analysis 22.2) compared to methotrexate (Wiebe 1999a).

Comparison 23: high-dose tamoxifen (160 mg)

Women administered high-dose tamoxifen (160 mg) might also experience slightly increased failure to achieve complete abortion (RR 1.96, 95% CI 0.93 to 4.15; 200 women; Analysis 23.1), reduced nausea (RR 0.78, 95% CI 0.54 to 1.10; 200 women; Analysis 23.2); similar vomiting (RR 0.65, 95% CI 0.28 to 1.53; 200 women; Analysis 23.2) and diarrhoea (RR 1.23, 95% CI 0.34 to 4.43; 200 women; Analysis 23.2) compared to methotrexate (Wiebe 1999b).

Combined regimen mifepristone/prostaglandin versus mifepristone/prostaglandin plus tamoxifen

Comparison 24

We included one study with 932 women (Wu 1993); the combined regimen without tamoxifen might lead to higher failure to achieve complete abortion (RR 1.29, 95% CI 0.82 to 2.02; Analysis 24.1).

Other comparisons

Wang 2000 compared mifepristone 25 mg a day over seven days (total dose of 250 mg) followed by oral misoprostol 200 mg a day over three days (total dose of 1200 µg) to mifepristone 150 mg on day one followed by oral misoprostol 600 µg on day three. The doses and regimens in the two groups make it difficult to draw any meaningful conclusion from this comparison. Koopersmith 1996 compared misoprostol alone to misoprostol/tamoxifen and misoprostol/laminaria. The sample size was very small, which precludes drawing any meaningful conclusions from this study. We included these two studies in the additional tables. Additionally, Blanchard 2005 used various doses, routes and timings of misoprostol administered alone in a very small sample of women.

DISCUSSION

The literature on different medical abortion methods is vast, but contains relatively few RCTs that compare the different regimens. The studies included in this review were all conducted after the mifepristone/misoprostol regimen was licensed for sale in the UK and France and sought to determine if a lower dose and less costly regimen could be as effective as the licensed one. Grimes 1997 and Bygdeman 2002 in their reviews mentioned the different aspects to be considered when using medical abortion methods. Medical methods used are mostly combined regimens and many different types of combinations are described. To facilitate synthesising the data, studies were grouped into comparisons, as listed above (see Data synthesis). The focus was mainly on primary outcomes, such as effectiveness, complications, side effects and acceptability.

Meta-analysis was complicated by the use of different pharmaceutical agents, different doses and different routes of application; therefore, most meta-analyses contain only a small number of reasonably comparable studies. The review also focused on unwanted effects.

Summary of main results

These data support that the most common combined regimen (mifepristone/misoprostol) is an effective and safe method for pregnancy termination in the first trimester. The effect of mifepristone is not decreased by lowering the dose from previously recommended 600 mg to 200 mg when combined with at least 400 µg of misoprostol. In earlier studies, it was demonstrated that the linear dose-response effect of mifepristone does not occur in doses above 100 mg (Beaulieu 1997). A combination regimen with a prostaglandin is more effective than use of prostaglandin alone. Similarly, mifepristone alone is less effective than when combined with prostaglandin.

Different prostaglandins have been used for medical abortion, but misoprostol has superior attributes; misoprostol is at least as effective as gemeprost and is less costly, does not require refrigeration and offers different routes of administration. Of the different routes of misoprostol administration, vaginal appears to be superior to oral administration in terms of efficacy in the meta-analysis and majority of studies, and has fewer side effects when compared to oral, buccal or sublingual routes.

Medical abortion administered by doctors in hospital seems as effective as that managed by women themselves at home, however, might be better administered by a doctor than a nurse within the healthcare system. However, the certainty of the evidence is low, and more data are needed to confirm this finding.

In regards to the role of gestational age, when comparing abortions at seven weeks' gestational age or less to those at seven weeks or more, there was insufficient data to evaluate whether effects differed according to gestational age in any of the included analyses.

Methotrexate, combined with a prostaglandin, has been used in some studies with an effectiveness of mostly greater than 90%. We did not identify any new studies for the current, updated review that compared mifepristone/prostaglandin with methotrexate/prostaglandin. Letrozole used prior to misoprostol might increase the efficacy of misoprostol in achieving complete abortion for women with missed abortion.

An important aspect of this review is the overall very low rate of major complications reported among the various medical abortion regimens. The most severe complication is the need for blood transfusion (see table Characteristics of included studies). The reported self-limiting side effects of medical abortion regimens are mainly due to the prostaglandins (nausea, vomiting, diarrhoea, abdominal pain). The dose, route and type of prostaglandin used may influence the occurrence of side effects, as higher doses and oral administration are associated with an increase in nausea and vomiting.

Overall completeness and applicability of evidence

The included studies have addressed the objectives of the review. The generalisability of these results to some settings may be limited, as most studies considered in the review had strict inclusion criteria: intrauterine pregnancy was confirmed by ultrasound, emergency back-up facilities were available and follow-up was high. Fortunately, an increasing number of studies are focusing on the provision of medical abortion outside these particular constructs. We have included studies relating to medical abortion administered by women themselves at home or by nurses in hospital in the current update of the review. Additional barriers to the introduction of medical abortion may include the relatively high cost and need for registration of mifepristone.

Acceptability of medical abortion methods is often associated with the success of the abortion, and may decrease with higher gestational ages (Honkanen 2002; von Hertzen 2003; Winikoff 1997). Whether acceptability of different application routes is linked to age, parity or cultural differences is not well established. The difference in time intervals between mifepristone and methotrexate and the administration of prostaglandin, or their use outside the healthcare setting may also play a role in the acceptability of one method over the other.

Other comparisons, such as the combination of tamoxifen/prostaglandin have not been evaluated extensively enough to draw firm conclusions. Some outcomes such as number of days of bleeding with the procedure, pain, time to return of menstruation, cost or acceptability have not been assessed sufficiently (Sjöström 2016).

Quality of the evidence

Overall, the certainty of the evidence is low to moderate. The methodology of included studies was adequate for most included studies, especially for the randomisation method. We included 99 RCTs in the current review relating to different interventions and different inclusion criteria of women; therefore the consistency of evidence should be evaluated for each subgroup but not for the whole review. All the included studies directly assessed the primary outcome of this review, complete abortion rate, using different methods. However, some studies also included a few women in the second trimester, and not all the included studies detailed the results of reported side effects. The sample size was adequate for most included studies to address the research objectives.

Potential biases in the review process

The risk of bias within the evidence was moderately high (Figure 2; Figure 3; Characteristics of included studies). Not all studies described their random sequence generation or allocation concealment methods, few studies reported blinding, and most studies randomly assigned only limited numbers of women.

Heterogeneity, especially clinical heterogeneity, was high in the included studies, including different dosage and type, different route, different interval, and different duration of follow-up.

We believe that we identified all relevant studies. However, attempted contact with primary authors of 10 records for full texts was unsuccessful. Imputation of individual values for attrition was undertaken by us. Subjective judgements are involved in the assessment of risk of bias. This potential limitation is minimised by following the procedures in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2017), with review authors independently assessing studies and resolving any disagreement through discussion, and if required involving a third review author in the decision.

Agreements and disagreements with other studies or reviews

Medical abortion for the first trimester has also been systematically reviewed by the following systematic reviews.

Being consistent with the results of the current review, other systematic reviews found that the combined regimen of mifepristone and misoprostol is more effective than misoprostol alone; misoprostol provided vaginally is better compared to oral administration; and adverse events are rare (Abubeker 2020; Al Wattar 2019; Kapp 2018). Baiju 2019 also found no significant difference in complete abortion rates between self-assessment and routine clinic follow-up. Therefore, the option of expulsion at home after misoprostol has been taken is also advised to women before 10 gestational weeks (Schmidt-Hansen 2020). Early medical abortion with mifepristone 200 mg followed by misoprostol is also effective and safe (Raymond 2013). But the evidence is also limited for the inconsistency among included studies.

However, being inconsistent with the present review, other systematic reviews found that incomplete abortion is equally effective and acceptable when performed by non-doctor providers as doctors (Sjöström 2017). Barnard 2015 found no difference between mid-level providers (such as midwives, nurses and other non-physician providers) and doctors for first trimester abortion. There was no difference in complete abortion between vaginal misoprostol with expectant care for women with incomplete abortion before 13 weeks' gestation Kim 2017. The efficacy of letrozole prior to misoprostol is controversial in Nash 2018. A 24-hour time interval between mifepristone and buccal misoprostol administration is slightly less effective than a 24-hour to 48-hour interval (Chen 2015b).

AUTHORS' CONCLUSIONS

Implications for practice

The available data from this review demonstrates that the combination mifepristone/misoprostol is probably the most effective abortion method in the first trimester. Effectiveness is probably not reduced by lowering the currently licensed dose of 600 mg of mifepristone to 200 mg. Vaginal misoprostol is probably more effective than oral administration, and may have fewer side effects than sublingual or buccal misoprostol. This review does not address introducing medical abortion where back-up facilities are not available and women are less likely to attend for the follow-up.

Implications for research

Letrozole in combination with a prostaglandin may be an alternative to the mifepristone/prostaglandin regimen in places where mifepristone is either unaffordable or unavailable. However, further research should be conducted to compare the letrozole/prostaglandin combination regimen with the standard mifepristone/prostaglandin regimen.

Some important outcomes relating to efficacy and safety of medical abortion should be reported in greater detail, including conceptus expulsion time, bleeding volume, and number of women needing additional uterotonics in each intervention group.

One of the most common and severe complication of medication abortion is blood loss requiring a blood transfusion, this important outcome should be reported in all future studies on medical abortion.

Time and cost expenditure, and time to return to menstruation are not reported by previously published studies; these outcomes are important to women and should be further evaluated in future studies.

There are scarce data on issues such as which method is preferable when addressing specific side effects, bleeding patterns, acceptability or the financial impact of the different methods.

Good-quality acceptability studies are important to investigate the components of medical abortion regimens that affect acceptability in different settings.

There are still few studies on medical abortion in settings where back-up facilities are not available and women are less likely to attend for follow-up. For example, medical abortion in low-income countries, where women may be less likely to follow up to evaluate efficacy and safety after they have received abortion medicine.

The follow-up duration for measuring outcomes of medical abortion was heterogeneous among the included studies. Further research is needed to determine the most clinically meaningful time point for measuring study outcomes, and also to understand the benefits, risks, and time and cost expenditures associated with required clinical follow-up visits following medical abortion.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Abbasalizadeh 2018

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** LET+ MS • Mean age (years): 29.21 ± 4.08 • Mean pregnancy age based on LMP (weeks): 7.74 ± 0.95 • Mean pregnancy age based on ultrasound (weeks): 8.09 ± 0.90 • Missed abortion (%): 17.2% MS • Mean age (years): 28.53 ± 5.24 • Mean pregnancy age based on LMP (weeks): 8.52 ± 1.29 • Mean pregnancy age based on ultrasound (weeks): 8.80 ± 1.17 • Missed abortion (%): 14.1% Overall • Mean age (years): unreported • Mean pregnancy age based on LMP (weeks): unreported • Mean pregnancy age based on ultrasound (weeks): unreported

Abbasalizadeh 2018 *(Continued)*

	• Missed abortion (%): unreported **Inclusion criteria:** minimum age of 18 years for mother, first trimester of pregnancy (< 12 weeks based on LMP), non-viable fetus (missed abortion or blighted ovum), lack of any maternal diseases such as: heart disease, asthma, history of thromboembolism, cancer, renal failure, and liver diseases, and consent of women and her spouse to participate in the study **Exclusion criteria:** medical problem in women which would interfere and emergency treatment, history of allergy to misoprostol or LET drugs, women who had spontaneous abortion before taking misoprostol **Pretreatment:** no **Study duration:** June 2013 to April 2015
Interventions	**Intervention characteristics** LET + MS vs MS • LET + MS: 10 mg oral LET for 3 days, 600 µg single dose oral MS • MS: placebo of LET like intervention group and 600 µg single dose oral MS
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: change from baseline Surgical evacuation • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Nausea • Outcome type: adverse event • Reporting: fully reported • Direction: lower is better • Data value: endpoint Vomiting • Outcome type: adverse event • Reporting: fully reported • Direction: lower is better • Data value: endpoint Diarrhoea • Outcome type: adverse event • Reporting: fully reported • Direction: lower is better • Data value: endpoint Dizziness • Outcome type: adverse event • Reporting: fully reported

Abbasalizadeh 2018 *(Continued)*

- Direction: lower is better
- Data value: endpoint

Headache

- Outcome type: adverse event
- Reporting: fully reported
- Direction: lower is better
- Data value: endpoint

Abdominal pain

- Outcome type: adverse event
- Reporting: fully reported
- Direction: lower is better
- Data value: endpoint

Fever

- Outcome type: adverse event
- Reporting: fully reported
- Direction: lower is better
- Data value: endpoint

Chills

- Outcome type: adverse event
- Reporting: fully reported
- Direction: lower is better
- Data value: endpoint

Haemoglobin level

- Outcome type: continuous outcome
- Reporting: fully reported
- Unit of measure: mg/dL
- Direction: Higher is better
- Data value: endpoint

Bleeding duration

- Outcome type: adverse event
- Reporting: fully reported
- Direction: lower is better
- Data value: endpoint

Identification	**Sponsorship source:** Women's Reproductive Health Research Center; pharmaceutical companies (Iran Hormone and Samisaz) **Country:** Iran **Setting:** Department of Gynecology and Obstetrics, Women's Reproductive Health Research Center, Tabriz University of Medical Science **Author(s):** Fatemeh Abbasalizadeh, M.D. 1; Farnaz Sahhaf, M.D. 1; Paria Sadeghi-Shabestari, M.D. 1; Mohammad Mirza-Aghazadeh-Attari, M.D. 2; Mohammad Naghavi-Behzad, M.D. **Institution:** 1 Department of Gynecology and Obstetrics, Women's Reproductive Health Research Center, Tabriz University of Medical Sciences, Tabriz, Iran; 2 Medical Philosophy and History Research Center, Tabriz University of Medical Sciences, Tabriz, Iran

Abbasalizadeh 2018 *(Continued)*

	Email: dr.naghavii@gmail.com **Address:** Daneshgah Street, Tabriz, Eastern Azerbaijan, Iran
Notes	**Follow up duration:** 14 days

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "randomly selected in two intervention and control groups using Rand list (version 1.2) software." Judgement comment: computer randomisation adequate
Allocation concealment (selection bias)	High risk	Judgement comment: unreported by authors
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "Through the study single blinding was applied and patients were not aware of studied groups" Judgement comment: single (participants), placebo of LET
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: single blinding (participants)
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: none lost to follow-up
Selective reporting (reporting bias)	Low risk	Quote: "IRCT2014030114293N1" Judgement comment: both efficacy and side effects were reported.
Other bias	Unclear risk	Quote: "Acknowledgements: Present study was conducted in cooperation with two pharmaceutical companies (Iran Hormone and Samisaz), and we sincerely appreciate the..." Judgement comment: pharmaceutical company support

Abdelshafy 2019

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS sublingual • Age (years): 27.6 ± 4.8 • Pregnancy age (weeks): 8.6 ± 1.4 • BMI (kg/m^2): 27.25 ± 3.1 • Parity: 2.17 ± 1.61

Abdelshafy 2019 *(Continued)*

	• Primigravida: 31 MS vaginal • Age (years): 28.6 ± 5.5 • Pregnancy age (weeks): 8.8 ± 1.3 • BMI (kg/m^2): 27.11 ± 2.1 • Parity: 2.03 ± 1.44 • Primigravida: 29 Overall • Age (years): unreported • Pregnancy age (weeks): unreported • BMI (kg/m^2): unreported • Parity: unreported • Primigravida: unreported **Inclusion criteria**: all women above 18 years of age; < 12 weeks of gestation; pregnancy is confirmed by pregnancy test or ultrasound scan; missed abortion; normal general and gynaecological examination; the size of the uterus on pelvic examination was compatible with the estimated duration of pregnancy **Exclusion criteria**: haemodynamically unstable; suspected sepsis with temperature 38 °C; concurrent medical illness e.g. haematological, cardiovascular, thromboembolism, respiratory illnesses, recent liver disease or pruritus of pregnancy; presence of IUD; suspect or proven ectopic pregnancy; failed medical or surgical evacuation before presentation; known allergy to MS **Pretreatment:** no **Study duration :** 1 February 2016 to 31 January 2017
Interventions	**Intervention characteristics** The treatment for each group comprised three doses of misoprostol 800 µg administered sublingually or vaginally every four hours. Each dose of misoprostol comprised four 200 µg tablets. MS sublingual • 1st at hospital, 2nd and 3rd doses at home; omit the 3rd dose of MS if the products of conception were passed out; women with a retained gestational sac were given a 2nd course 7 days later MS vaginal • 1st at hospital, 2nd and 3rd doses at home; omit the 3rd dose of MS if the products of conception were passed out; women with a retained gestational sac were given a 2nd course 7 days later
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome • Notes: ultrasound examination Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event

Abdelshafy 2019 *(Continued)*

	Diarrhoea • Outcome type: adverse event Need for analgesia • Outcome type: adverse event Fever • Outcome type: adverse event Chills • Outcome type: adverse event Haemoglobin level • Outcome type: continuous outcome • Unit of measure: g/L • Direction: higher is better • Data value: endpoint Bleeding duration • Outcome type: continuous outcome • Unit of measure: days • Direction: lower is better • Data value: endpoint • Notes: PBLAC to quantify the amount of bleeding Length of induction expulsion interval • Outcome type: continuous outcome • Reporting: fully reported • Unit of measure: hours • Direction: lower is better • Data value: endpoint Induction expulsion interval • Outcome type: continuous outcome • Unit of measure: days • Direction: lower is better • Data value: endpoint Side effects • Outcome type: adverse event • Reporting: fully reported • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Ain Shams Maternity Hospital **Country:** Egypt **Setting**: outpatient clinic of the maternity hospital of Ain Shams University **Author(s):** Ahmed Abdelshafy, Hassan Awwad, Amgad Abo-Gamra, Ahmed Alanwar, Ahmed M. Elkotb, Mohamed Shahin, Maya Abd El-Razek & Ahmed M. Abbas

Abdelshafy 2019 *(Continued)*

	Institution: Ain Shams Maternity Hospital, Faculty of Medicine, Ain Shams University, Cairo, Egypt; Women's Health Hospital, Faculty of Medicine, Assiut University, Assiut, Egypt **Email:** eladwar@hotmail.com **Address:** Ain Shams Maternity Hospital, Faculty of Medicine, Ain Shams University, Ramses Street (Ahmed Lotfy Elsayed Street), Abbassia, 11566 Cairo, Egypt **ClinicalTrials.gov NCT02686840**
Notes	**Follow-up duration:** 30 days

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "computer generated random table prepared by a statistician not otherwise involved in the study." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote:"Allocation concealment was done using serially numbered closed opaque envelopes. Each envelope was labelled with a serial number and had a card noting the intervention type. Allocation was never changed after opening the closed envelopes." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Quote: "only the study outcomes' assessor (one of the study investigators) was blinded to the participants' group." Judgement comment: adequate
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement comment: 11 participants lost to follow-up, 4 in sublingual and 7 in vaginal; ITT analysis was not performed by primary authors
Selective reporting (reporting bias)	Low risk	Judgement comment: outcomes according to the published protocol (NCT02686840)
Other bias	Low risk	Judgement comment: no funding supporting

Allameh 2020

Study characteristics	
Methods	Double-blind, placebo-controlled RCT
Participants	Women with missed abortion in the 1st trimester of pregnancy, from 1 January 2016 to 31 December 2018 **Inclusion criteria**: age > 18 years, 1st-trimester pregnancy (gestational age < 12 weeks) with missed abortion confirmed by ultrasonography and haemoglobin level > 12 mg/dL

Allameh 2020 *(Continued)*

	Exclusion criteria: abnormalities in blood tests; history or clinical evidence of any thromboembolic impairment or deep venous thrombosis; having an IUD; current or previous use of corticosteroids; history of any malignancy; existing cardiovascular disease contraindicating MS or LET; and scars on uterus; drug sensitivity leading to drug discontinuation, non-referral for evaluation of complications and consequences of abortion, abnormal vital signs, uncontrolled or severe vaginal bleeding at follow-up; adopted other treatments	
Interventions	**Intervention group:** 10 mg of LET daily for 3 days, followed by 2 doses of 800 µg of vaginal MS at a 4-hour interval **Control group:** 500 mg of calcium carbonate (Razavi Pharmaceutical Service Institute, Mashhad, Iran) daily for 3 consecutive days as a placebo, followed by 2 doses of 800 µg of vaginal MS at a 4-h interval. All women also received 100 mg of doxycycline every 12 h for 1 week	
Outcomes	Complete abortion (sonography was performed to check the abortion status); induction duration (hour)	
Identification	-	
Notes	**Follow-up duration:** 7 days	
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The women enrolled in the study were randomly allocated to the 2 study groups by using Random-Maker Software Random Allocation
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Used placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Used placebo
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	3 women in each group were lost to follow-up and were excluded from the analysis. No ITT analysis
Selective reporting (reporting bias)	Unclear risk	No published protocol
Other bias	Unclear risk	-

Arvidsson 2005

Study characteristics	
Methods	Computer randomisation

Arvidsson 2005 *(Continued)*

Participants	**Inclusion criteria:** 100 women randomised; age and gestational age averages not given; included gestational age up to 49 days confirmed by ultrasound **Exclusion criteria:** contraindications for medical abortion Setting: Karolinska Hospital, Sweden
Interventions	MF 600 mg (all) followed 36-48 h later by: • Group 1 MS 400 µg oral • Group 2 MS 800 µg vaginal
Outcomes	Experience of pain, occurrence of side-effects, duration of bleeding
Identification	
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer randomisation
Allocation concealment (selection bias)	High risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No blinding was reported
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No blinding was reported
Incomplete outcome data (attrition bias) All outcomes	High risk	10 women could not be reached by phone 3-7 weeks after abortion; 30 women did not agree or weren't asked to be called during this time frame
Selective reporting (reporting bias)	Low risk	Both efficacy and side-effects were reported
Other bias	Low risk	The study was supported by grants from the Swedish Medical Research Council (6392)

Baird 1995

Study characteristics

Methods	Computer-generated random numbers for the 1st 300 women, envelopes were shuffled in batches of 20 and numbered consecutively for the reminders No blinding for clinical staff
Participants	800 pregnant women ≤ 63 days of amenorrhoea in Edinburgh, Scotland

Baird 1995 *(Continued)*

Interventions	MF 200 mg (all) followed by: • Group 1 GP 0.5 mg vaginal and 3 tablets placebo after 48 h • Group 2: MS 600 µg oral and vaginal examination after 48 h
Outcomes	Complete, incomplete and missed abortion Ongoing pregnancy Side effects
Identification	
Notes	Power calculation (80% to detect 5% difference) Placebos were not identical to MS 1 woman needed blood transfusion (group 2)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Computer-generated random numbers only for the 1st 300 women
Allocation concealment (selection bias)	Unclear risk	Envelopes were shuffled in batches of 20 and numbered consecutively for the reminders
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No blinding for clinical staff, placebos were not identical to MS
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No blinding for clinical staff, placebos were not identical to MS
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	23 participants were excluded, but the ITT analysis was not performed
Selective reporting (reporting bias)	Low risk	Efficacy and side effects were reported
Other bias	Low risk	No funding was reported

Bartley 2001

Study characteristics

Methods	Computer-generated random numbers
Participants	999 pregnant women, < 63 days of gestation, confirmed by ultrasound if necessary, at the Royal Infirmary Hospital, Edinburgh **Inclusion criteria**: aged ≥ 16 years, available for follow-up within 2 weeks **Exclusion criteria:** ectopic pregnancy, active asthma, liver or renal disease, adrenal insufficiency, anaemia, haemolytic disease, treatment with anticoagulants, smoking > 20 cigarettes/day
Interventions	MF 200 mg (all) followed by:

Bartley 2001 *(Continued)*

	• Group 1: GP 0.5 mg/vaginal • Group 2: MS 800 µg/vaginal
Outcomes	Complete, incomplete abortion, ongoing pregnancy, duration of bleeding, side effects
Identification	
Notes	Double-blinded 2 women required blood transfusions (1 in each group)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Double-blinded
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Double-blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition between groups
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	No funding

Behroozi Lak 2018

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** LET + MS • Age (years): 29.56 (6.09) • Gravidity 2.46 (1.48) • Parity: 1.03 • Pregnancy age based on ultrasound (weeks): 7.6 (2.1) MS

Behroozi Lak 2018 *(Continued)*

	• Age (years): 28.31 • Gravidity: 2.46 (1.21) • Parity: 1.03 (1.16) • Pregnancy age based on ultrasound (weeks): 9.1 (2.3) Overall • Age (years): unreported • Gravidity: unreported • Parity: unreported • Pregnancy age based on ultrasound (weeks): NR **Inclusion criteria:** good general health; over the age of legal consent; gestational age < 14 weeks as confirmed by ultrasound; haemoglobin > 10 g/L, DBP < 95 mmHg **Exclusion criteria:** history of evidence of adrenal pathology, steroid-dependent, cancer, porphyria, DBP > 95 mmHg, bronchial asthma, arterial hypertension, thromboembolism, liver disease, breast feeding, anomaly of fetus, regular use of prescription drugs before admission to the study, abnormal values in pretreatment blood tests **Pretreatment**: gestational age in the MS group is more than LET + MS group **Missed abortion:** death of the fetus was confirmed by 2 separate ultrasounds
Interventions	**Intervention characteristics** LET + MS • 10 mg LET orally on days 1-3 followed by vaginal MS each 12 h x 3 times MS • vaginal MS each 12 h x 3 times; The initial dose of misoprostol was 800 μg per 12 h and transvaginally
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome The interval from misoprostol administration to abortion • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint • Notes: the interval from MS administration to abortion Severe vaginal haemorrhage • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Side effects (nausea, vomiting, diarrhoea, fatigue, dizziness, headache, abdominal pain, fever, rash, shivering) • Outcome type: adverse event • Reporting: NR

Behroozi Lak 2018 *(Continued)*

	• Direction: lower is better • Data value: endpoint • Notes: 'not significantly different between groups' reported
Identification	**Sponsorship source**: Reproductive Health Research Center, Department of Infertility, Urmia University of Medical Sciences, Urmia **Country:** Iran **Setting**: Reproductive Health Research Center, Department of Infertility **Author(s):** T Behroozi-Lak; S Derakhshan-Aydenloo; F Broomand **Institution:** a Reproductive Health Research Center, Department of Infertility, Urmia University of Medical Sciences, Urmia, Iran; Urmia University of Medical Sciences, Urmia, Iran **Email:** t.behrooz2@yahoo.com
Notes	The total administration of LET and MS was not the same in participants. 58.97% patients received only LET because of abortion occurring.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "The participants were randomized into the two groups of 39 patients each by the hospital pharmacists according to computer-generated random numbers." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Judgement comment: randomised by pharmacists
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "Until the completion of the study, both patients and the clinicians were blinded to the group assigned." Judgement comment: double-blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: double - clinicians and participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced
Selective reporting (reporting bias)	Unclear risk	Judgement comment: side effects were only reported as no significant difference
Other bias	Low risk	Judgement comment: no funding

Birgerson 1988

Study characteristics	
Methods	Random allocation, not specified

Birgerson 1988 *(Continued)*

Participants	153 women, ≤ 49 days of amenorrhoea, confirmed by positive pregnancy test and pelvic examination Uppsala, Sweden
Interventions	• Group 1: MF 10 mg/twice daily for 7 days • Group 2: MF 25 mg/twice daily for 7 days • Group 3: MF 50 mg/twice daily for 7 days (Group 1 vs Group 3)
Outcomes	Complete, incomplete abortion Ongoing pregnancy Bleeding pattern Side effects
Identification	
Notes	No mention of major complications

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Random allocation, not specified
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	NR

Blanchard 2005

Study characteristics

Methods	Random numbers generated in SPSS; numbered opaque envelopes
Participants	**Inclusion criteria:** > 18 years old in good general health and willing to return for follow-up and living < 1 h from clinic; ≤ 56 days of gestation

Blanchard 2005 *(Continued)*

	Exclusion criteria: < 18 years old, suspected ectopic pregnancy Study conducted in India and Vietnam
Interventions	MS only • Group 1: 4 x 400 μg/3 h/oral • Group 2: 2 x 800 μg/oral • Group 3: 1 x 600 μg/vaginal • Group 4: 2 x 800 μg/3 h/oral • Group 5: 1 x 800 μg/vaginal
Outcomes	Complete/incomplete abortion, ongoing pregnancy
Identification	
Notes	Initially, women were randomised between the first 3 regimens. Subsequent review of their low efficacy resulted in changing the regimens, and from that point on, women were randomised to treatment group 4 or 5.

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random numbers generated in SPSS
Allocation concealment (selection bias)	Low risk	Numbered opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	No funding

Blum 2012

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group

Blum 2012 *(Continued)*

Participants	**Baseline characteristics** MS • Mean age (years): 29 (6.2) • Primigravida: 21.2% • Gravidity: 2.98 (1.6) MF + MS • Mean age (years): 29 (6.5) • Primigravida: 27.5% • Gravidity: 2.74 (1.6) Overall • Mean age (years): no • Primigravida: no • Gravidity: no **Inclusion criteria**: pregnant women presenting for early medical abortion up to 63 days since LMP; live or work within a reasonable distance from the study hospital; have an intrauterine pregnancy; present in general good health; willing to provide informed consent and return for follow-up at the hospital **Exclusion criteria**: contraindications to MF and/or MS; allergy to MS Study duration: From 5 May 2009 to 20 July 2010 10 and 3 participants withdrew or lost to follow-up in MF + MS and MS group respectively. ITT analyses were conducted. Follow up duration: 7 days
Interventions	**Intervention characteristics** MS • placebo on day 1 and 1600 µg of misoprostol (2 doses of 800 µg, given 3 hours apart) on day 2 MF + MS • 200 mg of mifepristone on day 1 and 800 µg buccal misoprostol followed by placebo 3 hours later on day Participants were told to take all of the medications even if they believed that the abortion was already complete.
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome • Woman's abortion status would be assessed by clinical examination and/or transvaginal ultrasonography as the provider deemed necessary. Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event

Blum 2012 *(Continued)*

	Diarrhoea • Outcome type: adverse event Abdominal pain (more than expected) • Outcome type: adverse event • Notes: pain more than expected Fever • Outcome type: adverse event Chills • Outcome type: adverse event Bleeding (more than expected) • Outcome type: adverse event • Direction: lower is better • Data value: endpoint Ongoing pregnancy • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Participant's satisfaction (satisfied or very satisfied) • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: higher is better • Data value: endpoint Failure to achieve complete abortion(< 49 days) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Failure to achieve complete abortion(50-63 days) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Ongoing pregnancy (< 49 days) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Ongoing pregnancy (50-63 days) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Gynuity Health Projects, New York, USA

Blum 2012 *(Continued)*

	Country: Tunisia; Vietnam **Setting:** 2 large maternity hospitals: either the Centre de Maternite et Neonatologie de la Rabta in Tunisia (n = 193) or Hung Vuong Hospital, Ho Chi Minh City, Vietnam (n = 248) **Comments:** NCT00680394; **Author(s):** Jennifer Blum **Institution:** Gynuity Health Projects **Email:** jblum@gynuity.org **Address:** 15 E. 26th Street, Suite 801, New York, NY 10012, USA
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Treatment allocation was assigned in blocks of 10 using a computer-generated random sequence created by staff at Gynuity Health Projects (New York, USA)." Judgement comment: adequate, block randomisation
Allocation concealment (selection bias)	Low risk	Quote: "all participants received an envelope containing 3 packets of pills" Quote: "All medication packets were prepared by Gynuity Health Projects staff;" Judgement comment: medication packets were prepared by staff
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "providers and participants were blinded to study treatment." Judgement comment: double-blinding
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement comment: participants and providers were blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "Does not include 7 women lost to follow-up in the mifepristone–misoprostol group and 6 women who withdrew from study and did not take any study medication (3 in the mifepristone–misoprostol group; 3 in the misoprostol-only group)." Judgement comment: ITT analysis was not performed by study authors
Selective reporting (reporting bias)	Low risk	Quote: "Clinical trials.gov registration number: NCT00680394." Judgement comment: efficacy and side effects were reported
Other bias	Unclear risk	Quote: "The present study was funded by an anonymous donor." Judgement comment: unclear funder

Cameron 1986

Study characteristics	
Methods	Random allocation, not specified
Participants	**Inclusion criteria:** 45 pregnant women < 56 days amenorrhoea, confirmed by pregnancy test, pelvic examination and ultrasound **Exclusion criteria:** multiple pregnancy, spontaneous abortion, cardiovascular or pulmonary disease, allergy, epilepsy
Interventions	• Group 1: MF 150 mg/daily for 4 days • Group 2: MF 150 mg and GP 1-2 mg vaginal after 48 h
Outcomes	Complete abortion, treatment failure, complications, side effects, pain, bleeding pattern
Identification	
Notes	5 women receiving GP 2 mg were excluded from the analysis 1 woman received blood transfusion (Group 1); 1 woman had emergency evacuation due to heavy bleeding (Group 1)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Random allocation, not specified
Allocation concealment (selection bias)	Unclear risk	not reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	not reported
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	not reported
Incomplete outcome data (attrition bias) All outcomes	High risk	Unbalanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	Ru486 was kindly supplied by Roussel Laboratories, Ltd

Carbonell 1997b

Study characteristics	
Methods	Computer randomisation; sealed, opaque envelopes were numbered by a person unrelated to the study

Carbonell 1997b *(Continued)*

Participants	**Inclusion criteria:** 300 pregnant women, ≤ 63 days of amenorrhoea confirmed by ultrasound **Exclusion criteria:** previous use of vitamins/folates, white blood cell count < 3000/µL, platelet count < 100 000/µL, haemoglobin < 10.0 mg/dL, aspartate aminotransferase > 2 times normal or active liver disease, serum creatinine > 1.5 mg/dL or active renal disease, inflammatory bowel disease, intolerance to the medication
Interventions	MTX 50 mg/m^2 IM on recruitment day and MS 800 µg vaginal (self-administered) on: • Group 1: day 3 • Group 2: day 4 • Group 3: day 5 Additional 800 µg MS in 48 h interval (up to 4 doses)
Outcomes	Complete, incomplete abortion (complete expulsion with additional doses of MS), treatment failure, bleeding pattern, blood parameters, side effects
Identification	
Notes	Power calculation (85% power, significance level of 0.05) No major complications occurred

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer randomisation
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes were numbered by a person unrelated to the study
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	13 participants dropped out
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	No funding

Carbonell 1998

Study characteristics

Carbonell 1998 *(Continued)*

Methods	Computer randomisation; sealed, opaque envelopes were numbered by a person unrelated to the study
Participants	**Inclusion criteria:** 315 pregnant women, ≤ 63 days of amenorrhoea confirmed by ultrasound **Exclusion criteria:** previous use of vitamins/folates, white blood cell count < 3000/μL, platelet count < 100 000/μL, haemoglobin < 10.0 mg/dL, aspartate aminotransferase > 2 times normal or active liver disease, serum creatinine > 1.5 mg/dL or active renal disease, inflammatory bowel disease, intolerance to the medication
Interventions	MTX 50 mg oral on recruitment day and MS 800 μg vaginal (self-administered) on: • Group 1: day 3 • Group 2: day 4 • Group 3: day 5 Additional 800 μg MS in 48 h interval (up to 4 doses)
Outcomes	Complete, incomplete abortion (complete expulsion with additional doses of MS), treatment failure, bleeding pattern, blood parameters, side effects
Identification	
Notes	Power calculation (80% power, significance level of 0.05) No major complications occurred

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer randomisation
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes were numbered by a person unrelated to the study
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	15 participants dropped out
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Chai 2013

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (sublingual) • Mean age (years): 26.7 (7.2) • Mean pregnancy age (days): 49.6 (7.2) • Weight (kg): 52.0 (6.7) • Previous abortion: 13 (23.9) MF + MS (buccal) • Mean age (years): 28.6 (7.3) • Mean pregnancy age (days): 50.3 (7.5) • Weight (kg): 55.3 (8.3) • Previous abortion: 15 (33.3) Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Weight (kg): NR • Previous abortion: NR **Inclusion criteria:** good general health; over the age of legal consent (i.e. > 18 years old); requesting medical abortion and eligible for abortion; on Day 1 of the study (day of MF administration) the duration of pregnancy not more than 63 days as confirmed by pelvic ultrasound examination; intrauterine pregnancy (intrauterine amniotic sac seen in ultrasound); willing to use other than hormonal or intra-uterine contraception until the 1st menses after termination of pregnancy; if treatment fails she agrees to termination of pregnancy with the surgical method; willing and able to participate after the study has been explained; haemoglobin > 10g/L **Exclusion criteria:** a history or evidence of adrenal pathology, steroid-dependent cancer, porphyria, DBP > 95 mmHg, bronchial asthma, arterial hypotension; a history or evidence of thromboembolism, severe or recurrent liver disease or pruritus of pregnancy; the regular use of prescription drugs before admission to the study; the presence of an IUD in utero; breastfeeding; multiple pregnancies; heavy smokers; **Pretreatment:** none lost to follow-up: 2 in buccal group
Interventions	**Intervention characteristics** MF + MS (sublingual) • 200 mg MF, 48 h later, 800 µg MS was given; four MS (sublingually) and four placebo tablets (buccally) MF + MS (buccal) • 200 mg MF, 48 h later, 800 µg MS was given; four placebo tablets (sublingual) and four MS tablets (buccal)
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome

Chai 2013 *(Continued)*

Surgical evacuation

- Outcome type: dichotomous outcome

Nausea

- Outcome type: adverse event

Vomiting

- Outcome type: adverse event

Diarrhoea

- Outcome type: adverse event

Dizziness

- Outcome type: adverse event

Headache

- Outcome type: adverse event

Abdominal pain

- Outcome type: adverse event

Fever

- Outcome type: adverse event

Chills

- Outcome type: adverse event

Haemoglobin level

- Outcome type: continuous outcome
- Unit of measure: g/dL
- Direction: higher is better
- Data value: endpoint

Bleeding on Day 15

- Outcome type: adverse event
- Direction: lower is better
- Data value: endpoint

Failure to achieve complete abortion (< 49 days)

- Outcome type: dichotomous outcome
- Direction: lower is better
- Data value: endpoint

Failure to achieve complete abortion (50-63 days)

- Outcome type: dichotomous outcome
- Reporting: fully reported
- Direction: lower is better
- Data value: endpoint

Ongoing pregnancy

- Outcome type: dichotomous outcome

Chai 2013 *(Continued)*

	• Direction: lower is better • Data value: endpoint Ongoing pregnancy (< 49 days) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Ongoing pregnancy (50-63 days) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Induction to abortion interval • Outcome type: continuous outcome • Unit of measure: hours • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Family Planning Association in Hong Kong **Country:** China **Setting:** Family Planning Association in Hong Kong **Comments:** ClinicalTrials.Gov (NCT01156688) **Author(s):** Joyce Chai **Institution:** Department of Obstetrics and Gynaecology, University of Hong Kong, Hong Kong Special Administration Region, China **Email:** jchai@hkucc.hku.hk **Address:** Department of Obstetrics and Gynaecology, 6/F, Professorial Block, Queen Mary Hospital, 102 Pokfulam Road, Hong Kong
Notes	**follow-up duration:** 2 and 6 weeks

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Randomization assignment was made by the research nurse using a computer program to allocate the study subjects into two groups:" Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Based on the randomization schedule, the concealed packages of misoprostol and placebo tablets were prepared by the pharmacy. The placebo tablets were identical in appearance to the misoprostol tablets." Judgement comment: pharmacy prepared packages
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "We used placebo tablets so that both the study subjects and the investigators were blind to the allocation to avoid reporting and observation bias." Judgement comment: adequate

Chai 2013 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: both study participants and investigators were blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "Two women in the buccal group did not return for follow-up on day 43, and we tried to contact them by phone without success." Judgement comment: although 2 participants were lost to follow-up in buccal group, ITT analyses were performed in present study
Selective reporting (reporting bias)	Low risk	Quote: "(NCT01156688)." Judgement comment: both efficacy and side effects were reported
Other bias	Low risk	Judgement comment: no other bias was found

Chen 2015a

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + TCM (Shenghua decoction) + MS • Mean age (years): NR • Mean pregnancy age based on LMP (weeks): NR • Gestational sac: NR MF + MS + TCM (Shenghua decoction) • Mean age (years): NR • Mean pregnancy age based on LMP (weeks): NR • Gestational sac: NR MF + MS • Mean age (years): NR • Mean pregnancy age based on LMP (weeks): NR • Gestational sac: NR Overall • Mean age (years): NR • Mean pregnancy age based on LMP (weeks): NR • Gestational sac: NR **Inclusion criteria:** < 84 days' gestation from LMP; missed abortion; the diameter of gestational sac < 3 cm **Exclusion criteria:** acute inflammation in reproductive system; contraindications of MF or MS; with IUD **Pretreatment:** no

Chen 2015a *(Continued)*

	Study duration: June 2013 to September 2014
Interventions	**Intervention characteristics** MF + TCM (Shenghua decoction) + MS • Shenghua decoction from day 1; MF and MS as the control group MF + MS + TCM (Shenghua decoction) • Shenghua decoction from day 4; MF and MS as the control group MF + MS • MF 150 mg, oral, on day 3; MS 600 µg, vaginal, on day 4
Outcomes	Bleeding duration • Outcome type: continuous outcome • Unit of measure: days • Direction: lower is better • Data value: endpoint Time until passing of conceptus • Outcome type: continuous outcome • Unit of measure: hours • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Maternal and Child Health Hospital of Jianxi Province **Country:** China **Setting:** Maternal and Child Health Hospital of Jianxi Province **Comments:** TCM **Author(s):** Lingyan Chen **Institution:** Maternal and Child Health Hospital of Jianxi Province **Email:** no **Address:** Maternal and Child Health Hospital of Jianxi Province, Nanchang 330006, China
Notes	The effects were classified as 4 categories: cure, apparent efficacious; efficacious; no effect. Follow-up: 2 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Judgement comment: adequate, random number table
Allocation concealment (selection bias)	High risk	Judgement comment: no allocation concealment

Chen 2015a *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no participants dropped out or withdrew
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no registration protocol was achieved and some side effects were not reported, such as nausea, vomiting
Other bias	Low risk	Judgement comment: none

Cheng 1994

Study characteristics	
Methods	Double-blind, randomisation generated centrally; sealed, opaque envelopes
Participants	151 women, ≤ 49 days of amenorrhoea confirmed by ultrasound at Shanghai Medical University without medical disorders, contraindication for the study medication or IUD in situ

Interventions	• Group 1:
	  ◦ day 1-3: TP 100 mg/IM/day
	  ◦ day 4: PGE1 ester (ONO 802) 1 mg/vaginally/6-hourly for a maximum of 4 doses
	• Group 2:
	  ◦ day 1-3: placebo injections
	  ◦ day 4: PGE1 ester (ONO 802) 1 mg/vaginally/6-hourly for a maximum of 4 doses

Outcomes	Complete, incomplete abortion, ongoing pregnancy, blood transfusion, duration of bleeding
Identification	
Notes	No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation generated centrally
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Double-blind

Cheng 1994 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	Low risk	Double-blind
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Chong 2012

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (400 µg) • Mean age (years): 27.4 • Mean pregnancy age (days): 48 • Gravidity: 3.9 • Parity: 1.2 • Previous abortion: 58.3% MF + MS (800 µg) • Mean age (years): 27.9 • Mean pregnancy age (days): 48.2 • Gravidity: 4.2 • Parity: 1.2 • Previous abortion: 61.8% Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Gravidity: NR • Parity: NR • Previous abortion: NR **Inclusion criteria:** gestations up to 63 days since LMP **Exclusion criteria:** anticoagulant therapy, concurrent long-term corticosteroid therapy, adrenal insufficiency, inherited porphyria, suspicion of ectopic pregnancy, and known allergy to MF or MS **Pretreatment:** none **Study duration:** From May 2007 to September 2009
Interventions	**Intervention characteristics**

Chong 2012 *(Continued)*

	MF + MS (400 µg) • MF (200 mg, oral) + 2 x 200 µg MS (U-miso; U-Liang Ltd, Taiwan) pills and 2 placebo pill MF + MS (800 µg) • MF (200 mg, oral) + four 200 µg MS pill
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Weakness • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Satisfied • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint Ongoing pregnancy • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Failure to achieve complete abortion (< 49 days) • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Failure to achieve complete abortion (50-63 days) • Outcome type: dichotomous outcome • Reporting: fully reported

Chong 2012 *(Continued)*

	• Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Gynuity Health Projects, New York **Country:** Georgia and Vietnam **Setting:** 3 clinics in Tbilisi, the Republic of Georgia; Hoc Mon Hospital in Vietnam **Comments:** no **Author(s):** Erica Chong **Institution:** Gynuity Health Projects, New York **Email:** echong@gynuity.org **Address:** Gynuity Health Projects, New York, NY 10010, USA
Notes	Follow-up visits occurred 12-15 days after MF administration

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "a random code generated in blocks of 10 by Gynuity Health Projects in New York," Quote: "Randomization was stratified by study site." Judgement comment: adequate, block and stratified
Allocation concealment (selection bias)	Low risk	Quote: "staff packed the pills and organized them in sequential, sealed envelopes." Judgement comment: central allocation concealment
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "assignments were maintained at the Gynuity office in New York and were not disclosed to the participants or the providers." Judgement comment: adequate
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: participants and providers were blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "six women who were lost to follow-up and one woman who did not take misoprostol" Judgement comment: balanced between groups
Selective reporting (reporting bias)	Low risk	Judgement comment: both efficacy and side effects reported
Other bias	Low risk	Judgement comment: none

Coyaji 2007

Study characteristics	
Methods	Computer-generated random sequence; consecutive numbered opaque envelopes
Participants	≥ 18 years; 300 women randomised; gestational age < eight weeks; study conducted between January 2004-June 2005; no contraindications to study medication; lived or worked within one hour of the study site, agreed to provide an address and telephone number and return for a follow-up visit Study site: India (Pune and Mumbai)
Interventions	MF 200 mg followed after 48 hour by: • Group 1 400 µg oral MS and placebo 3 hour later • Group 2 400 µg MS repeated once 3 hour later
Outcomes	Complete abortion, side effects
Identification	
Notes	ITT done

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Consecutive numbered opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Placebo
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT done
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	No funding

Creinin 1994

Study characteristics	
Methods	Randomisation according to computer-generated random number table

Creinin 1994 *(Continued)*

	Numbered, sealed, opaque envelopes
Participants	63 pregnant women, ≤ 56 days of amenorrhoea, confirmed by ultrasound, San Francisco General Hospital **Exclusion criteria**: previous use of vitamins/folates, hematocrit ≤ 0.30, white blood cell count < 3000/µL, platelet count < 100 000/µL, haemoglobin < 10.0 mg/dL, aspartate aminotransferase > 2 times normal or active liver disease, serum creatinine > 1.5 mg/dL or active renal disease, inflammatory bowel disease, asthma, intolerance to the medication
Interventions	• Group 1: MTX 50 mg/m^2 IM and MS 800 µg/vaginal after 3 days • Group 2: MS 800 µg/vaginal
Outcomes	Complete abortion, duration of vaginal bleeding, side effects, change in beta-HCG levels
Identification	
Notes	Power calculation (80% power, significance level of 0.05) based on 95% success with MTX and 75% success with MS alone. The required sample size was 98. No mention of major complications

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation according to computer-generated random number table
Allocation concealment (selection bias)	Low risk	Numbered, sealed, opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Creinin 1995

Study characteristics

Methods	Randomisation according to computer-generated random number table Numbered, sealed, opaque envelopes No blinding

Creinin 1995 *(Continued)*

Participants	86 pregnant women, ≤ 56 days of amenorrhoea, confirmed by ultrasound, San Francisco General Hospital **Exclusion criteria**: previous use of vitamins/folates, hematocrit ≤ 0.30, white blood cell count < 3000/µL, platelet count < 100 000/µL, haemoglobin < 10.0 mg/dL, aspartate aminotransferase > 2 times normal or active liver disease, serum creatinine > 1.5 mg/dL or active renal disease, inflammatory bowel disease, asthma, intolerance to the medication
Interventions	MTX 50 mg/m^2 IM followed by: • Group 1: MS 800 µg/vaginal after 3 days • Group 2: MS 800 µg/vaginal after 7 days
Outcomes	Complete abortion, duration of vaginal bleeding, side effects, change in beta-HCG levels
Identification	
Notes	Power calculation (80% power, significance level of 0.05) No major complications occurred

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation according to computer-generated random number table
Allocation concealment (selection bias)	Low risk	Numbered, sealed, opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Creinin 1996a

Study characteristics

Methods	Randomisation according to random number tables Sealed, opaque envelopes were numbered by a by a person unrelated to the study No blinding
Participants	20 pregnant women, ≤ 49 days, confirmed by ultrasound, Magee-Women's Hospital, Pennsylvania, USA

Creinin 1996a *(Continued)*

	Exclusion criteria: previous use of vitamins/folates, haemoglobin < 10.0 mg/dL, aspartate aminotransferase > 2 times normal or active liver disease, serum creatinine > 1.5 mg/dL or active renal disease, inflammatory bowel disease, intolerance to the medication
Interventions	• Group 1: MTX 25 mg/orally followed by MS 800 µg/vaginal after 7 days • Group 2: MTX 50 mg/orally followed by MS 800 µg/vaginal after 7 days
Outcomes	Complete abortion, duration of vaginal bleeding, side effects, change in haemoglobin/aspartate transferase
Identification	
Notes	No major complications occurred

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation according to random number tables
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes were numbered by a person unrelated to the study
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Creinin 1997

Study characteristics	
Methods	Randomisation according to computer-generated random number table Numbered, sealed, opaque envelopes prepared by a person unrelated to the study No blinding
Participants	20 pregnant women, ≤ 49 days, confirmed by ultrasound, Magee-Women's Hospital, Pennsylvania, USA **Exclusion criteria**: previous use of vitamins/folates, hematocrit < 37%, white blood cell count < 3000/µL, platelet count < 100 000/µL, haemoglobin < 10.0 mg/dL, aspartate aminotransferase > 2 times normal or active liver disease, serum creatinine > 1.5 mg/dL or active renal disease, inflammatory bowel disease, asthma, intolerance to the medication

Creinin 1997 *(Continued)*

Interventions	• Group 1: MTX 50 mg/m^2 followed by MS 800 µg/vaginal after 7 days • Group 2: MTX 60 mg/m^2 followed by MS 800 µg/vaginal after 7 days
Outcomes	Complete abortion, time to passing of conceptus, side effects, MTX levels, change in haemoglobin/aspartate transferase
Identification	
Notes	No blinding No major complications were reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation according to computer-generated random number table
Allocation concealment (selection bias)	Low risk	Numbered, sealed, opaque envelopes prepared by a person unrelated to the study
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Creinin 2001a

Study characteristics

Methods	Random number tables in blocks of 10, sealed opaque envelopes prepared by person not involved in the trial
Participants	80 pregnant women, ≤ 49 days pregnant, single pregnancy, confirmed by ultrasound, at the University hospital Pittsburgh, USA **Exclusion criteria**: contraindication to MF/MS administration, haemoglobin < 10 gm/dL, cardiovascular disease, coagulopathies, IUD in situ, breastfeeding
Interventions	MF 100 mg (all) After 2 days, home administration: • Group 1: MS 400 µg oral

Creinin 2001a *(Continued)*

	• Group 2: MS 800 μg vaginal
Outcomes	Complete abortion, onset of bleeding and cramping, duration of bleeding, side effects
Identification	
Notes	Power calculation (80% power, significance level of 0.05) No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number tables in blocks of 10
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes prepared by person not involved in the trial
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	Departmental Research Fund, Magee-Women's Research Institute Family Planning Research Fund

Creinin 2001b

Study characteristics

Methods	Random number tables, sealed opaque envelopes
Participants	86 pregnant women, ≥ 18 years, ≤ 49 days pregnant, single pregnancy, at the University hospital Pittsburgh, USA **Exclusion criteria:** contraindication to MF/MS administration, haemoglobin < 10 gm/dL, cardiovascular disease, coagulopathies, IUD in situ, breastfeeding
Interventions	MF 600 mg (all) • Group 1: MS 400 μg after 6-8 h/oral • Group 2: MS 400 μg after 48 h/oral
Outcomes	Complete abortion, onset and duration of bleeding, side effects

Creinin 2001b *(Continued)*

Identification	
Notes	No blinding No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number tables
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Creinin 2004

Study characteristics

Methods	Computer-generated randomisation, permuted blocks, stratified by centre; sequentially numbered opaque envelopes
Participants	26 years old; 1080 women randomised. No more than 63 days' gestation confirmed by ultrasound; average gestational age of 51 days; willing to have surgical procedure and had a telephone; conducted in 2002-2003 in USA Magee-Women's Hospital in Pittsburgh, Pennsylvania, Columbia University, NY, Boston University, Massachusetts University of Rochester, NY
Interventions	MF 200 mg followed by MS 800 µg vaginal: • Group1: administered 6-8 h after MF • Group 2: administered 23-25 h after MF
Outcomes	Complete abortion, side-effects, bleeding, acceptability
Identification	
Notes	

Creinin 2004 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation, permuted blocks, stratified by centre
Allocation concealment (selection bias)	Low risk	Sequentially numbered opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT done
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Creinin 2007

Study characteristics	
Methods	Computer-generated random numbers, randomisation centrally, permuted block design with varying block sizes; randomisation after taking MF; no blinding; opaque envelopes
Participants	1128 women enrolled; mean age 27 years, women with no more than 63 days' gestation (mean gestational age 51-52 days' gestation) and willing to follow up and with access to a telephone **Exclusion criteria:** contraindications to MF or MS, haemoglobin < 10, IUD in place, on anticoagulants or with coagulopathy, active cervicitis or currently breastfeeding. Gestational age confirmed by ultrasound. 4 academic centres in the USA; University of Pittsburgh, Oregon Health and Science University, Northwestern University, University of Southern California
Interventions	MF 200 mg followed by: • Group 1: within 15 min, 800 µg MS vaginal • Group 2: 23-25 h later, 800 µg MS vaginal
Outcomes	Complete abortion; side-effects; bleeding; acceptability
Identification	
Notes	

Creinin 2007 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers, randomisation centrally, permuted block design with varying block sizes; randomisation after taking MF
Allocation concealment (selection bias)	Low risk	Opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT done
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Unclear risk	Dr. Creinin receives compensation from Danco Laboratories, LLC, the distributor of MF in the USA, for providing 3rd-party telephone consults to clinicians who call for expert advice on MF.

Dahiya 2011

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (sublingual) • Mean age (years): 27.51 ± 5.08 • Mean pregnancy age (days): 45.6 ± 7.79 • Parity: 2 MF + MS (oral) • Mean age (years): 27.04 ± 4.44 • Mean pregnancy age (days): 45.31 ± 8.23 • Parity: 2 Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Parity: NR

Dahiya 2011 *(Continued)*

	Inclusion criteria: intrauterine pregnancy < 56 days based on menstrual history and clinical examination, willing for follow-up visits **Exclusion criteria**: bronchial asthma, glaucoma, sickle cell anaemia, adrenal disease, hypertension, heart disease, haemoglobin < 10 g/dL, history or evidence of thromboembolism, liver disease, known allergy to prostaglandins and suspected or proven ectopic pregnancy **Pretreatment**: none
Interventions	**Intervention characteristics** MF + MS (sublingual) • MF (200 mg, oral) + MS (400 µg sublingual), 24 h interval MF + MS (oral) • MF (200 mg, oral) + MS (400 µg oral), 24 h interval
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Tiredness • Outcome type: adverse event • Direction: lower is better • Data value: endpoint Cramping • Outcome type: adverse event • Direction: lower is better • Data value: endpoint Flush • Outcome type: adverse event Ongoing pregnancy

Dahiya 2011 *(Continued)*

	• Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Department of Obstetrics and Gynecology, Pt. B.D. Sharma PGIMS Rohtak **Country:** India **Setting:** Postpartum Center at PGIMS Rohtak **Comments:** no **Author(s):** Krishna Dahiya **Institution:** Department of Obstetrics and Gynecology **Email:** krishnadahiya@rediffmail.com **Address:** Pt. B.D. Sharma PGIMS Rohtak, 74-R, Model Town, Rohtak 124001, Haryana, India
Notes	**Follow-up duration:** 7 days; without detailed data between groups for bleeding and satisfaction

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "Group assignment was done in a randomized fashion by computer generated random tables." Judgement comment: adequate
Allocation concealment (selection bias)	High risk	Judgement comment: none
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "prospective, open label, randomized" Judgement comment: open
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: open
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement comment: detailed outcome data between groups for bleeding and satisfaction NR
Selective reporting (reporting bias)	Unclear risk	Judgement comment: registered protocol was not found
Other bias	Low risk	Judgement comment: none

El-Refaey 1994

Study characteristics

Methods	Sealed, opaque envelopes

El-Refaey 1994 *(Continued)*

	Random assignment before MS administration
Participants	150 pregnant women ≤ 56 days of amenorrhoea, confirmed by ultrasound
Interventions	• Group 1: MF 200 mg and MS 800 µg/oral after 48 h • Group 2: MF 200 mg and MS 400 µg after 48 h plus 400 µg 2 h later/oral
Outcomes	Changes in blood pressure, pulse rate and temperature Complete and incomplete abortion Ongoing pregnancy Side effects Bleeding pattern
Identification	
Notes	Power calculation (5% significance level to detect a 20% reduction in incidence of side effects) No mention of major complications

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random assignment before MS administration
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

El-Refaey 1995

Study characteristics

Methods	Computer-generated random assignment before MS administration, sealed opaque envelopes
Participants	270 women ≤ 63 days of amenorrhoea, confirmed by ultrasound **Exclusion criteria**: contraindication for the use of MF and/or MS
Interventions	• Group 1: MF 600 mg and MS 800 µg/orally after 48 h

El-Refaey 1995 *(Continued)*

	• Group 2: MF 600 mg and MS 800 µg/vaginally (self-administration) after 48 h
Outcomes	Complete, incomplete and missed abortion Ongoing pregnancy Expulsion within 4 h Expulsion without need for surgery Side effects
Identification	
Notes	Power calculation (5% significance level to detect difference of 10% in the incidence of women aborting within 4 h vaginal MS by self-administration) 1 woman received a blood transfusion (Group 2)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random assignments
Allocation concealment (selection bias)	Low risk	Numbered, sealed, opaque envelopes contained the computer-generated random assignments.
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Fekih 2010

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (sublingual) • Mean age (years): 29.89 (5.83) • Mean pregnancy age (days): 45.47 (3.9) • BMI: 23.44 (6.5)

Fekih 2010 *(Continued)*

	• Gravidity: 3.5 (1.9) • Previous abortion: 0.6 (0.98) • Haemoglobin (g/dL): 12.19 (0.96) MF + MS (oral) • Mean age (years): 30.42 (5.5) • Mean pregnancy age (days): 44.66 (3.7) • BMI: 23.9 (4.5) • Gravidity: 3.35 (1.47) • Previous abortion: 0.46 (0.79) • Haemoglobin (g/dL): 12.27 (0.86) Overall • Mean age (years): NR • Mean pregnancy age (days): NR • BMI: NR • Gravidity: NR • Previous abortion: NR • Haemoglobin (g/dL): NR **Inclusion criteria**: intrauterine pregnancy of ≤ 56 days from their last menstrual period, determined by vaginal probe ultrasound and a maximum embryonic length of 17 mm **Exclusion criteria**: multiple pregnancy; < 20 years of age; haemolytic disorders; active bronchial asthma; inability to reach the hospital in < 1 h; more than 3 scars in the uterus; and valvulopathy **Pretreatment:** none **Study duration:** From 2007 to January 2008
Interventions	**Intervention characteristics** MS (sublingual) • 800 µg of sublingual MS repeated every 4 h for up to a maximum of 3 doses MF + MS (oral) • 200 mg of oral MF followed by 400 µg of oral MS
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Nausea • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Abdominal pain

Fekih 2010 *(Continued)*

	• Outcome type: adverse event Fever • Outcome type: adverse event Severe bleeding • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Bleeding duration • Outcome type: continuous outcome • Unit of measure: days • Direction: lower is better • Data value: endpoint Induction expulsion interval • Outcome type: continuous outcome • Unit of measure: minutes • Direction: lower is better • Data value: endpoint Satisfaction • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint
Identification	**Sponsorship source:** Department of Obstetrics and Gynecology, Farhat Hached Teaching Hospital, Sousse, Tunisia **Country:** Tunisia **Setting:** Department of Obstetrics and Gynecology, Farhat Hached Teaching Hospital **Comments:** no **Author(s):** Myriam Fekih **Institution:** Department of Obstetrics and Gynecology, Farhat Hached Teaching Hospital, Sousse, Tunisia **Email:** Fekihm2002@yahoo.com **Address:** Hached Teaching Hospital, 4000, Sousse, Tunisia
Notes	Follow-up: 2 weeks; none lost to follow-up Detailed outcome data between groups NR for hematocrit and haemoglobin

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "We used a computer-generated randomization sequence to assign the participants to one of 2 groups." Judgement comment: adequate

Fekih 2010 *(Continued)*

Allocation concealment (selection bias)	Low risk	Quote: "The assigned treatment group was written on a card and sealed in opaque envelopes that were consecutively numbered and opened immediately before the first drug dose was administered." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: none
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: none
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: none lost to follow-up
Selective reporting (reporting bias)	Low risk	Judgement comment: efficacy and side effects were reported, but no registration information
Other bias	Low risk	Judgement comment: none

Garg 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (buccal) • Mean age (years): 29.96 (4.54) • Mean pregnancy age (weeks): 6.13 (0.58) • BMI (kg/m^2): 24.51 (2) MF + MS (vaginal) • Mean age (years): 30.2 (4.69) • Mean pregnancy age (weeks): 5.98 (0.57) • BMI (kg/m^2): 22.88 (3.65) Overall • Mean age (years): NR • Mean pregnancy age (weeks): NR • BMI (kg/m^2): NR **Inclusion criteria:** age ≥ 18 years; requesting an elective termination of pregnancy; intrauterine pregnancy of ≤ 49 days; willing to undergo required follow-up and surgical management **Exclusion criteria:** contraindication to MF or MS; prior intervention in the present pregnancy; extrauterine pregnancy; pelvic infection; haemoglobin< 10 g/dL; clotting defect; anticoagulation therapy; cardiovascular disease; breastfeeding

Garg 2015 *(Continued)*

	Pretreatment: none
Interventions	**Intervention characteristics** MF + MS (buccal) • 200 mg MF orally, 48 h later 800 µg MS buccal MF + MS (vaginal) • 200 mg MF orally, 48 h later 800 µg MS vaginal
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Vomiting • Outcome type: adverse event Fever • Outcome type: adverse event Chills • Outcome type: adverse event Satisfied • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint
Identification	**Sponsorship source:** Department of Obstetrics and Gynecology, Government Multispeciality Hospital **Country:** India **Setting:** Department of Obstetrics and Gynecology, Government Multispeciality Hospital **Comments:** no **Author(s):** Garg Geetika **Institution:** Department of Obstetrics and Gynecology, Government Multispeciality Hospital **Email:** geetika_smiles@yahoo.co.in **Address:** Sector 16, Chandigarh, India
Notes	Follow-up 2 and 6 weeks

Risk of bias

Garg 2015 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "random number tables." Judgement comment: adequate
Allocation concealment (selection bias)	High risk	Judgement comment: no concealment
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no dropouts
Selective reporting (reporting bias)	Unclear risk	Judgement comment: protocol was not available
Other bias	Low risk	Judgement comment: none

Goel 2011

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (simultaneously) • Mean age (years): 25.65 (2.41) • Mean pregnancy age (days): 36.52 (3.03) • Previous abortion: 37.5% MF + MS (24 h interval) • Mean age (years): 24.92 (2.45) • Mean pregnancy age (days): 35.3 (4.08) • Previous abortion: 45% Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Previous abortion: NR **Inclusion criteria:** requesting an elective abortion and had a single intrauterine pregnancy of < 7 weeks (49 days) of gestation

Goel 2011 *(Continued)*

	Exclusion criteria: conceived with an IUD in situ or with a history of > 2 cesarean sections; history of allergy to prostaglandins, bronchial asthma, hypertension, coronary artery disease, arrhythmias, renal or hepatic dysfunction, chronic adrenal failure and patients on anticoagulants and corticosteroids **Pretreatment:** none **Study duration:** From October 2009 to July 2010
Interventions	**Intervention characteristics** MF + MS (simultaneously) • 200 mg of MF orally and 400 µg of MS vaginally simultaneously MF + MS (24 h interval) • 200 mg of MF orally and 400 µg of MS vaginally at 24-h interval
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Severe bleeding • Outcome type: adverse event Interval from MS administration to abortion • Outcome type: continuous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Bleeding duration • Outcome type: continuous outcome • Reporting: fully reported

Goel 2011 *(Continued)*

	• Unit of measure: days • Direction: lower is better • Data value: endpoint Ongoing pregnancy • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Satisfactory • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: higher is better • Data value: endpoint Repeat MS • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Obstetrics and Gynaecology Department, MMIMSR **Country:** India **Setting:** Obstetrics and Gynaecology Department **Comments:** no **Author(s):** Anupama Goel **Institution:** Obstetrics and Gynaecology Department, MMIMSR **Email:** hiyagoel@yahoo.com **Address:** Street no. 1/2, Shastri Nagar, Jagraon (Distt. Ludhiana), Punjab 142026, India
Notes	**Follow-up duration:** 14 days

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "Women were randomized in blocks of eight using a random number table" Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "person not linked to the clinical study performed the randomization sequence and prepared the envelopes. Women were asked to open the next sequentially numbered sealed envelope and assigned to a group accordingly." Judgement comment: adequate
Blinding of participants and personnel (performance bias)	High risk	Quote: "The researchers were not blinded to group assignment, and randomization was performed solely to avert any recruitment bias."

Goel 2011 *(Continued)*

All outcomes		Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: all participants were followed up
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol
Other bias	Low risk	Judgement comment: none

Guest 2007

Study characteristics	
Methods	Computer-generated fixed blocks of 20; 1:1 randomisation; sealed opaque envelopes
Participants	450 women aged 24-26 years; no more than 63 days' gestation confirmed by ultrasound (average 51 days of gestation) **Exclusion criteria:** contraindications for study medication, breastfeeding, haemoglobin < 10, coagulopathy or treatment with anticoagulants, IUD in situ, presence of cardiovascular disease, ectopic pregnancy; study conducted between September 2003-March 2005
Interventions	MF 200 mg followed by: • Group 1: 800 µg vaginal MS after 6 h • Group 2: 800 µg of vaginal MS after 36-48 h later
Outcomes	Complete abortion, side effects, acceptability
Identification	
Notes	ITT analysis

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated fixed blocks of 20; 1:1 randomisation; sealed opaque envelopes
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias)	High risk	No blinding

Guest 2007 *(Continued)*

All outcomes		
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Hamoda 2005

Study characteristics	
Methods	Random number tables; sealed opaque envelopes
Participants	340 women, average age 24 years randomised; average gestational age 65-68 days **Exclusion criteria:** < 16 years, severe asthma, haemorrhagic disorders and treatment with anticoagulants, known allergy to prostaglandins, history of cardiac disease, smoking over the age of 35 years with ECG abnormalities, breastfeeding Study conducted at Aberdeen Royal Infermary, UK, from July 2002-October 2003
Interventions	MF 200 mg, followed 36-48 h after by: • Group 1: 600 µg sublingual MS, followed 3 h later by 400 µg (if 9-13 weeks' gestation, a 3rd dose of 400 µg was administered) • Group 2: 800 µg vaginal MS, followed 3 h later by 400 µg (if 9-13 weeks' gestation, a 3rd dose of 400 µg was administered)
Outcomes	Complete/incomplete abortion, missed abortion, continuing pregnancy
Identification	
Notes	No ITT; loss to follow-up identical (13) in each group (total 26)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number tables
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding

Hamoda 2005 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No ITT
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Iyengar 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (home) • Mean age (years): 26.98 (4.78) • Primiparity: 5% • Previous induced abortion: 31% MF + MS (clinic) • Mean age (years): 27.22 (4.9) • Primiparity: 5% • Previous induced abortion: 34% Overall • Mean age (years): NR • Primiparity: NR • Previous induced abortion: NR **Inclusion criteria:** ≥ 18 years; women presenting with a positive urine test and uterine size equivalent to or up to 9 + 0 weeks (63 days) of gestation; opting for medical abortion; residing in an area where follow-up is feasible; woman agrees for a follow-up contact at 10-14 days **Exclusion criteria:** women with contraindications to medical abortions; haemoglobin level < 85; age < 18 years **Pretreatment:** none **Study duration:** From April 23, 2013 to May 15, 2014
Interventions	**Intervention characteristics** MF + MS (home) • 200 mg MF and 800 µg MS (oral, vaginal or sublingual): a pictorial instruction sheet, a low-sensitivity urine pregnancy test at home, contacted by a home visit or telephone call MF + MS (clinic)

Iyengar 2015 *(Continued)*

	• 200 mg MF and 800 µg MS (oral, vaginal or sublingual): doctor or nurse assessed the abortion outcome, did a low-sensitivity urine pregnancy test; research assistants undertook the follow-up interviews
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Side effects • Outcome type: adverse event Bleeding • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Repeated medical abortion • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Hospitalisation • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Blood transfusion • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Satisfaction • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint
Identification	**Sponsorship source:** Swedish Research Council and Swedish International Development Agency **Country:** India **Setting:** 6 health centres (3 rural, 3 urban) in Rajasthan **Comments:** low-resource setting **Author(s):** Kirti Iyengar **Institution:** Department of Women's and Children's Health, Karolinska Institutet/University Hospital, WHO Collaborating Centre

Iyengar 2015 *(Continued)*

	Email: kirti.iyengar@ki.se **Address:** SE-17176 Stockholm, Sweden **NCT01827995:** NCT01827995
Notes	Follow-up: 14 days, 30 days ITT analysis was performed by study authors.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "randomly assigned (1:1) to one of two groups by a research assistant: clinic follow-up or home assessment. Randomisation was done with a computer- generated randomisation sequence, with a block size of six. The sequence was generated at the coordinating centre based in Udaipur, India." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Sealed opaque envelopes containing the random allocation were numbered consecutively by an independent staff member at Udaipur, and were sent to the study sites." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "Blinding of the groups from research assistants and clinical staff was not possible," Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "home visits or telephone calls. 13 women (six in the clinic follow-up group and seven in the home-assessment group) did not use MS and were excluded from the analysis, and 18 women (11 in the clinic follow-up group and seven in the home- assessment group) were lost to follow-up (figure 2). 700 women (clinic follow-up, n=336;" Judgement comment: ITT analysis was performed by study authors.
Selective reporting (reporting bias)	Low risk	Quote: "This study is registered with ClinicalTrials.gov, number NCT01827995." Judgement comment: efficacy and side effects were all reported
Other bias	Low risk	Judgement comment: none

Jain 1999

Study characteristics

Methods	Randomisation using random number tables
Participants	150 women pregnant ≤ 56 days confirmed by ultrasound

Jain 1999 *(Continued)*

	Exclusion criteria: cervical dilatation, anaemia, pelvic inflammatory disease, uterine bleeding, uterine leiomyomata, serious medical problems, allergy or contraindications to the study medication
Interventions	• Group 1: TM 20 mg/twice daily and MS 800 µg/vaginally after 48 h • Group 2: placebo twice daily and MS 800 µg/vaginally after 48 h
Outcomes	Complete/incomplete abortion, ongoing pregnancy, complications, side effects
Identification	
Notes	Treatment and placebo were placed in identical capsules No major complications were reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation using random number tables
Allocation concealment (selection bias)	Low risk	Opaque capsule
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Double-blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Double-blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Jain 2002

Study characteristics

Methods	Computer-generated random table, opaque vials
Participants	250 healthy women, ≤ 56 days of amenorrhoea, confirmed by ultrasound **Exclusion criteria**: evidence of threatened spontaneous abortion, uterine infection, anaemia, bleeding disorders, cardiovascular or cerebrovascular disease, uterine leiomyomata, allergy against the study medication
Interventions	• Group 1: MF 200 mg, MS 800 µg/vaginally on day 3, repeated on day 4 if gestational sac present • Group 2: placebo, MS 800 µg/vaginally on day 3, repeated on day 4 if gestational sac present

Jain 2002 *(Continued)*

Outcomes	Complete abortion, side effects
Identification	
Notes	Placebos were vitamin C tablets (not identical); opaque vials were used to blind the investigator Power calculation (5% significance level to detect a 5% difference in success rates between the 2 study groups) No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random table
Allocation concealment (selection bias)	Low risk	Opaque vials
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Investigator and participants were blinded
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Investigator was blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Javadi 2015

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** LET + MS • Mean age (years): 26 ± 5.7 MS • Mean age (years): 27.7 ± 5.6 Overall

Javadi 2015 *(Continued)*

	• Mean age (years): NR **Inclusion criteria:** therapeutic abortion with gestational age of < 12 weeks; missed abortion; > 18 years; haemoglobin concentration (Hb) ≥ 10 g/dL. **Exclusion criteria**: history of previous cesarean section; history of adrenal disease, steroid-dependent cancer, liver disease, severe or recurrent, bronchial asthma, evidence or history of thromboembolism, history of renal disease, blood pressure > 95 mmHg, history of smoking **Pretreatment:** none
Interventions	**Intervention characteristics** LET + MS • LET 10 mg, oral, 3 days; 800 µg MS on the 3rd day, vaginal, in a single dose MS • 4 placebo tablets; 800 µg MS on the 3rd day, vaginal, in a single dose
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Induction to abortion interval • Outcome type: continuous outcome • Reporting: fully reported • Unit of measure: h • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Children Growth Research Center, Qazvin University of Medical Sciences **Country:** Iran **Setting:** Kowsar hospital of Qazvin **Author(s):** Ameneh Barikani **Institution:** Children Growth Research Center, Qazvin University of Medical Sciences, Qazvin, Iran **Email:** barikani.a@gmail.com **Address:** Children Growth Research Center, Qazvin University of Medical Sciences, Qazvin, Iran **IRCT2015050119037N8:** IRCT2015050119037N8
Notes	Follow-up duration was unclear; without detailed data on side effects

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "randomized" Judgement comment: without detailed method
Allocation concealment (selection bias)	High risk	Judgement comment: none reported

Javadi 2015 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "double-blind study" Judgement comment: double-blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: double
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no dropouts
Selective reporting (reporting bias)	Unclear risk	Judgement comment: detailed data for side effects NR
Other bias	Low risk	Judgement comment: none

Klingberg Allvin 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (midwives) • Mean age (years): 26.1 (6.5) • Mean pregnancy age (weeks): 8.7 (2.3) • Number of pregnancies: 3.4 (2.4) MS (physicians) • Mean age (years): 26.5 (6.5) • Mean pregnancy age (weeks): 8.8 (2.1) • Number of pregnancies: 3.5 (2.2) Overall • Mean age (years): 26.3 (6.5) • Mean pregnancy age (weeks): 8.8 (2.2) • Number of pregnancies: 3.4 (2.3) **Inclusion criteria:** women with signs of incomplete abortion i.e. contractions, pain and vaginal bleeding during pregnancy, an open cervical os and sometimes partial expulsion of products of conception **Exclusion criteria:** a uterine size of > 12 weeks of gestation; complete abortion; suspected ectopic pregnancy; unstable haemodynamic status and shock; signs of pelvic infection or sepsis; and a known allergy to MS **Pretreatment:** significantly more women in the midwife group reported to have induced the current abortion (14.6% vs 9.7%, P < 0.05) Study duration: April 2013 to June 2014

Klingberg Allvin 2015 *(Continued)*

Interventions	**Intervention characteristics** MS (midwives) • 1 single dose of 600 µg MS orally provided by midwives MS (physicians) • 1 single dose of 600 µg MS orally provided by physicians
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Chills • Outcome type: adverse event Bleeding duration • Outcome type: continuous outcome • Unit of measure: days • Direction: lower is better • Data value: endpoint Satisfaction • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint Pain (VAS) • Outcome type: continuous outcome • Reporting: fully reported • Range: 0-10 • Direction: lower is better • Data value: change from baseline Severe bleeding • Outcome type: dichotomous outcome • Direction: lower is better

Klingberg Allvin 2015 *(Continued)*

	• Data value: endpoint Unscheduled visit • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** The Swedish Research Council (521-2009-2605), Karolinska Institutet, and Dalarna University **Country:** Uganda **Setting:** 3 hospitals and 3 healthcare centres level IV in rural, peri-urban and urban settings **Comments:** ClinicalTrials.org, NCT 01844024 **Author(s):** Amanda Cleeve **Institution:** Department of Women's and Children's Health, Karolinska Institutet, Stockholm, Sweden **Email:** amanda.cleeve@ki.se **Address:** Department of Women's and Children's Health, Karolinska Institutet, Stockholm, Sweden **ClinicalTrials.org, NCT 01844024** **Study duration:** April 2013 to June 2014
Notes	ITT analyses were performed by review authors .

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "The randomisation (1:1) was done in blocks of 12 and was stratified for study site. We used a computer random number generator to generate a list of codes from 1 to 994 and each code was linked to one of the two study groups." Judgement comment: adequate; computer/block/stratified randomisation
Allocation concealment (selection bias)	Low risk	Quote: "Sequentially numbered, opaque, sealed envelopes, each containing a random allocation, were prepared at the coordinating centre, and later opened in consecutive order by the research assistants after obtaining written consent." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "The study was not masked to the study participants or providers." Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding

Klingberg Allvin 2015 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement comment: 34 and 23 participants were excluded or lost to follow-up after allocation in 2 groups respectively, just per-protocol analyses were conducted
Selective reporting (reporting bias)	Low risk	Judgement comment: NCT01844024
Other bias	Low risk	Judgement comment: none

Koopersmith 1996

Study characteristics	
Methods	Randomisation into 3 groups Randomisation procedure not stated
Participants	58 women, pregnant ≤ 10 weeks, confirmed by ultrasound, University Hospital Los Angeles, USA **Exclusion criteria**: uterine infection, prior uterine bleeding, cervical dilatation, anaemia, cardiovascular or cerebral disease, allergy to MS
Interventions	• Group A: MS 100 µg/vaginally/ 8-hourly to a maximum of 6 doses • Group B: MS 100 µg/vaginally/ 8-hourly to a maximum of 6 doses and TM 10 mg/orally after the 1st dose of MS • Group C: MS 100 µg/vaginally/ 8-hourly to a maximum of 6 doses and laminaria/intracervical immediately before the 1st dose of MS The dose of MS was increased after the success rate was unsatisfactory after the first 26 women
Outcomes	Complete abortion, failure rate, side effects, mean number of doses of MS used, time until passing of conceptus
Identification	
Notes	No mention of major complications

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No detailed information about randomisation
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias)	Low risk	No attrition

Koopersmith 1996 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Kopp Kallner 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** Termination of pregnancy (midwives) • Mean age (years): 27 • Mean pregnancy age (days): 45 • Gravidity: 2 • Parity: 0 • Miscarriage: 0 • Caesarian section: 0 Termination of pregnancy (physicians) • Mean age (years): 27 • Mean pregnancy age (days): 45 • Gravidity: 2 • Parity: 0 • Miscarriage: 0 • Caesarian section: 0 Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Gravidity: NR • Parity: NR • Miscarriage: NR • Caesarian section: NR **Inclusion criteria:** women seeking early medical termination of pregnancy; pregnancy of ≤ 63 days; ≥ 18 years; in good general health with no continuing medication for chronic disease **Exclusion criteria:** contraindication for medical termination of pregnancy **Pretreatment:** none **Study duration:** 28 February 2011 to 25 July 2012
Interventions	**Intervention characteristics** Termination of pregnancy (midwives)

Kopp Kallner 2015 *(Continued)*

	• Termination of pregnancy (MF 200 mg; 800 µg of MS vaginally 24-48 h after MF; if no bleeding occurred within 3 h, additional dose of 400 µg of MS orally) Termination of pregnancy (physicians) • Termination of pregnancy (MF 200 mg; 800 µg of MS vaginally 24-48 h after MF; if no bleeding occurred within 3 h, additional dose of 400 µg of MS orally)
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Blood transfusion • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Acceptability • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint Complication • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better • Data value: endpoint Unscheduled visits • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Completion • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint
Identification	**Sponsorship source:** Department of Women's and Children's Health, Karolinska Institutet, Karolinska University Hospital; Swedish Research Council; the Swedish Council for Working Life and Social Research; Stockholm County Council and Karolinska Institutet **Country:** Sweden **Setting**: outpatient family planning clinic of the Karolinska University Hospital **Author(s):** K Gemzell-Danielsson **Institution:** Division of Obstetrics and Gynaecology, Department of Women's and Children's Health, Karolinska Institutet, Karolinska University Hospital **Email:** Kristina.Gemzell@ki.se

Kopp Kallner 2015 *(Continued)*

	Address: 171 77 Stockholm, Sweden **NCT01612923:** NCT01612923
Notes	54 and 76 participants were lost to follow-up, and ITT analyses were performed by review authors but not the study authors.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "computer-generated randomisation allocation code, in blocks of ten women, at a ratio of 1: 1." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "consecutive opening of numbered opaque and sealed envelopes" Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "study was not blinded." Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	High risk	Quote: "In the nurse-midwife group 54 women were lost to follow–up, compared with 76 women in the standard-care group (130 women in total, P = 0.038). The flow of patients is presented in Figure 1. Analysis was performed per protocol." Judgement comment: 130 participants were lost to follow-up, but study authors did not perform ITT analysis
Selective reporting (reporting bias)	Low risk	Quote: "NCT01612923," Judgement comment: all outcomes were reported as per the protocol
Other bias	Low risk	Judgement comment: none

Lee 2011

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** LET + MS • Mean age (years): 30.4 (7.2) • Mean pregnancy age (days): 49.6 (6.5) • Primigravida: 24 (28.6)

Lee 2011 *(Continued)*

	• Miscarriage: 13 (15.5) • Height (cm): 159.2 (4.9) • Weight (kg): 55.85 (9) MS • Mean age (years): 30.6 (6.9) • Mean pregnancy age (days): 49.4 (6.2) • Primigravida: 21 (25) • Miscarriage: 11 (13.1) • Height (cm): 158.5 (5) • Weight (kg): 53.8 (7.7) Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Primigravida: NR • Miscarriage: NR • Height (cm): NR • Weight (kg): NR **Inclusion criteria:** up to 63 days of gestation; good general health; > the age of legal consent (i.e. > 18 years); haemoglobin > 10 g/L **Exclusion criteria:** adrenal pathology, steroid-dependent cancer, porphyria, DBP > 95 mmHg, bronchial asthma, arterial hypotension; thromboembolism, severe or recurrent liver disease or pruritus of pregnancy; regular use of prescription drugs before admission to the study; presence of an IUD; any abnormal values in pretreatment blood tests, namely complete blood picture, renal and liver function tests (which include serum urea, creatinine, electrolytes, albumin, globulin, and liver enzymes **Pretreatment:** none **Study duration:** September 1, 2008 to September 30, 2009
Interventions	**Intervention characteristics** LET + MS • LET 10 mg daily for 3 days followed by 800 µg of vaginal MS MS • placebo for 3 days followed by 800 µg of vaginal MS
Outcomes	Failure to achieve complete abortion • **O**utcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea

Lee 2011 *(Continued)*

- Outcome type: adverse event

Dizziness

- Outcome type: adverse event

Headache

- Outcome type: adverse event

Abdominal pain

- Outcome type: adverse event

Fever

- Outcome type: adverse event

Chills

- Outcome type: adverse event

Bleeding

- Outcome type: continuous outcome
- Unit of measure: days
- Direction: lower is better
- Data value: endpoint

Interval from MS administration to abortion

- Outcome type: continuous outcome
- Unit of measure: h
- Direction: lower is better
- Data value: endpoint

Ongoing pregnancy

- Outcome type: dichotomous outcome
- Direction: lower is better
- Data value: endpoint

Failure to achieve complete abortion (< 49 days)

- Outcome type: dichotomous outcome
- Direction: lower is better
- Data value: endpoint

Failure to achieve complete abortion (50-63 days)

- Outcome type: dichotomous outcome
- Direction: lower is better
- Data value: endpoint

Ongoing pregnancy (< 49 days)

- Outcome type: dichotomous outcome
- Direction: lower is better
- Data value: endpoint

Ongoing pregnancy (50-63 days)

- Outcome type: dichotomous outcome

Lee 2011 *(Continued)*

	• Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** General Research Fund of the Research Grants Council of Hong Kong **Country:** China **Setting:** Department of Obstetrics and Gynecology, The University of Hong Kong **Author(s):** Vivian Chi Yan Lee **Institution:** Department of Obstetrics and Gynecology, The University of Hong Kong, Queen Mary Hospital **Email:** v200lee@hku.hk **Address:** Pokfulam Road, Hong Kong HKU 765508M, www.hkclinicaltrials.com
Notes	**Follow-up duration:** 2 weeks and 6 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "The participants were randomized into the two groups by the hospital pharmacists according to computer-generated random numbers." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "The packages of letrozole (10 mg daily for 3 days) and packages of similar number of placebo tablets, which had the same appearance and taste, were prepared by the hospital pharmacy according to the randomization schedule." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "Until the completion of the study, both patients and the clinicians were blinded to the group assigned." Judgement comment: adequate
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: adequate
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: ITT analysis was performed by study authors
Selective reporting (reporting bias)	Low risk	Quote: "HKClinicalTrials.com, http://www.hkclinicaltrials.com, HKCTR-349." Judgement comment: efficacy and side effects were both reported
Other bias	Low risk	Judgement comment: none

Li 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF (150) + MS • Mean age (years): 27.2 (6.8) • Amenorrhea (days): 31.1 (2.3) • Menstrual cycle (days): 29.1 (1.4) • HCG (IU/L): 756.4 (123.2) MF (125) + MS • Mean age (years): 27.1 (7.6) • Amenorrhea (days): 30.6 (2.8) • Menstrual cycle (days): 29 (1.5) • HCG (IU/L): 692.3 (152.3) MF (100) + MS • Mean age (years): 27.2 (8.3) • Amenorrhea (days): 31.2 (1.8) • Menstrual cycle (days): 29.2 (1.3) • HCG (IU/L): 683.2 (168.6) MF (75) + MS • Mean age (years): 27.1 (5.2) • Amenorrhea (days): 30.6 (1.5) • Menstrual cycle (days): 29 (1.3) • HCG (IU/L): 629.7 (181.1) MF (50) + MS • Mean age (years): 27.3 (7.5) • Amenorrhea (days): 30.7 (1.8) • Menstrual cycle (days): 29 (1.1) • HCG (IU/L): 620.4 (162) Overall • Mean age (years): NR • Amenorrhea (days): NR • Menstrual cycle (days): NR • HCG (IU/L): NR **Inclusion criteria:** early pregnancy and regular menstrual cycle and amenorrhoea lasting no more than 35 days **Exclusion criteria:** significant end organ diseases, hormone therapy, haematologic diseases, allergy to MF or MS, pregnancy with IUD in situ, and threatened or spontaneous abortion **Pretreatment:** none **Study duration:** July 2011 to June 2013

Li 2015 *(Continued)*

Interventions	**Intervention characteristics** MF (150) + MS • 150 mg oral MF (6 pills of MF only) and 200 µg oral MS 24 h later MF (125) + MS • 125 mg oral MF (5 pills of MF plus 1 pill of placebo) and 200 µg oral MS 24 h later MF (100) + MS • 100 mg oral MF (4 pills of MF plus 2 pills of placebo) and 200 µg oral MS 24 h later MF (75) + MS • 75 mg oral MF (3 pills of MF plus 3 pills of placebo) and 200 µg oral MS 24 h later MF (50) + MS • 50 mg oral MF (2 pills of MF plus 4 pills of placebo) and 200 µg oral MS 24 h later
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Induction expulsion interval (> 6 h) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Induction expulsion interval *(*< 6 h) • Outcome type: dichotomous outcome

Li 2015 *(Continued)*

	• Direction: lower is better • Data value: endpoint Bleeding duration • Outcome type: continuous outcome • Unit of measure: days • Direction: lower is better • Data value: endpoint Prolonged menstruation • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Third affiliated hospital of Guangzhou Medical University **Country:** China **Setting:** Third Affiliated Hospital of Guangzhou Medical University and 3 other affiliated teaching hospitals of the University in the same city **Author(s):** Cui-Lan Li **Institution:** Third Affiliated Hospital of Guangzhou Medical University & Key Laboratory for Major Obstetric Diseases of Guangdong Province **Email:** ninacuilanli@gmail.com **Address:** 63 Duobao Road, Liwan District, Guangzhou 510145, China
Notes	Unclear follow-up duration

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "Enrolled patients were randomly assigned to 5 groups and sequentially prescribed medications."
Allocation concealment (selection bias)	Low risk	Quote: "Nurses No. 5 and 6 sealed the medications randomly in numbered containers group by group and made records in computer at the same time." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "All nurses involved in the process were all blinded to the study. The author L. P. Song who was in charge of the medications allocation was blinded to the medication regimens." Judgement comment: adequate
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Quote: "All investigators and researchers were blinded to the treatment administrated." Judgement comment: adequate
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "79 patients dropped from the study for personal reasons, but there were no demographic or other differences between these patients and the recruited patients."

Li 2015 *(Continued)*

		Judgement comment: balanced dropouts
Selective reporting (reporting bias)	Unclear risk	Judgement comment: protocol unavailable
Other bias	Low risk	Judgement comment: none

Li 2017

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (self-administration) • Mean age (years): 27.1 (6.3) • Amenorrhea (days): 32.1 (3.2) • Menstrual cycle (days): 29.2 (1.3) • HCG (IU/L): 724.6 (152.7) MF + MS (hospital administration) • Mean age (years): 26.8 (4.9) • Amenorrhea (days): 32.4 (2.7) • Menstrual cycle (days): 29.1 (1.4) • HCG (IU/L): 705.6 (131.8) Overall • Mean age (years): NR • Amenorrhea (days): NR • Menstrual cycle (days): NR • HCG (IU/L): NR **Inclusion criteria:** women with ultra-early pregnancy (amenorrhoea < 35 days) and regular menstrual cycles who sought medical abortion **Exclusion criteria:** end-stage organ failure, hormone replacement treatment, haematological diseases, allergy to MS or MF, ectopic pregnancy, spontaneous or threatened abortion, and pregnancy occurring during use of an IUD **Pretreatment:** none **Study duration:** February 2012 to May 2015
Interventions	**Intervention characteristics** MF + MS (self-administration) • 75 mg oral MF; followed by oral 400 µg MS 24 h later: 3 urine HCG self-detections were scheduled at 24 h, 1 week, and 2 weeks after MS; 2 telephone calls were scheduled MF + MS (hospital administration)

Li 2017 *(Continued)*

	• 75 mg oral MF; followed by oral 400 µg MS 24 h later: follow-up visits to the hospital involved serum b-hCG detection and vaginal ultrasonic examination
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Unscheduled re-attendance • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Ongoing pregnancy • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Irregular menstruation • Outcome type: dichotomous outcome Bleeding duration • Outcome type: continuous outcome • Unit of measure: days Satisfied • Outcome type: dichotomous outcome Time spent in hospital • Outcome type: continuous outcome • Unit of measure: hours • Direction: lower is better • Data value: endpoint Cost expenditures in hospital • Outcome type: continuous outcome • Unit of measure: USD

Li 2017 *(Continued)*

	• Direction: lower is better • Data value: change from baseline
Identification	**Sponsorship source:** Third Affiliated Hospital of Guangzhou Medical University **Country:** China **Setting:** an institute of obstetrics and gynaecology and 4 hospitals **Author(s):** Cui-Lan Li **Institution:** Key Laboratory for Major Obstetric Diseases of Guangdong Province and Key Laboratory for Reproduction and Genetics of Guangdong Higher-Education Institutes, The Third Affiliated Hospital, Guangzhou Institute of Obstetrics and Gynecology, Guangzhou Medical **Email:** cuilanli@gzhmu.edu.cn **Address:** 63 Duobao Road, Liwan District, Guangzhou 510145, People’s Republic of China
Notes	**Follow-up duration:** 14 days and 1 month

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Quote: "Equal numbers of enrolled participants were allocated randomly to the hospital administration and self-administration groups." Judgement comment: unclear method
Allocation concealment (selection bias)	High risk	Judgement comment: none
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "After the exclusion of 2 participants in the hospital administration group who failed to return to the hospital for misoprostol administration and 7 participants in the self-administration group who were lost during follow-up," Judgement comment: without ITT analysis
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol
Other bias	Low risk	Judgement comment: none

Liao 2004

Study characteristics

Liao 2004 *(Continued)*

Methods	Computer random table; use of identical-appearing packages and capsules/tablets from pharmacy; identical placebo tablets
Participants	480 women, average age 26 years; ≤ 49 days' gestation confirmed by ultrasound **Exclusion criteria**: abnormal menses, IUD in situ, contraindications for use of study medication; study conducted between November 2001-June 2002 in 3 hospitals affiliated to University of Beijing, China
Interventions	• Group 1: MF: 50 mg, then 12 h later 25 mg, then 12 h later 50 mg, and finally, 12 h later, 25 mg (total: 150 mg). 24 h after last dose 600 µg MS orally • Group 2: MF 30 mg, then 15 mg every 12 h for 3 doses (total: 75 mg). 24 h after last dose, 600 µg MS orally
Outcomes	Complete abortion
Identification	
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Sealed packs each containing: one 25 mg tablet of MF and one 5 mg placebo capsule or one 5 mg capsule MF and one 25 mg placebo tablet. The active tablets or capsules and the corresponding placebo tablets or capsules were identical in size, shape, taste, and colour.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Both participants and hospital staff were blind to the randomisation code until the last eligible participant had attended for the final visit
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Both participants and hospital staff were blind to the randomisation code until the last eligible participant had attended for the final visit
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	The manufacturer (Zizhu Pharmaceuticals Co Ltd., Beijing, China) supplied the 3 centres with medications

Marwah 2016

Study characteristics

Methods	**Study design:** RCT

Marwah 2016 *(Continued)*

	Study grouping: parallel-group
Participants	**Baseline characteristics** MS (oral) • Mean age (years): 23.9 ± 3.7 • BMI (kg/m^2): 22.155 ± 1.8281 • Primigravida: 68% • Previous abortion: 34% • Gestational age (days): 68.74 ± 1.348 MS (vaginal) • Mean age (years): 23.7 ± 3.9 • BMI (kg/m^2): 21.713 ± 2.2017 • Primigravida: 72% • Previous abortion: 36% • Gestational age (days): 68.84 ± 1.468 Overall • Mean age (years): 23.7 ± 3.9 • BMI (kg/m^2): NR • Primigravida: NR • Previous abortion: NR • Gestational age (days): NR **Inclusion criteria:** 18-45 years; gestational age ≤ 12 weeks; missed abortion; mild vaginal bleeding or spotting per vaginum; closed cervix; haemoglobin ≥ 9 gm/dL; axillary temperature < 37.5 °C; no history of inflammatory bowel disease, asthma, liver disease or contraindication to use of MS; place of residence within 100 km from of the hospital; willingness and ability to give informed consent, abstain from intercourse for 1st 14 days of study, comply with follow-up schedule **Exclusion criteria:** cervical dilatation; excessive uterine bleeding; haemodynamic instability; blood pressure ≥ 160/90 mmHg; poor general health of any cause; deranged coagulation profile (defined as prothrombin time index ≤ 85%); infection; asthma or cardiac disease or cerebral disease; anticoagulants or any bleeding disorder; allergy to MS; active lactation; any prior medical or surgical treatment to interrupt current pregnancy; twin gestation sac; molar pregnancy; inability or refusal of patient to adhere to follow-up. **Pretreatment:** none **Study duration:** July 2013 to June 2014
Interventions	**Intervention characteristics** MS (oral) • 400 µg of oral MS, repeated every 6 h for a maximum of 3 doses MS (vaginal) • 400 µg of MS intravaginally into posterior fornix (soaked in normal saline solution), repeated 6-hourly up to a maximum of 3 doses
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Severe bleeding

Marwah 2016 *(Continued)*

	• Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Satisfied • Outcome type: dichotomous outcome • Direction: higher is better • Data value: endpoint Interval from MS administration to abortion • Outcome type: continuous outcome
Identification	**Sponsorship source:** Department of Obstetrics and Gynaecology, VMMC and Safdarjung Hospital **Country:** India **Setting:** Government Multi-Specialty Hospital, Chandigarh **Comments:** no **Author(s):** Sheeba Marwah **Institution:** Department of Obstetrics and Gynaecology, VMMC and Safdarjung Hospital **Email:** sheebamarwah@yahoo.co.in **Address:** New Delhi-110029, India
Notes	**Follow-up duration:** 2 and 6 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Judgement comment: adequate
Allocation concealment (selection bias)	High risk	Judgement comment: none

Marwah 2016 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: none
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: none
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no dropouts
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol
Other bias	Low risk	Judgement comment: none

McKinley 1993

Study characteristics	
Methods	Identical envelopes, shuffled and numbered consecutively
Participants	220 pregnant women, ≤ 63 days of amenorrhoea, University Hospital Edinburgh, Scotland
Interventions	• Group 1: MF 200 mg and MS 600 µg/orally after 48 h • Group 2: MF 600 mg and MS 600 µg/orally after 48 h
Outcomes	Complete and incomplete abortion, time until passing of conceptus, side effects, bleeding pattern, analgesia use
Identification	
Notes	Blinding for outcome assessment No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unclear
Allocation concealment (selection bias)	Unclear risk	220 identical envelopes, which had been shuffled and numbered consecutively
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	The treatment codes were only available to the investigators at the completion of the study. Although the nurses and the women were aware of the dose of MF, this information was withheld from the doctor who subsequently made the decision as to whether the abortion was complete or not.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The treatment codes were only available to the investigators at the completion of the study

McKinley 1993 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	No

Middleton 2005

Study characteristics	
Methods	Computer-generated randomisation in blocks of 8; sealed envelopes
Participants	442 women < 56 days randomised; mean age 26 years; mean gestational age 47 days; study conducted between December 2001-June 2004 at 2 clinics at University of Rochester; USA
Interventions	1-2 days after MF 200 mg: • Group 1: MS 800 µg buccal; 2 tablets placed inside each cheek and remainders swallowed after 30 minutes • Group 2: MS 800 µg vaginal; all 4 tablets placed into the vagina with 1 finger
Outcomes	Complete abortion, side effects, acceptability
Identification	
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random, randomised in blocks of 8 using a scheme created by study staff
Allocation concealment (selection bias)	Low risk	Sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label
Incomplete outcome data (attrition bias) All outcomes	Low risk	Although 21 women were lost to follow-up because they did not return to the clinic per protocol, successful outcome data and survey information were captured via telephone on 8 of these women. Thus, only 13 cases are reported as lost to follow-up, 7 in the buccal arm and 6 in the vaginal arm. All data are reported on the basis of randomisation and ITT

Middleton 2005 *(Continued)*

Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	This study was funded by the Population Council of New York City

Mizrachi 2017

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (repeated dose) • Mean age (years): 32.1 (6.5) • BMI (kg/m^2): 24.8 (4.3) • Mean pregnancy age (weeks): 6.4 (1.3) • Parity: 1.3 (1.4) MS (single dose) • Mean age (years): 33.6 (5.5) • BMI (kg/m^2): 24.3 (5.3) • Mean pregnancy age (weeks): 6.5 (1.4) • Parity: 1.7 (1.5) Overall • Mean age (years): NR • BMI (kg/m^2): NR • Mean pregnancy age (weeks): NR • Parity: NR **Inclusion criteria:** early pregnancy loss, < 12 weeks' gestation **Exclusion criteria:** incomplete or septic abortion, need for urgent surgical evacuation due to heavy bleeding, haemodynamic instability, anaemia (haemoglobin level < 9 g/dL), a history of bleeding disorder, failed medical abortion, multiple pregnancies and contraindication to MS **Pretreatment:** higher parity in single-dose group **Study duration:** August 2015 to June 2016
Interventions	**Intervention characteristics** MS (repeated dose) • 800 µg of MS vaginally on day 1; follow-up visit on day 4; a 2nd dose of 800 µg of MS vaginally if not completed expulsion; another follow-up visit on day 8 MS (single dose) • 800 µg of MS vaginally on day 1; follow-up visit on day 8
Outcomes	Failure to achieve complete abortion

Mizrachi 2017 *(Continued)*

	• Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Fever • Outcome type: adverse event Chills • Outcome type: adverse event Bleeding • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Sackler Faculty of Medicine, Tel Aviv University **Country:** Israel **Setting:** single tertiary hospital **Author(s):** Yossi Mizrachi **Institution:** Department of Obstetrics & Gynecology, Edith Wolfson Medical Center, Holon **Email:** mizrachi.yossi@gmail.com **Address:** 62 Halochamim St. POB 58100, Holon, Israel **ClinicalTrials.gov (NCT02515604)**
Notes	**Follow-up duration:** 1 and 6 weeks

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "subjects were randomly assigned to either a single-dose protocol or a repeat-dose protocol in a 1:1 ratio. A blocked randomization scheme was created using a computer- generated list of random numbers." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Treatment allocation was concealed by placing assignments in sequentially numbered opaque envelopes." Judgement comment: adequate

Mizrachi 2017 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "The physicians and the participants were not blinded to the study group allocation." Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "Five (2.7%) participants were lost to follow-up and two participants withdrew after randomization." Judgement comment: ITT analysis was performed
Selective reporting (reporting bias)	Low risk	Quote: "ClinicalTrials.gov (NCT02515604)." Judgement comment: according to protocol
Other bias	Low risk	Judgement comment: none

Ng 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (outpatient) • Mean age (years): 29 (27, 33.2) • Mean pregnancy age (weeks): 9 (8, 11) • Primiparity (%): 35.1% • Previous miscarriage (%): 13.5% MS (inpatient) • Mean age (years): 31 (28, 34) • Mean pregnancy age (weeks): 10 (9, 11) • Primiparity (%): 32.5% • Previous miscarriage (%): 9.1% Overall • Mean age (years): 30 (28, 34) • Mean pregnancy age (weeks): 10 (8, 11) • Primiparity (%): 33.8% • Previous miscarriage (%): 11.3% **Inclusion criteria:** 1st trimester incomplete miscarriage; > 18 years of age; confirmed by a positive urine pregnancy test or ultrasound scan; endometrial thickness of > 15 mm **Exclusion criteria**: haemodynamically unstable; suspected sepsis with temperature > 38 °C; concurrent medical illness; history of previous uterine surgery; failed medical or surgical evacuation prior to presentation; known allergy to MS; patients with difficult access to hospital; patients who live alone

Ng 2015 *(Continued)*

	Pretreatment: none **Study duration:** May 2012 to April 2013
Interventions	**Intervention characteristics** MS (outpatient) • intra-vaginal MS 800 µg, 8-hourly to a maximum of 3 doses; the 1st dose in the Post Abortion Care (PAC) room, 2nd or 3rd at home MS (inpatient) • all 3 doses of MS 800 µg were administered in the ward 8 h apart
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Reduction haemoglobin level • Outcome type: continuous outcome • Data value: change from baseline Bleeding duration • Outcome type: dichotomous outcome • Unit of measure: days
Identification	**Sponsorship source:** Universiti Kebangsaan Malaysia Medical Centre **Country:** Malaysia **Setting:** a tertiary hospital, Department of Obstetrics and Gynaecology, Universiti Kebangsaan Malaysia Medical Centre **Comments:** no **Author(s):** Beng Kwang Ng **Institution:** Department of Obstetrics and Gynaecology, Universiti Kebangsaan Malaysia Medical Centre **Email:** nbk_9955@yahoo.com; nbk9955@ppukm.ukm.edu.my **Address:** Jalan Yaacob Latif, Cheras, 56000 Kuala Lumpur, Malaysia
Notes	**Follow-up duration:** 1 week

Ng 2015 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Computer generated 'research randomiser' was used to randomise patients to either group A or B." Judgement comment: adequate
Allocation concealment (selection bias)	High risk	Judgement comment: none
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "Due to the nature of the intervention, both the researcher and patient were not blinded to the trial." Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "Three patients from the outpatient group did not complete the trial. One decided to withdraw from the study, as she wanted to wait for spontaneous expulsion. Another two patients opted to undergo further treatment at their hometown after first dose of misoprostol." Judgement comment: 3 participants dropped out in home group
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol available
Other bias	Low risk	Judgement comment: no

Ngoc 2011

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS • Mean age (years): 28 (6.2) • Married (%): 81.3% • Gravidity: 2.5 (1.3) • Primigravida (%): 27.8% • Previous surgical abortion: 0.41 (0.74) • Previous medical abortion: 0.13 (0.35) MF + MS • Mean age (years): 29 (6.3) • Married (%): 82.7% • Gravidity: 2.6 (1.5)

Ngoc 2011 *(Continued)*

	• Primigravida (%): 26.2% • Previous surgical abortion: 0.48 (0.81) • Previous medical abortion: 0.15 (0.38) Overall • Mean age (years): NR • Married (%): NR • Gravidity: NR • Primigravida (%): NR • Previous surgical abortion: NR • Previous medical abortion: NR **Inclusion criteria:** gestational ages up to 63 days seeking early medical abortion; living or working within 1 h from the hospital; intrauterine pregnancy, general good health, able to provide informed consent and willing to return for follow-up **Exclusion criteria:** allergy to either MF or MS, suspicion of ectopic pregnancy, chronic adrenal failure, concurrent long-term corticosteroid therapy, history of haemorrhagic disorders, concurrent anticoagulant therapy or inherited porphyria **Pretreatment:** none **Study duration:** August 2007 to March 2008
Interventions	**Intervention characteristics** MS • placebo followed by 2 x 800 µg buccal MS repeated 24 and 48 h later (1600 µg total) MF + MS • 200 mg MF followed 24 h later by 800 µg buccal MS followed by placebo 24 h later
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Repeated medical abortion • Outcome type: dichotomous outcome Unscheduled visit • Outcome type: dichotomous outcome Abdominal pain

Ngoc 2011 *(Continued)*

	• Outcome type: adverse event Fever • Outcome type: adverse event Chills • Outcome type: adverse event Satisfied • Outcome type: dichotomous outcome Bleeding (more than expected) • Outcome type: adverse event • Direction: higher is better Time of abortion procedure (longer than expected) • Outcome type: dichotomous outcome • Direction: lower is better • Data value: endpoint Failure to achieve complete abortion (< 49 days) • Outcome type: dichotomous outcome Failure to achieve complete abortion (50-63 days) • Outcome type: dichotomous outcome Ongoing pregnancy (< 49 days) • Outcome type: dichotomous outcome Ongoing pregnancy (50-63 days) • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** Gynuity Health Projects **Country:** Vietnam **Setting:** Center for Research and Consultancy in Reproductive Health, Ho Chi Minh City **Author(s):** Jennifer Blumb **Institution:** Gynuity Health Projects, New York **Email:** jblum@gynuity.org **Address:** NY 10011, USA **NCT00680394:** NCT00680394
Notes	**Follow-up duration:** 1 week

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Treatment group was assigned by a computer-generated random sequence in blocks of 10 created at Gynuity Health Projects in New York."

Ngoc 2011 *(Continued)*

		Judgement comment: adequate
Allocation concealment (selection bias)	Unclear risk	Judgement comment: NR
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "double-blind"
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "double-blind"
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Quote: "placebo"
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "Two women allocated to misoprostol-only and no woman given mifepristone+misoprostol were lost to follow-up." Judgement comment: ITT not used
Selective reporting (reporting bias)	Low risk	Quote: "NCT00680394." Judgement comment: efficacy and side effects were reported
Other bias	Unclear risk	Quote: "The study was stopped early due to the unexpectedly high number of ongoing pregnancies which caused concern in the study team, particularly among the Vietnamese providers. An interim analysis was conducted examining outcomes by study arm after this high number of ongoing pregnancies. This article presents data from 400 women enrolled in Vietnam before stopping the study." Judgement comment: study stopped early for high incidence of ongoing pregnancy

Olavarrieta 2015

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (nurse) • Mean age (years): 26.3 (6.3) • Gestational age from LMP (days): 53.2 (16.8) • Gestational age determined by ultrasound (days): 49.7 (14) MF + MS (physician) • Mean age (years): 25.7 (6) • Gestational age from LMP (days): 51.8 (14)

Olavarrieta 2015 *(Continued)*

	• Gestational age determined by ultrasound (days): 49.7 (13.3) Overall • Mean age (years): 26 (6.2) • Gestational age from LMP (days): 52.5 (16.1) • Gestational age determined by ultrasound (days): 49.7 (13.3) **Inclusion criteria:** pregnant women seeking abortion at a gestational duration of ≤ 70 days; aged ≥ 18 years, willing to provide contact information for follow-up **Exclusion criteria:** allergy to MF or MS, chronic systemic corticosteroid use, chronic adrenal failure, coagulopathy or current therapy with anticoagulants, inherited porphyria, chronic medical conditions including pre-existing heart, severe hepatic or renal disease and severe anaemia; received a medical abortion as part of the Mexico City legal abortion programme; women with an inserted IUD when pregnant; could not confirm gestational duration or intrauterine pregnancy **Pretreatment:** none **Study duration:** November 2012 to January 2013
Interventions	**Intervention characteristics** MF + MS (nurse) • 200 mg of oral MF taken on site followed by 800 µg of MS self–administered buccally at home 24 h later: medical abortion and contraceptive counselling provided by nurses MF + MS (physician) • 200 mg of oral MF taken on site followed by 800 µg of MS self–administered buccally at home 24 h later: medical abortion and contraceptive counselling provided by physicians
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Repeated medical abortion • Outcome type: dichotomous outcome Satisfaction • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** Department of Reproductive Health and Research including UNDP/UNFPA/UNICEF/WHO/the World Bank Special Programme of Research **Country:** Mexico **Setting:** 2 Mexico City Ministry of Health abortion clinics and 1 hospital **Comments:** side effects NR, ITT analysis performed by primary authors **Author(s):** Claudia Diaz Olavarrieta **Institution:** Instituto Nacional de Salud Pública **Email:** claudiadiazolavarrieta@gmail.com

Olavarrieta 2015 *(Continued)*

	Address: Septima Cerrada de Fray Pedro de Gante No 50, Col Seccion XVI, Tlalpan, Mexico City, 14000, Mexico **ACTRN12613001230741:** ACTRN12613001230741
Notes	**Follow-up duration:** 7-15 days

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "We generated a list of consecutive identification numbers and randomly allocated a physician or a nurse that should provide medical abortion to each number using R 3.1.2 software for Windows (R Foundation for Statistical Computing, Vienna, Austria). As women were recruited, they were assigned an identification number." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Allocation was concealed in a sealed envelope, which was only opened once the participant was considered eligible and consented to enrol." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "Forty-nine women did not return for follow-up. Attrition was similar in both groups" Judgement comment: ITT analysis was performed by study authors
Selective reporting (reporting bias)	Unclear risk	Judgement comment: protocol was retrospectively registered, Side effects NR
Other bias	Low risk	Judgement comment: none

Ozeren 1999

Study characteristics

Methods	Random number tables; sealed opaque envelopes, sequentially numbered
Participants	108 women ≤ 63 days of amenorrhoea confirmed by ultrasound, University hospital Trabzon, Turkey **Exclusion criteria**: haemoglobin < 100 g/L, leucocythaemia, active liver disease, active renal disease, inflammatory bowel disease, history of MTX/ MS intolerance
Interventions	• Group 1. MTX 50 mg/m^2/IM • Group 2: MS 800 μg/vaginally

Ozeren 1999 *(Continued)*

	• Group 3: MTX 50 mg/m^2/IM and MS 800 µg/vaginally after 3 days
Outcomes	Complete abortions, ongoing pregnancies, side effects
Identification	
Notes	No major complications were reported; 10/36 women in the MS-only group received additional MS on day 4

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants were randomised using a random number table
Allocation concealment (selection bias)	Low risk	Sequentially numbered opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Paritakul 2010

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (sublingual) • Mean age (years): 29.0 (6.7) • Multiparous (%): 56.2% • Mean pregnancy age (weeks): 10.5 (2.7) • Previous abortion (%): 21.9% • Previous curettage (%): 12.5% MS (oral)

Paritakul 2010 *(Continued)*

	• Mean age (years): 32.2 (7.4) • Multiparous (%): 43.8% • Mean pregnancy age (weeks): 10.4 (1.8) • Previous abortion (%): 21.9% • Previous curettage (%): 12.5% Overall • Mean age (years): NR • Multiparous (%): NR • Mean pregnancy age (weeks): NR • Previous abortion (%): NR • Previous curettage (%): NR **Inclusion criteria:** < 14 weeks' gestation; good health; volunteered to be enrolled in the study; incomplete miscarriage **Exclusion criteria:** haemodynamically unstable, suspected of septic abortion, had a history of allergy to MS, or with suspected ectopic pregnancy **Pretreatment:** none **Study duration:** July 2007 to August 2008
Interventions	**Intervention characteristics** MS (sublingual) • 600 µg MS sublingually MS (oral) • 600 µg MS orally
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Satisfaction

Paritakul 2010 *(Continued)*

	• Outcome type: dichotomous outcome Interval from MS administration to abortion • Outcome type: continuous outcome
Identification	
Notes	Follow-up: 48 h and 1 week

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "A randomization scheme was generated using a random number table. The co-investigator generated the allocation sequence, and study staff enrolled participants and assigned participants to their groups." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "a sequentially numbered opaque envelope which contained a ticket identifying the treatment group." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "Neither the provider nor the woman was blinded to the treatment regimens." Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement comment: no drop out
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol
Other bias	Low risk	none

Qian 2015

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (oral) • Mean age (years): 27 (5) • Mean pregnancy age (days): 74 (21) • Weight (kg): 55 (8)

Qian 2015 *(Continued)*

	• Height (cm): 162 (5) • Previous abortion: 0.8 (0.9) MF + MS (vaginal) • Mean age (years): 27 (6) • Mean pregnancy age (days): 70 (20) • Weight (kg): 54 (7) • Height (cm): 162 (4) • Previous abortion: 0.9 (0.9) Overall • Mean age (years): NR • Mean pregnancy age (days): NR • Weight (kg): NR • Height (cm): NR • Previous abortion: NR **Inclusion criteria:** 8-16 weeks' gestation; 18-40 years old; good health; regular menstruation **Exclusion criteria:** persistent vaginal bleeding; abnormal placenta; reproductive tumour; STD; contraindication to MF or MS **Pretreatment:** none **Study duration:** Jan 2011 to Oct 2012
Interventions	**Intervention characteristics** MF + MS (oral) • 200 mg oral MF followed by oral MS 400 µg every 3 h, no more than 4 doses (1600 µg) MF + MS (vaginal) • 200 mg oral MF followed by vaginal MS 400 µg every 6 h for a maximum of 4 doses 36-48 h later
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Bleeding volume • Outcome type: continuous outcome • Unit of measure: mL Interval from MS administration to abortion • Outcome type: continuous outcome • Unit of measure: hours • Direction: lower is better • Data value: endpoint
Identification	**Sponsorship source:** Fudan University **Country:** China **Setting:** 11 tertiary hospitals in China **Comments:** 8-16 weeks' gestation

Qian 2015 *(Continued)*

	Author(s): Huang Zirong **Institution:** Department of Family Planning, Obstetrics and Gynecology Hospital, Fudan University **Email:** zirong1130@hotmail.com **Address:** Shanghai 200011, China
Notes	Follow-up: 3 weeks

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: envelope with random number
Allocation concealment (selection bias)	Unclear risk	Judgement comment: NR
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "Of the 625 participants, 4 were dropouts after taking MF and were not included in the outcomes analyses" Judgement comment: without ITT analysis
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol and most side effect without detailed outcome data
Other bias	Unclear risk	Judgement comment: including 8-16 weeks' gestation and stratified by gestational weeks

Raghavan 2009

Study characteristics

Methods	Random code generated in blocks of 10; sequentially numbered, sealed envelopes
Participants	480 women; ≤ 63 days' gestation. gestational age confirmed by ultrasound if needed **Exclusion criteria**: ectopic pregnancy, contraindications to study medication, treatment with anticoagulants, lived > 1 h away from hospital; study conducted between July 2005-November 2006 at University Hospital Chisinau, Moldova
Interventions	MF 200 mg followed 24 h later by: • Group 1: MS 400 µg sublingual • Group 2: MS 400 µg oral

Raghavan 2009 *(Continued)*

	For sublingual: tablet for 30 min under the tongue and swallow rest after; no repeat doses of MS offered
Outcomes	Complete abortion, side effects, acceptability
Identification	
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random code generated in blocks of 10 and printed on slips by Gynuity Health Projects in New York
Allocation concealment (selection bias)	Low risk	Sequential sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition between groups
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	This study was funded by an anonymous donor

Raghavan 2010

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (sublingual) • Mean age (years): 24.8 (18-43) • Previous live birth: 0.61 • Gravidity: 2.6 • Previous abortion (%): 48% MF + MS (buccal) • Mean age (years): 24.9 (18-40) • Previous live birth: 0.52

Raghavan 2010 *(Continued)*

	• Gravidity: 2.4 • Previous abortion (%): 52% Overall • Mean age (years): NR • Previous live birth: NR • Gravidity: NR • Previous abortion (%): NR **Inclusion criteria:** presenting for abortion with gestational ages through 63 days **Exclusion criteria:** contraindications to use of MF–MS abortion **Pretreatment:** none **Study duration:** July 2007 to March 2009
Interventions	**Intervention characteristics** MF + MS (sublingual) • 400 µg of MS sublingually 24 h after ingestion of 200 mg of MF MF + MS (buccal) • 400 µg of MS buccal 24 h after ingestion of 200 mg of MF
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Severe bleeding

Raghavan 2010 *(Continued)*

	• Outcome type: adverse event Satisfaction • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome Failure to achieve complete abortion (< 49 days) • Outcome type: dichotomous outcome Failure to achieve complete abortion (50-63 days) • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** Gynuity Health Projects **Country:** the Republic of Moldova **Setting:** University Clinic, Municipal Clinical Hospital No.1, in Chisinau **Comments:** 9 weeks' gestation **Author(s):** Sheila Raghavan **Institution:** Gynuity Health Projects, New York **Email:** sraghavan@gynuity.org **Address:** Gynuity Health Projects, New York
Notes	**Follow-up duration:** 2 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "computer-generated random code." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Providers instructed women on the route of misoprostol administration by opening sealed envelopes in sequential order indicating assignment of route. The envelopes were prepared by Gynuity Health Projects staff in New York" Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "5 in the buccal arm and 6 in the sublingual arm, were lost-to-follow-up"

Raghavan 2010 *(Continued)*

		Judgement comment: no ITT analysis
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol
Other bias	Low risk	Judgement comment: none

Rodger 1989

Study characteristics	
Methods	Randomisation not stated
Participants	120 pregnant women, < 56 days of amenorrhoea, Gynaecological Outatient Department, Royal Infirmary Hospital, Edinburgh, Scotland
Interventions	MF 600 mg (all) combined with GP • Group 1: GP 0.5 mg/vaginally after 48 h • Group 2: GP 1 mg/vaginally after 48 h
Outcomes	Complete, incomplete abortion, onset and duration of bleeding, side effects, haemoglobin levels
Identification	
Notes	1 woman received blood transfusion (Group 2)

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No detailed randomisation method
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Neither the participants nor the investigators knew the dose of GP given
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Neither the participants nor the investigators knew the dose of GP given
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	Quote: "The mifepristone was kindly supplied by MS Angela Davey of Roussel Laboratories Ltd"

Sandstrom 1999

Study characteristics	
Methods	Randomly allocated; using sealed envelopes
Participants	64 pregnant women, ≤ 56 days, Hillerod Hospital, Denmark **Exclusion criteria**: previous uterine surgery, previous abnormal vaginal bleeding, concomitant medication, IUD in situ, contraindication to one of the study drugs
Interventions	All: MF 600 mg combined with GP • Group1: GP 1 mg/vaginally after 24 h • Group 2: GP 1 mg/vaginally after 48 h
Outcomes	Complete, incomplete abortion, side effects
Identification	
Notes	1 woman needed blood transfusion, not mentioned which group

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	The women were randomly allocated into 2 parallel groups by drawing a sealed envelope
Allocation concealment (selection bias)	Unclear risk	Sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	MF and GP was kindly supplied by Roussel-Uclaf

Sang 1994

Study characteristics	
Methods	Random number tables

Sang 1994 *(Continued)*

Participants	600 women, ≤ 49 days of pregnancy, multicentre trial in 5 hospitals in Shanghai, China; pregnancy confirmed by gynaecological examination, urine pregnancy test or ultrasound. Women were included if there was no history of medical disorders, no IUD in situ and no contraindication for the study medication
Interventions	• Group 1: MF 150 mg divided into 5 doses, orally, within 3 days; MS 600 µg orally 36-48 h later • Group 2: MF 150 mg divided into 5 doses /orally, within 3 days; PGF2alpha vaginally 36-48 h later • Group 3: MF 200 mg orally; MS 600 µg/orally after 36-48 h
Outcomes	Complete, incomplete abortion, duration of bleeding, time of resumption of menses, side effects
Identification	
Notes	No mention of major complications

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Sang 1999

Study characteristics

Methods	Randomisation was generated centrally and women were randomised within centres; sealed opaque envelopes
Participants	Multicentre trial, 78 hospitals and family planning clinics from 8 provinces in China; 17,542 pregnant women, ≤ 49 days of amenorrhoea, pregnancy confirmed by gynaecological examination, urine pregnancy test or ultrasound. Women were included if there was no history of medical disorders, no IUD in situ and no contraindication for the study medication

Sang 1999 *(Continued)*

Interventions	MF 150 mg divided into 5 doses taken orally within 3 days • Group 1: PGF2alpha 1 mg vaginally 36-48 h after 1st dose of MF • Group 2: MS 600 µg orally 36-48 h after 1st dose of MF
Outcomes	Complete, incomplete abortion, duration of vaginal bleeding, time to resumption of menses, side effects, women's satisfaction with the procedure
Identification	
Notes	1 woman had an allergic shock after MS (Group 2)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was generated centrally and women were randomised within centres
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	The medication was provided by the pharmacy company

Schaff 2000

Study characteristics

Methods	Computer-generated random assignment, allocation, randomisation stratified by sites; allocation was "concealed" 53 women used repeat dose of MS - number of women per group receiving additional MS not described
Participants	Multicenter trial (16 centres), 2295 women with pregnancies ≤ 56 days confirmed by ultrasound; from 16 primary care and referral abortion facilities, USA; routine inclusion and exclusion criteria
Interventions	All women received MF 200 mg on day 1 • Group 1: MS 800 µg vaginally next day at home • Group 2: MS 800 µg vaginally 2 days later at home

Schaff 2000 *(Continued)*

	• Group 3: MS 800 µg vaginally 3 days later at home
Outcomes	Complete abortion, acceptability, adverse effects
Identification	
Notes	2 women received blood transfusion (not mentioned which group)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random assignment, allocation, randomisation stratified by sites
Allocation concealment (selection bias)	Low risk	Allocation was "concealed"
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	53 women used repeat dose of MS - number of women per group receiving additional MS not described

Schaff 2001

Study characteristics

Methods	Computer-generated random assignment, open-label
Participants	Multicenter trial at 15 sites in the USA, including hospitals, non-profit abortion facilities, private family practice and gynaecologist offices 1168 women, ≤ 63 days pregnant confirmed by ultrasound, without clinical or haematological abnormalities or contraindication to the trial medication
Interventions	All women received MF 200 mg on day 1 • Group 1: MS 800 µg orally minimum 24 h after at home • Group 2: MS 800 µg vaginally minimum 24 h after at home
Outcomes	Complete, incomplete abortion, time to bleeding, side effects

Schaff 2001 *(Continued)*

Identification		
Notes	Open-labelled study, power calculation to detect a 5% difference from 95% to 90% efficacy No hospitalisations and no blood transfusions	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomised assignments
Allocation concealment (selection bias)	Unclear risk	Unclear
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	19 were lost to follow-up and 5 used the MS contrary to their assignment, unbalanced
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Low risk	None

Schreiber 2018

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS • Mean age (years): 30.7 ± 6.3 • Previous miscarriage (%): 35.6% MS • Mean age (years): 30.2 ± 6.0 • Previous miscarriage (%): 34.4% Overall • Mean age (years): NR

Schreiber 2018 *(Continued)*

	• Previous miscarriage (%): NR **Inclusion criteria**: healthy women ≥ 18 years; nonviable intrauterine pregnancy between 5 and 12 completed weeks of gestation **Exclusion criteria:** incomplete or inevitable abortion; contraindication to MF or MS; ectopic pregnancy, haemoglobin level < 9.5 g/dL; clotting defect; receiving anticoagulants, pregnancy with an IUD in place; unwilling to adhere to the trial protocol **Pretreatment:** none **Study duration:** May 2014 to April 2017
Interventions	**Intervention characteristics** MF + MS • 800 µg vaginal MS, preceded by 200 mg oral MF 24 h prior MS • 800 µg of vaginal MS
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Fever • Outcome type: adverse event Chills • Outcome type: adverse event Cramping • Outcome type: dichotomous outcome Fatigue • Outcome type: dichotomous outcome

Schreiber 2018 *(Continued)*

	Blood transfusion • Outcome type: dichotomous outcome Pelvic infection • Outcome type: dichotomous outcome Satisfaction • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** National Institute of Child Health and Human Development **Country:** USA **Setting:** Department of Obstetrics and Gynecology, University of Pennsylvania **Author(s):** Courtney A. Schreiber **Institution:** Department of Obstetrics and Gynecology, University of Pennsylvania **Email:** schreibe@ upenn.edu **Address:** Philadelphia, PA 19104 **NCT02012491:** NCT02012491
Notes	Follow-up timepoint: 3 days, 1 and 4 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "Participants were randomly assigned in permuted blocks of two to eight, stratified according to trial site, with the use of Research Electronic Data Capture" Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "software (REDCap, Vanderbilt University)."
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "the women in the misoprostol-alone group did not receive placebo. 23"
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Quote: "an investigator who was unaware of the treatment-group assignments assessed the outcome by means of endo-vaginal ultrasonography." Judgement comment: for primary outcome
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "The primary outcome was assessed among all women who had at least one follow-up visit according to a preplanned modified intention-to-treat principle." Judgement comment: ITT analysis
Selective reporting (reporting bias)	Low risk	Quote: "ClinicalTrials.gov number, NCT02012491.)"

Schreiber 2018 *(Continued)*

Other bias	Low risk	Quote: "the manufacturer had no other role in the trial. The protocol, including the statistical analysis plan, is available with the full text of this article at NEJM.org."

Shannon 2006

Study characteristics	
Methods	Computer-generated random numbers in group of 15; MS tablets were provided in sealed, opaque envelopes after administration of MF
Participants	971 women, mean age 28 years, < 56 days of gestation; mean gestational age of 44 days **Exclusion criteria**: haemoglobin < 9.5 g/dL, active hepatic or renal disease, type I diabetes mellitus, adrenal insufficiency, glaucoma, sickle cell anaemia, coagulopathy, uncontrolled seizure disorder, severe cardiovascular disease, allergy or intolerance to study medication, use of chronic oral steroid medications or anticoagulants Study conducted in 2001 at University of British Columbia; University of Sherbrooke; Laval University; University of Toronto; Canada
Interventions	MF 200 mg followed 24-28 h later by: • Group 1: MS 400 µg oral • Group 2: MS 600 µg oral • Group 3: MS 800 µg vaginal MS self-administered at home. Participants were advised to take a 2nd dose of MS in case bleeding was less than normal menstruation. Ultrasound after 7 days. If ongoing pregnancy: MS 800 µg vaginally
Outcomes	Complete abortion, acceptability, side effects
Identification	
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers in group of 15
Allocation concealment (selection bias)	Low risk	MS tablets were provided in sealed, opaque envelopes after administration of MF
Blinding of participants and personnel (performance bias) All outcomes	High risk	Not blinded
Blinding of outcome assessment (detection bias) All outcomes	High risk	Not blinded

Shannon 2006 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Unclear risk	Packard Foundation support for this project

Sheldon 2019

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (sublingual) • Mean age (years): 26.9 (7.3) • Nulliparous: 27.6% • Previous abortion (%): 23% MS (buccal) • Mean age (years): 27.2 (7) • Nulliparous: 29.2% • Previous abortion (%): 28.7% Overall • Mean age (years): NR • Nulliparous: NR • Previous abortion (%): NR **Inclusion criteria:** seeking abortion with pregnancy duration ≤ 10 weeks **Exclusion criteria:** contraindications to MS (IUD in place or history of allergies to prostaglandins), confirmed or suspected ectopic or molar pregnancy, and anyone not willing to provide contact information and return for follow-up **Pretreatment:** none **Study duration:** April 2015 to September 2016
Interventions	**Intervention characteristics** MS (sublingual) • 3 doses of MS 800 µg sublingually, repeated every 3 h MS (buccal) • 3 doses of MS 800 µg buccally, repeated every 3 h
Outcomes	Failure to achieve complete abortion

Sheldon 2019 *(Continued)*

	• Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Ongoing pregnancy • Outcome type: dichotomous outcome Repeated medical abortion • Outcome type: dichotomous outcome Satisfaction • Outcome type: dichotomous outcome Fever • Outcome type: adverse event Chills • Outcome type: adverse event Failure to achieve complete abortion (< 49 days) • Outcome type: dichotomous outcome Failure to achieve complete abortion (50-63 days) • Outcome type: dichotomous outcome Failure to achieve complete abortion (64-70 days) • Outcome type: dichotomous outcome Ongoing pregnancy (< 49 days) • Outcome type: dichotomous outcome Ongoing pregnancy (50-63 days) • Outcome type: dichotomous outcome Ongoing pregnancy (64-70 days) • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** Gynuity Health Projects **Country:** USA **Setting:** 6 health clinics located in urban and peri-urban settings of 2 Latin American countries

Sheldon 2019 *(Continued)*

	Author(s): Wendy R. Sheldon **Institution:** Gynuity Health Projects, New York **Email:** wsheldon@gynuity.org **Address:** 220 East 42nd Street, Suite 710, New York, NY, 10017 **ClinicialTrials.gov NCT02299401**	
Notes	Follow-up: 7-14 days after MS	
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "computer- generated randomization that was stratified by site with a block size of 10." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "masked allocation using sealed, consecutively numbered opaque envelopes, each containing a card revealing the participant's study group allocation (sublingual or buccal misoprostol)." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Judgement comment: single blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Quote: "To blind study staff, providers and participants were instructed not to ask about or reveal group allocation until the participant's exit interview." Judgement comment: study staff blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "Ten women were lost to follow-up;" Judgement comment: ITT analysis was performed by study authors, but not all participants lost to follow-up were analysed
Selective reporting (reporting bias)	Low risk	Quote: "NCT02299401" Judgement comment: according to the protocol
Other bias	Low risk	Judgement comment: none

Shrestha 2014

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (home)

Shrestha 2014 *(Continued)*

	• Mean age (years): 27.4 (4.9) • Gravidity: 3.04 (1.6) • Parity: 1.6 (1.2) • Gestational age (days): 45.5 (7) MF + MS (hospital) • Mean age (years): 27.3 (5) • Gravidity: 2.9 (1.2) • Parity: 1.5 (0.9) • Gestational age (days): 44.4 (7.6) Overall • Mean age (years): NR • Gravidity: NR • Parity: NR • Gestational age (days): NR **Inclusion criteria:** requesting legal termination of pregnancy at a gestation up to 63 days; healthy, > 18 years, agreed to surgical termination of pregnancy if the treatment failed **Exclusion criteria:** indication of serious past or present illness; allergic to MF or MS; contraindication to the use of MF **Pretreatment:** none **Study duration:** 1st April 2011 to 15th August 2012
Interventions	**Intervention characteristics** MF + MS (home) • self-administration of vaginal MS (800 µg) at home 24 h after oral 200 mg MF MF + MS (hospital) • vaginal MS (800 µg) by hospital administration 24 h after oral 200 mg MF
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Dizziness • Outcome type: adverse event Headache

Shrestha 2014 *(Continued)*

	• Outcome type: adverse event Sweating • Outcome type: adverse event Satisfaction • Outcome type: dichotomous outcome Bleeding • Outcome type: continuous outcome
Identification	**Sponsorship source:** Chitwan Medical College teaching hospital **Country:** Nepal **Setting:** Department of Obstetrics and Gynecology, Chitwan Medical College teaching hospital **Author(s):** Anju Shrestha **Institution:** Department of Obstetrics and Gynaecology, Chitwan Medical College Teaching Hospital **Email:** anjushr2002@yahoo.co.in **Address:** Chitwan, Nepal
Notes	**Follow-up duration:** 2 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "Eligible women were allocated randomly to two groups using a computer generated randomization sequence in blocks of variable size:" Judgement comment: adequate
Allocation concealment (selection bias)	High risk	Judgement comment: none
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no dropouts
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol available
Other bias	Low risk	Judgement comment: none

Sinha 2018

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS • Mean age (years): 24.69 (3.67) • Mean pregnancy age based on LMP (days): 72.44 (10.58) • BMI (kg/m^2): 21.98 (1.49) • Parity: 1 • Mean pregnancy age based on ultrasound (days): 48.78 (7.78) MS • Mean age (years): 25.74 (4.42) • Mean pregnancy age based on LMP (days): 71.36 (9.72) • BMI (kg/m^2): 22.1 (1.53) • Parity: 1 • Mean pregnancy age based on ultrasound (days): 51.62 (8.55) Overall • Mean age (years): NR • Mean pregnancy age based on LMP (days): NR • BMI (kg/m^2): NR • Parity: NR • Mean pregnancy age based on ultrasound (days): NR **Inclusion criteria:** missed abortion; < 12 weeks of gestation **Exclusion criteria:** incomplete abortion, inevitable abortion, haemodynamic instability, haemoglobin < 8 g%, bleeding disorder, obvious infection, and known allergy to MF/MS **Pretreatment:** none **Study duration:** October 2011 to April 2013
Interventions	**Intervention characteristics** MF + MS • 200 mg MF orally and 800 µg MS vaginally 48 h later MS • placebo and 800 µg MS vaginally 48 h later If no expulsion occurred within 4 h, repeat doses of 400 µg MS were given orally at 3-hourly interval to a maximum of 2 doses
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation

Sinha 2018 *(Continued)*

	• Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Ongoing pregnancy • Outcome type: dichotomous outcome Repeated medical abortion • Outcome type: dichotomous outcome Satisfied • Outcome type: dichotomous outcome Interval from MS administration to abortion • Outcome type: continuous outcome
Identification	**Sponsorship source:** University College of Medical Sciences and Guru Teg Bahadur Hospital **Country:** India **Setting:** Department of Obstetrics and Gynaecology at University College of Medical Sciences and Guru Teg Bahadur Hospital **Comments:** no **Author(s):** Richa Aggarwal **Institution:** Department of Obstetrics and Gynecology, University College of Medical Sciences and Guru Teg Bahadur Hospital **Email:** richa.agg77@gmail.com **Address:** Delhi 110095, India **CTRI 2013/03/003492:** CTRI 2013/03/003492
Notes	Jing Zhang on 19 February 2020 00:49 **Population** < 12 weeks' gestation; missed abortion Jing Zhang on 19 February 2020 00:53 **Outcomes** Follow-up 2 weeks

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement comment: computer, adequate
Allocation concealment (selection bias)	Low risk	Quote: "The sealed packets were numbered from 1 to 92 by simple randomization using computer generated random tables on a master list which was available with the third party." Judgement comment: adequate

Sinha 2018 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "both the treating obstetrician and the patient were blinded regarding the nature of the drug." Judgement comment: double-blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: double-blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "One patient in each group was lost to follow-up during study period" Judgement comment: balanced dropout and ITT analysis
Selective reporting (reporting bias)	Low risk	Quote: "(CTRI 2013/03/003492)." Judgement comment: according to protocol
Other bias	Low risk	Judgement comment: none

Song 2018

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (home) • Mean age (years): 26.7 (8.6) • Mean pregnancy age based on LMP (days): 44.6 (4.9) • Gravidity: 0.5 (0.6) • Parity: 0 MF + MS (hospital) • Mean age (years): 27.6 (7.1) • Mean pregnancy age based on LMP (days): 45.2 (5.1) • Gravidity: 0.5 (0.7) • Parity: 0 Overall • Mean age (years): NR • Mean pregnancy age based on LMP (days): NR • Gravidity: NR • Parity: NR **Inclusion criteria:** pregnant woman in good general health; 18–40 years of age; regular menstrual cycles of 25- to 35-day duration over the past 6 months; < 49-day interval from LMP; access to a telephone **Exclusion criteria:** allergy to MF or MS; history of ectopic pregnancy; haematological disease (haemolytic illness, coagulation disease or thrombotic disorders) or end-stage organ (e.g. heart, lung, liver and kidney) failure; having received hormone replacement therapy; haemoglobin < 90 g/L; having become pregnant while using an IUD

Song 2018 *(Continued)*

	Pretreatment: none Study duration: July of 2013 to December of 2016
Interventions	**Intervention characteristics** MF + MS (home) • 100 mg of oral MF in hospital followed by 200 µg of sublingual MS 24 h later at home MF + MS (hospital) • 100 mg of oral MF in hospital followed by 200 µg of sublingual MS 24 h later in hospital
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Ongoing pregnancy • Outcome type: dichotomous outcome Dizziness • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Satisfied • Outcome type: dichotomous outcome Bleeding duration • Outcome type: continuous outcome
Identification	**Sponsorship source:** Third Affiliated Hospital, Guangzhou Medical University **Country:** China **Setting:** Obstetrics and Gynecology Department **Author(s):** Cui-Lan Li **Institution:** Third Affiliated Hospital, Guangzhou Medical University **Email:** ninacuilanli@gmail.com

Song 2018 *(Continued)*

	Address: 63 Duobao Road, Liwan District, Guangzhou 510145, China
Notes	Unclear follow-up

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "The participants were divided randomly into two groups:" Judgement comment: no detailed method
Allocation concealment (selection bias)	High risk	Judgement comment: no concealment
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	High risk	Judgement comment: participants dropped out but without ITT analysis
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol available
Other bias	Low risk	Judgement comment: none

Sonsanoh 2014

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (sublingual) • Mean age (years): 29.5 (6.2) • Mean pregnancy age (days): 67.9 (13.7) • BMI (kg/m^2): 22.9 (4) • Primigravida: 18.82% • Previous abortion: 8.3% MS (vaginal) • Mean age (years): 29.4 (7.3) • Mean pregnancy age (days): 69 (15.4) • BMI (kg/m^2): 22.5 (4.4)

Sonsanoh 2014 *(Continued)*

	• Primigravida: 34.9% • Previous abortion: 11.6% Overall • Mean age (years): NR • Mean pregnancy age (days): NR • BMI (kg/m^2): NR • Primigravida: NR • Previous abortion: NR **Inclusion criteria:** women with early pregnancy failure; > 18 years old **Exclusion criteria:** women with suspected ectopic pregnancy, heavy vaginal bleeding, and maternal history of asthma or cardiac disease, open internal cervix on speculum examination, history of allergy to prostaglandins, vital signs unstable and an inability to confirm pregnancy failure or intrauterine gestational sac location **Pretreatment:** none **Study duration:** August 2012 to August 2013
Interventions	**Intervention characteristics** MS (sublingual) • 4 tablets of 200 µg MS sublingually MS (vaginal) • 4 tablets of 200 µg MS with 2-3 drops of normal saline placed in the posterior vaginal fornix by digital insertion
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Ongoing pregnancy • Outcome type: dichotomous outcome Diarrhoea • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event

Sonsanoh 2014 *(Continued)*

	Bleeding (more than expected) • Outcome type: dichotomous outcome Interval from MS administration to abortion • Outcome type: continuous outcome
Identification	**Sponsorship source:** Department of Obstetrics and Gynecology, Chonburi Hospital **Country:** Thailand **Setting:** Department of Obstetrics and Gynecology, Chonburi Hospital **Author(s):** Areerat Sonsanoh **Institution:** Department of Obstetrics and Gynecology, Chonburi Hospital **Email:** none **Address:** Chonburi 20000, Thailand
Notes	**Follow-up duration:** 2 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "We used and assigned blocks of four randomizations to two groups of participants." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Cards labeled with the assigned route were placed in sealed, opaque envelopes which were filled and labeled in accordance with the list of randomizations. The allocation was concealed by the use of sealed number of treatments." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: all participants were followed up
Selective reporting (reporting bias)	Unclear risk	Judgement comment: protocol unavailable
Other bias	Low risk	Judgement comment: none

Souizi 2020

Study characteristics	
Methods	Open-label, RCT with 3 parallel groups
Participants	1st trimester of pregnancy from August 2014-August 2016, Shahidan Mobini Hospital in Sabzevar, Iran **Inclusion criteria**: gestational age < 12 weeks with ultrasound-confirmed missed abortion, confirmation of the need for termination of the pregnancy by a gynaecologist, absence or slight vaginal bleeding, and axillary temperature < 37.5°C **Exclusion criteria**: emergency status of the patient and urgent need for termination of pregnancy by surgical evacuation; anaemia (haemoglobin < 9 g/dL); coagulation disorders or history of current anticoagulant therapy; history of hepatic, renal, and cardiovascular diseases; and history of prostaglandin allergy, smoking, inflammatory bowel disease, IUD implantation, adnexal mass suspicious for ectopic pregnancy, uterine abnormalities, and lactation
Interventions	A maximum of 4 doses of 600-µg MS were administered by vaginal, sublingual, or oral administration every 6 h until uterine evacuation or vaginal bleeding began
Outcomes	Complete uterine evacuation; induction abortion interval in hours; severity and volume of blood loss; nausea, vomiting, diarrhoea, dizziness, fatigue, low abdominal pain, headache, chills, and fever
Identification	IRCT2014101015905N2
Notes	3 groups

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Permuted block randomisation was performed with random block sizes of 6 using allocation software provided by the statistics consultant
Allocation concealment (selection bias)	Low risk	Sequentially numbered opaque sealed envelopes that contained a group assignment for the participants were shuffled and distributed among the participants
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	No blinding
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Balanced attrition
Selective reporting (reporting bias)	Low risk	Protocol was published
Other bias	Unclear risk	3 groups

Swahn 1989

Study characteristics	
Methods	Randomly allocated
Participants	42 pregnant women, ≤ 49 days of amenorrhoea, confirmed by ultrasound
Interventions	All: MF 25 mg/twice daily/ for 4 days (= 200 mg in total) and: • Group 1: 1 placebo morning and evening, orally • Group 2: PGE2 (minprostin) 1 mg morning and placebo evening, orally • Group 3: PGE2 1 mg morning and evening, orally
Outcomes	Complete, incomplete abortion Failures, complaints, hormone levels (E2 prostaglandin, beta-HCG, prolactin) Bleeding pattern
Identification	
Notes	Originally planned sample size was 120: study was discontinued due to interim analysis, which showed no difference between placebo and PGE2 in the complete abortion rate No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Randomiszed
Allocation concealment (selection bias)	Unclear risk	Double-blind, placebo-controlled without detailed method
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Double-blind, placebo-controlled without detailed method
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Double-blind, placebo-controlled
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	This trial was discontinued prematurely when an interim analysis indicated that the rate of complete abortion in participants receiving the adjuvant PGE2 therapy was not more than that obtained with RU 486 alone
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Unclear risk	Financial support from the Special Programme of Research, Development and Research Training in Human Reproduction of the WHO. RU 486 used in this study was kindly donated by Roussel-Uclaf, Paris, France

Tang 2002

Study characteristics

Tang 2002 *(Continued)*

Methods	Computer-generated random table
Participants	150 pregnant women, ≤ 63 days of amenorrhoea, confirmed by ultrasound at the University Hospital Hong Kong **Inclusion criteria:** good health, willing to use barrier methods for contraception until 1st menses after termination, haemoglobin level > 110g/L **Exclusion criteria:** significant past or present illness, allergy/contraindication towards study medication, IUD, heavy smoker, breastfeeding
Interventions	MF 200 mg for all women • Group A: MS 800 µg orally and MS 400 µg twice/day orally for day 4-10 • Group B: MS 800 µg vaginally on day 3 and MS 400 µg twice/day orally for day 4-10 • Group C: MS 800 µg vaginally on day 3 and placebo tablets on day 4-10
Outcomes	Complete, incomplete, missed abortion, ongoing pregnancy, blood loss, haemoglobin levels
Identification	
Notes	No major complications were reported

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random table
Allocation concealment (selection bias)	Unclear risk	Unclear method
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Double-blind, placebo-controlled but without detailed method
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Double-blind, placebo-controlled but without detailed method
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unbalanced attrition but without ITT
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	Support of the Family Planning Association of Hong Kong in the recruitment of participants. This study was supported by the Special Programme of Research, Development and Research Training in Human Reproduction, WHO, Geneva, Switzerland

Tang 2003

Study characteristics

Tang 2003 *(Continued)*

Methods	Computer-generated random numbers; double-blinded: women received placebo for vaginal application in the sublingual group; and placebo tablets for sublingual application in the vaginal group
Participants	224 women, average age 23 years, ≤ 9 weeks of gestation; average gestational age 7.7 weeks; gestational age confirmed by ultrasound **Exclusion criteria**: using prescription drugs regularly, IUD in situ, breastfeeding, multiple pregnancies and heavy smoking. Study conducted at University of Hong Kong
Interventions	MF 200 mg followed 48 h later by: • Group1: MS 800 µg sublingual • Group 2: MS 800 µg vaginal
Outcomes	Complete/incomplete abortion, ongoing pregnancy; haemoglobin concentration; days of bleeding; induction-abortion interval, side effects
Identification	
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Double-blinded: women received placebo for vaginal application in the sublingual group; and placebo tablets for sublingual application in the vaginal group
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Double-blinded: women received placebo for vaginal application in the sublingual group; and placebo tablets for sublingual application in the vaginal group
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	This study was fully supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No: HKU 7244/01M). WHO for the supply of MF tablets used in this study

Tanha 2010

Study characteristics

Tanha 2010 *(Continued)*

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS (sublingual) • Mean age (years): 28.49 • Mean pregnancy age (weeks): 10.55 • Gravidity: 2.85 • Previous abortion: 0.71 • Parity: 0.23 MS (vaginal) • Mean age (years): 29.05 • Mean pregnancy age (weeks): 10.76 • Gravidity: 2.87 • Previous abortion: 0.6 • Parity: 0.44 Overall • Mean age (years): NR • Mean pregnancy age (weeks): NR • Gravidity: NR • Previous abortion: NR • Parity: NR **Inclusion criteria:** 1st trimester silent miscarriage < 13 weeks' gestation; no known contraindications to MS, general good health and no vaginal bleeding **Exclusion criteria:** incomplete abortion, inevitable abortion (products of gestation bulging from the cervix), suspicion of an extra uterine pregnancy, drug or alcohol abuse as reported by the woman, and abnormal blood count tests obtained routinely **Pretreatment:** none **Study duration:** January 2005 to February 2007
Interventions	**Intervention characteristics** MS (sublingual) • MS 400 µg tablets every 6 h sublingually MS (vaginal) • MS 400 µg tablets every 6 h vaginally
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome

Tanha 2010 *(Continued)*

	Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Satisfaction • Outcome type: dichotomous outcome Blood transfusion • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Severe bleeding • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** Tehran University of Medical Sciences **Country:** Iran **Setting:** Mirza Kochak Khan Hospital **Author(s):** Fateme Davari Tanha **Institution:** Department of Obstetrics and Gynecology, Tehran University of Medical Sciences, Mirza Kochak Khan Hospital **Email:** fatedavari@yahoo.com **Address:** Ostadnejatollahi Shomaly Avenue, Poule Karim Khane Zand-Villa Street, Valiasr Reproductive Health Center, Tehran 15978, Iran
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "This randomization scheme was created by Population Council staff, using a computer-generated code." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "The study investigator opened the next sequentially numbered randomized envelope to determine the treatment arm." Judgement comment: adequate, central allocation
Blinding of participants and personnel (performance bias) All outcomes	High risk	Quote: "Neither the investigator nor the woman was blinded to the treatment assignment." Judgement comment: no blinding

Tanha 2010 *(Continued)*

Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: all participants were followed
Selective reporting (reporting bias)	Unclear risk	Judgement comment: no protocol
Other bias	Low risk	Judgement comment: none

Teimoori 2019

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS • Mean age (years): NR • Mean pregnancy age (days): NR MS + estradiol valerate • Mean age (years): NR • Mean pregnancy age (days): NR Overall • Mean age (years): 28 ± 7 • Mean pregnancy age (days): 52 ± 13 **Inclusion criteria:** missed abortions under 14 weeks of gestational age **Exclusion criteria:** cesarean section history, adrenal disease, renal or hepatic diseases, and prostaglandin allergy **Pretreatment:** none
Interventions	**Intervention characteristics** MS • vaginal dose of 800 µg MS, up to 3 times at 8-h intervals MS + E • 1st dose of 800 µg MS with the oral estradiol valerate (2 mg), up to 3 times at 8-h intervals
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation

Teimoori 2019 *(Continued)*

	• Outcome type: dichotomous outcome Repeated medical abortion • Outcome type: dichotomous outcome Interval from MS administration to abortion • Outcome type: continuous outcome
Identification	**Sponsorship source:** Zahedan University of Medical Sciences **Country:** Iran **Setting:** Ali Ibn-e Abitalib hospital of Zahedan **Author(s):** Batool Teimoori **Institution:** Department of Obstetrics and Gynecology, Infectious Diseases and Tropical Medicine Research Center, School of Medicine, Zahedan University of Medical Sciences **Email:** Farahnaz1826@yahoo.com **Address:** 1Department of Obstetrics and Gynecology, Health Promotion Research Center, School of Medicine, Zahedan University of Medical Sciences, Zahedan, Iran IRCT2017070934981N1
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Quote: "randomized" Judgement comment: unclear method
Allocation concealment (selection bias)	High risk	Judgement comment: NR
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: NR
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no dropouts
Selective reporting (reporting bias)	High risk	Quote: "IRCT2017070934981N1;" Judgement comment: side effects were NR
Other bias	Low risk	Judgement comment: none

Torky 2018

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** LET + MS • Mean age (years): 26.52 (3.87) • Mean pregnancy age based on LMP (days): 50.9 (7.78) • BMI: 25.12 (3.3) MS • Mean age (years): 26.62 (4.3) • Mean pregnancy age based on LMP (days): 48.83 (8) • BMI: 25.81 (2.7) Overall • Mean age (years): NR • Mean pregnancy age based on LMP (days): NR • BMI: NR **Inclusion criteria:** with unexplained 1st trimester miscarriage **Exclusion criteria:** previous attempt to terminate the pregnancy, abnormal uterine lesions such as fibroids or congenital malformations, pregnancy despite an IUD (a known cause of miscarriage), medical disorders such as cardiac or haemorrhagic disease, and known hypersensitivity to any of the medications used **Pretreatment:** none **Study duration:** July 2015 to June 2016
Interventions	**Intervention characteristics** LET + MS • LET 10 mg twice daily for 3 days followed by 800 µg MS administered vaginally MS • placebo tablets twice daily for 3 days, followed by 800 µg of MS vaginally on the 4th day of enrolment
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Severe bleeding • Outcome type: dichotomous outcome Abdominal pain • Outcome type: adverse event

Torky 2018 *(Continued)*

	Fever • Outcome type: adverse event Interval from MS administration to abortion • Outcome type: continuous outcome • Unit of measure: days
Identification	**Sponsorship source:** October 6th University and As-Salam International Hospital **Country:** Egypt **Setting:** Al-Azhar University Hospitals, Al-Galaa Teaching Hospital, October 6th University Hospital and Airforce Specialized Hospital **Author(s):** Haitham Torky **Institution:** Department of Obstetrics & Gynecology, October 6th University and As-Salam International Hospital **Email:** haithamtorky@yahoo.com **Address:** Cornisch E–Nile-Maadi, Cairo, Egypt PACTR201701001973403; www.pactr.org
Notes	**Follow-up duration:** 1 week

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Patients allocated into the study were randomized into either of two groups using a computer-generated list at a 1: 1 ratio." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Concealment was achieved using opaque envelopes" Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement comment: placebo
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement comment: placebo
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: all participants were followed
Selective reporting (reporting bias)	Low risk	Quote: "PACTR201701001973403 Pan African Clinical Trials Registry www.pactr.org"
Other bias	Low risk	Judgement comment: none

Vahid Roudsari 2010

Study characteristics	
Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MS • Mean age (years): 27 (5) • Mean pregnancy age (weeks): 10 (2) MS + MTX • Mean age (years): 27 (2) • Mean pregnancy age (weeks): 9 (1) Overall • Mean age (years): NR • Mean pregnancy age (weeks): NR **Inclusion criteria:** complete awareness of patients from both medical and surgical methods for pregnancy termination and its side-effects. An intrauterine pregnancy < 13 weeks on the basis of LMP or abdominal ultrasonography. Specific reasons for pregnancy termination (missed abortion, blighted ovum and therapeutic abortion) **Exclusion criteria:** hypersensitivity to MS, severe anaemia (haemoglobin < 10g/dL), coagulopathy disorders or the use of anticoagulant drugs, acute liver and adrenal disease, cardiovascular diseases, uncontrolled seizure, and the use of corticosteroids **Pretreatment:** none
Interventions	**Intervention characteristics** MS • 800 µg vaginal MS in the posterior fornix of vagina, another 800 µg vaginal dose of MS would be administered if no expulsion MS + MTX • IM administration of 50 mg/m^2 of MTX at 1st visit and returned after 72 h (2nd visit) for vaginal administration of 800 µg MS. At the 3rd visit (24 h later), if expulsion did not occur, 800 µg of vaginal MS was again inserted
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** medical faculty of Mashhad University **Country:** Iran **Setting:** Teaching hospitals related to Mashhad University of Medical Sciences **Author(s):** Sedigheh Ayati **Institution:** Department of Obstetrics and Gynecology, Mashhad University of Medical Sciences, Women's Health Research Center

Vahid Roudsari 2010 *(Continued)*

	Email: ayatis@mums.ac.ir **Address:** Ghaem Hospital, Mashhad, Iran
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Quote: "The patients were randomisedly divided into two groups of 100 patients." Judgement comment: unclear method
Allocation concealment (selection bias)	Unclear risk	Judgement comment: NR
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: no blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement comment: no dropouts
Selective reporting (reporting bias)	Unclear risk	Judgement comment: unavailable protocol
Other bias	Low risk	Judgement comment: none

Verma 2011

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (24-h interval) • Mean age (years): • Mean pregnancy age (weeks): 5.93 • Nulliparous: 1 MF + MS (48-h interval) • Mean age (years): 27.92 • Mean pregnancy age (weeks): 5.94 • Nulliparous: 2

Verma 2011 *(Continued)*

	Overall • Mean age (years): • Mean pregnancy age (weeks): • Nulliparous: **Inclusion criteria:** **Exclusion criteria:** **Pretreatment:** **Study duration:** August 2007 to July 2008
Interventions	**Intervention characteristics** MF + MS (24-h interval) • 200 mg of oral MF; 400 µg MS at 24 h, vaginal MF + MS (48-h interval) • 200 mg of oral MF; 400 µg MS at 48 h, vaginal
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome • Notes: 2 participants in control group withdrew from the study, but the gestational ages were unclear, so the ITT data in the article were applied by review author Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Failure to achieve complete abortion (< 49 days) • Outcome type: dichotomous outcome • Reporting: fully reported • Direction: lower is better Failure to achieve complete abortion (50-63 days) • Outcome type: dichotomous outcome • Reporting: partially reported • Direction: lower is better
Identification	**Sponsorship source:** Department of Obstetrics and Gynecology, Chathrapati Shahuji Maharaj Medical University, Lucknow, UttarPradesh **Country:** India **Setting:** Family planning outdoor patient department, Department of Obstetrics and Gynecology, Chathrapati Shahuji Maharaj Medical University, Lucknow, UttarPradesh **Comments:** no **Author(s):** Manju Lata Verma **Institution:** Department of Obstetrics and Gynecology, CSMMU, Chowk

Verma 2011 *(Continued)*

	Email: gaganmlv@gmail.com **Address:** Lucknow-226003, Uttar Pradesh, India CTRI No.2010/091/001422
Notes	Follow-up: 14 days. 8 and 12 participants were lost to follow-up or dropout, but ITT analysis were performed by study authors and review authors .

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Sequential randomization was done using" Quote: "allocation ratio of 1:1" Judgement comment: without detailed method
Allocation concealment (selection bias)	High risk	Judgement comment: none
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: none
Blinding of outcome assessment (detection bias) All outcomes	High risk	Judgement comment: none
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "18 subjects did not turn up for follow-up. Two subjects refused for follow-up ultrasound." Judgement comment: ITT analyses were performed by authors. Only mean data were reported for abortion induction time
Selective reporting (reporting bias)	Low risk	Quote: "(CTRI No. 2010/091/001422)." Judgement comment: both efficacy and side effects were reported
Other bias	Low risk	Judgement comment: none

Verma 2017

Study characteristics	
Methods	Prospective, single-centre, parallel-group, open-labelled RCT
Participants	**Inclusion criteria**: an intrauterine pregnancy up to 63 d of period of gestation; willingness to comply with schedule; willingness to have surgical abortion if indicated **Exclusion criteria**: ectopic pregnancy; systemic steroid therapy; adrenal insufficiency; bronchial asthma; glaucoma; poorly controlled seizures; haemoglobin < 10 g/dL; sickle cell anaemia; known coagulopathy: rheumatic heart disease; patients on anticoagulants; pregnancy with IUD in utero; acute cervicitis on examination; and ultrasound demonstrating early pregnancy failure

Verma 2017 *(Continued)*

Interventions	• Group A: 200 mg oral MF and 400 µg vaginal MS concurrently • Group B: 200 mg oral MF followed by 400 µg vaginal MS after 48 h
Outcomes	Success rate of complete expulsion; induction abortion interval, side effects and compliance
Identification	-
Notes	Follow-up duration 14 days

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Participants recruited in the study were randomised in 2 groups using computer software
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	High risk	The researcher and participants were aware of the intervention carried out
Blinding of outcome assessment (detection bias) All outcomes	High risk	The researcher and participants were aware of the intervention carried out
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analyses were performed by study authors
Selective reporting (reporting bias)	Unclear risk	No protocol
Other bias	Unclear risk	-

von Hertzen 2003

Study characteristics

Methods	computer generated random sequence; packing company prepared bags containing the medication according to the randomisation sequence
Participants	**Inclusion criteria:** </= 63 days of amenorrhoea; single intrauterine pregnancies, haemoglobin > 100 g/L **Exclusion criteria:** medical contraindications or allergy for either mifepristone or misoprostol; past or present thromboembolism; liver disease, pruritus of pregnancy; previous surgery of uterine cervix; presence of an intrauterine device; suspected or proven ectopic pregnancy; smoking > 10 cigarettes/day; risk factor for cardiovascular disease; breastfeeding;
Interventions	MF 200mg followed 36-48 hours later by: • group 1: MS 800 µg orally followed by MS 400 µg twice/day for 6 days oral • group 2: MS 800 µg vaginally followed by MS 400 µg twice/day for 6 days oral

von Hertzen 2003 *(Continued)*

	• group 3: MS 800 µg vaginally followed by placebo tablets twice/day for 6 days oral
Outcomes	side-effects and acceptability
Identification	
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	computer generated random sequence
Allocation concealment (selection bias)	Low risk	packing company prepared bags containing the medication according to the randomisation sequence
Blinding of participants and personnel (performance bias) All outcomes	Low risk	double blinding
Blinding of outcome assessment (detection bias) All outcomes	Low risk	double blinding
Incomplete outcome data (attrition bias) All outcomes	Low risk	balanced
Selective reporting (reporting bias)	Low risk	both the efficacy and side effects were reported
Other bias	Low risk	none

von Hertzen 2007

Study characteristics

Methods	Centrally; random permuted blocks of 10; sealed, sequentially labelled envelopes
Participants	2066 women; average age 27 years **Inclusion criteria:** haemoglobin > 95g/L, gestational age ≤ 63 days, willing to have surgical procedure in case of failure, no serious illnesses, no contraindications for use of study medication, no uterine or cervical scars, no: uncontrolled asthma, hypertension, valvular heart disease, IUD in situ, history of thromboembolism or haemolytic disease, sickle cell anaemia or liver disease. Gestational age confirmed by ultrasound Study conducted at 11 obstetrics and gynaecologic teaching departments in 6 countries (Armenia, Cuba, Georgia, India, Mongolia, Vietnam)
Interventions	MS 3 doses of 800 µg each, in the manner of one of the following: • Group 1: sublingual every 3 h • Group 2: sublingual every 12 h

von Hertzen 2007 *(Continued)*

	• Group 3: vaginal every 3 h • Group 4: vaginal every 12 h
Outcomes	Complete, incomplete abortion, side effects
Identification	
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation sequence. Every centre received assignments by randomly permuted blocks with a fixed block size of 8. Randomisation was stratified by centre and gestational age.
Allocation concealment (selection bias)	Low risk	Allocation was concealed with sealed, sequentially numbered envelopes, which were filled and labelled in accordance with the list of randomisation for each centre
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Every dose consisted of 4 sublingual tablets (active or placebo) and 4 vaginal tablets (active or placebo) used at the same time. Placebo tablets were of similar shape and colour to MS (Cytotec, Searle, UK) tablets but differed by taste and were without the name of the manufacturer. Thus, the route of administration and interval were known
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Every dose consisted of four sublingual tablets (active or placebo) and four vaginal tablets (active or placebo) used at the same time. Placebo tablets were of similar shape and colour to MS (Cytotec, Searle, UK) tablets but differed by taste and were without the name of the manufacturer. Thus, the route of administration and interval were known
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No ITT
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

von Hertzen 2009

Study characteristics	
Methods	Computer-generated, central randomisation, random permutation in groups of 8, stratified by gestational age; sealed, opaque, sequentially numbered envelopes
Participants	2181 women; gestational age ≤ 63 days, inclusion criteria: haemoglobin > 100g/L, willing to have surgical abortion for failure, agreed to return for follow-up. Gestational age confirmed by ultrasound. Study conducted between 2003-2005 at 13 departments of obstetrics and gynaecology in nine countries (China, Hungary, India, Mongolia, Romania, Slovenia, South Africa, Viet Nam, Serbia)

von Hertzen 2009 *(Continued)*

	Exclusion criteria: ill health, contraindications to study medication, severe uncontrolled asthma, porphyria, valvular heart disease, smoking and other risk for CV disease, glaucoma, thromboembolism, liver disease, IUD in situ, breastfeeding, haemolytic disorders.
Interventions	• Group 1: MF 100 mg and 24 h later MS 800 µg vaginal • Group 2: MF 100 mg and 48 h later MS 800 µg vaginal • Group 3: MF 200 mg and 24 h later MS 800 µg vaginal • Group 4: MF 200 mg and 48 h later MS 800 µg vaginal Follow-up at 2 and 6 weeks
Outcomes	Complete, incomplete, missed abortion, side effects
Identification	
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated, central randomisation, random permutation in groups of 8, stratified by gestational age
Allocation concealment (selection bias)	Low risk	Sealed, opaque sequentially numbered envelopes
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Placebo tablets were of similar shape and colour as MF tablets. The interval to MS administration was not blinded
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Placebo tablets were of similar shape and colour as MF tablets. The interval to MS administration was not blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	According to the protocol, a decision to discontinue a treatment regimen due to low efficacy was based on the comparison with the regimen of 200 mg of MF followed by 800 µg of MS vaginally 48 h later ITT was not performed
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Unclear risk	The UNDP/UNFPA/WHO/World Bank Special Programme of Research, Development and Research Training in Human Reproduction funded this study including MS tablets. MF and placebo tablets were donated by Hualian Pharmaceuticals Ltd. Neither the donors and sponsors of the Programme nor Hualian Pharmaceuticals Ltd had a role in the study design, data collection, data analysis, data interpretation, writing of the report or the decision to submit the paper for publication.

von Hertzen 2010

Study characteristics

von Hertzen 2010 *(Continued)*

Methods	**Study design:** RCT **Study grouping:** parallel-group
Participants	**Baseline characteristics** MF + MS (400 µg, sublingual) • Mean age (years): 27 (6.3) • Weight (kg): 55.3 (10) • Haemoglobin (g/L): 121.6 (12.2) • Parous: 52.9% • Previous abortion: 42.9% MF + MS (800 µg, sublingual) • Mean age (years): 26.6 (6) • Weight (kg): 55.3 (10.3) • Haemoglobin (g/L): 122.1 (12) • Parous: 54.5% • Previous abortion: 42.3% MF + MS (400 µg, vaginal) • Mean age (years): 26.7 (6.1) • Weight (kg): 55.5 (10.4) • Haemoglobin (g/L): 122.6 (11.8) • Parous: 54.2% • Previous abortion: 45.9% MF + MS (800 µg, vaginal) • Mean age (years): 26.6 (5.8) • Weight (kg): 55.3 (10.3) • Haemoglobin (g/L): 121.9 (12) • Parous: 52.2% • Previous abortion: 39.9% Overall • Mean age (years): 26.7 (6.1) • Weight (kg): 55.4 (10.2) • Haemoglobin (g/L): 122.1 (12) • Parous: 53.4% • Previous abortion: 42.8% **Inclusion criteria:** healthy, > the age of legal consent, had haemoglobin concentration ≥ 90 g/L, had a single intrauterine pregnancy with a duration of ≤ 63 days verified by ultrasound, agreed to surgical termination of pregnancy should the treatment fail and were willing to participate in the study **Exclusion criteria:** indication of serious past or present illness; allergic to MF or MS; DBP > 90 mmHg, SBP < 90 mmHg or with a medical condition or disease that required special treatment, care or precaution; with an IUD; breastfeeding; molar or nonviable pregnancy **Pretreatment:** none **Study duration:** April 2007 to June 2008
Interventions	**Intervention characteristics**

von Hertzen 2010 *(Continued)*

	MF + MS (400 µg, sublingual) • MF + MS (different dose and route): 200 mg MF orally followed by 4 tablets administered vaginally (placebo) and 4 tablets administered sublingually (2 active and 2 placebo) 24 h later MF + MS (800 µg, sublingual) • MF + MS (different dose and route): 200 mg MF orally followed by 4 tablets administered vaginally (placebo) and 4 tablets administered sublingually (active) 24 h later MF + MS (400 µg, vaginal) • MF + MS (different dose and route): 200 mg MF orally followed by 4 tablets administered vaginally (2 active and 2 placebo) and 4 tablets administered sublingually (placebo) 24 h later MF + MS (800 µg, vaginal) • MF + MS (different dose and route): 200 mg MF orally followed by 4 tablets administered vaginally (active) and 4 tablets administered sublingually (placebo) 24 h later
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome Nausea • Outcome type: adverse event Vomiting • Outcome type: adverse event Diarrhoea • Outcome type: adverse event Headache • Outcome type: adverse event Abdominal pain • Outcome type: adverse event Fever • Outcome type: adverse event Chills • Outcome type: adverse event Satisfaction • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** WHO **Country:** China; Havana, Cuba; Tbilisi, Georgia; Mumbai, New Delhi and Trivandrum, India; Ulaanbaatar, Mongolia; Cluj Napoca, Romania; Ljubljana, Slovenia; Stockholm, Sweden; Bangkok, Thailand; and Hanoi and Ho Chi Minh City

von Hertzen 2010 *(Continued)*

	Setting: 15 obstetrics/gynaecology departments in 10 countries **Comments:** 4 arms **Author(s):** H von Hertzen **Institution:** Concept Foundation **Email:** helena.vonhertzen@gmail.com **Address:** Ch Louis Dunant 17, 1202 Geneva, Switzerland **ISRCTN87811512:** ISRCTN87811512
Notes	Follow-up after 2 and 6 weeks

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "This was a controlled, randomised, 2 · 2 factorial, noninferiority trial, stratified by centre and by length of gestation. At each of the participating centres, eligible women were allocated randomly to the four treatment groups using a computer-generated randomisation sequence in blocks of variable size." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Allocation was concealed using sealed, sequentially numbered envelopes, which were filled and labelled by Labatec (Geneva, Switzerland) according to the randomisation for each centre." Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Quote: "placebo tablets (Labatec-Pharma SA, Geneva, Switzerland) were packed in blisters and were of similar shape and colour but differed in taste."
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Quote: "placebo tablets did not have the label of the manufacturer on them."
Incomplete outcome data (attrition bias) All outcomes	Low risk	Quote: "The main efficacy analysis included all women for whom the treatment outcome was known, in the groups as randomised."
Selective reporting (reporting bias)	Low risk	Balanced
Other bias	Low risk	Judgement comment: none
Other bias	Low risk	Judgement comment: none

Wang 2000

Study characteristics	
Methods	Women were randomly divided into 2 groups by 2:1 ratio

Wang 2000 *(Continued)*

Participants	Multicentre trial in 9 hospitals in Hebei,China; 1612 pregnant women ≤ 49 days of amenorrhoea, confirmed by ultrasound; without clinical or haematological abnormalities, contraindication for the study medication or IUD in situ
Interventions	• Group 1: ∘ day 1: MF 50 mg orally 12 h apart (= total of 100 mg) ∘ day 2-day 7: MF 25 mg orally daily (= total of 250 mg) ∘ day 3: MS 600 µg orally ∘ day 4-day 6: MS 200 µg daily (= total of 600 µg) • Group 2: ∘ day 1: MF 50 mg orally then 25 mg 12-hourly, 4 times (= total of 150 mg) ∘ day 3: MS 600 µg orally
Outcomes	Complete/incomplete abortion, duration of bleeding, resumption of menses, side effects
Identification	
Notes	Post-randomisation exclusion, protocol deviation, loss to follow-up not mentioned No mention of major complications

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No detailed methods reported
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Unclear
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Unclear
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unclear
Selective reporting (reporting bias)	Unclear risk	Side effects were not reported in detail
Other bias	Low risk	None

Warriner 2011

Study characteristics

Methods	**Study design:** RCT **Study grouping:** parallel-group

Warriner 2011 *(Continued)*

Participants	**Baseline characteristics** MF + MS (nurse) • Mean age (years): 28.1 (6) • Mean pregnancy age based on LMP (weeks): 6.6 (1.1) • Height (cm): 151.5 (5.4) • Weight (kg): 48.8 (8) • Livebirths: 2.3 (1.1) • Previous abortions: 0.2 (0.7) • Mean pregnancy age based on clinical examination (weeks): 6.9 (1) MF + MS (physician) • Mean age (years): 28 (5.8) • Mean pregnancy age based on LMP (weeks): 6.6 (1.1) • Height (cm): 151.8 (5.3) • Weight (kg): 49.5 (8.4) • Livebirths: 2.2 (1.1) • Previous abortions: 0.3 (0.7) • Mean pregnancy age based on clinical examination (weeks): 6.6 (1) Overall • Mean age (years): NR • Mean pregnancy age based on LMP (weeks): NR • Height (cm): NR • Weight (kg): NR • Livebirths: NR • Previous abortions: NR • Mean pregnancy age based on clinical examination (weeks): NR **Inclusion criteria:** < 9 weeks (≤ 63 days) of gestation; > 16 years, resided no more than 90 min journey from the study clinic, and were willing to be randomly assigned to a provider, to return to the clinic 10–14 days after the start of treatment, and to provide written informed consent **Exclusion criteria:** any contraindication to medical abortion: previous allergic reaction to any of the drugs in the medical abortion regimen; known or suspected ectopic pregnancy or undiagnosed adnexal mass; inherited porphyria; chronic adrenal failure; long-term corticosteroid therapy; haemorrhagic disorder or anticoagulant therapy; or an IUD that could not be removed before administration of MF **Pretreatment:** none **Study duration:** April 15, 2009,and March 17, 2010
Interventions	**Intervention characteristics** MF + MS (nurse) • mid-level provider for oral administration of 200 mg MF followed by 800 µg MS vaginally 2 days later MF + MS (physician) • doctor provider for oral administration of 200 mg MF followed by 800 µg MS vaginally 2 days later
Outcomes	Failure to achieve complete abortion • Outcome type: dichotomous outcome Surgical evacuation

Warriner 2011 *(Continued)*

	• Outcome type: dichotomous outcome Ongoing pregnancy • Outcome type: dichotomous outcome Abdominal pain • Outcome type: adverse event Bleeding (more than expected) • Outcome type: dichotomous outcome Satisfied • Outcome type: dichotomous outcome
Identification	**Sponsorship source:** UNDP/UNFPA/WHO/World Bank Special Programme of Research, Development and Research Training in Human Reproduction (HRP), Department of Reproductive Health and Research (RHR), WHO **Country:** Nepal **Setting:** five district hospitals in Nepal **Comments:** nurse vs physician **Author(s):** I K Warriner **Institution:** Special Programme of Research, Development and Research Training in Human Reproduction (HRP), WHO **Email:** warrineri@who.int **Address:** 20 Avenue Appia, 1211 Geneva 27, Switzerland **NCT01186302:** NCT01186302
Notes	**Follow-up duration:** 10-14 days

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "We used a computer-generated randomisation scheme stratified by study centre with a block size of six generated at RHR, WHO in Geneva, Switzerland." Judgement comment: adequate
Allocation concealment (selection bias)	Low risk	Quote: "Sealed opaque envelopes containing the random allocation were consecutively numbered and were opened and assigned sequentially to women by a research assistant once" Judgement comment: adequate
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement comment: no blinding
Blinding of outcome assessment (detection bias)	High risk	Judgement comment: no blinding

Warriner 2011 *(Continued)*

All outcomes		
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Quote: "Loss to follow-up of women receiving treatment was 24 of 542 (4%) in the midlevel health-care providers' group and 19 of 535 (4%) in the doctors' group." Judgement comment: although ITT analysis was performed, not all participants who dropped out were analysed
Selective reporting (reporting bias)	Unclear risk	Judgement comment: side effects not reported in detail
Other bias	Low risk	Judgement comment: none

WHO 1989

Study characteristics	
Methods	Randomly allocated 10/261 post-randomisation exclusions: 2: cycle length < 25 days 6: > 49 days pregnant 1: pregnancy not confirmed 1: wrongly randomised 1 woman was lost to follow-up (group 2)
Participants	Multicentre, hospitals in Aberdeen, Milan, New Delhi, Shanghai, Singapore, Stockholm, Szeged 261 pregnant women, ≤ 35 years ≤ 49 days of amenorrhoea confirmed by ultrasound and beta-HCG if ultrasound inconclusive **Inclusion criteria:** regular cycles (25-35 days) for last 3 months **Exclusion criteria**: unsure about dates, IUD in situ, hormonal contraception during last cycle and intention to start hormonal contraception before 1st period after abortion
Interventions	• Group 1: MF 25 mg twice daily for 3 days and sulprostone 0.25 mg IM on morning of 3rd day • Group 2: MF 25 mg twice daily for 4 days and sulprostone 0.25 mg IM on morning of 4th day
Outcomes	Complete, and incomplete abortion Failure (intact amniotic sac on follow-up at 2 weeks) Undetermined outcome Hormone levels (beta-HCG, estradiol, prolactin, cortisol, prostaglandin)
Identification	
Notes	2 women received blood transfusion; not mentioned which group
Risk of bias	

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No detailed method
Allocation concealment (selection bias)	Unclear risk	Unclear

WHO 1989 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Outcomes were all reported
Other bias	Unclear risk	RU 486 and sulprostone used in this study were kindly donated by Roussel-Uclaf (Paris, France) and Schering AG (West-Berlin), respectively

WHO 1991

Study characteristics	
Methods	Randomisation at WHO, using random permutation block technique with block size of 8, random numbers were provided to each centre in a sealed envelope
Participants	Multicentre; 10 mostly academic hospitals: Aberdeen, Havana, Hong Kong, Ljubljana, Milan, Shanghai, Singapore, Stockholm, Szeged, Wuhan. 385 women were randomised **Inclusion criteria:** amenorrhoea ≤ 49 days, regular cycles (25-35 days) for last 3 months **Exclusion criteria**: unsure about dates, IUD in situ, hormonal contraception during last cycle and intention to start hormonal contraception before 1st period after abortion
Interventions	• Group 1: MF 25 mg 12-hourly 5 doses and GP 1 mg vaginally 60 h after the start of the treatment • Group 2: MF 600 mg single dose and GP 1 mg vaginally 60 h after the start of the treatment
Outcomes	Complete, incomplete, missed abortion, continuing pregnancy, side effects, bleeding pattern, haemoglobin and hormone levels
Identification	
Notes	1 woman received blood transfusion; not mentioned which group

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation at WHO, using random permutation block technique with block size of 8
Allocation concealment (selection bias)	Low risk	Random numbers were provided to each centre in a sealed envelope
Blinding of participants and personnel (performance bias)	Unclear risk	NR

WHO 1991 *(Continued)*

All outcomes		
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No ITT
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	Supported by the Special Programme of Research, Development and Research Training in Human Reproduction, WHO, Geneva, Switzerland. MF used in this study was kindly provided by RousselUclaf (Paris, France), and May and Baker (Dagenham, United Kingdom)

WHO 1993

Study characteristics	
Methods	Randomisation at WHO, using random permutation block technique with block size of 9, tablets were disposed into labelled bottles, placebos were added to women receiving the lower dose so that all received 3 tablets)
Participants	Multicentre, hospitals in Aberdeen, Edinburgh, Havana, Hong Kong, Ljubljana, Milan, Shanghai, Stockholm, Szeged, Tianjin, Wuhan 1182 pregnant women with a menstrual delay of 7-28 days **Inclusion criteria:** regular cycles (25-35 days) for last 3 months, pregnancy confirmed by ultrasound **Exclusion criteria**: unsure about dates, IUD in situ, hormonal contraception during last cycle and intention to start hormonal contraception before 1st period after abortion, contraindication to MF/MS, regular use of prescribed drugs
Interventions	• Group 1: MF 200 mg/oral • Group 2: MF 400 mg/oral • Group 3: MF 600 mg/oral All: prostaglandin 1 mg/vaginally after 48 h
Outcomes	Complete, incomplete, missed abortion, continuing pregnancy, haemoglobin levels, side effects
Identification	
Notes	3 women received blood transfusion; not mentioned which group

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation at WHO, using random permutation block technique with block size of 9
Allocation concealment (selection bias)	Low risk	Tablets were disposed into labelled bottles

WHO 1993 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Tablets were disposed into labelled bottles, placebos were added to women receiving the lower dose so that all received 3 tablets
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Tablets were disposed into labelled bottles, placebos were added to women receiving the lower dose so that all received 3 tablets
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None

WHO 2000

Study characteristics	
Methods	Computer-generated random numbers
Participants	Multicentre trial: Beijing, Havana, Helsinki, Ho Chi Min City, Hong Kong, Ljubljana, Melbourne, Moscow, Mumbai, Shanghai, Stockholm, St Petersburg, Szeged, Tbilisi, Tianjin, Tunis, Yerevan 1589 women ≤ 63 days of amenorrhoea, with positive pregnancy test and uterine size consistent with menstrual history **Exclusion criteria**: contraindications for study drug use, history of thromboembolism, liver disease, regular use of prescription drugs, IUD, suspected ectopic pregnancy, heavy cigarette smoking, breastfeeding, irregular menses
Interventions	• Group 1: MF 200 mg orally • Group 2: MF 600 mg orally Both groups received MS 400 μg orally after 48 h
Outcomes	Complete/incomplete/missed/unclassified failed abortion, side effects
Identification	
Notes	Identical placebos, identical pill bottles; power calculation (90% power, significance level of 0.05) No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Low risk	The allocation sequence was concealed from investigators and participants by using identically appearing pill bottles

WHO 2000 *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	The allocation sequence was concealed from investigators and participants by using identically appearing pill bottles
Blinding of outcome assessment (detection bias) All outcomes	Low risk	The allocation sequence was concealed from investigators and participants by using identically appearing pill bottles
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis was done
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None

WHO 2001a

Study characteristics	
Methods	Computer-generated sequence of random numbers in block of 10, identical placebo tablets
Participants	Multicentre trial, 10 centres: Chandigarh, Edinburgh, Havana, Hong Kong, Ljubljana, Shanghai, Stockholm, Szeged, Tbilisi, Tianjin 896 women, at 57-63 days of gestation with regular menstrual cycles, pregnancy confirmed clinically or by ultrasound **Exclusion criteria**: contraindication to the study drugs, chronic respiratory, digestive, endocrine, genitourinary, neurological or cardiovascular disease, severe liver disease, history of thromboembolism, IUD in situ, breastfeeding
Interventions	• Group 1: MF 200 mg • Group 2: MF 600 mg All: GP 1 mg after 48 h
Outcomes	Complete, incomplete, missed abortion, time to onset of bleeding, duration of bleeding, time to return to menses, bleeding before GP, time of expulsion
Identification	
Notes	Power calculation (80% power at a significant level of 0.05) ITT analysis 2 women received blood transfusion, not mentioned which group

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated sequence of random numbers in block of 10
Allocation concealment (selection bias)	Low risk	The allocation sequence was concealed from both investigators and participants by use of a central pharmacy

WHO 2001a *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Identical placebo tablets
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Identical placebo tablets
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None

WHO 2001b

Study characteristics

Methods	Computer-generated number sequence
Participants	Multicentre trial, 13 centres: Aberdeen, Chandigarh, Edinburgh, Havana, Hong Kong, Ljubljana, Lusaka, Shanghai, Singapore, Stockholm, Szeged, Tbilisi, Tianjin 1224 women < 57 days pregnant **Inclusion criteria:** regular cycles, no hormonal contraception or IUD use before 1st menses after abortion **Exclusion criteria**: medical contraindication for the study medication, history of thromboembolism, liver disease, pruritus in pregnancy, IUD in situ, breastfeeding, heavy smokers
Interventions	• Group 1: MF 50 mg orally and GP 0.5 mg vaginally on day 3 • Group 2: MF 50 mg orally and GP 1.0 mg vaginally on day 3 • Group 3: MF 200 mg orally and GP 0.5 mg vaginally on day 3 • Group 4: MF 200 mg orally and GP 1.0 mg vaginally on day 3
Outcomes	Complete /incomplete/missed abortion, side effects
Identification	
Notes	Group 1 was discontinued as interim analysis showed below cut-off results No blinding for GP 7 women received blood transfusion (2 Group 1, 2 Group 2, 1 Group 3, 2 Group 4)

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated number sequence
Allocation concealment (selection bias)	Low risk	The allocation sequence was concealed from investigators and participants by using sealed, opaque envelopes labelled with the study number

WHO 2001b *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Participants were blinded as to the therapy and physicians to the dose of MF. Clinicians were kept unaware of the MF dose
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Clinicians were kept unaware of the MF dose. Analysts were not blinded as to treatment
Incomplete outcome data (attrition bias) All outcomes	Low risk	The analyses were by ITT including participants eventually found to have been ineligible for the study
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Unclear risk	Group 1 was discontinued as interim analysis showed below cut-off results. The MF tablets (50 mg or 200 mg) and the corresponding placebo tablets supplied by Roussel-Uclaf (Romainville, France)

Wiebe 1999a

Study characteristics	
Methods	Computer-generated list of random numbers, sealed, opaque envelopes
Participants	398 women, ≤ 7 weeks pregnant confirmed by ultrasound, University Hospital Vancouver, Canada **Exclusion criteria**: abnormal haematologic parameters
Interventions	Phase 1 • Group 1: TM 40 mg orally and 800 µg MS vaginally > 48 h • Group 2: MTX 50 mg/m^2 and MS 800 µg vaginally > 96 h Phase 2 • Group 1: TM 40 mg/day for 4 days (= total dose of 160 mg) and MS 800 µg vaginally > 48 h • Group 2: MTX 50 mg/m^2 and MS 800 µg/vaginally > 96 h
Outcomes	Failure rate, side effects, women's preference
Identification	
Notes	No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated list of random numbers
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes

Wiebe 1999a *(Continued)*

Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unclear
Selective reporting (reporting bias)	Low risk	All reported
Other bias	Low risk	None

Wiebe 1999b

Study characteristics	
Methods	Computer-generated list of random numbers, sealed, opaque envelopes
Participants	100 women, ≤ 7 weeks pregnant confirmed by ultrasound, University Hospital Vancouver, Canada **Exclusion criteria**: abnormal haematologic parameters, systemic disease, intolerance to study medication
Interventions	• Group 1: MTX 50 mg/m^2 orally and MS 600 µg vaginally > 96 h • Group 2: MTX 50 mg/m^2 IM and MS 600 µg vaginally > 96 h
Outcomes	Complete, incomplete abortion, side effects
Identification	
Notes	Only data from phase 1 are included, phase 2 was non-random No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated list of random numbers
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias)	Unclear risk	NR

Wiebe 1999b *(Continued)*

All outcomes		
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No ITT
Selective reporting (reporting bias)	Low risk	All reported
Other bias	Low risk	None

Wiebe 2004

Study characteristics	
Methods	Computer-generated random list; sealed opaque envelopes
Participants	309 women at ≤ 7 weeks of gestation confirmed by ultrasound; average age 27 years; average gestational age 42 days **Exclusion criteria**: haemoglobin < 9.5 g/L, seizure disease, active liver disease, renal insufficiency, allergy/intolerance to study medication Study conducted at University of British Colombia, Canada
Interventions	MTX 50 mg/m^2 followed 72-44 h later by: • Group 1: MS 600 µg buccal (insert between their cheeks and leave for 1 h) • Group 2: MS 600 µg vaginal Both groups instructed to repeat the dose 24 h later if no heavy bleeding had occurred
Outcomes	Successful abortion; side-effects; acceptability
Identification	
Notes	Women with ongoing pregnancy at day 8 follow-up received 1-2 more doses of MS

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated list of random numbers
Allocation concealment (selection bias)	Low risk	Sealed numbered opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR

Wiebe 2004 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None

Wiebe 2006

Study characteristics	
Methods	"Randomised"
Participants	300 women with ≤ 7 weeks' gestation confirmed by ultrasound. Patient characteristics NR ("similar between groups")
Interventions	• Group 1: MTX 50 mg/m^2 followed > 72 h by MS 400 µg vaginally • Group 2: MS 400 µg sublingual and 400 µg vaginal
Outcomes	Complete abortion, side effects, acceptability
Identification	
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No detailed method
Allocation concealment (selection bias)	Unclear risk	Unclear
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	Reproductive Health and the Department of Family Practice of the University of British Columbia provided financial support for this study

Winikoff 2008

Study characteristics	
Methods	Computer-generated random assignment; random blocks of 8
Participants	966 women > 18 years old were enrolled. no contraindication to study medication, ≤ 63 days since LMP, access to telephone and emergency transportation. Gestational age confirmed by ultrasound if needed. Between September 2006-May 2007. Study conducted at 7 family planning centres: New York, Chicago, Pittsburgh, Waco, Austin, Boston; USA
Interventions	MF 200 mg followed 24-26 h later at home by: • Group 1: MS 800 µg orally • Group 2: MS 800 µg buccal (2 x 200 mg in each cheek to keep for 30 min and swallow the remnants)
Outcomes	
Identification	
Notes	Results presented as per protocol analysis; successful abortion was defined as: without need for surgical intervention, regardless of how many doses of MS needed. 14 women in the buccal group and 13 in the oral group received a 2nd dose of MS

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random assignment; random blocks of 8
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label
Blinding of outcome assessment (detection bias) All outcomes	High risk	Open-label
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analyses
Selective reporting (reporting bias)	Low risk	All outcomes reported
Other bias	Low risk	None

Wu 1993

Study characteristics

Wu 1993 *(Continued)*

Methods	Randomisation sequence generated centrally
Participants	Multicentre trial in 5 hospitals in Beijing, China 990 women ≤ 49 days of amenorrhoea, pregnancy confirmed by ultrasound, without medical disorders, contraindication for the study medication and IUD in situ
Interventions	• Group 1: ∘ day 1: MF 200 mg and TM 40 mg orally ∘ day 2: TM 40 mg orally ∘ day 3: PGF2alpha vaginally • Group 2: ∘ day 1: MF 200 mg and placebo orally ∘ day 2: placebo orally ∘ day 3: PGF2 alpha vaginally
Outcomes	Complete, incomplete abortion, duration of bleeding, resumption of menses, side effects
Identification	
Notes	58/990 women were excluded post-randomisation due to protocol violation No major complications were reported

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Randomisation sequence generated centrally but without detailed information
Allocation concealment (selection bias)	Unclear risk	Unclear
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Without ITT analysis
Selective reporting (reporting bias)	Low risk	All outcomes were reported
Other bias	Low risk	None

Zheng 1989

Study characteristics

Zheng 1989 *(Continued)*

Methods	Publication includes 4 studies, 1 of them is a randomised trial, randomisation procedure not stated
Participants	192 women, ≤ 49 days of pregnancy seeking abortion in China Inclusion/exclusion criteria not stated Follow-up on day 8 or day 14
Interventions	• Group 1: MF 600 mg • Group 2: MF 600 mg and prostaglandin F2alpha 1 mg vaginally
Outcomes	Complete and incomplete abortion, ongoing pregnancy, time until passing of conceptus
Identification	
Notes	Only data from trial 4 are included No mention of major complications

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation method NR
Allocation concealment (selection bias)	Unclear risk	NR
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	NR
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	NR
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition
Selective reporting (reporting bias)	Low risk	Both efficacy and side effects were reported
Other bias	Unclear risk	RU 486 was supplied by Roussel Uclaf Ltd, France

BMI: body mass index; **DBP:** diastolic blood pressure; **E:** estradiol valerate; **ECG:** electrocardiogram; **GP:** gemeprost; **HCG:** human chorionic gonadotropin; **IM:** intramuscular; **ITT:** intention-to-treat; **IUD:** intrauterine (contraceptive) device; **LMP:** last menstrual period; **LET:** letrozole; **MF:** mifepristone; **MS:** misoprostol; **MTX:** methotrexate; **NR:** not reported; **PBLAC:** pictorial blood loss assessment chart; **PGE1:** prostaglandin E1 analogue; **PGF2alpha:** prostaglandin F2 alpha; **RCT:** randomised controlled trial; **SBP:** systolic blood pressure; **SPSS:** Statistical Package for the Social Sciences; **STD:** sexually transmitted disease; **TCM:** traditional Chinese medicine; **TM:** tamoxifen; **TP:** testosterone propionate; **UNFPA:** United Nations Population Fund; **UNDP:** United Nations Development Programme; **WHO:** World Health Organization

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
ACTRN12612001254886	Uncompleted study
ACTRN12613001230741	Duplication
Aggarwal 2017	Ineligible patient population
Ashok 2002	Single cohort, no comparison group
Aubeny 2000	Randomisation by day of admission
Chelly 2010	Duplication
Cheng 1999	Women up to 16 weeks of gestation are included
ChiCTR1900025763	Uncompleted study
ChiCTR-TRC-11001438	Duplication
Creinin 1996b	Single cohort, no comparison group
CTRI/2011/07/001894	Ineligible study design
Dabash 2012	Duplication
Dalenda 2010	Quasi-RCT
Davis 1999	Data for 1 group (MTX) was reported for all (randomised and non-randomised) women together
De 2019	Ineligible intervention
Dehbashi 2016	Quasi-RCT
De Nonno 2000	Not RCT
Derakhshan Aydenloo 2018	Duplication
Dey 2013	Ineligible intervention
Duhan 2011	Ineligible intervention
Elbareg 2018	Duplication
ICMR 2000	Allocation concealment and randomisation not stated
IRCT201207259491N3	Ineligible intervention
IRCT2013090414563N1	Ineligible study design
IRCT2014030114293N1	Duplication
IRCT2014070711020N4	Ineligible intervention
IRCT2014110812789N9	Ineligible patient population

Study	Reason for exclusion
IRCT201412111760N37	Duplication
IRCT2017011526962N3	Duplication
IRCT2017040633255N1	Duplication
IRCT201704179014N159	Duplication
IRCT20180109038284N1	Ineligible study design
Jacobson 1990	This study was not designed to achieve abortion: only to test an existing regimen for treatment of ulcer and its effect on early pregnancy
JapicCTI-195008	Ineligible study design
Lee 2011a	Ineligible outcomes
Lee 2011b	Ineligible study design
Lee 2012	Ineligible intervention
Li 2012	Ineligible study design
Martin 1998	Intervention not in the scope of the review (oral contraceptives or MTX to shorten the duration of bleeding)
Mostafa Gharebaghi 2010	Ineligible patient population
NCT01156688	Duplication
NCT01173003	Ineligible study design
NCT01387256	Duplication
NCT01475318	Ineligible study design
NCT01615211	Uncompleted study
NCT01615224	Uncompleted study
NCT01811056	Ineligible study design
NCT01818414	Ineligible intervention
NCT01827995	Duplication
NCT01856985	Ineligible study design
NCT01920022	Ineligible intervention
NCT01966874	Ineligible study design
NCT02012491	Duplication
NCT02018796	Ineligible study design

Study	Reason for exclusion
NCT02141555	Uncompleted study
NCT02191774	Ineligible study design
NCT02299401	Duplication
NCT02314754	Ineligible study design
NCT02342002	Uncompleted study
NCT02363556	Ineligible patient population
NCT02401425	Duplication
NCT02480543	Ineligible intervention
NCT02515604	Duplication
NCT02522078	Ineligible intervention
NCT02614781	Ineligible study design
NCT02686840	Duplication
NCT02707653	Ineligible study design
NCT02720991	Ineligible study design
NCT02957305	Ineligible intervention
NCT03140384	Uncompleted study
NCT03320057	Ineligible study design
NCT03727308	Ineligible study design
Ngai 2000	Intervention not in the scope of the review (water and MS compared to MS alone)
NL6366	Duplication
Norman 1992	Non-randomised and randomised outcomes presented together
Parveen 2011	Ineligible intervention
Prabhu 2015	Quasi-RCT
Prine 2010	Duplication
Rezai 2014	No English full text
Rouzi 2014	Ineligible study design
Seervi 2014	Quasi-RCT
Swahn 1994	Single cohort, no comparison group

Study	Reason for exclusion
Swica 2013	Ineligible study design
Tang 1999	Intervention not in the scope of the review (oral contraceptives vs placebo for effectiveness, bleeding duration)
Tehranian 2010	No English full text
Torre 2012	Ineligible patient population
Tsereteli 2010	Duplication
Varona Sanchez 2010	Ineligible study design
Wedisinghe 2010	Ineligible study design
Wen 2010	Ineligible study design
Wiebe 2001	Review
Winikoff 2012	Ineligible study design
Yung 2016	Ineligible outcomes

MS: misoprostol; **MTX:** methotrexate; **RCT:** randomised controlled trial

Characteristics of studies awaiting classification *[ordered by study ID]*

Ayati 2013

Methods	RCT
Participants	80 women at first trimester termination of pregnancy in patients with previous uterus surgery who referred for pregnancy termination
Interventions	2 groups of MS 800 µg vaginal and 800 µg rectal
Outcomes	The complication and outcomes of abortion (complete evacuation, fever, nausea, diarrhoea, chills, sedation, need for transfusion, uterus rupture)
Notes	Ghaem hospital of Mashhad, Iran in 2010

Ibrahim 2019

Methods	RCT
Participants	120 women with missed miscarriage at 6-13 weeks' gestation for medical termination
Interventions	**LET + MS:** 60 women received 7.5 mg of LET orally for 2 days, and MS vaginally 600 µg 2 days later **TM + MS:** 60 mg of TM for 2 days, and 600 µg MS vaginally 2 days later If they had no vaginal bleeding a second dose of 800 microgram MS was given after 24 h

Ibrahim 2019 *(Continued)*

Outcomes	Surgical evacuation, satisfaction
Notes	Study duration: from January 2015-November 2015 Study setting: Tuzkhurmatu General Hospital in Salah Al-Deen government (Iraq)

IRCT201403244686N11

Methods	Parallel RCT, triple-blinded
Participants	72 women aged 18-45 years with missed abortion **Inclusion criterion:** missed abortion according sonography. **Exclusion criteria:** severe anaemia; allergic to prostaglandin; coagulopathy; acute liver disease; heart disease
Interventions	**MS + castor oil:** MS 800 µg vaginally plus castor oil 60 mL orally; repeated MS up to 3 times with 24-h interval, if without response **MS + placebo**: MS 800 µg vaginally plus placebo 60 mL orally; repeated MS up to 3 times with 24-h interval, if without response
Outcomes	Complete abortion, side effects
Notes	irct.ir/trial/5024, IRCT201403244686N11

IRCT20150407021653N19

Methods	Parallel RCT, simple randomisation by random numbers table, no blinding
Participants	30 women who attend to undergo outpatient treatment for the excretion of pregnancy products. The gestational age should be < 14 weeks. Do not have a history of high-risk diseases (diabetes and blood pressure problems and cardiovascular diseases, etc.) **Exclusion criteria:** certain illnesses, fever, blood pressure, don't tolerate drug therapy, be tired of the treatment
Interventions	**Group 1:** oxytocin + MS **Group 2:** methyl ergonovin (metergin) + MS
Outcomes	Success in excretion of pregnancy products during the 9 days after abortion; the mean time for disposal of pregnancy products after spontaneous abortion
Notes	irct.ir/trial/34519, IRCT20150407021653N19

IRCT2015112421506N3

Methods	Parallel RCT, single-blinded

IRCT2015112421506N3 *(Continued)*

Participants	100 pregnant women with gestational age ≤ 91 days based on sonography and menstruation that refereed to Akbar Abadi hospital and were candidates for the termination of pregnancies (blighted ovum, missed abortion) or medical abortion license included this study
Interventions	**Group 1:** vaginal administration of isonicotinic acid hydrazide (INH) (1800 mg) + MS (800 µg sublingual 8 h later), repeated every 3 h up to 3 times in 12 h as needed **Gourp 2:** 800 µg sublingual MS given that in the absence of pain and vaginal bleeding occurs every 3 hours. (Maximum of 3 doses in 12 h)
Outcomes	Abortion time, vomiting, diarrhoea, fever, nausea
Notes	irct.ir/trial/18840, IRCT2015112421506N3

IRCT2017042533635N1

Methods	Parallel RCT, double-blinded
Participants	140 pregnant women from 18-35 years old **Inclusion:** singleton pregnancy, gravid 2-4, without cervical dilatation and bleeding or tissue passing, gestational age < 14 weeks, BMI 19.8-26 kg/m^2 **Exclusion:** background diseases related with study such as overt diabetes, chronic hypertension, systemic lupus erythematosus, fever or chorioamnionitis signs, using unusual drugs in pregnant women, addiction, etc
Interventions	**Rectal MS:** MS 600 µg, then 2 doses of 300 every 6 h rectally **Oral MS:** MS 600 µg, then 2 doses of 300 every 6 h orally
Outcomes	Complete abortion during hospitalisation using uterine sonography; incomplete abortion; cervical ripening; nausea; vomiting; fever; time of drug effect
Notes	irct.ir/trial/25896, IRCT2017042533635N1

IRCT2017070934981N1

Methods	Parallel RCT, double-blinded
Participants	**Inclusion criteria:** 100 pregnant women aged 6-14 weeks' gestation, missed abortion **Exclusion criteria:** history of caesarean section, history of adrenal, kidney or liver disease, pulmonary disease, coagulation disorders, evidence of thromboembolic events, prostaglandin sensitivity, smoking, water repellent, vaginal bleeding, history of thromboembolism, oestrogen-dependent cancers (breast and genital system), intrauterine device, any abnormal amount of blood
Interventions	**MS:** 800 µg every 8 h (4 pills 200 µg) will be placed vaginally in the posterior foreskin (up to a maximum dose of 2400 µg) **MS+ oestrogen priming:** 800 µg every 8 h (4 pills 200 µg) will be placed vaginally in the posterior foreskin (up to a maximum dose of 2400 µg); and oral oestrogen valerima with a dose of 2 mg
Outcomes	Complete abortion 4 h after disposal of pregnancy based on the amount of pregnancy residues in ultrasound after excretion of pregnancy products

IRCT2017070934981N1 *(Continued)*

	Side effects based on completed patient questionnaire
Notes	irct.ir/trial/26580, IRCT2017070934981N1

Jha 2013

Methods	Prospective RCT
Participants	150 women for early abortion
Interventions	**MF + MS (1000 µg):** MF 200 mg on day 1, 400 µg MS 48 h later, 600 µg MS 51 h later (3 tablets of 200 µg MS) **MF + MS (400 µg):** MF 200 mg on day 1, 400 µg MS 48 h later, 3 tablets of placebo
Outcomes	The time of expulsion, completeness of the process, amount of bleeding encountered, side-effects
Notes	RG Kar Medical College and hospital

NCT01508143

Methods	Parallel RCT, double-blinded
Participants	**Inclusion criteria** • All women who were candidates for early pregnancy termination because of fetal death or other medical conditions • Before 14th week of gestation calculated according to menstruation or first trimester sonography **Exclusion criteria** • Chorioamnionitis • Hypersensitivity to prostaglandins • Past medical history of cardiovascular, kidney or liver or lung diseases • Positive history for uterus pathologies • Suspicious to extra-uterus pregnancy • Signs and symptoms of uterus infection • Molar pregnancy
Interventions	Group 1: MS 400 µg, each 6 h, up to 8 doses Group 2: MS 800 µg, each 12 h, up to 4 doses
Outcomes	Complete abortion, duration of abortion, adverse effects, need for surgery
Notes	NCT01508143

NCT03487354

Methods	Parallel RCT, quadruple-blinded (participant, care provider, investigator, outcomes assessor)
Participants	Woman 18-40 years

NCT03487354 *(Continued)*

	Inclusion criteria • Maternal age > 18 years old (age of legal consent) • Gestational age < 13 weeks • Haemoglobin > 10 g/dL • BMI between 25 kg/m^2 and 35 kg/m^2 • Missed abortion • Living fetus with multiple congenital malformations incompatible with life **Exclusion criteria** • Maternal age < 18 years old • Gestational age > 12 weeks • Haemoglobin < 10 g/dL • Anencephaly • Fibroid uterus • BMI < 25 kg/m^2 and > 35 kg/m^2 • Coagulopathy • History or evidence of adrenal pathology • Previous attempts for induction of abortion in the current pregnancy • Allergy to MS or LET • Medical disorder that contraindicates induction of abortion (e.g. heart failure)
Interventions	MS: single daily dose vs multiple daily doses
Outcomes	Complete miscarriage (Time frame: 1 week): assessed by transvaginal ultrasound where endometrial thickness < 15 mm is considered complete miscarriage
Notes	NCT03487354

BMI: body mass index; **LET:** letrozole; **MF:** mifepristone; **MS:** misoprostol; **RCT:** randomised controlled trial; TM: tamoxifen

Characteristics of ongoing studies *[ordered by study ID]*

CTRI201606006980

Study name	Medicines for abortion between 10-20 weeks
Methods	Unknown
Participants	Unknown
Interventions	Unknown
Outcomes	Unknown
Starting date	Unknown
Contact information	Unknown
Notes	CTRI/2016/06/006980

EUCTR2017-004083-35-FR

Study name	A double-blind, randomized, multicenter study evaluating 200 mg versus 600 mg of MF on pain in voluntary abortion by drug prior to 7 SA. DoMy Study
Methods	Double-blind, randomised, multicentre study
Participants	Women ≥ 18 years, wanting an abortion with medication before 7 weeks of amenorrhoea Single intrauterine pregnancy, the term of which is < 7 weeks on the day of Mifegyne intake, estimated by ultrasound with a cranio-caudal length measurement ≤ 10 mm Seeking medical abortion in hospital Signed a written informed consent and agrees to abide by the protocol
Interventions	Evaluating 200 mg versus 600 mg of MF on pain in voluntary abortion by drug prior to 7 weeks of amenorrhoea
Outcomes	Efficacy in pain reduction • Pain within days of taking MS • Pain between taking Mifegyne and MS • Failed method • Additional consultations and actions following consultation • Tolerance of drug-induced abortion • Abortion experience documented by self-reported questionnaire • Impact on the degree of anxiety of the participants by anxiety questionnaire • Patient satisfaction with a scale and questionnaire
Starting date	28 March 2018
Contact information	Assistance Publique Hôpitaux de Marseille
Notes	clinicaltrialsregister.eu/ctr-search/trial/2017-004083-35/FR

EUCTR2018-003675-35-SE

Study name	Efficacy of very early medical abortion - a randomized controlled non-inferiority trial
Methods	Non-inferiority RCT
Participants	• Women ≥ 18 years eligible for medical abortion • No signs of pathological pregnancy • No confirmed intrauterine pregnancy • Willing and able to participate after the study has been explained • Signed informed consent
Interventions	Group 1: very early medical abortion of vaginal MS Group 2: routine administration (delayed) of vaginal MS
Outcomes	To investigate if the efficacy of very early medical abortion is non-inferior within a non-inferiority margin of 3% to delayed abortion treatment initiated when an intrauterine pregnancy can be confirmed on ultrasound • Safety: proportions of adverse events and serious adverse events (includes bleeding, genital infections and complications to surgical interventions) will be compared between the treatment groups.

EUCTR2018-003675-35-SE *(Continued)*

	• Ectopic and other pathologic pregnancies (time point at diagnosis, treatment, numbers, percentage ruptured/unruptured). Defined as "outcome of special interest". • Any additional medical treatment of incomplete abortion or ongoing pregnancy (bleeding (days), pain (VAS score), surgical treatment for ongoing pregnancy, incomplete abortion or any other indications, unscheduled contacts and visits, acceptability with the time of the abortion treatment (early/delayed, would recommend to a friend yes/no) and initiation of postabortion contraception (yes/no, type))
Starting date	1 November 2018
Contact information	Kristina Gemzell Danielsson, kristina.gemzell@ki.se
Notes	clinicaltrialsregister.eu/ctr-search/trial/2018-003675-35/SE

NCT02704481

Study name	Non-surgical alternatives to treatment of failed medical abortion
Methods	Multi-site, double-blind RCT
Participants	Women who are diagnosed with ongoing pregnancy ≤ 77 days' gestational age at 1 week follow-up after medical abortion. The sample will be stratified in 2 cohorts: women with ongoing pregnancies ≤ 56 days of gestation and women with ongoing pregnancies 57-77 days of gestation
Interventions	A repeat MF-MS regimen and with a 2-dose MS-alone regimen • Group 1: 200 mg MF followed in 24-48 h by 800 µg buccal MS, followed by 4 MS placebo pills 3-12 h later • Group 2: 1 MF placebo pill, followed by 800 µg buccal MS 24-48 h later and another 800 µg dose repeated in 3-12 h
Outcomes	Primary outcome measures • The proportion of women in each study group, by gestational age cohort, who have a successful abortion without recourse to surgical intervention for any reason (time frame: 1 week after taking 1st study medication) Secondary outcome measures • The proportion of women in each arm by gestational age cohort with resolved ongoing pregnancies following study treatment, regardless of surgical intervention (time frame: 1 week after taking 1st study medication) • The proportion of women who found medical treatment to be an acceptable method to treat ongoing pregnancy, as measured by a follow-up questionnaire (time frame: 1 week after taking 1st study medication)
Starting date	June 2016
Contact information	Ilana Dzuba, MHSGynuity Health Projects
Notes	ClinicalTrials.gov/show/NCT02704481

NCT03065660

Study name	Mifepristone and misoprostol versus misoprostol alone in the medical management of missed miscarriage
Methods	A randomised, parallel-group, double-blind, placebo-controlled multicentre study
Participants	**Inclusion criteria** • Women diagnosed with missed miscarriage by pelvic ultrasound scan in the 1st 13 + 6 weeks of pregnancy that choose to have medical management of miscarriage • Age ≥ 16 years • Willing and able to give informed consent **Exclusion criteria** • Women opting for alternative methods of miscarriage management (expectant or surgical) • Diagnosis of incomplete miscarriage • Life-threatening bleeding • Contraindications to MF or MS use for example chronic adrenal failure, known hypersensitivity to either drug, haemorrhagic disorders and anticoagulant therapy, prosthetic heart valve or history of endocarditis, existing cardiovascular disease, severe asthma uncontrolled by therapy or inherited porphyria • Participation in any other blinded, placebo-controlled studies of investigational medicinal products in pregnancy • Previous participation in the MifeMiso study • Woman not able to attend for day 6-7 ultrasound scan

Interventions	• Group 1: active comparator: MF
	  ◦ A single dose of oral MF 200 mg, followed by a single dose of vaginal, oral or sublingual MS 800 μg 2 days later
	• Group 2: placebo comparator: placebo
	  ◦ Oral placebo tablet followed by a single dose of vaginal, oral or sublingual MS 800 μg 2 days later

Outcomes	Primary outcome measures • Failure to spontaneously pass the gestational sac within 7 days after randomisation (time frame: within 7 days after randomisation) Secondary outcome measures • Surgical intervention to resolve the miscarriage • Surgical intervention to resolve the miscarriage up to and including day 7 post-randomisation • Surgical intervention to resolve the miscarriage after day 7 post-randomisation to discharge • Need for further doses of MS up to day 7 post-randomisation • Need for further doses of MS up to discharge from EPU care • Overall patient satisfaction score (measured using the questionnaire and collected upon discharge) • Patient quality of life (index value and overall health status measured using the questionnaire • Duration of bleeding reported by woman (days). (Collected up to discharge) • Diagnosis of infection associated with miscarriage requiring outpatient antibiotic treatment (collected up to discharge) • Diagnosis of infection associated with miscarriage requiring inpatient antibiotic treatment (collected up to discharge) • Negative pregnancy test result 21 days (± 2 days) after randomisation • Time from randomisation to discharge from EPU care (described using summary statistics only) • Blood transfusion required (collected up to discharge) • Side effects (collected up to discharge)

NCT03065660 *(Continued)*

	• Death (collected up to discharge) • Any serious complications (collected up to discharge)
Starting date	20 September 2017
Contact information	Arri Coomarasamy, University of Birmingham
Notes	ClinicalTrials.gov/show/NCT03065660

NCT03440866

Study name	Comparison of the effectiveness of treatment with mifepristone and misoprostol at the same time compared to the administration of drugs at a 48-h interval for medical abortion
Methods	Prospective RCT
Participants	**Inclusion criteria** • Intrauterine singleton pregnancy of < 49 days, confirmed by ultrasound examination • Woman who desires to terminate their pregnancy and have approval of the committee for termination of pregnancy • Woman who gave their consent to have a surgical abortion eventual if needed **Exclusion criteria** • Contraindication to MF or MS • Suspected ectopic pregnancy • Systemic treatment with steroids • Adrenal insufficiency • Heart and blood vessels disease • Coagulopathy or use of antithrombotic agents • Uncontrolled asthma • Liver or kidney insufficiency • Anorexia • IUD • Breastfeeding
Interventions	Oral administration of MF 600 mg and oral MS 400 mcg in a time interval of 48 h
Outcomes	Primary outcome measures • Success rate of complete abortion without any other surgical intervention Secondary outcome measures • Assessment of bleeding amount and duration by the participant and haemoglobin level before and 2 weeks after the procedure • VAS pain severity with scale from 0-10, with score 10 representing the maximum level of pain during those 2 weeks • Post-operative women's satisfaction with scale from 0-10, with score 10 representing the maximum level of satisfaction. The participant will be asked if she would select this method next time if necessary
Starting date	1 March 2018

NCT03440866 *(Continued)*

Contact information	Principal Investigator: Meirav Braverman, MDHa Emek Medical Center, Afula, Israel
Notes	ClinicalTrials.gov/show/NCT03440866

NCT03628625

Study name	Letrozole pretreatment with misoprostol induction of abortion in first-trimester missed miscarriage
Methods	RCT, quadruple blinding (participant, care provider, investigator, outcomes assessor)
Participants	**80 participants** **Inclusion criteria** • Gestational age < 64 days' gestation (< 9 weeks) • Haemoglobin > 10 g/dL • BMI between 18.5 kg/m^2 and 25 kg/m^2 • Missed abortion **Exclusion criteria** • Molar pregnancy • Fibroid uterus • Uterine anomalies • Coagulopathy • Medical disorder that contraindicate induction of abortion (e.g. heart failure) • Previous attempts for induction of abortion in the current pregnancy • Allergy to MS or LET

Interventions	• Group 1: experimental study group
	  ◦ 3 tablets of LET 2.5 mg will be given as single daily dose, 7.5 mg/day for 2 days at home and the 3rd dose will be given on admission to hospital on day 3 and will be followed by vaginal MS 800 µg every 3 h up to maximum 2 doses
	• Group 2: placebo comparator: control group
	  ◦ 3 tablets of placebo will be given as a single daily dose, for 2 days at home and the 3rd dose will be given on admission to hospital on day 3 and continue treatment with vaginal MS 800 µg every 3 h up to maximum 2 doses

Outcomes	Primary outcome measures • Incidence of complete miscarriage Secondary outcome measures • Need for surgical evacuation of the products of conception • Incidence of septic abortion
Starting date	14 August 2018
Contact information	Dr Dina Yahia Mansour, Ain Shams Maternity Hospital
Notes	clinicaltrials.gov/ct2/show/NCT03628625?cond=Abortion

NCT03989869

Study name	Very early medical abortion
Methods	Allocation: randomised Intervention model: parallel assignment Masking: none (open label)
Participants	**Inclusion criteria** 1. Women ≥ 18 years opting for medical abortion with a pregnancy estimated by gynaecological history and LMP (if known) to be < 6 weeks 2. No signs of ectopic pregnancy, miscarriage or other pathological pregnancy 3. Transvaginal ultrasound (part of the clinical protocol at the abortion visit) shows no confirmed intrauterine pregnancy 4. Willing and able to return to the clinic for possible delayed treatment and at 1-2 weeks after the start of treatment for follow-up 5. Capable of giving their informed consent to participate **Exclusion criteria**: 1. Women with visible (confirmed) intrauterine pregnancy 2. Women with contraindications to medical abortion including diagnosed pathological pregnancy at the initial examination 3. Inability to give informed consent

Interventions	• Group 1: experimental: very early medical abortion
	  ◦ Immediate medical abortion treatment
	• Group 2: no intervention: standard care
	  ◦ Delayed care until an intrauterine pregnancy has been confirmed with ultrasound

Outcomes	Primary outcome measures • Complete abortion rate (efficacy) Secondary outcome measures • Complication rate (time frame: within 30 days of treatment initiation) • Duration of post-abortion bleeding (time frame: within 30 days of treatment initiation) • Acceptability of method (time frame: within 30 days of treatment initiation) • VAS pain score (time frame: within 30 days of treatment initiation)
Starting date	18 June 2019
Contact information	John Reynolds-Wright, University of Edinburgh
Notes	clinicaltrials.gov/ct2/show/NCT03989869?term=NCT03989869&draw=2&rank=1

NCT04215835

Study name	Letrozole pretreatment with misoprostol for induction of abortion in first-trimester missed miscarriage among women with one or more previous cesarean deliveries: a randomized controlled trial
Methods	RCT, quadruple blinding (participant, care provider, investigator, outcomes assessor)

NCT04215835 *(Continued)*

Participants	200 participants Ages eligible for study: ≥ 18 years (adult, older adult) Sexes eligible for study: female Accepts healthy volunteers: yes **Inclusion criteria** • Gestational age < 64 days' gestation (< 9 weeks) • Haemoglobin >10 g/dL • BMI between 18.5 kg/m^2 and 25 kg/m^2 • Missed abortion • Previous ≥ 1 caesarean deliveries **Exclusion criteria** • Molar pregnancy • Fibroid uterus • Uterine anomalies • Coagulopathy • Medical disorder that contraindicates induction of abortion (e.g. heart failure) • Previous attempts for induction of abortion in the current pregnancy • Allergy to MS or LET

Interventions	• Group 1: experimental study group
	  ◦ 3 tablets of LET 2.5 mg will be given as single daily dose, 7.5 mg/day for 2 days at home and the 3rd dose will be given on admission to hospital on day 3 and will be followed by vaginal MS 800 µg every 3 h up to maximum 2 doses.
	• Group 2: placebo comparator: control group
	  ◦ 3 tablets of placebo will be given as a single daily dose, for 2 days at home and the 3rd dose will be given on admission to hospital on day 3 and continue treatment with MS 800 µg every 3 h up to maximum 2 doses

Outcomes	Primary outcome measures • Incidence of complete miscarriage (time frame: 6 h) • Induction to abortion time (time frame: 6 h) Secondary outcome measures • Need for surgical evacuation of the products of conception (time frame: 6 h)
Starting date	2 January 2020
Contact information	Contact: Ahmed Samy, +201100681167, ahmedsamy8233@gmail.com , Cairo University
Notes	ClinicalTrials.gov/show/NCT 04215835

TCTR20180825005

Study name	Comparison efficacy for termination of early pregnancy failure by mifepristone followed by misoprostol with misoprostol alone: a randomized double-blind placebo- controlled trial
Methods	Double-blind, phase 2/phase 3, placebo-controlled RCT

TCTR20180825005 *(Continued)*

Participants	Pregnancy gestational age < 12 weeks; pregnancy failure; vital sign stable; accepted to follow this research
Interventions	MF then followed by MS, MS only
Outcomes	Complete abortion, satisfaction of participant
Starting date	12 July 2018
Contact information	Sawanwattanaku@gmail.com
Notes	www.thaiclinicaltrials.org

Van den Berg 2019

Study name	The comparison of two drug treatments for miscarriage
Methods	Multi-centred, prospective, 2-armed, randomised, double-blinded and placebo controlled. 464 women will be randomised in a 1:1 ratio, stratified by centre
Participants	Women with confirmed EPF by ultrasonography (6–14 weeks), managed expectantly for at least 1 week
Interventions	• Group 1: pre-treatment with oral MF (600 mg) or oral placebo (identical in appearance). • Group 2: pre-treatment with oral placebo (identical in appearance). In both arms pre-treatment will be followed by oral MS, which will start 36–48 h later consisting of 2 doses 400 µg (4 h apart), repeated after 24 h if no tissue is lost
Outcomes	Complete or incomplete evacuation Secondary outcome measures: patient satisfaction, complications, side effects and costs
Starting date	Unknown
Contact information	l.hamel@cwz.nl; lotte.hamel@radboudumc.nl; triplem.studie@gmail.com
Notes	MF and MS versus MS alone for uterine evacuation after early pregnancy failure: a pilot study - M&M trial/2016

BMI: body mass index; **IUD:** intrauterine (contraceptive) device; **LMP:** last menstrual period; **LET:** letrozole; **MF:** mifepristone; **MS:** misoprostol; **RCT:** randomised controlled trial; **VAS:** visual analogue scale;

DATA AND ANALYSES

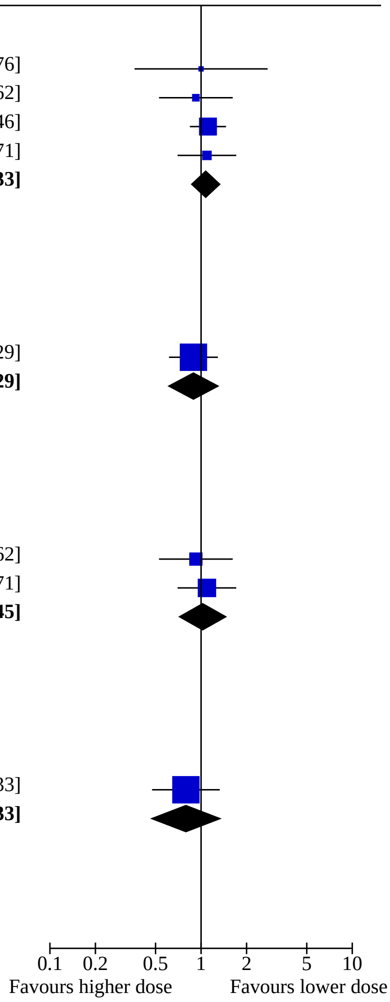
Comparison 1. Combined regimen mifepristone/prostaglandin: dose of mifepristone

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Failure to achieve complete abortion	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1.1 600 mg vs 200 mg	4	3494	Risk Ratio (M-H, Random, 95% CI)	1.07 [0.87, 1.33]
1.1.2 200 mg vs 100 mg (> 49 days)	1	1182	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.61, 1.29]
1.1.3 600 mg vs 200 mg with gemeprost 1 mg vaginal	2	1685	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.72, 1.45]
1.1.4 200 mg vs 100 mg (≤ 49 days)	1	941	Risk Ratio (M-H, Random, 95% CI)	0.79 [0.47, 1.33]
1.2 Side effects	2		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.2.1 Nausea	2	2432	Risk Ratio (M-H, Fixed, 95% CI)	1.02 [0.95, 1.09]
1.2.2 Vomiting	1	1584	Risk Ratio (M-H, Fixed, 95% CI)	1.02 [0.87, 1.19]
1.2.3 Diarrhoea	1	1584	Risk Ratio (M-H, Fixed, 95% CI)	0.80 [0.58, 1.09]
1.3 Time until passing of conceptus > 3 to 6 hours	2	1116	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.69, 1.10]
1.4 Ongoing pregnancy	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

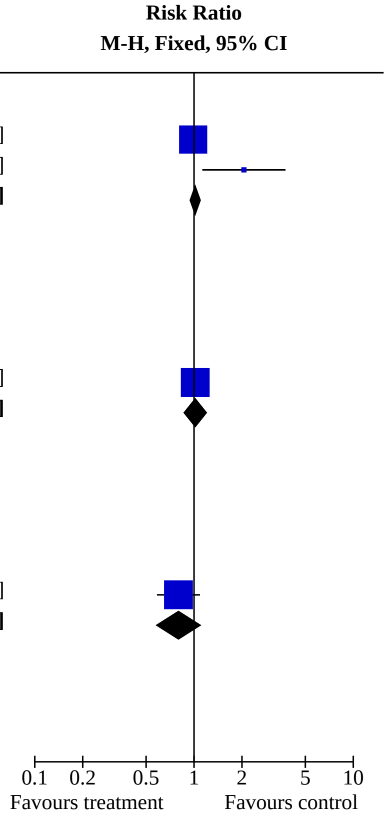
Analysis 1.1. Comparison 1: Combined regimen mifepristone/prostaglandin: dose of mifepristone, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
1.1.1 600 mg vs 200 mg						
McKinley 1993	7	110	7	110	4.4%	1.00 [0.36 , 2.76]
WHO 1993	22	393	24	396	14.2%	0.92 [0.53 , 1.62]
WHO 2000	95	797	85	792	59.0%	1.11 [0.84 , 1.46]
WHO 2001a	37	447	34	449	22.4%	1.09 [0.70 , 1.71]
Subtotal (95% CI)		**1747**		**1747**	**100.0%**	**1.07 [0.87 , 1.33]**
Total events:	161		150			
Heterogeneity: Tau² = 0.00; Chi² = 0.36, df = 3 (P = 0.95); I² = 0%						
Test for overall effect: Z = 0.65 (P = 0.51)						
1.1.2 200 mg vs 100 mg (> 49 days)						
von Hertzen 2009	47	579	55	603	100.0%	0.89 [0.61 , 1.29]
Subtotal (95% CI)		**579**		**603**	**100.0%**	**0.89 [0.61 , 1.29]**
Total events:	47		55			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.61 (P = 0.54)						
1.1.3 600 mg vs 200 mg with gemeprost 1 mg vaginal						
WHO 1993	22	393	24	396	38.8%	0.92 [0.53 , 1.62]
WHO 2001a	37	447	34	449	61.2%	1.09 [0.70 , 1.71]
Subtotal (95% CI)		**840**		**845**	**100.0%**	**1.02 [0.72 , 1.45]**
Total events:	59		58			
Heterogeneity: Tau² = 0.00; Chi² = 0.21, df = 1 (P = 0.65); I² = 0%						
Test for overall effect: Z = 0.13 (P = 0.89)						
1.1.4 200 mg vs 100 mg (≤ 49 days)						
von Hertzen 2009	25	482	30	459	100.0%	0.79 [0.47 , 1.33]
Subtotal (95% CI)		**482**		**459**	**100.0%**	**0.79 [0.47 , 1.33]**
Total events:	25		30			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.88 (P = 0.38)						



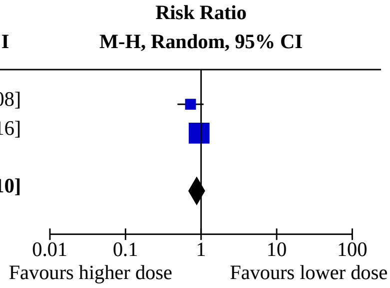
Analysis 1.2. Comparison 1: Combined regimen mifepristone/ prostaglandin: dose of mifepristone, Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
1.2.1 Nausea						
WHO 2000	527	794	531	790	97.3%	0.99 [0.92 , 1.06]
WHO 2001a	31	425	15	423	2.7%	2.06 [1.13 , 3.75]
Subtotal (95% CI)		**1219**		**1213**	**100.0%**	**1.02 [0.95 , 1.09]**
Total events:	558		546			
Heterogeneity: Chi² = 5.95, df = 1 (P = 0.01); I² = 83%						
Test for overall effect: Z = 0.46 (P = 0.64)						
1.2.2 Vomiting						
WHO 2000	224	794	219	790	100.0%	1.02 [0.87 , 1.19]
Subtotal (95% CI)		**794**		**790**	**100.0%**	**1.02 [0.87 , 1.19]**
Total events:	224		219			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.22 (P = 0.83)						
1.2.3 Diarrhoea						
WHO 2000	65	794	81	790	100.0%	0.80 [0.58 , 1.09]
Subtotal (95% CI)		**794**		**790**	**100.0%**	**0.80 [0.58 , 1.09]**
Total events:	65		81			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.42 (P = 0.16)						



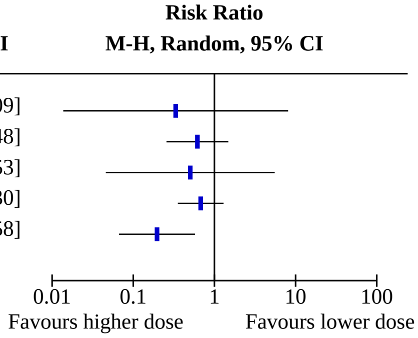
Analysis 1.3. Comparison 1: Combined regimen mifepristone/prostaglandin: dose of mifepristone, Outcome 3: Time until passing of conceptus > 3 to 6 hours

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
McKinley 1993	29	110	40	110	28.3%	0.72 [0.49 , 1.08]
WHO 2001a	121	447	129	449	71.7%	0.94 [0.76 , 1.16]
Total (95% CI)		**557**		**559**	**100.0%**	**0.87 [0.69 , 1.10]**
Total events:	150		169			
Heterogeneity: Tau² = 0.01; Chi² = 1.30, df = 1 (P = 0.25); I² = 23%						
Test for overall effect: Z = 1.13 (P = 0.26)						
Test for subgroup differences: Not applicable						



Analysis 1.4. Comparison 1: Combined regimen mifepristone/prostaglandin: dose of mifepristone, Outcome 4: Ongoing pregnancy

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
McKinley 1993	0	110	1	110	0.33 [0.01 , 8.09]
von Hertzen 2009	8	1089	13	1092	0.62 [0.26 , 1.48]
WHO 1993	1	393	2	396	0.50 [0.05 , 5.53]
WHO 2000	15	797	22	792	0.68 [0.35 , 1.30]
WHO 2001b	4	650	18	574	0.20 [0.07 , 0.58]



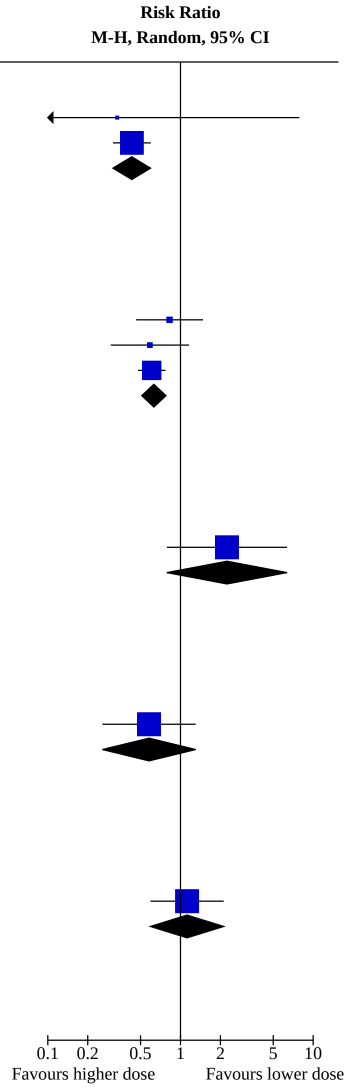
Comparison 2. Combined regimen mifepristone/prostaglandin: dose of prostaglandin

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Failure to achieve complete abortion	6		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.1.1 Combined dosage comparison: gemeprost 1 mg versus 0.5 mg with different dosage of mifepristone	2	1034	Risk Ratio (M-H, Random, 95% CI)	0.43 [0.31, 0.60]
2.1.2 Misoprostol 800 µg vs 400 µg all	3	4424	Risk Ratio (M-H, Random, 95% CI)	0.63 [0.51, 0.78]
2.1.3 Misoprostol 800 µg vs 400 µg ≤ 49 days	1	545	Risk Ratio (M-H, Random, 95% CI)	2.24 [0.79, 6.36]
2.1.4 Misoprostol 800 µg vs 400 µg > 49 days	1	570	Risk Ratio (M-H, Random, 95% CI)	0.58 [0.26, 1.30]
2.1.5 Misoprostol vaginal 800 µg vs oral 400 µg	1	637	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.59, 2.12]
2.2 Surgical evacuation	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.2.1 Misoprostol 800 µg vs 400 µg	2	934	Risk Ratio (M-H, Random, 95% CI)	0.82 [0.44, 1.55]
2.3 Ongoing pregnancy	4	5060	Risk Ratio (M-H, Random, 95% CI)	0.41 [0.26, 0.65]
2.3.1 Misoprostol 800 µg vs 400 µg	3	4424	Risk Ratio (M-H, Random, 95% CI)	0.41 [0.26, 0.65]
2.3.2 Misoprostol vaginal 800 µg vs oral 400 µg	1	636	Risk Ratio (M-H, Random, 95% CI)	Not estimable
2.4 Nausea	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.4.1 Misoprostol 800 µg vs 400 µg	3	4424	Risk Ratio (M-H, Random, 95% CI)	0.99 [0.94, 1.05]
2.4.2 Misoprostol vaginal 800 µg vs oral 400 µg	2	734	Risk Ratio (M-H, Random, 95% CI)	0.93 [0.67, 1.28]
2.5 Vomiting	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.5.1 Misoprostol 800 µg vs 400 µg	3	4424	Risk Ratio (M-H, Random, 95% CI)	1.10 [0.77, 1.57]
2.5.2 Misoprostol vaginal 800 µg vs oral 400 µg	2	734	Risk Ratio (M-H, Random, 95% CI)	0.82 [0.32, 2.15]
2.6 Diarrhoea	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.6.1 Misoprostol 800 µg vs 400 µg	2	3302	Risk Ratio (M-H, Random, 95% CI)	1.70 [1.43, 2.03]
2.6.2 Misoprostol vaginal 800 µg vs oral 400 µg	2	734	Risk Ratio (M-H, Random, 95% CI)	1.00 [0.59, 1.73]
2.7 Abdominal pain	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.7.1 Misoprostol 800 µg vs 400 µg	2	4127	Risk Ratio (M-H, Random, 95% CI)	1.01 [0.89, 1.15]
2.8 Women's dissatisfaction with the procedure	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.8.1 Misoprostol 800 µg vs 400 µg	3	4420	Risk Ratio (M-H, Random, 95% CI)	0.75 [0.60, 0.93]
2.8.2 Misoprostol vaginal 800 µg vs oral 400 µg	2	735	Risk Ratio (M-H, Random, 95% CI)	1.31 [0.89, 1.92]

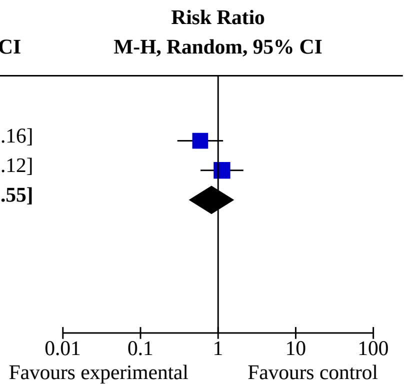
Analysis 2.1. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.1.1 Combined dosage comparison: gemeprost 1 mg versus 0.5 mg with different dosage of mifepristone						
Rodger 1989	0	30	1	30	1.1%	0.33 [0.01 , 7.87]
WHO 2001b	57	650	66	324	98.9%	0.43 [0.31 , 0.60]
Subtotal (95% CI)		**680**		**354**	**100.0%**	**0.43 [0.31 , 0.60]**
Total events:	57		67			
Heterogeneity: Tau² = 0.00; Chi² = 0.02, df = 1 (P = 0.87); I² = 0%						
Test for overall effect: Z = 5.07 (P < 0.00001)						
2.1.2 Misoprostol 800 µg vs 400 µg all						
Chong 2012	20	563	24	559	13.1%	0.83 [0.46 , 1.48]
Coyaji 2007	12	150	20	147	9.6%	0.59 [0.30 , 1.16]
von Hertzen 2010	99	1503	163	1502	77.3%	0.61 [0.48 , 0.77]
Subtotal (95% CI)		**2216**		**2208**	**100.0%**	**0.63 [0.51 , 0.78]**
Total events:	131		207			
Heterogeneity: Tau² = 0.00; Chi² = 0.98, df = 2 (P = 0.61); I² = 0%						
Test for overall effect: Z = 4.30 (P < 0.0001)						
2.1.3 Misoprostol 800 µg vs 400 µg ≤ 49 days						
Chong 2012	11	270	5	275	100.0%	2.24 [0.79 , 6.36]
Subtotal (95% CI)		**270**		**275**	**100.0%**	**2.24 [0.79 , 6.36]**
Total events:	11		5			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.52 (P = 0.13)						
2.1.4 Misoprostol 800 µg vs 400 µg > 49 days						
Chong 2012	9	290	15	280	100.0%	0.58 [0.26 , 1.30]
Subtotal (95% CI)		**290**		**280**	**100.0%**	**0.58 [0.26 , 1.30]**
Total events:	9		15			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.32 (P = 0.19)						
2.1.5 Misoprostol vaginal 800 µg vs oral 400 µg						
Shannon 2006	19	318	17	319	100.0%	1.12 [0.59 , 2.12]
Subtotal (95% CI)		**318**		**319**	**100.0%**	**1.12 [0.59 , 2.12]**
Total events:	19		17			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.35 (P = 0.72)						



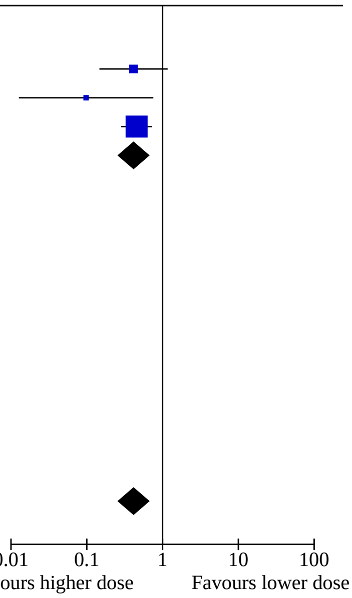
Analysis 2.2. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 2: Surgical evacuation

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.2.1 Misoprostol 800 µg vs 400 µg						
Coyaji 2007	12	150	20	147	48.2%	0.59 [0.30 , 1.16]
Shannon 2006	19	318	17	319	51.8%	1.12 [0.59 , 2.12]
Subtotal (95% CI)		**468**		**466**	**100.0%**	**0.82 [0.44 , 1.55]**
Total events:	31		37			
Heterogeneity: Tau² = 0.10; Chi² = 1.85, df = 1 (P = 0.17); I² = 46%						
Test for overall effect: Z = 0.61 (P = 0.54)						
Test for subgroup differences: Not applicable						



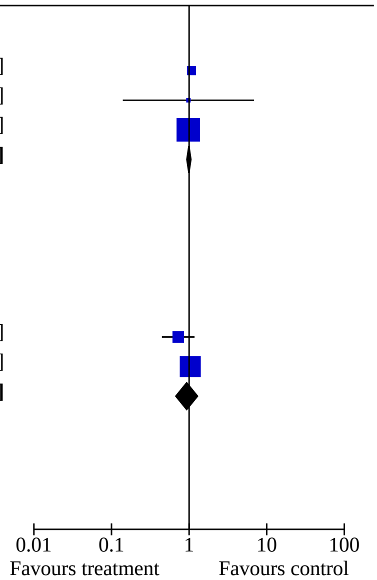
Analysis 2.3. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 3: Ongoing pregnancy

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.3.1 Misoprostol 800 µg vs 400 µg						
Chong 2012	5	563	12	559	18.4%	0.41 [0.15 , 1.17]
Coyaji 2007	1	150	10	147	4.9%	0.10 [0.01 , 0.76]
von Hertzen 2010	25	1503	55	1502	76.6%	0.45 [0.28 , 0.72]
Subtotal (95% CI)		**2216**		**2208**	**100.0%**	**0.41 [0.26 , 0.65]**
Total events:	31		77			
Heterogeneity: Tau² = 0.01; Chi² = 2.10, df = 2 (P = 0.35); I² = 5%						
Test for overall effect: Z = 3.79 (P = 0.0002)						
2.3.2 Misoprostol vaginal 800 µg vs oral 400 µg						
Shannon 2006	0	318	0	318		Not estimable
Subtotal (95% CI)		**318**		**318**		**Not estimable**
Total events:	0		0			
Heterogeneity: Not applicable						
Test for overall effect: Not applicable						
Total (95% CI)		**2534**		**2526**	**100.0%**	**0.41 [0.26 , 0.65]**
Total events:	31		77			
Heterogeneity: Tau² = 0.01; Chi² = 2.10, df = 2 (P = 0.35); I² = 5%						
Test for overall effect: Z = 3.79 (P = 0.0002)						
Test for subgroup differences: Not applicable						



Analysis 2.4. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 4: Nausea

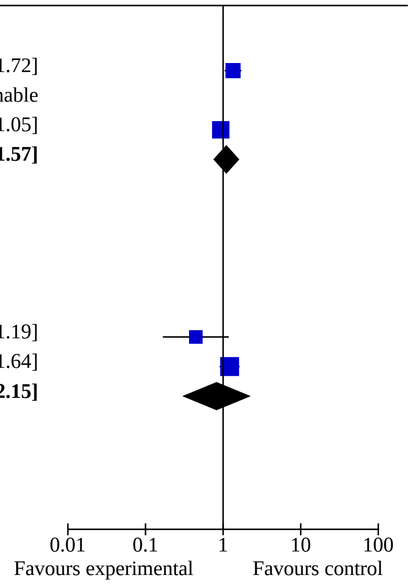
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.4.1 Misoprostol 800 µg vs 400 µg						
Chong 2012	254	563	235	559	19.7%	1.07 [0.94 , 1.23]
Coyaji 2007	2	150	2	147	0.1%	0.98 [0.14 , 6.87]
von Hertzen 2010	800	1503	820	1502	80.2%	0.97 [0.91 , 1.04]
Subtotal (95% CI)		**2216**		**2208**	**100.0%**	**0.99 [0.94 , 1.05]**
Total events:	1056		1057			
Heterogeneity: Tau² = 0.00; Chi² = 1.61, df = 2 (P = 0.45); I² = 0%						
Test for overall effect: Z = 0.21 (P = 0.83)						
2.4.2 Misoprostol vaginal 800 µg vs oral 400 µg						
Arvidsson 2005	17	49	23	48	29.9%	0.72 [0.45 , 1.18]
Shannon 2006	128	318	124	319	70.1%	1.04 [0.85 , 1.25]
Subtotal (95% CI)		**367**		**367**	**100.0%**	**0.93 [0.67 , 1.28]**
Total events:	145		147			
Heterogeneity: Tau² = 0.03; Chi² = 1.81, df = 1 (P = 0.18); I² = 45%						
Test for overall effect: Z = 0.44 (P = 0.66)						
Test for subgroup differences: Chi² = 0.16, df = 1 (P = 0.69), I² = 0%						



Analysis 2.5. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 5: Vomiting

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.5.1 Misoprostol 800 µg vs 400 µg						
Chong 2012	119	563	88	559	45.3%	1.34 [1.05 , 1.72]
Coyaji 2007	0	150	0	147		Not estimable
von Hertzen 2010	377	1503	404	1502	54.7%	0.93 [0.83 , 1.05]
Subtotal (95% CI)		**2216**		**2208**	**100.0%**	**1.10 [0.77 , 1.57]**
Total events:	496		492			
Heterogeneity: Tau² = 0.06; Chi² = 6.66, df = 1 (P = 0.010); I² = 85%						
Test for overall effect: Z = 0.53 (P = 0.60)						
2.5.2 Misoprostol vaginal 800 µg vs oral 400 µg						
Arvidsson 2005	5	49	11	48	38.8%	0.45 [0.17 , 1.19]
Shannon 2006	75	318	62	319	61.2%	1.21 [0.90 , 1.64]
Subtotal (95% CI)		**367**		**367**	**100.0%**	**0.82 [0.32 , 2.15]**
Total events:	80		73			
Heterogeneity: Tau² = 0.37; Chi² = 3.70, df = 1 (P = 0.05); I² = 73%						
Test for overall effect: Z = 0.40 (P = 0.69)						
Test for subgroup differences: Chi² = 0.31, df = 1 (P = 0.58), I² = 0%						

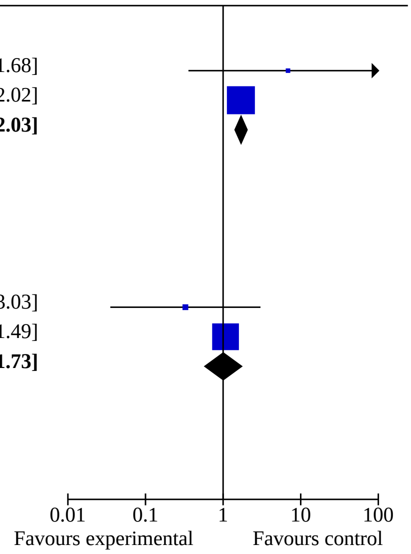
Risk Ratio
M-H, Random, 95% CI



Analysis 2.6. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 6: Diarrhoea

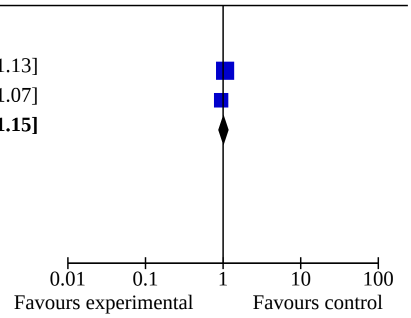
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.6.1 Misoprostol 800 µg vs 400 µg						
Coyaji 2007	3	150	0	147	0.4%	6.86 [0.36 , 131.68]
von Hertzen 2010	285	1503	168	1502	99.6%	1.70 [1.42 , 2.02]
Subtotal (95% CI)		**1653**		**1649**	**100.0%**	**1.70 [1.43 , 2.03]**
Total events:	288		168			
Heterogeneity: Tau² = 0.00; Chi² = 0.86, df = 1 (P = 0.35); I² = 0%						
Test for overall effect: Z = 5.92 (P < 0.00001)						
2.6.2 Misoprostol vaginal 800 µg vs oral 400 µg						
Arvidsson 2005	1	49	3	48	5.7%	0.33 [0.04 , 3.03]
Shannon 2006	60	318	56	319	94.3%	1.07 [0.77 , 1.49]
Subtotal (95% CI)		**367**		**367**	**100.0%**	**1.00 [0.59 , 1.73]**
Total events:	61		59			
Heterogeneity: Tau² = 0.05; Chi² = 1.08, df = 1 (P = 0.30); I² = 7%						
Test for overall effect: Z = 0.02 (P = 0.99)						
Test for subgroup differences: Chi² = 3.31, df = 1 (P = 0.07), I² = 69.8%						

Risk Ratio
M-H, Random, 95% CI



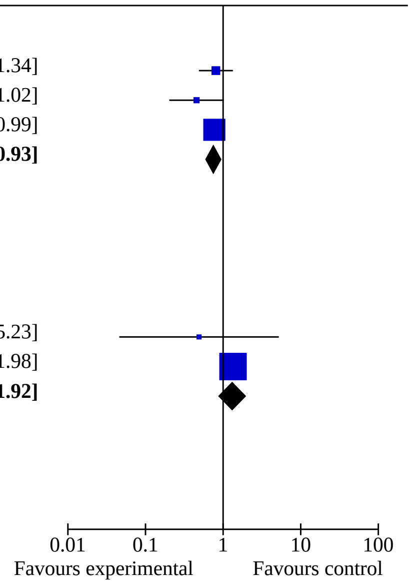
Analysis 2.7. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 7: Abdominal pain

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.7.1 Misoprostol 800 µg vs 400 µg						
Chong 2012	437	563	409	559	58.1%	1.06 [0.99 , 1.13]
von Hertzen 2010	354	1503	375	1502	41.9%	0.94 [0.83 , 1.07]
Subtotal (95% CI)		**2066**		**2061**	**100.0%**	**1.01 [0.89 , 1.15]**
Total events:	791		784			
Heterogeneity: Tau² = 0.01; Chi² = 3.48, df = 1 (P = 0.06); I² = 71%						
Test for overall effect: Z = 0.15 (P = 0.88)						
Test for subgroup differences: Not applicable						



Analysis 2.8. Comparison 2: Combined regimen mifepristone/prostaglandin: dose of prostaglandin, Outcome 8: Women's dissatisfaction with the procedure

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
2.8.1 Misoprostol 800 µg vs 400 µg						
Chong 2012	26	563	32	559	18.3%	0.81 [0.49 , 1.34]
Coyaji 2007	8	149	17	144	7.1%	0.45 [0.20 , 1.02]
von Hertzen 2010	101	1503	131	1502	74.6%	0.77 [0.60 , 0.99]
Subtotal (95% CI)		**2215**		**2205**	**100.0%**	**0.75 [0.60 , 0.93]**
Total events:	135		180			
Heterogeneity: Tau² = 0.00; Chi² = 1.60, df = 2 (P = 0.45); I² = 0%						
Test for overall effect: Z = 2.64 (P = 0.008)						
2.8.2 Misoprostol vaginal 800 µg vs oral 400 µg						
Arvidsson 2005	1	49	2	48	2.6%	0.49 [0.05 , 5.23]
Shannon 2006	51	319	38	319	97.4%	1.34 [0.91 , 1.98]
Subtotal (95% CI)		**368**		**367**	**100.0%**	**1.31 [0.89 , 1.92]**
Total events:	52		40			
Heterogeneity: Tau² = 0.00; Chi² = 0.68, df = 1 (P = 0.41); I² = 0%						
Test for overall effect: Z = 1.36 (P = 0.17)						
Test for subgroup differences: Chi² = 6.13, df = 1 (P = 0.01), I² = 83.7%						



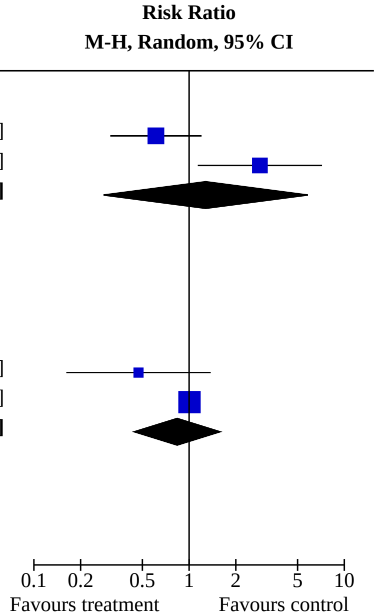
Comparison 3. Combined regimen mifepristone/prostaglandin: type of prostaglandin

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Failure to achieve complete abortion	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
3.1.1 Gemeprost vs misoprostol	2	1687	Risk Ratio (M-H, Random, 95% CI)	1.28 [0.28, 5.81]
3.1.2 PGF2α vs misoprostol	2	17974	Risk Ratio (M-H, Random, 95% CI)	0.84 [0.44, 1.58]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.2 Side effects	1		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.2.1 Vomiting	1	910	Risk Ratio (M-H, Fixed, 95% CI)	1.49 [1.06, 2.10]
3.2.2 Diarrhoea	1	910	Risk Ratio (M-H, Fixed, 95% CI)	2.66 [1.35, 5.26]
3.3 Ongoing pregnancy	2	1687	Risk Ratio (M-H, Fixed, 95% CI)	0.64 [0.28, 1.48]
3.4 Time until passing of conceptus > 3 to 6 hours	1	910	Risk Ratio (M-H, Fixed, 95% CI)	0.97 [0.77, 1.23]

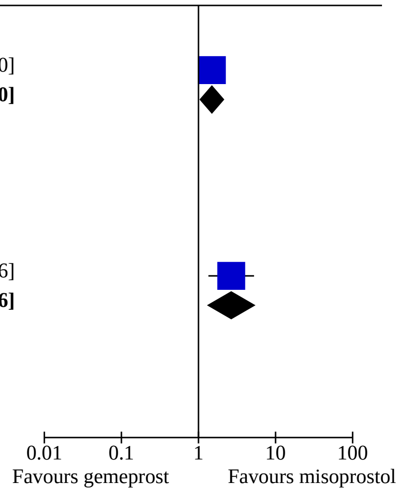
Analysis 3.1. Comparison 3: Combined regimen mifepristone/prostaglandin: type of prostaglandin, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
3.1.1 Gemeprost vs misoprostol						
Baird 1995	13	391	21	386	52.1%	0.61 [0.31 , 1.20]
Bartley 2001	17	453	6	457	47.9%	2.86 [1.14 , 7.18]
Subtotal (95% CI)		**844**		**843**	**100.0%**	**1.28 [0.28 , 5.81]**
Total events:	30		27			
Heterogeneity: Tau² = 1.03; Chi² = 7.03, df = 1 (P = 0.008); I² = 86%						
Test for overall effect: Z = 0.32 (P = 0.75)						
3.1.2 PGF2α vs misoprostol						
Sang 1994	4	150	17	301	24.2%	0.47 [0.16 , 1.38]
Sang 1999	674	9934	512	7589	75.8%	1.01 [0.90 , 1.12]
Subtotal (95% CI)		**10084**		**7890**	**100.0%**	**0.84 [0.44 , 1.58]**
Total events:	678		529			
Heterogeneity: Tau² = 0.14; Chi² = 1.89, df = 1 (P = 0.17); I² = 47%						
Test for overall effect: Z = 0.55 (P = 0.58)						



Analysis 3.2. Comparison 3: Combined regimen mifepristone/prostaglandin: type of prostaglandin, Outcome 2: Side effects

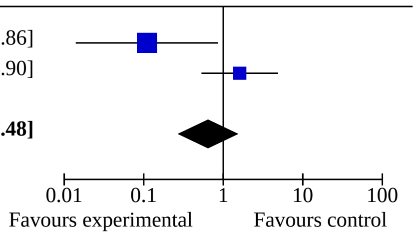
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
3.2.1 Vomiting						
Bartley 2001	71	453	48	457	100.0%	1.49 [1.06 , 2.10]
Subtotal (95% CI)		**453**		**457**	**100.0%**	**1.49 [1.06 , 2.10]**
Total events:	71		48			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.29 (P = 0.02)						
3.2.2 Diarrhoea						
Bartley 2001	29	453	11	457	100.0%	2.66 [1.35 , 5.26]
Subtotal (95% CI)		**453**		**457**	**100.0%**	**2.66 [1.35 , 5.26]**
Total events:	29		11			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.81 (P = 0.005)						



Analysis 3.3. Comparison 3: Combined regimen mifepristone/prostaglandin: type of prostaglandin, Outcome 3: Ongoing pregnancy

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Baird 1995	1	391	9	386	64.5%	0.11 [0.01 , 0.86]
Bartley 2001	8	453	5	457	35.5%	1.61 [0.53 , 4.90]
Total (95% CI)		**844**		**843**	**100.0%**	**0.64 [0.28 , 1.48]**
Total events:	9		14			

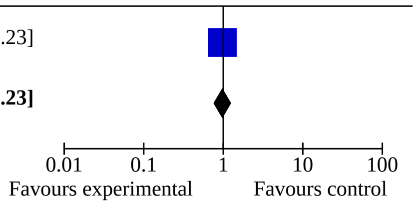
Heterogeneity: Chi² = 5.47, df = 1 (P = 0.02); I² = 82%
Test for overall effect: Z = 1.04 (P = 0.30)
Test for subgroup differences: Not applicable



Analysis 3.4. Comparison 3: Combined regimen mifepristone/prostaglandin: type of prostaglandin, Outcome 4: Time until passing of conceptus > 3 to 6 hours

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Bartley 2001	107	453	111	457	100.0%	0.97 [0.77 , 1.23]
Total (95% CI)		**453**		**457**	**100.0%**	**0.97 [0.77 , 1.23]**
Total events:	107		111			

Heterogeneity: Not applicable
Test for overall effect: Z = 0.24 (P = 0.81)
Test for subgroup differences: Not applicable



Comparison 4. Combined regimen mifepristone/prostaglandin: timing of prostaglandin

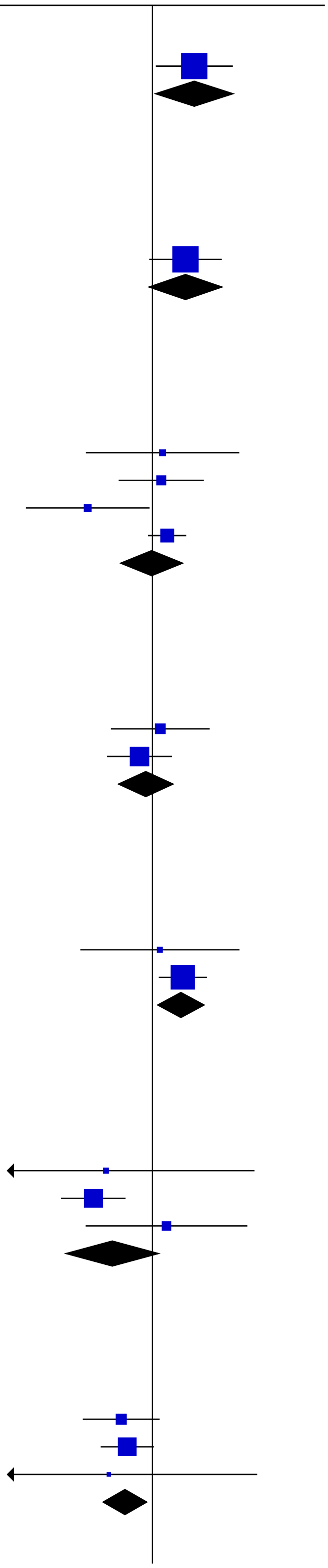
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1 Failure to achieve complete abortion	10		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
4.1.1 Day 3 vs day 1	1	1489	Risk Ratio (M-H, Random, 95% CI)	1.94 [1.05, 3.58]
4.1.2 Day 3 vs day 2	1	1521	Risk Ratio (M-H, Random, 95% CI)	1.69 [0.95, 3.01]
4.1.3 Day 2 vs day 1 (all)	4	3887	Risk Ratio (M-H, Random, 95% CI)	0.99 [0.61, 1.61]
4.1.4 Day 2 vs day 1 (≤ 49 days)	2	1116	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.58, 1.38]
4.1.5 Day 2 vs day 1 (> 49 days)	2	1207	Risk Ratio (M-H, Random, 95% CI)	1.57 [1.09, 2.27]
4.1.6 Day 2 vs day 0	3	711	Risk Ratio (M-H, Random, 95% CI)	0.53 [0.25, 1.09]
4.1.7 Day 1 vs day 0 (all)	3	2236	Risk Ratio (M-H, Random, 95% CI)	0.65 [0.46, 0.91]
4.1.8 Day 1 vs day 0 (≤ 49 days)	3	1078	Risk Ratio (M-H, Random, 95% CI)	0.65 [0.38, 1.11]
4.1.9 Day 1 vs day 0 (> 49 days)	2	1158	Risk Ratio (M-H, Random, 95% CI)	0.67 [0.42, 1.06]
4.2 Side effects	8		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
4.2.1 Nausea day 3 vs day 1	1	1358	Risk Ratio (M-H, Random, 95% CI)	1.05 [0.96, 1.14]
4.2.2 Nausea day 3 vs day 2	1	1384	Risk Ratio (M-H, Random, 95% CI)	0.98 [0.91, 1.06]
4.2.3 Nausea day 2 vs day 1	1	1434	Risk Ratio (M-H, Random, 95% CI)	1.07 [0.98, 1.16]
4.2.4 Nausea day 2 vs day 0	3	644	Risk Ratio (M-H, Random, 95% CI)	0.75 [0.58, 0.98]
4.2.5 Vomiting day 3 vs day 1	1	1358	Risk Ratio (M-H, Random, 95% CI)	1.01 [0.86, 1.19]
4.2.6 Vomiting day 3 vs day 2	1	1384	Risk Ratio (M-H, Random, 95% CI)	0.97 [0.83, 1.13]
4.2.7 Vomiting day 2 vs day 1	2	1634	Risk Ratio (M-H, Random, 95% CI)	1.05 [0.91, 1.22]
4.2.8 Vomiting day 2 vs day 0	3	644	Risk Ratio (M-H, Random, 95% CI)	0.95 [0.66, 1.38]
4.2.9 Diarrhoea day 3 vs day 1	1	1358	Risk Ratio (M-H, Random, 95% CI)	1.21 [0.99, 1.48]
4.2.10 Diarrhoea day 3 vs day 2	1	1384	Risk Ratio (M-H, Random, 95% CI)	1.16 [0.95, 1.42]
4.2.11 Diarrhoea day 2 vs day 1	2	1634	Risk Ratio (M-H, Random, 95% CI)	0.51 [0.09, 2.77]
4.2.12 Diarrhoea day 2 vs day 0	3	644	Risk Ratio (M-H, Random, 95% CI)	0.66 [0.42, 1.02]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.2.13 Nausea day 1 vs day 0	3	2217	Risk Ratio (M-H, Random, 95% CI)	1.03 [0.81, 1.32]
4.2.14 Vomiting day 1 vs day 0	3	2217	Risk Ratio (M-H, Random, 95% CI)	1.16 [0.82, 1.63]
4.2.15 Diarrhoea day 1 vs day 0	3	2217	Risk Ratio (M-H, Random, 95% CI)	0.85 [0.60, 1.21]
4.2.16 Abdominal pain day 1 vs day 0	1	80	Risk Ratio (M-H, Random, 95% CI)	0.67 [0.12, 3.78]
4.3 Surgical evacuation	6		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
4.3.1 Day 3 vs day 1	1	1489	Risk Ratio (M-H, Random, 95% CI)	1.49 [0.78, 2.83]
4.3.2 Day 3 vs day 2	1	1521	Risk Ratio (M-H, Random, 95% CI)	1.80 [0.92, 3.52]
4.3.3 Day 2 vs day 1	2	3681	Risk Ratio (M-H, Random, 95% CI)	1.06 [0.68, 1.67]
4.3.4 Day 2 vs day 0	2	511	Risk Ratio (M-H, Random, 95% CI)	0.40 [0.19, 0.86]
4.3.5 Day 1 vs day 0	2	1208	Risk Ratio (M-H, Random, 95% CI)	0.65 [0.36, 1.16]
4.4 Ongoing pregnancy	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.4.1 Day 3 vs day 1	1	1489	Risk Ratio (M-H, Fixed, 95% CI)	1.56 [0.51, 4.73]
4.4.2 Day 3 vs day 2	1	1521	Risk Ratio (M-H, Fixed, 95% CI)	2.71 [0.72, 10.16]
4.4.3 Day 2 vs day 1 (all)	2	3681	Risk Ratio (M-H, Fixed, 95% CI)	0.92 [0.45, 1.90]
4.4.4 Day 1 vs day 0	3	2288	Risk Ratio (M-H, Fixed, 95% CI)	0.34 [0.07, 1.66]
4.5 Women's dissatisfaction with the procedure	2	1429	Risk Ratio (M-H, Fixed, 95% CI)	0.99 [0.80, 1.23]
4.6 Time until passing of conceptus	1	80	Mean Difference (IV, Fixed, 95% CI)	-0.55 [-1.27, 0.17]
4.6.1 Day 1 vs day 0	1	80	Mean Difference (IV, Fixed, 95% CI)	-0.55 [-1.27, 0.17]
4.7 Days of bleeding	1	80	Mean Difference (IV, Fixed, 95% CI)	0.33 [-0.64, 1.30]
4.7.1 Day 1 vs day 0	1	80	Mean Difference (IV, Fixed, 95% CI)	0.33 [-0.64, 1.30]
4.8 Additional uterotonics used	1	80	Risk Ratio (M-H, Fixed, 95% CI)	0.50 [0.05, 5.30]

Analysis 4.1. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 1: Failure to achieve complete abortion

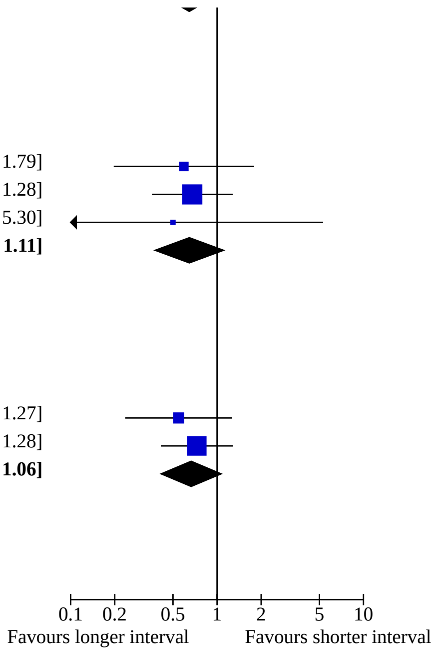
Study or Subgroup	longer interval Events	longer interval Total	shorter interval Events	shorter interval Total	Weight	Risk Ratio M-H, Random, 95% CI
4.1.1 Day 3 vs day 1						
Schaff 2000	30	755	15	734	100.0%	1.94 [1.05 , 3.58]
Subtotal (95% CI)		**755**		**734**	**100.0%**	**1.94 [1.05 , 3.58]**
Total events:	30		15			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.13 (P = 0.03)						
4.1.2 Day 3 vs day 2						
Schaff 2000	30	755	18	766	100.0%	1.69 [0.95 , 3.01]
Subtotal (95% CI)		**755**		**766**	**100.0%**	**1.69 [0.95 , 3.01]**
Total events:	30		18			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.79 (P = 0.07)						
4.1.3 Day 2 vs day 1 (all)						
Sandstrom 1999	5	33	4	31	12.3%	1.17 [0.35 , 3.98]
Schaff 2000	18	766	15	734	26.4%	1.15 [0.58 , 2.26]
Verma 2011	5	100	14	100	16.9%	0.36 [0.13 , 0.95]
von Hertzen 2009	88	1066	69	1057	44.4%	1.26 [0.93 , 1.71]
Subtotal (95% CI)		**1965**		**1922**	**100.0%**	**0.99 [0.61 , 1.61]**
Total events:	116		102			
Heterogeneity: Tau² = 0.12; Chi² = 5.82, df = 3 (P = 0.12); I² = 48%						
Test for overall effect: Z = 0.05 (P = 0.96)						
4.1.4 Day 2 vs day 1 (≤ 49 days)						
Verma 2011	12	90	10	85	30.1%	1.13 [0.52 , 2.48]
von Hertzen 2009	25	476	30	465	69.9%	0.81 [0.49 , 1.36]
Subtotal (95% CI)		**566**		**550**	**100.0%**	**0.90 [0.58 , 1.38]**
Total events:	37		40			
Heterogeneity: Tau² = 0.00; Chi² = 0.48, df = 1 (P = 0.49); I² = 0%						
Test for overall effect: Z = 0.48 (P = 0.63)						
4.1.5 Day 2 vs day 1 (> 49 days)						
Verma 2011	3	10	4	15	8.4%	1.13 [0.32 , 3.99]
von Hertzen 2009	63	590	39	592	91.6%	1.62 [1.11 , 2.38]
Subtotal (95% CI)		**600**		**607**	**100.0%**	**1.57 [1.09 , 2.27]**
Total events:	66		43			
Heterogeneity: Tau² = 0.00; Chi² = 0.29, df = 1 (P = 0.59); I² = 0%						
Test for overall effect: Z = 2.42 (P = 0.02)						
4.1.6 Day 2 vs day 0						
Creinin 2001b	1	44	2	42	8.7%	0.48 [0.04 , 5.07]
Guest 2007	18	215	45	210	67.0%	0.39 [0.23 , 0.65]
Verma 2017	5	100	4	100	24.3%	1.25 [0.35 , 4.52]
Subtotal (95% CI)		**359**		**352**	**100.0%**	**0.53 [0.25 , 1.09]**
Total events:	24		51			
Heterogeneity: Tau² = 0.14; Chi² = 2.72, df = 2 (P = 0.26); I² = 26%						
Test for overall effect: Z = 1.72 (P = 0.09)						
4.1.7 Day 1 vs day 0 (all)						
Creinin 2004	16	531	26	525	31.8%	0.61 [0.33 , 1.12]
Creinin 2007	33	546	50	554	66.1%	0.67 [0.44 , 1.02]
Goel 2011	1	40	2	40	2.1%	0.50 [0.05 , 5.30]
Subtotal (95% CI)		**1117**		**1119**	**100.0%**	**0.65 [0.46 , 0.91]**
Total events:	50		78			
Heterogeneity: Tau² = 0.00; Chi² = 0.11, df = 2 (P = 0.95); I² = 0%						

Risk Ratio
M-H, Random, 95% CI



Analysis 4.1. (Continued)

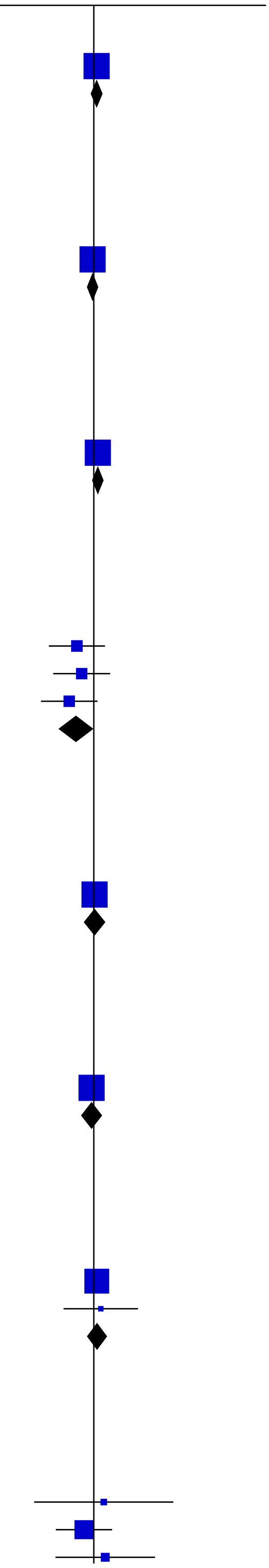
Total events:	50		78			
Heterogeneity: Tau² = 0.00; Chi² = 0.11, df = 2 (P = 0.95); I² = 0%						
Test for overall effect: Z = 2.49 (P = 0.01)						
4.1.8 Day 1 vs day 0 (≤ 49 days)						
Creinin 2004	5	258	8	245	23.6%	0.59 [0.20 , 1.79]
Creinin 2007	14	229	24	266	71.3%	0.68 [0.36 , 1.28]
Goel 2011	1	40	2	40	5.2%	0.50 [0.05 , 5.30]
Subtotal (95% CI)		**527**		**551**	**100.0%**	**0.65 [0.38 , 1.11]**
Total events:	20		34			
Heterogeneity: Tau² = 0.00; Chi² = 0.09, df = 2 (P = 0.96); I² = 0%						
Test for overall effect: Z = 1.59 (P = 0.11)						
4.1.9 Day 1 vs day 0 (> 49 days)						
Creinin 2004	8	273	15	280	31.1%	0.55 [0.24 , 1.27]
Creinin 2007	20	317	25	288	68.9%	0.73 [0.41 , 1.28]
Subtotal (95% CI)		**590**		**568**	**100.0%**	**0.67 [0.42 , 1.06]**
Total events:	28		40			
Heterogeneity: Tau² = 0.00; Chi² = 0.30, df = 1 (P = 0.58); I² = 0%						
Test for overall effect: Z = 1.70 (P = 0.09)						



Analysis 4.2. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 2: Side effects

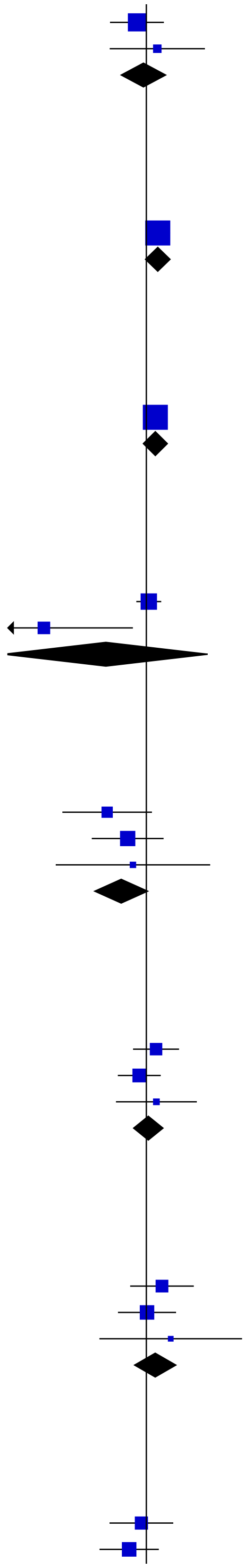
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
4.2.1 Nausea day 3 vs day 1						
Schaff 2000	414	654	426	704	100.0%	1.05 [0.96 , 1.14]
Subtotal (95% CI)		**654**		**704**	**100.0%**	**1.05 [0.96 , 1.14]**
Total events:	414		426			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.06 (P = 0.29)						
4.2.2 Nausea day 3 vs day 2						
Schaff 2000	414	654	471	730	100.0%	0.98 [0.91 , 1.06]
Subtotal (95% CI)		**654**		**730**	**100.0%**	**0.98 [0.91 , 1.06]**
Total events:	414		471			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.47 (P = 0.64)						
4.2.3 Nausea day 2 vs day 1						
Schaff 2000	471	730	426	704	100.0%	1.07 [0.98 , 1.16]
Subtotal (95% CI)		**730**		**704**	**100.0%**	**1.07 [0.98 , 1.16]**
Total events:	471		426			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.57 (P = 0.12)						
4.2.4 Nausea day 2 vs day 0						
Creinin 2001b	18	43	23	42	33.9%	0.76 [0.49 , 1.20]
Guest 2007	27	171	36	188	32.8%	0.82 [0.52 , 1.30]
Verma 2017	23	100	34	100	33.3%	0.68 [0.43 , 1.06]
Subtotal (95% CI)		**314**		**330**	**100.0%**	**0.75 [0.58 , 0.98]**
Total events:	68		93			
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.38$, df = 2 (P = 0.83); $I^2 = 0\%$						
Test for overall effect: Z = 2.14 (P = 0.03)						
4.2.5 Vomiting day 3 vs day 1						
Schaff 2000	205	654	218	704	100.0%	1.01 [0.86 , 1.19]
Subtotal (95% CI)		**654**		**704**	**100.0%**	**1.01 [0.86 , 1.19]**
Total events:	205		218			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.15 (P = 0.88)						
4.2.6 Vomiting day 3 vs day 2						
Schaff 2000	205	654	237	730	100.0%	0.97 [0.83 , 1.13]
Subtotal (95% CI)		**654**		**730**	**100.0%**	**0.97 [0.83 , 1.13]**
Total events:	205		237			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.45 (P = 0.66)						
4.2.7 Vomiting day 2 vs day 1						
Schaff 2000	237	730	218	704	93.8%	1.05 [0.90 , 1.22]
Verma 2011	19	100	17	100	6.2%	1.12 [0.62 , 2.02]
Subtotal (95% CI)		**830**		**804**	**100.0%**	**1.05 [0.91 , 1.22]**
Total events:	256		235			
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.04$, df = 1 (P = 0.84); $I^2 = 0\%$						
Test for overall effect: Z = 0.68 (P = 0.50)						
4.2.8 Vomiting day 2 vs day 0						
Creinin 2001b	6	43	5	42	11.0%	1.17 [0.39 , 3.55]
Guest 2007	28	171	36	188	67.4%	0.86 [0.55 , 1.34]
Verma 2017	12	100	10	100	21.6%	1.20 [0.54 , 2.65]

Risk Ratio
M-H, Random, 95% CI



Analysis 4.2. (Continued)

Guest 2007	28	171	36	188	67.4%	0.86 [0.55 , 1.34]
Verma 2017	12	100	10	100	21.6%	1.20 [0.54 , 2.65]
Subtotal (95% CI)		**314**		**330**	**100.0%**	**0.95 [0.66 , 1.38]**
Total events:	46		51			
Heterogeneity: Tau² = 0.00; Chi² = 0.68, df = 2 (P = 0.71); I² = 0%						
Test for overall effect: Z = 0.26 (P = 0.80)						
4.2.9 Diarrhoea day 3 vs day 1						
Schaff 2000	155	654	138	704	100.0%	1.21 [0.99 , 1.48]
Subtotal (95% CI)		**654**		**704**	**100.0%**	**1.21 [0.99 , 1.48]**
Total events:	155		138			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.83 (P = 0.07)						
4.2.10 Diarrhoea day 3 vs day 2						
Schaff 2000	155	654	149	730	100.0%	1.16 [0.95 , 1.42]
Subtotal (95% CI)		**654**		**730**	**100.0%**	**1.16 [0.95 , 1.42]**
Total events:	155		149			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.47 (P = 0.14)						
4.2.11 Diarrhoea day 2 vs day 1						
Schaff 2000	149	730	138	704	59.1%	1.04 [0.85 , 1.28]
Verma 2011	2	100	11	100	40.9%	0.18 [0.04 , 0.80]
Subtotal (95% CI)		**830**		**804**	**100.0%**	**0.51 [0.09 , 2.77]**
Total events:	151		149			
Heterogeneity: Tau² = 1.25; Chi² = 5.31, df = 1 (P = 0.02); I² = 81%						
Test for overall effect: Z = 0.78 (P = 0.44)						
4.2.12 Diarrhoea day 2 vs day 0						
Creinin 2001b	8	43	15	42	34.6%	0.52 [0.25 , 1.10]
Guest 2007	16	171	24	188	53.8%	0.73 [0.40 , 1.33]
Verma 2017	4	100	5	100	11.6%	0.80 [0.22 , 2.89]
Subtotal (95% CI)		**314**		**330**	**100.0%**	**0.66 [0.42 , 1.02]**
Total events:	28		44			
Heterogeneity: Tau² = 0.00; Chi² = 0.59, df = 2 (P = 0.74); I² = 0%						
Test for overall effect: Z = 1.87 (P = 0.06)						
4.2.13 Nausea day 1 vs day 0						
Creinin 2004	52	523	44	520	40.5%	1.18 [0.80 , 1.72]
Creinin 2007	51	544	58	550	46.5%	0.89 [0.62 , 1.27]
Goel 2011	13	40	11	40	13.1%	1.18 [0.60 , 2.32]
Subtotal (95% CI)		**1107**		**1110**	**100.0%**	**1.03 [0.81 , 1.32]**
Total events:	116		113			
Heterogeneity: Tau² = 0.00; Chi² = 1.27, df = 2 (P = 0.53); I² = 0%						
Test for overall effect: Z = 0.26 (P = 0.79)						
4.2.14 Vomiting day 1 vs day 0						
Creinin 2004	30	523	23	520	41.7%	1.30 [0.76 , 2.20]
Creinin 2007	31	544	31	550	50.0%	1.01 [0.62 , 1.64]
Goel 2011	6	40	4	40	8.3%	1.50 [0.46 , 4.91]
Subtotal (95% CI)		**1107**		**1110**	**100.0%**	**1.16 [0.82 , 1.63]**
Total events:	67		58			
Heterogeneity: Tau² = 0.00; Chi² = 0.66, df = 2 (P = 0.72); I² = 0%						
Test for overall effect: Z = 0.85 (P = 0.40)						
4.2.15 Diarrhoea day 1 vs day 0						
Creinin 2004	25	523	27	520	43.6%	0.92 [0.54 , 1.56]
Creinin 2007	26	544	35	550	50.4%	0.75 [0.46 , 1.23]



Analysis 4.2. (Continued)

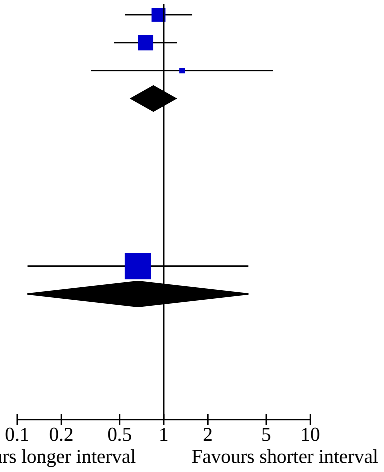
Creinin 2004	25	523	27	520	43.6%	0.92 [0.54 , 1.56]
Creinin 2007	26	544	35	550	50.4%	0.75 [0.46 , 1.23]
Goel 2011	4	40	3	40	6.0%	1.33 [0.32 , 5.58]
Subtotal (95% CI)		**1107**		**1110**	**100.0%**	**0.85 [0.60 , 1.21]**
Total events:	55		65			

Heterogeneity: Tau² = 0.00; Chi² = 0.71, df = 2 (P = 0.70); I² = 0%
Test for overall effect: Z = 0.91 (P = 0.36)

4.2.16 Abdominal pain day 1 vs day 0

Goel 2011	2	40	3	40	100.0%	0.67 [0.12 , 3.78]
Subtotal (95% CI)		**40**		**40**	**100.0%**	**0.67 [0.12 , 3.78]**
Total events:	2		3			

Heterogeneity: Not applicable
Test for overall effect: Z = 0.46 (P = 0.65)

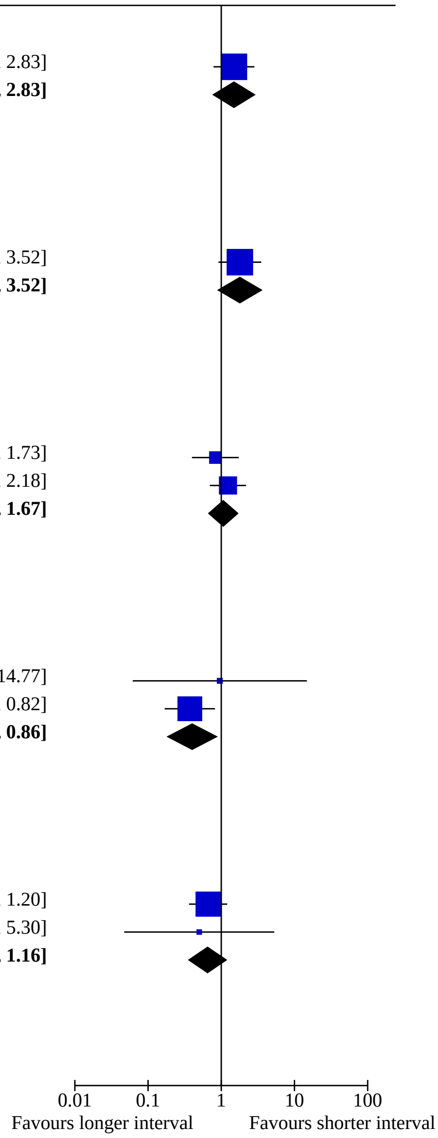


Analysis 4.3. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 3: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
4.3.1 Day 3 vs day 1						
Schaff 2000	23	755	15	734	100.0%	1.49 [0.78 , 2.83]
Subtotal (95% CI)		**755**		**734**	**100.0%**	**1.49 [0.78 , 2.83]**
Total events:	23		15			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.22 (P = 0.22)						
4.3.2 Day 3 vs day 2						
Schaff 2000	23	755	13	766	100.0%	1.80 [0.92 , 3.52]
Subtotal (95% CI)		**755**		**766**	**100.0%**	**1.80 [0.92 , 3.52]**
Total events:	23		13			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.70 (P = 0.09)						
4.3.3 Day 2 vs day 1						
Schaff 2000	13	766	15	734	37.4%	0.83 [0.40 , 1.73]
von Hertzen 2009	26	1092	21	1089	62.6%	1.23 [0.70 , 2.18]
Subtotal (95% CI)		**1858**		**1823**	**100.0%**	**1.06 [0.68 , 1.67]**
Total events:	39		36			
Heterogeneity: Tau² = 0.00; Chi² = 0.70, df = 1 (P = 0.40); I² = 0%						
Test for overall effect: Z = 0.27 (P = 0.79)						
4.3.4 Day 2 vs day 0						
Creinin 2001b	1	44	1	42	7.7%	0.95 [0.06 , 14.77]
Guest 2007	8	215	21	210	92.3%	0.37 [0.17 , 0.82]
Subtotal (95% CI)		**259**		**252**	**100.0%**	**0.40 [0.19 , 0.86]**
Total events:	9		22			
Heterogeneity: Tau² = 0.00; Chi² = 0.42, df = 1 (P = 0.52); I² = 0%						
Test for overall effect: Z = 2.36 (P = 0.02)						
4.3.5 Day 1 vs day 0						
Creinin 2007	17	561	26	567	93.9%	0.66 [0.36 , 1.20]
Goel 2011	1	40	2	40	6.1%	0.50 [0.05 , 5.30]
Subtotal (95% CI)		**601**		**607**	**100.0%**	**0.65 [0.36 , 1.16]**
Total events:	18		28			
Heterogeneity: Tau² = 0.00; Chi² = 0.05, df = 1 (P = 0.82); I² = 0%						
Test for overall effect: Z = 1.45 (P = 0.15)						

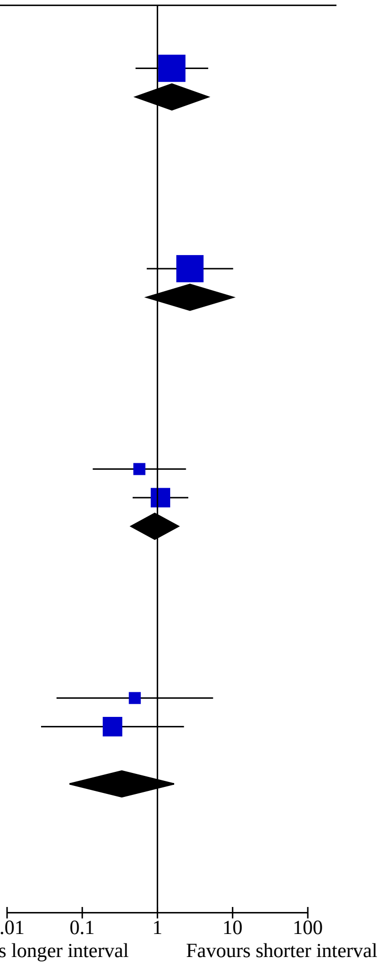
Test for subgroup differences: Chi² = 12.12, df = 4 (P = 0.02), I² = 67.0%

Risk Ratio
M-H, Random, 95% CI



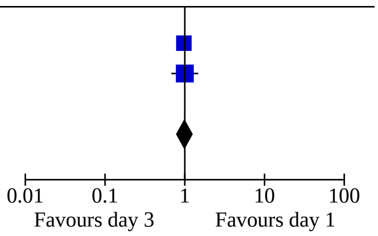
Analysis 4.4. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 4: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
4.4.1 Day 3 vs day 1						
Schaff 2000	8	755	5	734	100.0%	1.56 [0.51 , 4.73]
Subtotal (95% CI)		**755**		**734**	**100.0%**	**1.56 [0.51 , 4.73]**
Total events:	8		5			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.78 (P = 0.44)						
4.4.2 Day 3 vs day 2						
Schaff 2000	8	755	3	766	100.0%	2.71 [0.72 , 10.16]
Subtotal (95% CI)		**755**		**766**	**100.0%**	**2.71 [0.72 , 10.16]**
Total events:	8		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.47 (P = 0.14)						
4.4.3 Day 2 vs day 1 (all)						
Schaff 2000	3	766	5	734	33.8%	0.57 [0.14 , 2.40]
von Hertzen 2009	11	1092	10	1089	66.2%	1.10 [0.47 , 2.57]
Subtotal (95% CI)		**1858**		**1823**	**100.0%**	**0.92 [0.45 , 1.90]**
Total events:	14		15			
Heterogeneity: Chi² = 0.58, df = 1 (P = 0.45); I² = 0%						
Test for overall effect: Z = 0.22 (P = 0.82)						
4.4.4 Day 1 vs day 0						
Creinin 2004	1	540	2	540	33.5%	0.50 [0.05 , 5.50]
Creinin 2007	1	561	4	567	66.5%	0.25 [0.03 , 2.25]
Goel 2011	0	40	0	40		Not estimable
Subtotal (95% CI)		**1141**		**1147**	**100.0%**	**0.34 [0.07 , 1.66]**
Total events:	2		6			
Heterogeneity: Chi² = 0.17, df = 1 (P = 0.68); I² = 0%						
Test for overall effect: Z = 1.34 (P = 0.18)						



Analysis 4.5. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 5: Women's dissatisfaction with the procedure

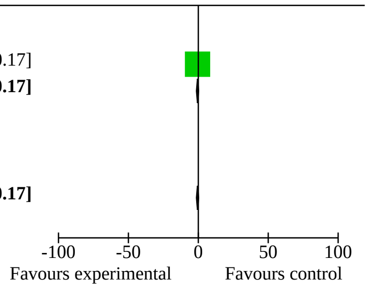
Study or Subgroup	Day 3 Events	Day 3 Total	Day 1 Events	Day 1 Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Goel 2011	38	40	39	40	45.1%	0.97 [0.89 , 1.06]
Schaff 2000	46	653	49	696	54.9%	1.00 [0.68 , 1.47]
Total (95% CI)		**693**		**736**	**100.0%**	**0.99 [0.80 , 1.23]**
Total events:	84		88			
Heterogeneity: Chi² = 0.11, df = 1 (P = 0.74); I² = 0%						
Test for overall effect: Z = 0.10 (P = 0.92)						
Test for subgroup differences: Not applicable						



Analysis 4.6. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 6: Time until passing of conceptus

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Fixed, 95% CI
4.6.1 Day 1 vs day 0								
Goel 2011	5.95	1.81	40	6.5	1.48	40	100.0%	-0.55 [-1.27 , 0.17]
Subtotal (95% CI)			**40**			**40**	**100.0%**	**-0.55 [-1.27 , 0.17]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 1.49 (P = 0.14)								
Total (95% CI)			**40**			**40**	**100.0%**	**-0.55 [-1.27 , 0.17]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 1.49 (P = 0.14)								
Test for subgroup differences: Not applicable								

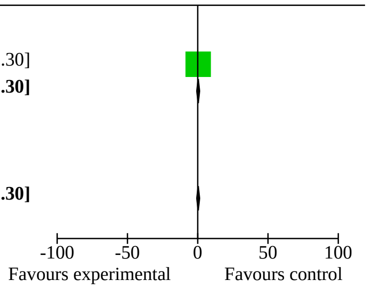
Mean Difference
IV, Fixed, 95% CI



Analysis 4.7. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 7: Days of bleeding

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Fixed, 95% CI
4.7.1 Day 1 vs day 0								
Goel 2011	8.23	2.14	40	7.9	2.3	40	100.0%	0.33 [-0.64 , 1.30]
Subtotal (95% CI)			**40**			**40**	**100.0%**	**0.33 [-0.64 , 1.30]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 0.66 (P = 0.51)								
Total (95% CI)			**40**			**40**	**100.0%**	**0.33 [-0.64 , 1.30]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 0.66 (P = 0.51)								
Test for subgroup differences: Not applicable								

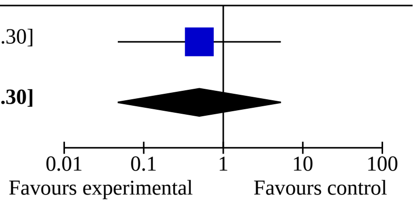
Mean Difference
IV, Fixed, 95% CI



Analysis 4.8. Comparison 4: Combined regimen mifepristone/prostaglandin: timing of prostaglandin, Outcome 8: Additional uterotonics used

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Fixed, 95% CI
Goel 2011	1	40	2	40	100.0%	0.50 [0.05 , 5.30]
Total (95% CI)		**40**		**40**	**100.0%**	**0.50 [0.05 , 5.30]**
Total events:	1		2			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.58 (P = 0.56)						
Test for subgroup differences: Not applicable						

Risk Ratio
M-H, Fixed, 95% CI



Comparison 5. Combined regimen mifepristone/prostaglandin: administration strategy

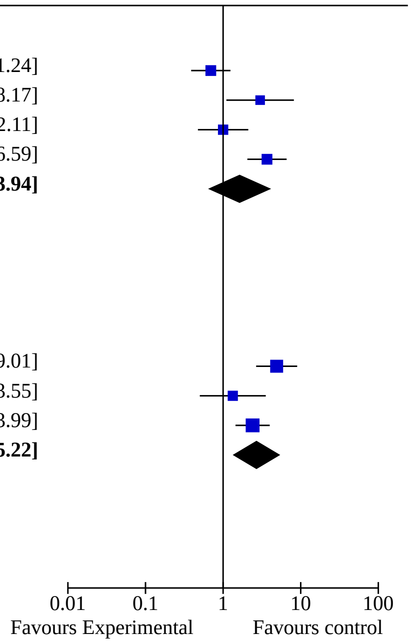
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1 Failure to achieve complete abortion	7		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1.1 Home vs hospital	4	2263	Risk Ratio (M-H, Random, 95% CI)	1.63 [0.68, 3.94]
5.1.2 Nurse vs doctor	3	3056	Risk Ratio (M-H, Random, 95% CI)	2.69 [1.39, 5.22]
5.2 Surgical evacuation	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
5.2.1 Home vs hospital	2	1331	Risk Ratio (M-H, Random, 95% CI)	1.71 [0.37, 7.87]
5.2.2 Nurse vs doctor	2	2172	Risk Ratio (M-H, Random, 95% CI)	3.36 [1.67, 6.78]
5.3 Ongoing pregnancy	4		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
5.4 Side effects	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
5.4.1 Nausea	3	1532	Risk Ratio (M-H, Random, 95% CI)	1.09 [0.74, 1.61]
5.4.2 Vomiting	3	1532	Risk Ratio (M-H, Random, 95% CI)	1.78 [0.97, 3.26]
5.4.3 Diarrhoea	2	932	Risk Ratio (M-H, Random, 95% CI)	1.40 [0.80, 2.45]
5.4.4 Abdominal pain	4	3071	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.95, 1.32]
5.5 Women's dissatisfaction with the procedure	6		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
5.5.1 Home vs hospital	4	2155	Risk Ratio (M-H, Random, 95% CI)	1.63 [0.95, 2.80]
5.5.2 Nurse vs doctor	2	1988	Risk Ratio (M-H, Random, 95% CI)	2.70 [0.16, 44.49]
5.6 Additional uterotonics used	2		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
5.7 Days of bleeding	3	1532	Mean Difference (IV, Random, 95% CI)	-0.01 [-0.25, 0.22]
5.8 Blood transfusion	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
5.8.1 Home vs hospital	1	731	Risk Ratio (M-H, Random, 95% CI)	0.33 [0.01, 8.18]
5.8.2 Nurse vs doctor	1	1068	Risk Ratio (M-H, Random, 95% CI)	Not estimable

Analysis 5.1. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
5.1.1 Home vs hospital						
Iyengar 2015	18	365	26	366	26.7%	0.69 [0.39 , 1.24]
Li 2017	15	372	5	372	21.7%	3.00 [1.10 , 8.17]
Shrestha 2014	12	94	12	94	24.8%	1.00 [0.47 , 2.11]
Song 2018	55	321	13	279	26.7%	3.68 [2.05 , 6.59]
Subtotal (95% CI)		**1152**		**1111**	**100.0%**	**1.63 [0.68 , 3.94]**
Total events:	100		56			
Heterogeneity: Tau² = 0.67; Chi² = 18.93, df = 3 (P = 0.0003); I² = 84%						
Test for overall effect: Z = 1.09 (P = 0.28)						
5.1.2 Nurse vs doctor						
Kopp Kallner 2015	59	535	12	533	36.0%	4.90 [2.66 , 9.01]
Olavarrieta 2015	9	434	7	450	24.3%	1.33 [0.50 , 3.55]
Warriner 2011	48	552	20	552	39.7%	2.40 [1.44 , 3.99]
Subtotal (95% CI)		**1521**		**1535**	**100.0%**	**2.69 [1.39 , 5.22]**
Total events:	116		39			
Heterogeneity: Tau² = 0.22; Chi² = 5.85, df = 2 (P = 0.05); I² = 66%						
Test for overall effect: Z = 2.93 (P = 0.003)						
Test for subgroup differences: Chi² = 0.79, df = 1 (P = 0.37), I² = 0%						

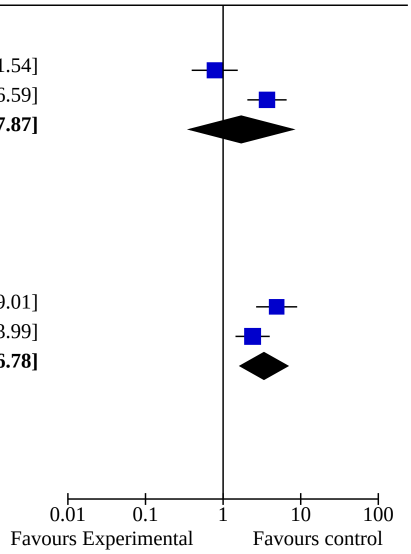
Risk Ratio M-H, Random, 95% CI



Analysis 5.2. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 2: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
5.2.1 Home vs hospital						
Iyengar 2015	14	365	18	366	49.3%	0.78 [0.39 , 1.54]
Song 2018	55	321	13	279	50.7%	3.68 [2.05 , 6.59]
Subtotal (95% CI)		**686**		**645**	**100.0%**	**1.71 [0.37 , 7.87]**
Total events:	69		31			
Heterogeneity: Tau² = 1.11; Chi² = 11.54, df = 1 (P = 0.0007); I² = 91%						
Test for overall effect: Z = 0.69 (P = 0.49)						
5.2.2 Nurse vs doctor						
Kopp Kallner 2015	59	535	12	533	47.2%	4.90 [2.66 , 9.01]
Warriner 2011	48	552	20	552	52.8%	2.40 [1.44 , 3.99]
Subtotal (95% CI)		**1087**		**1085**	**100.0%**	**3.36 [1.67 , 6.78]**
Total events:	107		32			
Heterogeneity: Tau² = 0.18; Chi² = 3.14, df = 1 (P = 0.08); I² = 68%						
Test for overall effect: Z = 3.38 (P = 0.0007)						
Test for subgroup differences: Chi² = 0.62, df = 1 (P = 0.43), I² = 0%						

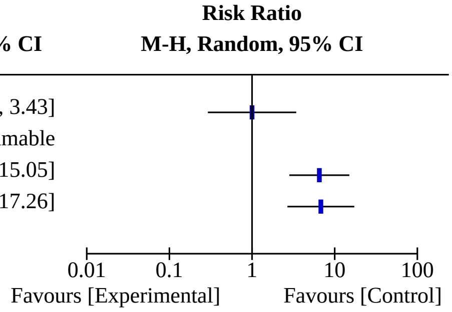
Risk Ratio M-H, Random, 95% CI



Analysis 5.3. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 3: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
Li 2017	5	372	5	372	1.00 [0.29 , 3.43]
Shrestha 2014	0	94	0	94	Not estimable
Song 2018	45	321	6	279	6.52 [2.82 , 15.05]
Warriner 2011	34	552	5	552	6.80 [2.68 , 17.26]

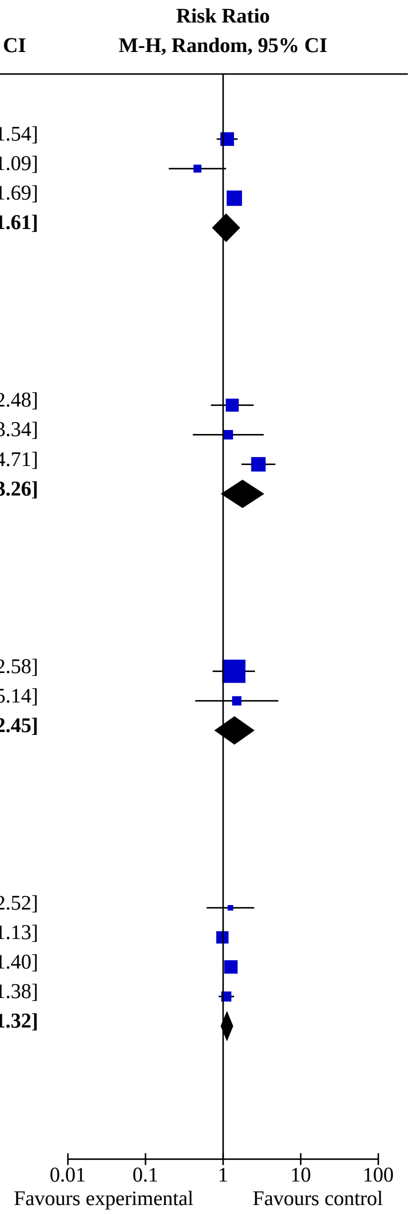
Test for subgroup differences: Not applicable



Analysis 5.4. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 4: Side effects

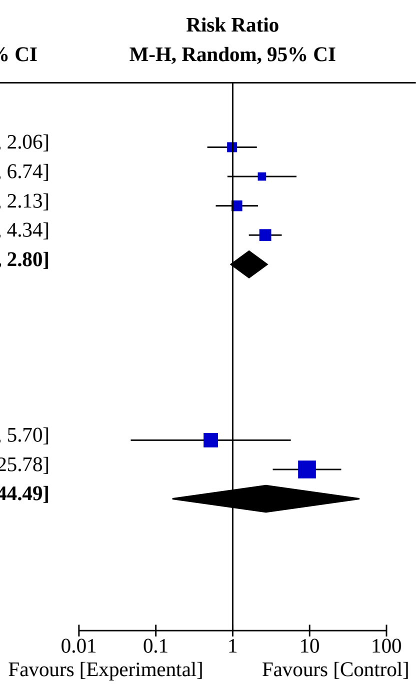
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
5.4.1 Nausea						
Li 2017	70	372	62	372	39.0%	1.13 [0.83 , 1.54]
Shrestha 2014	7	94	15	94	14.8%	0.47 [0.20 , 1.09]
Song 2018	159	321	99	279	46.1%	1.40 [1.15 , 1.69]
Subtotal (95% CI)		**787**		**745**	**100.0%**	**1.09 [0.74 , 1.61]**
Total events:	236		176			
Heterogeneity: Tau² = 0.08; Chi² = 6.94, df = 2 (P = 0.03); I² = 71%						
Test for overall effect: Z = 0.45 (P = 0.66)						
5.4.2 Vomiting						
Li 2017	21	372	16	372	36.2%	1.31 [0.70 , 2.48]
Shrestha 2014	7	94	6	94	21.4%	1.17 [0.41 , 3.34]
Song 2018	59	321	18	279	42.5%	2.85 [1.72 , 4.71]
Subtotal (95% CI)		**787**		**745**	**100.0%**	**1.78 [0.97 , 3.26]**
Total events:	87		40			
Heterogeneity: Tau² = 0.16; Chi² = 4.61, df = 2 (P = 0.10); I² = 57%						
Test for overall effect: Z = 1.86 (P = 0.06)						
5.4.3 Diarrhoea						
Li 2017	22	372	16	372	79.4%	1.38 [0.73 , 2.58]
Shrestha 2014	6	94	4	94	20.6%	1.50 [0.44 , 5.14]
Subtotal (95% CI)		**466**		**466**	**100.0%**	**1.40 [0.80 , 2.45]**
Total events:	28		20			
Heterogeneity: Tau² = 0.00; Chi² = 0.02, df = 1 (P = 0.90); I² = 0%						
Test for overall effect: Z = 1.18 (P = 0.24)						
5.4.4 Abdominal pain						
Iyengar 2015	19	349	12	274	4.8%	1.24 [0.61 , 2.52]
Li 2017	184	372	188	372	33.3%	0.98 [0.85 , 1.13]
Song 2018	258	321	178	279	38.0%	1.26 [1.14 , 1.40]
Warriner 2011	122	552	111	552	23.9%	1.10 [0.87 , 1.38]
Subtotal (95% CI)		**1594**		**1477**	**100.0%**	**1.12 [0.95 , 1.32]**
Total events:	583		489			
Heterogeneity: Tau² = 0.02; Chi² = 8.48, df = 3 (P = 0.04); I² = 65%						
Test for overall effect: Z = 1.37 (P = 0.17)						

Test for subgroup differences: Chi² = 2.61, df = 3 (P = 0.46), I² = 0%



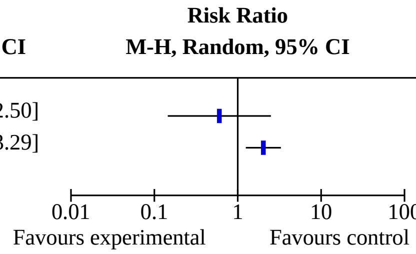
Analysis 5.5. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 5: Women's dissatisfaction with the procedure

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
5.5.1 Home vs hospital						
Iyengar 2015	15	349	12	274	23.9%	0.98 [0.47 , 2.06]
Li 2017	12	372	5	372	16.8%	2.40 [0.85 , 6.74]
Shrestha 2014	17	94	15	94	27.3%	1.13 [0.60 , 2.13]
Song 2018	58	321	19	279	32.0%	2.65 [1.62 , 4.34]
Subtotal (95% CI)		**1136**		**1019**	**100.0%**	**1.63 [0.95 , 2.80]**
Total events:	102		51			
Heterogeneity: Tau² = 0.17; Chi² = 7.28, df = 3 (P = 0.06); I² = 59%						
Test for overall effect: Z = 1.78 (P = 0.08)						
5.5.2 Nurse vs doctor						
Olavarrieta 2015	1	434	2	450	42.7%	0.52 [0.05 , 5.70]
Warriner 2011	37	552	4	552	57.3%	9.25 [3.32 , 25.78]
Subtotal (95% CI)		**986**		**1002**	**100.0%**	**2.70 [0.16 , 44.49]**
Total events:	38		6			
Heterogeneity: Tau² = 3.29; Chi² = 4.72, df = 1 (P = 0.03); I² = 79%						
Test for overall effect: Z = 0.70 (P = 0.49)						
Test for subgroup differences: Chi² = 0.12, df = 1 (P = 0.73), I² = 0%						



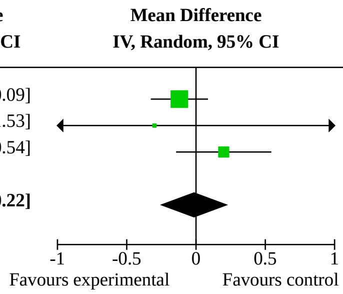
Analysis 5.6. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 6: Additional uterotonics used

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
Iyengar 2015	3	365	5	366	0.60 [0.14 , 2.50]
Olavarrieta 2015	45	434	23	450	2.03 [1.25 , 3.29]



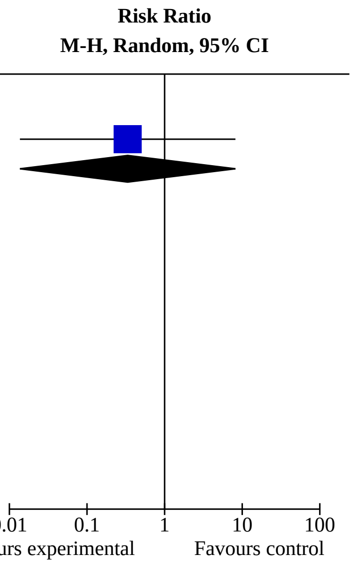
Analysis 5.7. Comparison 5: Combined regimen mifepristone/prostaglandin: administration strategy, Outcome 7: Days of bleeding

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Li 2017	5.72	1.38	372	5.84	1.49	372	64.6%	-0.12 [-0.33 , 0.09]
Shrestha 2014	9.6	6.1	94	9.9	6.7	94	1.6%	-0.30 [-2.13 , 1.53]
Song 2018	9.11	1.89	321	8.91	2.34	279	33.9%	0.20 [-0.14 , 0.54]
Total (95% CI)			**787**			**745**	**100.0%**	**-0.01 [-0.25 , 0.22]**
Heterogeneity: Tau² = 0.01; Chi² = 2.53, df = 2 (P = 0.28); I² = 21%								
Test for overall effect: Z = 0.12 (P = 0.90)								
Test for subgroup differences: Not applicable								



Analysis 5.8. Comparison 5: Combined regimen mifepristone/ prostaglandin: administration strategy, Outcome 8: Blood transfusion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
5.8.1 Home vs hospital						
Iyengar 2015	0	365	1	366	100.0%	0.33 [0.01 , 8.18]
Subtotal (95% CI)		**365**		**366**	**100.0%**	**0.33 [0.01 , 8.18]**
Total events:	0		1			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.67 (P = 0.50)						
5.8.2 Nurse vs doctor						
Kopp Kallner 2015	0	535	0	533		Not estimable
Subtotal (95% CI)		**535**		**533**		**Not estimable**
Total events:	0		0			
Heterogeneity: Not applicable						
Test for overall effect: Not applicable						



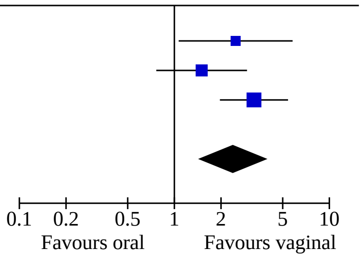
Comparison 6. Combined regimen mifepristone/prostaglandin: misoprostol oral vs vaginal

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.1 Failure to achieve complete abortion	3	1704	Risk Ratio (M-H, Random, 95% CI)	2.38 [1.46, 3.87]
6.2 Side effects	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
6.2.1 Nausea	2	1380	Risk Ratio (M-H, Random, 95% CI)	1.14 [1.03, 1.26]
6.2.2 Vomiting	2	1219	Risk Ratio (M-H, Random, 95% CI)	0.99 [0.51, 1.90]
6.2.3 Diarrhoea	2	1379	Risk Ratio (M-H, Random, 95% CI)	1.80 [1.49, 2.17]
6.3 Bleeding volume	1	297	Mean Difference (IV, Random, 95% CI)	8.00 [-2.94, 18.94]
6.4 Time until passing of conceptus	1	297	Mean Difference (IV, Fixed, 95% CI)	0.50 [-0.71, 1.71]

Analysis 6.1. Comparison 6: Combined regimen mifepristone/prostaglandin: misoprostol oral vs vaginal, Outcome 1: Failure to achieve complete abortion

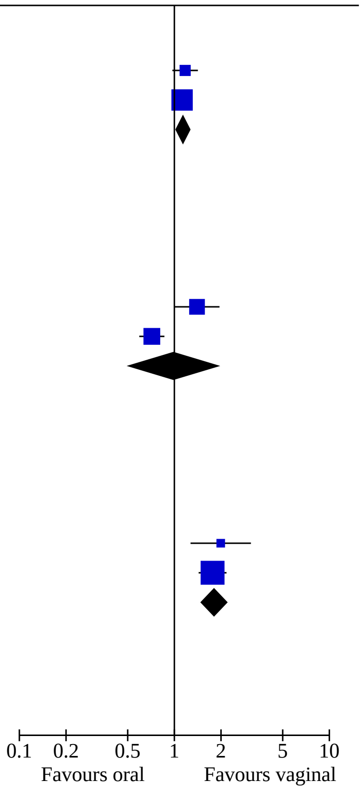
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
El-Refaey 1995	17	130	7	133	23.7%	2.48 [1.07 , 5.79]
Qian 2015	30	198	10	99	32.2%	1.50 [0.76 , 2.94]
Schaff 2001	57	548	19	596	44.1%	3.26 [1.97 , 5.41]
Total (95% CI)		**876**		**828**	**100.0%**	**2.38 [1.46 , 3.87]**
Total events:	104		36			

Heterogeneity: Tau² = 0.07; Chi² = 3.28, df = 2 (P = 0.19); I² = 39%
Test for overall effect: Z = 3.50 (P = 0.0005)
Test for subgroup differences: Not applicable



Analysis 6.2. Comparison 6: Combined regimen mifepristone/prostaglandin: misoprostol oral vs vaginal, Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
6.2.1 Nausea						
El-Refaey 1995	81	116	72	121	28.4%	1.17 [0.97 , 1.42]
Schaff 2001	282	548	273	595	71.6%	1.12 [1.00 , 1.26]
Subtotal (95% CI)		**664**		**716**	**100.0%**	**1.14 [1.03 , 1.26]**
Total events:	363		345			
Heterogeneity: Tau² = 0.00; Chi² = 0.16, df = 1 (P = 0.69); I² = 0%						
Test for overall effect: Z = 2.48 (P = 0.01)						
6.2.2 Vomiting						
El-Refaey 1995	51	116	38	121	47.8%	1.40 [1.00 , 1.96]
Schaff 2001	144	547	160	435	52.2%	0.72 [0.59 , 0.86]
Subtotal (95% CI)		**663**		**556**	**100.0%**	**0.99 [0.51 , 1.90]**
Total events:	195		198			
Heterogeneity: Tau² = 0.21; Chi² = 11.82, df = 1 (P = 0.0006); I² = 92%						
Test for overall effect: Z = 0.04 (P = 0.97)						
6.2.3 Diarrhoea						
El-Refaey 1995	42	116	22	121	17.6%	1.99 [1.27 , 3.12]
Schaff 2001	179	548	110	594	82.4%	1.76 [1.43 , 2.17]
Subtotal (95% CI)		**664**		**715**	**100.0%**	**1.80 [1.49 , 2.17]**
Total events:	221		132			
Heterogeneity: Tau² = 0.00; Chi² = 0.23, df = 1 (P = 0.63); I² = 0%						
Test for overall effect: Z = 6.14 (P < 0.00001)						

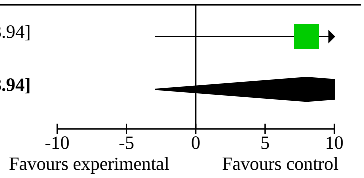


Analysis 6.3. Comparison 6: Combined regimen mifepristone/prostaglandin: misoprostol oral vs vaginal, Outcome 3: Bleeding volume

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Qian 2015	63	46	198	55	45	99	100.0%	8.00 [-2.94 , 18.94]
Total (95% CI)			**198**			**99**	**100.0%**	**8.00 [-2.94 , 18.94]**

Heterogeneity: Not applicable
Test for overall effect: Z = 1.43 (P = 0.15)
Test for subgroup differences: Not applicable

Mean Difference IV, Random, 95% CI

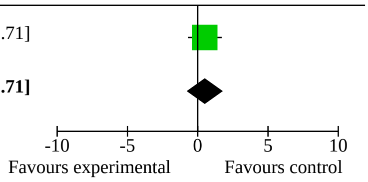


Analysis 6.4. Comparison 6: Combined regimen mifepristone/prostaglandin: misoprostol oral vs vaginal, Outcome 4: Time until passing of conceptus

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Fixed, 95% CI
Qian 2015	4.3	7.9	198	3.8	2.5	99	100.0%	0.50 [-0.71 , 1.71]
Total (95% CI)			**198**			**99**	**100.0%**	**0.50 [-0.71 , 1.71]**

Heterogeneity: Not applicable
Test for overall effect: Z = 0.81 (P = 0.42)
Test for subgroup differences: Not applicable

Mean Difference IV, Fixed, 95% CI



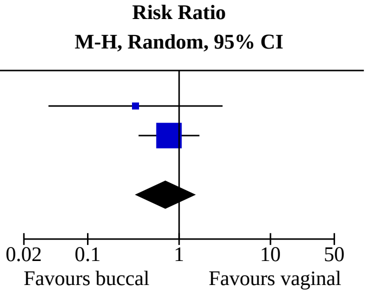
Comparison 7. Combined regimen mifepristone/prostaglandin: misoprostol buccal vs vaginal

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
7.1 Failure to achieve complete abortion	2	479	Risk Ratio (M-H, Random, 95% CI)	0.71 [0.34, 1.46]
7.2 Side effects	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
7.2.1 Nausea	2	479	Risk Ratio (M-H, Random, 95% CI)	1.11 [0.97, 1.27]
7.2.2 Vomiting	2	479	Risk Ratio (M-H, Random, 95% CI)	1.16 [0.89, 1.51]
7.2.3 Diarrhoea	2	479	Risk Ratio (M-H, Random, 95% CI)	1.51 [1.12, 2.03]
7.3 Surgical evacuation	1	50	Risk Ratio (M-H, Random, 95% CI)	0.33 [0.01, 7.81]
7.4 Women's dissatisfaction with the procedure	1	50	Risk Ratio (M-H, Random, 95% CI)	0.20 [0.01, 3.97]

Analysis 7.1. Comparison 7: Combined regimen mifepristone/prostaglandin: misoprostol buccal vs vaginal, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Garg 2015	1	25	3	25	10.9%	0.33 [0.04 , 2.99]
Middleton 2005	11	216	14	213	89.1%	0.77 [0.36 , 1.67]
Total (95% CI)		**241**		**238**	**100.0%**	**0.71 [0.34 , 1.46]**
Total events:	12		17			

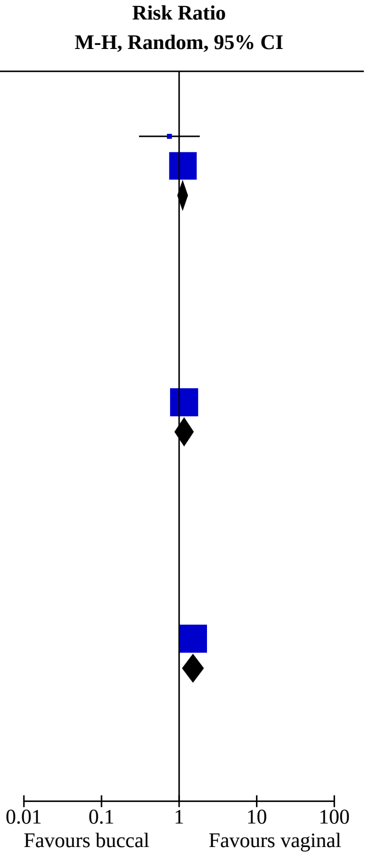
Heterogeneity: Tau² = 0.00; Chi² = 0.51, df = 1 (P = 0.48); I² = 0%
Test for overall effect: Z = 0.94 (P = 0.35)
Test for subgroup differences: Not applicable



Analysis 7.2. Comparison 7: Combined regimen mifepristone/prostaglandin: misoprostol buccal vs vaginal, Outcome 2: Side effects

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
7.2.1 Nausea						
Garg 2015	6	25	8	25	2.3%	0.75 [0.30 , 1.85]
Middleton 2005	150	216	132	213	97.7%	1.12 [0.98 , 1.29]
Subtotal (95% CI)		**241**		**238**	**100.0%**	**1.11 [0.97 , 1.27]**
Total events:	156		140			
Heterogeneity: Tau² = 0.00; Chi² = 0.76, df = 1 (P = 0.38); I² = 0%						
Test for overall effect: Z = 1.51 (P = 0.13)						
7.2.2 Vomiting						
Garg 2015	0	25	0	25		Not estimable
Middleton 2005	80	216	68	213	100.0%	1.16 [0.89 , 1.51]
Subtotal (95% CI)		**241**		**238**	**100.0%**	**1.16 [0.89 , 1.51]**
Total events:	80		68			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.11 (P = 0.27)						
7.2.3 Diarrhoea						
Garg 2015	0	25	0	25		Not estimable
Middleton 2005	78	216	51	213	100.0%	1.51 [1.12 , 2.03]
Subtotal (95% CI)		**241**		**238**	**100.0%**	**1.51 [1.12 , 2.03]**
Total events:	78		51			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.70 (P = 0.007)						

Test for subgroup differences: Chi² = 3.36, df = 2 (P = 0.19), I² = 40.5%

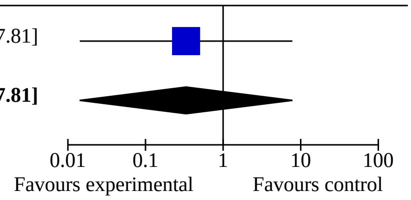


Analysis 7.3. Comparison 7: Combined regimen mifepristone/prostaglandin: misoprostol buccal vs vaginal, Outcome 3: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Garg 2015	0	25	1	25	100.0%	0.33 [0.01 , 7.81]
Total (95% CI)		**25**		**25**	**100.0%**	**0.33 [0.01 , 7.81]**
Total events:	0		1			

Heterogeneity: Not applicable
Test for overall effect: Z = 0.68 (P = 0.49)
Test for subgroup differences: Not applicable

Risk Ratio M-H, Random, 95% CI

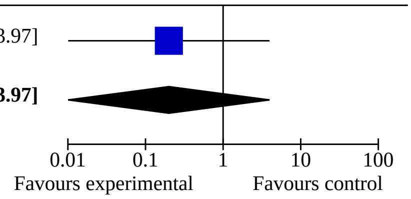


Analysis 7.4. Comparison 7: Combined regimen mifepristone/prostaglandin: misoprostol buccal vs vaginal, Outcome 4: Women's dissatisfaction with the procedure

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Garg 2015	0	25	2	25	100.0%	0.20 [0.01 , 3.97]
Total (95% CI)		**25**		**25**	**100.0%**	**0.20 [0.01 , 3.97]**
Total events:	0		2			

Heterogeneity: Not applicable
Test for overall effect: Z = 1.06 (P = 0.29)
Test for subgroup differences: Not applicable

Risk Ratio M-H, Random, 95% CI



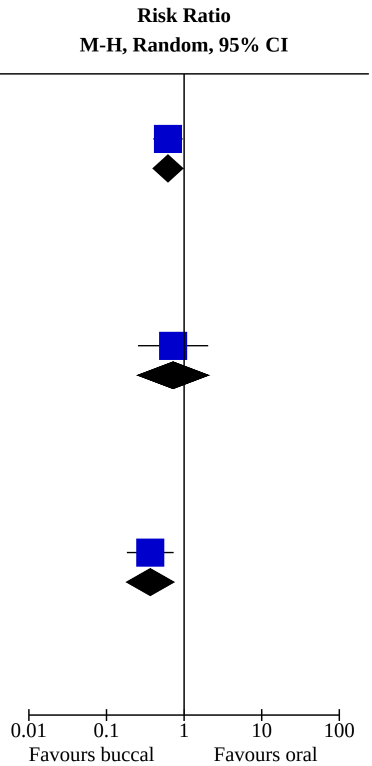
Comparison 8. Combined regimen mifepristone/prostaglandin: misoprostol buccal vs oral

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.1 Failure to achieve complete abortion	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
8.1.1 Failure to achieve complete abortion (all)	1	847	Risk Ratio (M-H, Random, 95% CI)	0.62 [0.40, 0.96]
8.1.2 Failure to achieve complete abortion ≤ 49 days	1	418	Risk Ratio (M-H, Random, 95% CI)	0.72 [0.25, 2.04]
8.1.3 Failure to achieve complete abortion > 49 days	1	429	Risk Ratio (M-H, Random, 95% CI)	0.37 [0.18, 0.73]
8.2 Ongoing pregnancy	1	847	Risk Ratio (M-H, Random, 95% CI)	0.31 [0.09, 1.08]
8.2.1 Ongoing pregnancy ≤ 49 days	1	418	Risk Ratio (M-H, Random, 95% CI)	0.64 [0.11, 3.80]
8.2.2 Ongoing pregnancy > 49 days	1	429	Risk Ratio (M-H, Random, 95% CI)	0.18 [0.04, 0.78]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.3 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
8.3.1 Nausea	1	830	Risk Ratio (M-H, Random, 95% CI)	1.10 [1.01, 1.19]
8.3.2 Vomiting	1	830	Risk Ratio (M-H, Random, 95% CI)	1.09 [0.94, 1.27]
8.3.3 Diarrhoea	1	830	Risk Ratio (M-H, Random, 95% CI)	1.11 [0.94, 1.31]
8.4 Women's dissatisfaction with the procedure	1	835	Risk Ratio (M-H, Random, 95% CI)	1.21 [0.76, 1.91]

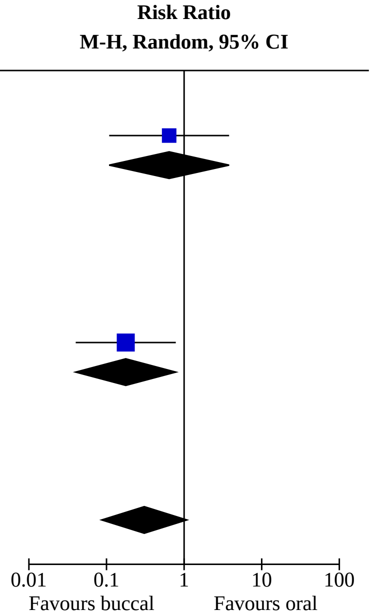
Analysis 8.1. Comparison 8: Combined regimen mifepristone/prostaglandin: misoprostol buccal vs oral, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
8.1.1 Failure to achieve complete abortion (all)						
Winikoff 2008	30	421	49	426	100.0%	0.62 [0.40 , 0.96]
Subtotal (95% CI)		**421**		**426**	**100.0%**	**0.62 [0.40 , 0.96]**
Total events:	30		49			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.16 (P = 0.03)						
8.1.2 Failure to achieve complete abortion ≤ 49 days						
Winikoff 2008	6	213	8	205	100.0%	0.72 [0.25 , 2.04]
Subtotal (95% CI)		**213**		**205**	**100.0%**	**0.72 [0.25 , 2.04]**
Total events:	6		8			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.61 (P = 0.54)						
8.1.3 Failure to achieve complete abortion > 49 days						
Winikoff 2008	10	208	29	221	100.0%	0.37 [0.18 , 0.73]
Subtotal (95% CI)		**208**		**221**	**100.0%**	**0.37 [0.18 , 0.73]**
Total events:	10		29			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.84 (P = 0.005)						



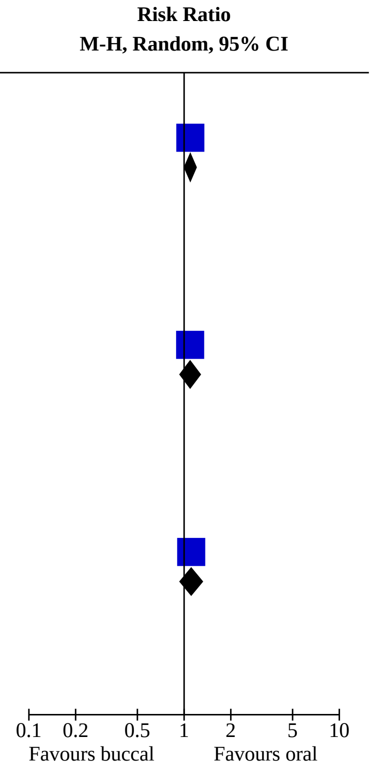
Analysis 8.2. Comparison 8: Combined regimen mifepristone/ prostaglandin: misoprostol buccal vs oral, Outcome 2: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
8.2.1 Ongoing pregnancy ≤ 49 days						
Winikoff 2008	2	213	3	205	42.6%	0.64 [0.11 , 3.80]
Subtotal (95% CI)		**213**		**205**	**42.6%**	**0.64 [0.11 , 3.80]**
Total events:	2		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.49 (P = 0.62)						
8.2.2 Ongoing pregnancy > 49 days						
Winikoff 2008	2	208	12	221	57.4%	0.18 [0.04 , 0.78]
Subtotal (95% CI)		**208**		**221**	**57.4%**	**0.18 [0.04 , 0.78]**
Total events:	2		12			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.28 (P = 0.02)						
Total (95% CI)		**421**		**426**	**100.0%**	**0.31 [0.09 , 1.08]**
Total events:	4		15			
Heterogeneity: Tau² = 0.15; Chi² = 1.21, df = 1 (P = 0.27); I² = 17%						
Test for overall effect: Z = 1.84 (P = 0.07)						
Test for subgroup differences: Chi² = 1.19, df = 1 (P = 0.28), I² = 15.7%						



Analysis 8.3. Comparison 8: Combined regimen mifepristone/ prostaglandin: misoprostol buccal vs oral, Outcome 3: Side effects

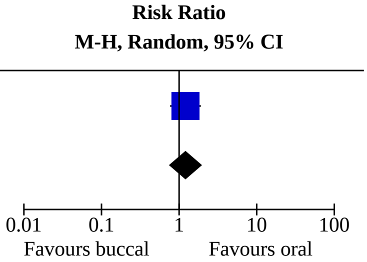
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
8.3.1 Nausea						
Winikoff 2008	311	414	285	416	100.0%	1.10 [1.01 , 1.19]
Subtotal (95% CI)		**414**		**416**	**100.0%**	**1.10 [1.01 , 1.19]**
Total events:	311		285			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.11 (P = 0.03)						
8.3.2 Vomiting						
Winikoff 2008	197	414	181	416	100.0%	1.09 [0.94 , 1.27]
Subtotal (95% CI)		**414**		**416**	**100.0%**	**1.09 [0.94 , 1.27]**
Total events:	197		181			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.18 (P = 0.24)						
8.3.3 Diarrhoea						
Winikoff 2008	178	414	161	416	100.0%	1.11 [0.94 , 1.31]
Subtotal (95% CI)		**414**		**416**	**100.0%**	**1.11 [0.94 , 1.31]**
Total events:	178		161			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.26 (P = 0.21)						



Analysis 8.4. Comparison 8: Combined regimen mifepristone/prostaglandin: misoprostol buccal vs oral, Outcome 4: Women's dissatisfaction with the procedure

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Winikoff 2008	37	415	31	420	100.0%	1.21 [0.76 , 1.91]
Total (95% CI)		**415**		**420**	**100.0%**	**1.21 [0.76 , 1.91]**
Total events:	37		31			

Heterogeneity: Not applicable
Test for overall effect: Z = 0.81 (P = 0.42)
Test for subgroup differences: Not applicable



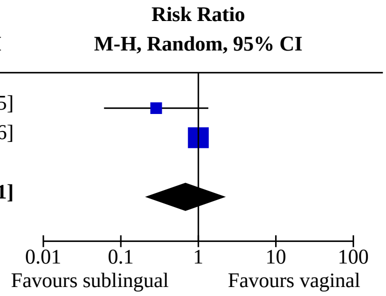
Comparison 9. Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs vaginal

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
9.1 Failure to achieve complete abortion	2	3229	Risk Ratio (M-H, Random, 95% CI)	0.68 [0.22, 2.11]
9.2 Surgical evacuation	1	327	Risk Ratio (M-H, Random, 95% CI)	0.80 [0.18, 3.53]
9.3 Ongoing pregnancy	2	3229	Risk Ratio (M-H, Random, 95% CI)	0.67 [0.13, 3.47]
9.4 Women's dissatisfaction with the procedure	2	3303	Risk Ratio (M-H, Random, 95% CI)	1.67 [0.80, 3.50]
9.5 Side effects	3		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
9.5.1 Diarrhoea	3	3543	Risk Ratio (M-H, Random, 95% CI)	1.83 [1.33, 2.50]
9.5.2 Nausea	3	3543	Risk Ratio (M-H, Random, 95% CI)	1.11 [0.93, 1.33]
9.5.3 Vomiting	3	3543	Risk Ratio (M-H, Random, 95% CI)	1.21 [0.88, 1.66]
9.5.4 Abdominal pain	1	3005	Risk Ratio (M-H, Random, 95% CI)	0.99 [0.87, 1.12]

Analysis 9.1. Comparison 9: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs vaginal, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Tang 2003	2	112	7	112	30.6%	0.29 [0.06 , 1.35]
von Hertzen 2010	131	1503	131	1502	69.4%	1.00 [0.79 , 1.26]
Total (95% CI)		**1615**		**1614**	**100.0%**	**0.68 [0.22 , 2.11]**
Total events:	133		138			

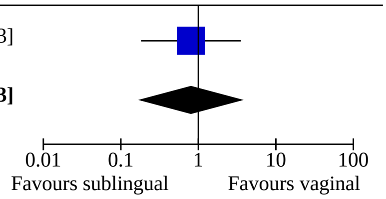
Heterogeneity: $Tau^2 = 0.47$; $Chi^2 = 2.46$, df = 1 (P = 0.12); $I^2 = 59\%$
Test for overall effect: Z = 0.66 (P = 0.51)
Test for subgroup differences: Not applicable



Analysis 9.2. Comparison 9: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs vaginal, Outcome 2: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Hamoda 2005	3	158	4	169	100.0%	0.80 [0.18 , 3.53]
Total (95% CI)		**158**		**169**	**100.0%**	**0.80 [0.18 , 3.53]**
Total events:	3		4			

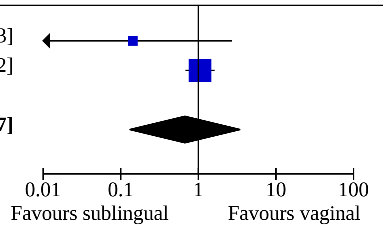
Heterogeneity: Not applicable
Test for overall effect: Z = 0.29 (P = 0.77)
Test for subgroup differences: Not applicable



Analysis 9.3. Comparison 9: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs vaginal, Outcome 3: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Tang 2003	0	112	3	112	22.5%	0.14 [0.01 , 2.73]
von Hertzen 2010	41	1503	39	1502	77.5%	1.05 [0.68 , 1.62]
Total (95% CI)		**1615**		**1614**	**100.0%**	**0.67 [0.13 , 3.47]**
Total events:	41		42			

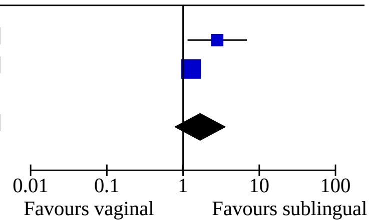
Heterogeneity: Tau² = 0.86; Chi² = 1.74, df = 1 (P = 0.19); I² = 43%
Test for overall effect: Z = 0.48 (P = 0.63)
Test for subgroup differences: Not applicable



Analysis 9.4. Comparison 9: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs vaginal, Outcome 4: Women's dissatisfaction with the procedure

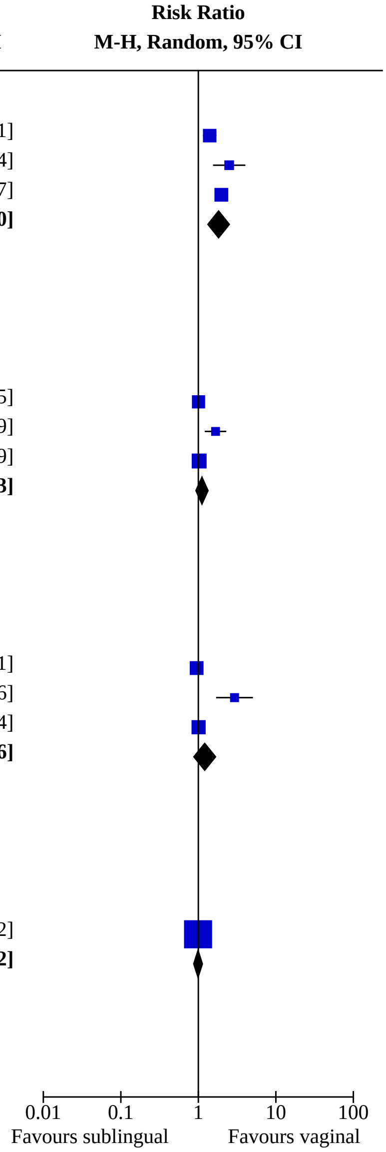
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Hamoda 2005	18	154	6	144	34.6%	2.81 [1.15 , 6.87]
von Hertzen 2010	130	1503	102	1502	65.4%	1.27 [0.99 , 1.63]
Total (95% CI)		**1657**		**1646**	**100.0%**	**1.67 [0.80 , 3.50]**
Total events:	148		108			

Heterogeneity: Tau² = 0.20; Chi² = 2.78, df = 1 (P = 0.10); I² = 64%
Test for overall effect: Z = 1.37 (P = 0.17)
Test for subgroup differences: Not applicable



Analysis 9.5. Comparison 9: Combined regimen mifepristone/ prostaglandin: misoprostol sublingual vs vaginal, Outcome 5: Side effects

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
9.5.1 Diarrhoea						
Hamoda 2005	105	158	74	156	38.4%	1.40 [1.15 , 1.71]
Tang 2003	45	112	18	112	22.2%	2.50 [1.55 , 4.04]
von Hertzen 2010	301	1503	152	1502	39.4%	1.98 [1.65 , 2.37]
Subtotal (95% CI)		**1773**		**1770**	**100.0%**	**1.83 [1.33 , 2.50]**
Total events:	451		244			
Heterogeneity: Tau² = 0.06; Chi² = 9.42, df = 2 (P = 0.009); I² = 79%						
Test for overall effect: Z = 3.73 (P = 0.0002)						
9.5.2 Nausea						
Hamoda 2005	115	158	113	156	37.0%	1.00 [0.88 , 1.15]
Tang 2003	60	112	36	112	18.6%	1.67 [1.21 , 2.29]
von Hertzen 2010	820	1503	800	1502	44.4%	1.02 [0.96 , 1.09]
Subtotal (95% CI)		**1773**		**1770**	**100.0%**	**1.11 [0.93 , 1.33]**
Total events:	995		949			
Heterogeneity: Tau² = 0.02; Chi² = 8.89, df = 2 (P = 0.01); I² = 78%						
Test for overall effect: Z = 1.19 (P = 0.23)						
9.5.3 Vomiting						
Hamoda 2005	104	158	108	156	39.7%	0.95 [0.82 , 1.11]
Tang 2003	41	112	14	112	19.2%	2.93 [1.69 , 5.06]
von Hertzen 2010	392	1503	389	1502	41.1%	1.01 [0.89 , 1.14]
Subtotal (95% CI)		**1773**		**1770**	**100.0%**	**1.21 [0.88 , 1.66]**
Total events:	537		511			
Heterogeneity: Tau² = 0.06; Chi² = 15.50, df = 2 (P = 0.0004); I² = 87%						
Test for overall effect: Z = 1.16 (P = 0.25)						
9.5.4 Abdominal pain						
von Hertzen 2010	363	1503	366	1502	100.0%	0.99 [0.87 , 1.12]
Subtotal (95% CI)		**1503**		**1502**	**100.0%**	**0.99 [0.87 , 1.12]**
Total events:	363		366			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.14 (P = 0.89)						



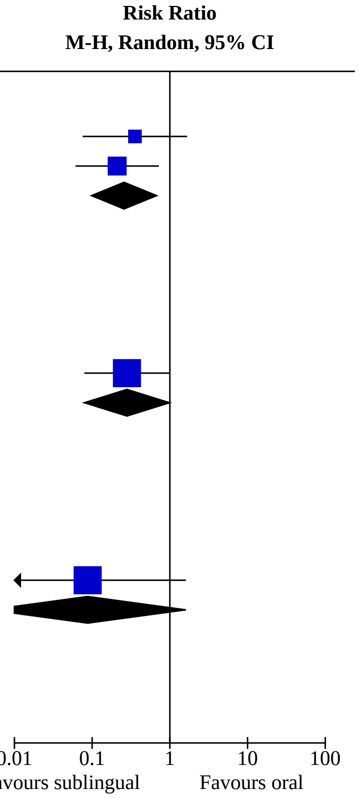
Comparison 10. Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs oral

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10.1 Failure to achieve complete abortion	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
10.1.1 All	2	564	Risk Ratio (M-H, Random, 95% CI)	0.26 [0.10, 0.68]
10.1.2 ≤ 49 days	1	422	Risk Ratio (M-H, Random, 95% CI)	0.28 [0.08, 0.99]
10.1.3 > 49 days	1	48	Risk Ratio (M-H, Random, 95% CI)	0.09 [0.00, 1.60]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
10.2 Ongoing pregnancy	1	93	Risk Ratio (M-H, Random, 95% CI)	0.36 [0.01, 8.50]
10.3 Surgical evacuation	1	93	Risk Ratio (M-H, Random, 95% CI)	0.36 [0.08, 1.67]
10.4 Women's dissatisfaction with the procedure	1	471	Risk Ratio (M-H, Random, 95% CI)	1.96 [0.94, 4.09]
10.5 Side effects	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
10.5.1 Nausea	2	564	Risk Ratio (M-H, Random, 95% CI)	0.62 [0.27, 1.41]
10.5.2 Vomiting	2	564	Risk Ratio (M-H, Random, 95% CI)	0.63 [0.25, 1.62]
10.5.3 Diarrhea	1	93	Risk Ratio (M-H, Random, 95% CI)	0.32 [0.09, 1.09]

Analysis 10.1. Comparison 10: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs oral, Outcome 1: Failure to achieve complete abortion

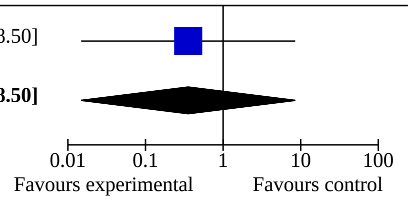
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
10.1.1 All						
Dahiya 2011	2	45	6	48	38.9%	0.36 [0.08 , 1.67]
Raghavan 2009	3	238	14	233	61.1%	0.21 [0.06 , 0.72]
Subtotal (95% CI)		**283**		**281**	**100.0%**	**0.26 [0.10 , 0.68]**
Total events:	5		20			
Heterogeneity: Tau² = 0.00; Chi² = 0.27, df = 1 (P = 0.60); I² = 0%						
Test for overall effect: Z = 2.76 (P = 0.006)						
10.1.2 ≤ 49 days						
Raghavan 2009	3	208	11	214	100.0%	0.28 [0.08 , 0.99]
Subtotal (95% CI)		**208**		**214**	**100.0%**	**0.28 [0.08 , 0.99]**
Total events:	3		11			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.97 (P = 0.05)						
10.1.3 > 49 days						
Raghavan 2009	0	30	3	18	100.0%	0.09 [0.00 , 1.60]
Subtotal (95% CI)		**30**		**18**	**100.0%**	**0.09 [0.00 , 1.60]**
Total events:	0		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.64 (P = 0.10)						



Analysis 10.2. Comparison 10: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs oral, Outcome 2: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Dahiya 2011	0	45	1	48	100.0%	0.36 [0.01 , 8.50]
Total (95% CI)		**45**		**48**	**100.0%**	**0.36 [0.01 , 8.50]**
Total events:	0		1			

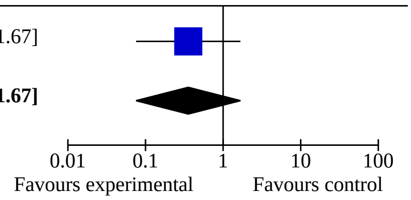
Heterogeneity: Not applicable
Test for overall effect: Z = 0.64 (P = 0.52)
Test for subgroup differences: Not applicable



Analysis 10.3. Comparison 10: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs oral, Outcome 3: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Dahiya 2011	2	45	6	48	100.0%	0.36 [0.08 , 1.67]
Total (95% CI)		**45**		**48**	**100.0%**	**0.36 [0.08 , 1.67]**
Total events:	2		6			

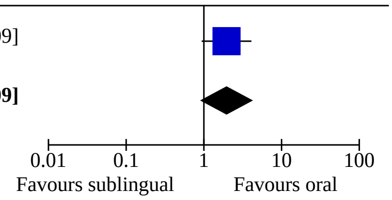
Heterogeneity: Not applicable
Test for overall effect: Z = 1.31 (P = 0.19)
Test for subgroup differences: Not applicable



Analysis 10.4. Comparison 10: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs oral, Outcome 4: Women's dissatisfaction with the procedure

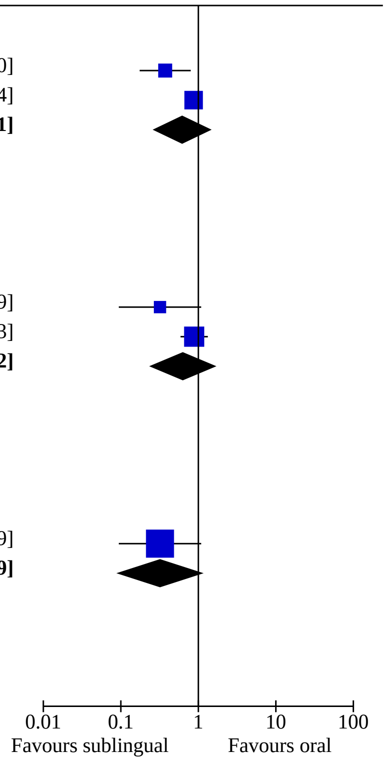
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Raghavan 2009	20	238	10	233	100.0%	1.96 [0.94 , 4.09]
Total (95% CI)		**238**		**233**	**100.0%**	**1.96 [0.94 , 4.09]**
Total events:	20		10			

Heterogeneity: Not applicable
Test for overall effect: Z = 1.79 (P = 0.07)
Test for subgroup differences: Not applicable



Analysis 10.5. Comparison 10: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs oral, Outcome 5: Side effects

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
10.5.1 Nausea						
Dahiya 2011	7	45	20	48	40.4%	0.37 [0.17 , 0.80]
Raghavan 2009	111	238	125	233	59.6%	0.87 [0.73 , 1.04]
Subtotal (95% CI)		**283**		**281**	**100.0%**	**0.62 [0.27 , 1.41]**
Total events:	118		145			
Heterogeneity: Tau² = 0.29; Chi² = 4.65, df = 1 (P = 0.03); I² = 78%						
Test for overall effect: Z = 1.14 (P = 0.25)						
10.5.2 Vomiting						
Dahiya 2011	3	45	10	48	33.3%	0.32 [0.09 , 1.09]
Raghavan 2009	37	238	41	233	66.7%	0.88 [0.59 , 1.33]
Subtotal (95% CI)		**283**		**281**	**100.0%**	**0.63 [0.25 , 1.62]**
Total events:	40		51			
Heterogeneity: Tau² = 0.30; Chi² = 2.40, df = 1 (P = 0.12); I² = 58%						
Test for overall effect: Z = 0.96 (P = 0.34)						
10.5.3 Diarrhea						
Dahiya 2011	3	45	10	48	100.0%	0.32 [0.09 , 1.09]
Subtotal (95% CI)		**45**		**48**	**100.0%**	**0.32 [0.09 , 1.09]**
Total events:	3		10			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.82 (P = 0.07)						



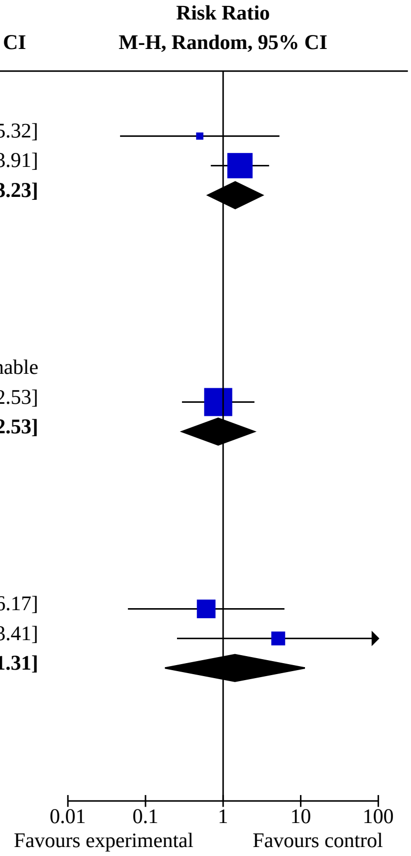
Comparison 11. Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
11.1 Failure to achieve complete abortion	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
11.1.1 All	2	640	Risk Ratio (M-H, Random, 95% CI)	1.43 [0.64, 3.23]
11.1.2 ≤ 49 days	2	512	Risk Ratio (M-H, Random, 95% CI)	0.86 [0.30, 2.53]
11.1.3 > 49 days	2	117	Risk Ratio (M-H, Random, 95% CI)	1.42 [0.18, 11.31]
11.2 Surgical evacuation	2	640	Risk Ratio (M-H, Random, 95% CI)	1.72 [0.75, 3.96]
11.3 Ongoing pregnancy	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
11.3.1 All	2	640	Risk Ratio (M-H, Random, 95% CI)	2.59 [0.88, 7.61]
11.3.2 ≤ 49 days	1	48	Risk Ratio (M-H, Random, 95% CI)	Not estimable
11.3.3 > 49 days	1	42	Risk Ratio (M-H, Random, 95% CI)	3.60 [0.16, 83.60]
11.4 Side effects	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
11.4.1 Vomiting	2	640	Risk Ratio (M-H, Random, 95% CI)	1.33 [1.01, 1.77]
11.4.2 Nausea	2	640	Risk Ratio (M-H, Random, 95% CI)	1.06 [0.88, 1.27]
11.4.3 Diarrhoea	2	640	Risk Ratio (M-H, Random, 95% CI)	2.19 [0.56, 8.51]
11.4.4 Abdominal pain	2	640	Risk Ratio (M-H, Random, 95% CI)	1.10 [0.79, 1.52]
11.5 Women's dissatisfaction with the procedure	1	550	Risk Ratio (M-H, Random, 95% CI)	2.54 [1.14, 5.66]
11.6 Haemoglobin level	1	90	Mean Difference (IV, Random, 95% CI)	0.00 [-0.40, 0.40]
11.7 Time until passing of conceptus	1	90	Mean Difference (IV, Random, 95% CI)	-0.88 [-2.44, 0.68]

Analysis 11.1. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 1: Failure to achieve complete abortion

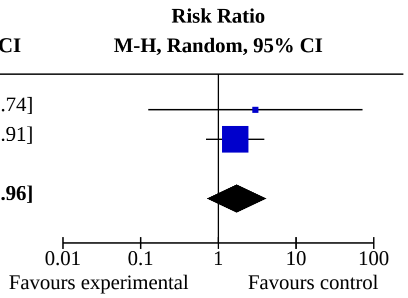
	Experimental		Control			Risk Ratio
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI
11.1.1 All						
Chai 2013	1	45	2	45	11.8%	0.50 [0.05 , 5.32]
Raghavan 2010	13	273	8	277	88.2%	1.65 [0.69 , 3.91]
Subtotal (95% CI)		**318**		**322**	**100.0%**	**1.43 [0.64 , 3.23]**
Total events:	14		10			
Heterogeneity: Tau² = 0.00; Chi² = 0.86, df = 1 (P = 0.35); I² = 0%						
Test for overall effect: Z = 0.87 (P = 0.39)						
11.1.2 ≤ 49 days						
Chai 2013	0	26	0	22		Not estimable
Raghavan 2010	6	231	7	233	100.0%	0.86 [0.30 , 2.53]
Subtotal (95% CI)		**257**		**255**	**100.0%**	**0.86 [0.30 , 2.53]**
Total events:	6		7			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.27 (P = 0.79)						
11.1.3 > 49 days						
Chai 2013	1	19	2	23	60.1%	0.61 [0.06 , 6.17]
Raghavan 2010	2	37	0	38	39.9%	5.13 [0.25 , 103.41]
Subtotal (95% CI)		**56**		**61**	**100.0%**	**1.42 [0.18 , 11.31]**
Total events:	3		2			
Heterogeneity: Tau² = 0.46; Chi² = 1.25, df = 1 (P = 0.26); I² = 20%						
Test for overall effect: Z = 0.33 (P = 0.74)						
Test for subgroup differences: Chi² = 0.56, df = 2 (P = 0.75), I² = 0%						



Analysis 11.2. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 2: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Chai 2013	1	45	0	45	6.9%	3.00 [0.13 , 71.74]
Raghavan 2010	13	273	8	277	93.1%	1.65 [0.69 , 3.91]
Total (95% CI)		**318**		**322**	**100.0%**	**1.72 [0.75 , 3.96]**
Total events:	14		8			

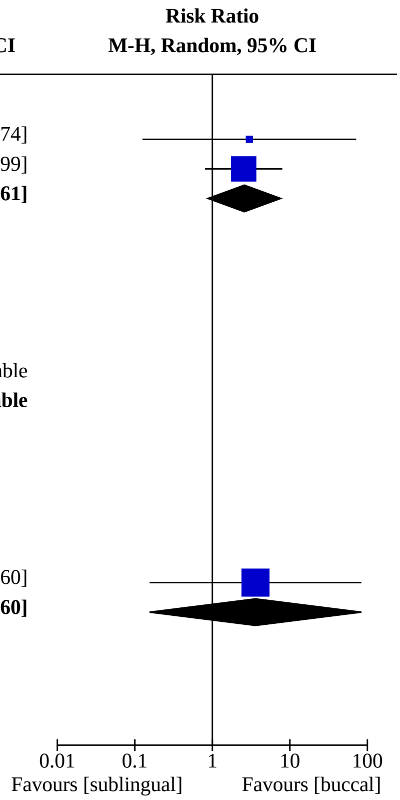
Heterogeneity: Tau² = 0.00; Chi² = 0.13, df = 1 (P = 0.72); I² = 0%
Test for overall effect: Z = 1.27 (P = 0.20)
Test for subgroup differences: Not applicable



Analysis 11.3. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 3: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
11.3.1 All						
Chai 2013	1	45	0	45	11.6%	3.00 [0.13 , 71.74]
Raghavan 2010	10	273	4	277	88.4%	2.54 [0.81 , 7.99]
Subtotal (95% CI)		**318**		**322**	**100.0%**	**2.59 [0.88 , 7.61]**
Total events:	11		4			
Heterogeneity: Tau² = 0.00; Chi² = 0.01, df = 1 (P = 0.92); I² = 0%						
Test for overall effect: Z = 1.73 (P = 0.08)						
11.3.2 ≤ 49 days						
Chai 2013	0	26	0	22		Not estimable
Subtotal (95% CI)		**26**		**22**		**Not estimable**
Total events:	0		0			
Heterogeneity: Not applicable						
Test for overall effect: Not applicable						
11.3.3 > 49 days						
Chai 2013	1	19	0	23	100.0%	3.60 [0.16 , 83.60]
Subtotal (95% CI)		**19**		**23**	**100.0%**	**3.60 [0.16 , 83.60]**
Total events:	1		0			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.80 (P = 0.42)						

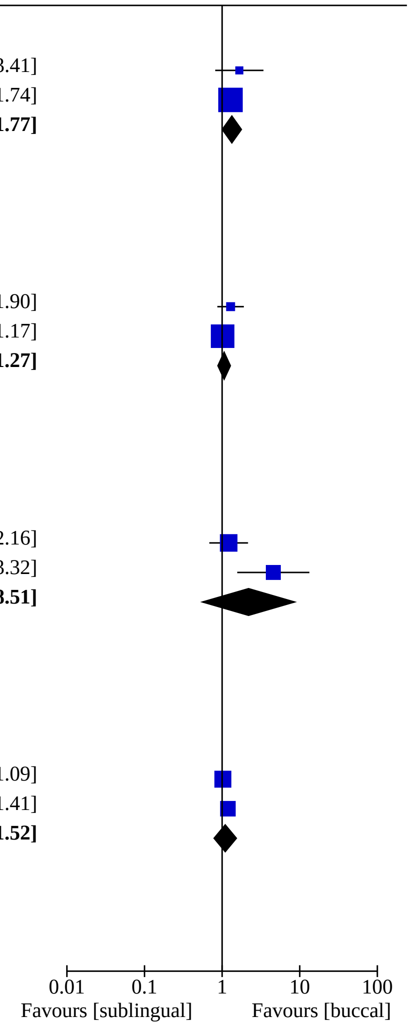
Test for subgroup differences: Chi² = 0.04, df = 1 (P = 0.85), I² = 0%



Analysis 11.4. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 4: Side effects

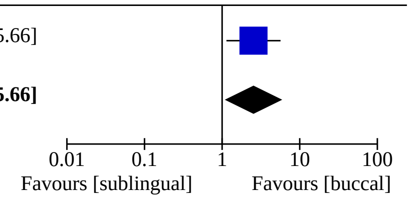
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
11.4.1 Vomiting						
Chai 2013	15	45	9	45	15.3%	1.67 [0.81 , 3.41]
Raghavan 2010	72	273	57	277	84.7%	1.28 [0.95 , 1.74]
Subtotal (95% CI)		**318**		**322**	**100.0%**	**1.33 [1.01 , 1.77]**
Total events:	87		66			
Heterogeneity: Tau² = 0.00; Chi² = 0.44, df = 1 (P = 0.51); I² = 0%						
Test for overall effect: Z = 2.02 (P = 0.04)						
11.4.2 Nausea						
Chai 2013	27	45	21	45	19.1%	1.29 [0.87 , 1.90]
Raghavan 2010	157	273	157	277	80.9%	1.01 [0.88 , 1.17]
Subtotal (95% CI)		**318**		**322**	**100.0%**	**1.06 [0.88 , 1.27]**
Total events:	184		178			
Heterogeneity: Tau² = 0.01; Chi² = 1.23, df = 1 (P = 0.27); I² = 19%						
Test for overall effect: Z = 0.64 (P = 0.52)						
11.4.3 Diarrhoea						
Chai 2013	17	45	14	45	55.5%	1.21 [0.68 , 2.16]
Raghavan 2010	18	273	4	277	44.5%	4.57 [1.57 , 13.32]
Subtotal (95% CI)		**318**		**322**	**100.0%**	**2.19 [0.56 , 8.51]**
Total events:	35		18			
Heterogeneity: Tau² = 0.78; Chi² = 5.05, df = 1 (P = 0.02); I² = 80%						
Test for overall effect: Z = 1.13 (P = 0.26)						
11.4.4 Abdominal pain						
Chai 2013	45	45	44	45	53.0%	1.02 [0.96 , 1.09]
Raghavan 2010	145	273	124	277	47.0%	1.19 [1.00 , 1.41]
Subtotal (95% CI)		**318**		**322**	**100.0%**	**1.10 [0.79 , 1.52]**
Total events:	190		168			
Heterogeneity: Tau² = 0.05; Chi² = 12.95, df = 1 (P = 0.0003); I² = 92%						
Test for overall effect: Z = 0.55 (P = 0.58)						

Test for subgroup differences: Chi² = 2.74, df = 3 (P = 0.43), I² = 0%



Analysis 11.5. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 5: Women's dissatisfaction with the procedure

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Raghavan 2010	20	273	8	277	100.0%	2.54 [1.14 , 5.66]
Total (95% CI)		**273**		**277**	**100.0%**	**2.54 [1.14 , 5.66]**
Total events:	20		8			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.27 (P = 0.02)						
Test for subgroup differences: Not applicable						

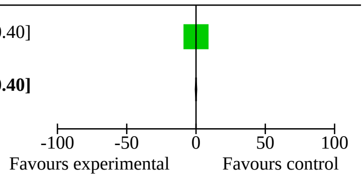


Analysis 11.6. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 6: Haemoglobin level

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Chai 2013	12.3	0.97	45	12.3	0.97	45	100.0%	0.00 [-0.40 , 0.40]
Total (95% CI)			**45**			**45**	**100.0%**	**0.00 [-0.40 , 0.40]**

Heterogeneity: Not applicable
Test for overall effect: Z = 0.00 (P = 1.00)
Test for subgroup differences: Not applicable

Mean Difference IV, Random, 95% CI

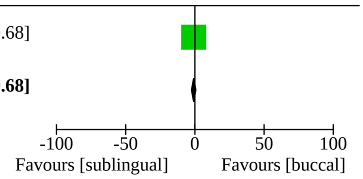


Analysis 11.7. Comparison 11: Combined regimen mifepristone/prostaglandin: misoprostol sublingual vs buccal, Outcome 7: Time until passing of conceptus

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Chai 2013	3.04	3.34	45	3.92	4.17	45	100.0%	-0.88 [-2.44 , 0.68]
Total (95% CI)			**45**			**45**	**100.0%**	**-0.88 [-2.44 , 0.68]**

Heterogeneity: Not applicable
Test for overall effect: Z = 1.10 (P = 0.27)
Test for subgroup differences: Not applicable

Mean Difference IV, Random, 95% CI



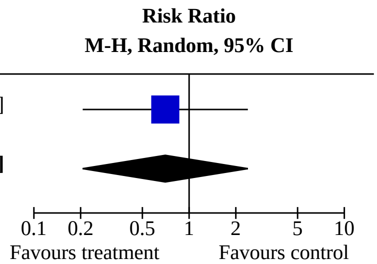
Comparison 12. Combined regimen mifepristone/prostaglandin: single vs split dose of prostaglandin

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
12.1 Failure to achieve complete abortion	1	154	Risk Ratio (M-H, Random, 95% CI)	0.70 [0.21, 2.39]
12.2 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
12.2.1 Nausea	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
12.2.2 Vomiting	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
12.2.3 Diarrhoea	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 12.1. Comparison 12: Combined regimen mifepristone/prostaglandin: single vs split dose of prostaglandin, Outcome 1: Failure to achieve complete abortion

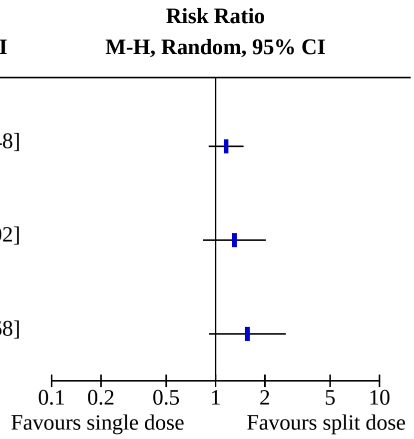
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
El-Refaey 1994	4	75	6	79	100.0%	0.70 [0.21 , 2.39]
Total (95% CI)		**75**		**79**	**100.0%**	**0.70 [0.21 , 2.39]**
Total events:	4		6			

Heterogeneity: Not applicable
Test for overall effect: Z = 0.57 (P = 0.57)
Test for subgroup differences: Not applicable



Analysis 12.2. Comparison 12: Combined regimen mifepristone/prostaglandin: single vs split dose of prostaglandin, Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
12.2.1 Nausea					
El-Refaey 1994	51	75	44	75	1.16 [0.91 , 1.48]
12.2.2 Vomiting					
El-Refaey 1994	30	75	23	75	1.30 [0.84 , 2.02]
12.2.3 Diarrhoea					
El-Refaey 1994	25	75	16	75	1.56 [0.91 , 2.68]



Comparison 13. Combined regimen mifepristone/prostaglandin:single vs continuous prostaglandin

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
13.1 Failure to achieve complete abortion	2		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
13.1.1 All oral vs vaginal and continuous oral	2	1581	Risk Ratio (M-H, Random, 95% CI)	1.48 [1.01, 2.16]
13.1.2 All oral vs single vaginal	2	1578	Risk Ratio (M-H, Random, 95% CI)	1.19 [0.83, 1.70]
13.1.3 Vaginal and continuous oral vs single vaginal	2	1579	Risk Ratio (M-H, Random, 95% CI)	0.80 [0.54, 1.19]
13.1.4 All oral vs vaginal & continuous oral ≤ 49 days	1	476	Risk Ratio (M-H, Random, 95% CI)	1.17 [0.57, 2.41]
13.1.5 All oral vs vaginal and continuous oral > 49 days	1	1004	Risk Ratio (M-H, Random, 95% CI)	1.60 [1.00, 2.57]

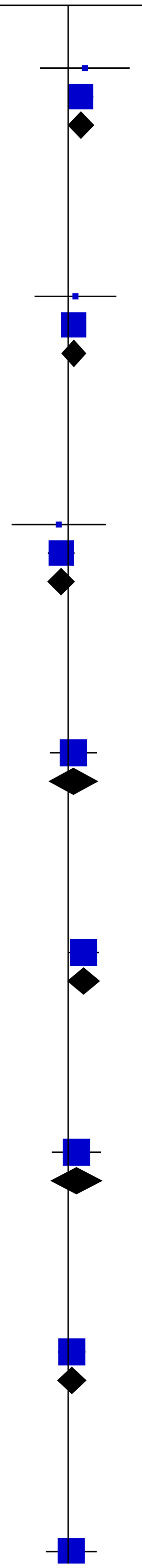
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
13.1.6 All oral vs single vaginal ≤ 49 days	1	459	Risk Ratio (M-H, Random, 95% CI)	1.29 [0.60, 2.74]
13.1.7 All oral vs single vaginal > 49 days	1	1014	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.73, 1.70]
13.1.8 Vaginal and continuous oral vs single vaginal ≤ 49days	1	463	Risk Ratio (M-H, Random, 95% CI)	1.10 [0.50, 2.40]
13.1.9 Vaginal and continuous oral vs single vaginal > 49 days	1	1010	Risk Ratio (M-H, Random, 95% CI)	0.70 [0.43, 1.13]
13.2 Diarrhoea	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
13.2.1 All oral vs vaginal and continuous oral	1	1481	Risk Ratio (M-H, Random, 95% CI)	1.83 [1.11, 3.01]
13.2.2 All oral vs single vaginal	1	1478	Risk Ratio (M-H, Random, 95% CI)	2.09 [1.24, 3.53]
13.2.3 Vaginal and continuous oral vs single vaginal	1	1479	Risk Ratio (M-H, Random, 95% CI)	1.15 [0.63, 2.07]
13.3 Nausea	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
13.3.1 All oral vs vaginal and continuous oral	1	1481	Risk Ratio (M-H, Random, 95% CI)	0.84 [0.65, 1.09]
13.3.2 All oral vs single vaginal	1	1478	Risk Ratio (M-H, Random, 95% CI)	0.94 [0.72, 1.24]
13.3.3 Vaginal and continuous oral vs single vaginal	1	1479	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.87, 1.45]
13.4 Vomiting	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
13.4.1 All oral vs vaginal and continuous oral	1	1481	Risk Ratio (M-H, Random, 95% CI)	0.80 [0.49, 1.30]
13.4.2 All oral vs single vaginal	1	1478	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.53, 1.43]
13.4.3 Vaginal and continuous oral vs single vaginal	1	1479	Risk Ratio (M-H, Random, 95% CI)	1.09 [0.68, 1.74]
13.5 Ongoing pregnancy at follow-up	2		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
13.5.1 All oral vs vaginal and continuous oral	2		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
13.5.2 All oral vs single vaginal	2		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
13.5.3 Vaginal and continuous oral vs single vaginal	2		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 13.1. Comparison 13: Combined regimen mifepristone/prostaglandin:single vs continuous prostaglandin, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
13.1.1 All oral vs vaginal and continuous oral						
Tang 2002	5	50	3	50	7.6%	1.67 [0.42 , 6.60]
von Hertzen 2003	57	740	39	741	92.4%	1.46 [0.99 , 2.17]
Subtotal (95% CI)		**790**		**791**	**100.0%**	**1.48 [1.01 , 2.16]**
Total events:	62		42			
Heterogeneity: Tau² = 0.00; Chi² = 0.03, df = 1 (P = 0.86); I² = 0%						
Test for overall effect: Z = 2.02 (P = 0.04)						
13.1.2 All oral vs single vaginal						
Tang 2002	5	50	4	50	8.0%	1.25 [0.36 , 4.38]
von Hertzen 2003	57	740	48	738	92.0%	1.18 [0.82 , 1.71]
Subtotal (95% CI)		**790**		**788**	**100.0%**	**1.19 [0.83 , 1.70]**
Total events:	62		52			
Heterogeneity: Tau² = 0.00; Chi² = 0.01, df = 1 (P = 0.94); I² = 0%						
Test for overall effect: Z = 0.96 (P = 0.34)						
13.1.3 Vaginal and continuous oral vs single vaginal						
Tang 2002	3	50	4	50	7.5%	0.75 [0.18 , 3.18]
von Hertzen 2003	39	741	48	738	92.5%	0.81 [0.54 , 1.22]
Subtotal (95% CI)		**791**		**788**	**100.0%**	**0.80 [0.54 , 1.19]**
Total events:	42		52			
Heterogeneity: Tau² = 0.00; Chi² = 0.01, df = 1 (P = 0.92); I² = 0%						
Test for overall effect: Z = 1.08 (P = 0.28)						
13.1.4 All oral vs vaginal & continuous oral ≤ 49 days						
von Hertzen 2003	15	236	13	240	100.0%	1.17 [0.57 , 2.41]
Subtotal (95% CI)		**236**		**240**	**100.0%**	**1.17 [0.57 , 2.41]**
Total events:	15		13			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.43 (P = 0.66)						
13.1.5 All oral vs vaginal and continuous oral > 49 days						
von Hertzen 2003	42	504	26	500	100.0%	1.60 [1.00 , 2.57]
Subtotal (95% CI)		**504**		**500**	**100.0%**	**1.60 [1.00 , 2.57]**
Total events:	42		26			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.95 (P = 0.05)						
13.1.6 All oral vs single vaginal ≤ 49 days						
von Hertzen 2003	15	236	11	223	100.0%	1.29 [0.60 , 2.74]
Subtotal (95% CI)		**236**		**223**	**100.0%**	**1.29 [0.60 , 2.74]**
Total events:	15		11			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.66 (P = 0.51)						
13.1.7 All oral vs single vaginal > 49 days						
von Hertzen 2003	42	504	38	510	100.0%	1.12 [0.73 , 1.70]
Subtotal (95% CI)		**504**		**510**	**100.0%**	**1.12 [0.73 , 1.70]**
Total events:	42		38			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.52 (P = 0.60)						
13.1.8 Vaginal and continuous oral vs single vaginal ≤ 49days						
von Hertzen 2003	13	240	11	223	100.0%	1.10 [0.50 , 2.40]

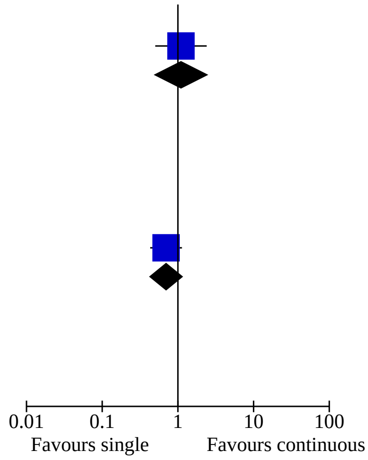
Risk Ratio M-H, Random, 95% CI



Analysis 13.1. (Continued)

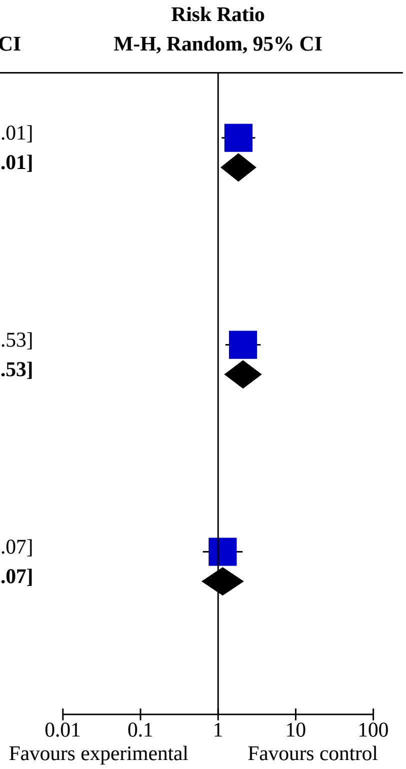
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
13.1.8 Vaginal and continuous oral vs single vaginal ≤ 49days						
von Hertzen 2003	13	240	11	223	100.0%	1.10 [0.50 , 2.40]
Subtotal (95% CI)		**240**		**223**	**100.0%**	**1.10 [0.50 , 2.40]**
Total events:	13		11			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.23 (P = 0.81)						
13.1.9 Vaginal and continuous oral vs single vaginal > 49 days						
von Hertzen 2003	26	500	38	510	100.0%	0.70 [0.43 , 1.13]
Subtotal (95% CI)		**500**		**510**	**100.0%**	**0.70 [0.43 , 1.13]**
Total events:	26		38			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.46 (P = 0.14)						

Test for subgroup differences: Chi² = 10.90, df = 8 (P = 0.21), I² = 26.6%



Analysis 13.2. Comparison 13: Combined regimen mifepristone/ prostaglandin:single vs continuous prostaglandin, Outcome 2: Diarrhoea

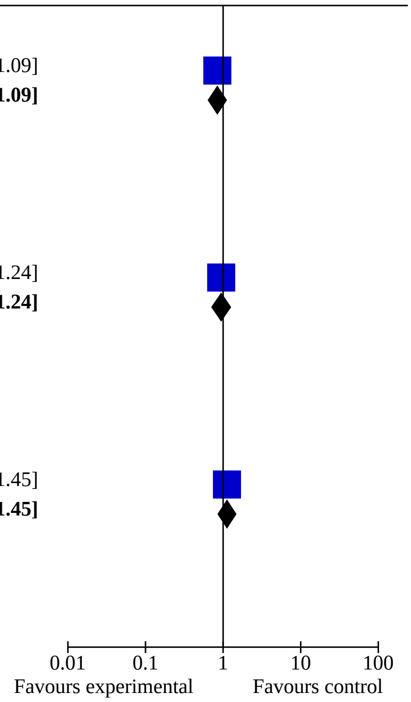
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
13.2.1 All oral vs vaginal and continuous oral						
von Hertzen 2003	42	740	23	741	100.0%	1.83 [1.11 , 3.01]
Subtotal (95% CI)		**740**		**741**	**100.0%**	**1.83 [1.11 , 3.01]**
Total events:	42		23			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.37 (P = 0.02)						
13.2.2 All oral vs single vaginal						
von Hertzen 2003	42	740	20	738	100.0%	2.09 [1.24 , 3.53]
Subtotal (95% CI)		**740**		**738**	**100.0%**	**2.09 [1.24 , 3.53]**
Total events:	42		20			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.77 (P = 0.006)						
13.2.3 Vaginal and continuous oral vs single vaginal						
von Hertzen 2003	23	741	20	738	100.0%	1.15 [0.63 , 2.07]
Subtotal (95% CI)		**741**		**738**	**100.0%**	**1.15 [0.63 , 2.07]**
Total events:	23		20			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.45 (P = 0.65)						



Analysis 13.3. Comparison 13: Combined regimen mifepristone/prostaglandin:single vs continuous prostaglandin, Outcome 3: Nausea

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
13.3.1 All oral vs vaginal and continuous oral						
von Hertzen 2003	90	740	107	741	100.0%	0.84 [0.65 , 1.09]
Subtotal (95% CI)		**740**		**741**	**100.0%**	**0.84 [0.65 , 1.09]**
Total events:	90		107			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.29 (P = 0.20)						
13.3.2 All oral vs single vaginal						
von Hertzen 2003	90	740	95	738	100.0%	0.94 [0.72 , 1.24]
Subtotal (95% CI)		**740**		**738**	**100.0%**	**0.94 [0.72 , 1.24]**
Total events:	90		95			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.41 (P = 0.68)						
13.3.3 Vaginal and continuous oral vs single vaginal						
von Hertzen 2003	107	741	95	738	100.0%	1.12 [0.87 , 1.45]
Subtotal (95% CI)		**741**		**738**	**100.0%**	**1.12 [0.87 , 1.45]**
Total events:	107		95			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.88 (P = 0.38)						

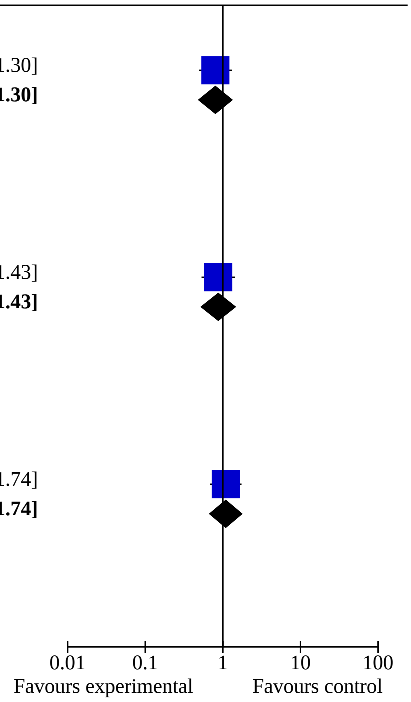
Test for subgroup differences: Chi² = 2.39, df = 2 (P = 0.30), I² = 16.2%



Analysis 13.4. Comparison 13: Combined regimen mifepristone/prostaglandin:single vs continuous prostaglandin, Outcome 4: Vomiting

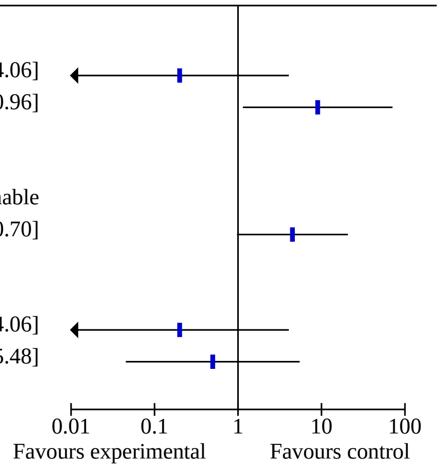
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
13.4.1 All oral vs vaginal and continuous oral						
von Hertzen 2003	28	740	35	741	100.0%	0.80 [0.49 , 1.30]
Subtotal (95% CI)		**740**		**741**	**100.0%**	**0.80 [0.49 , 1.30]**
Total events:	28		35			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.89 (P = 0.37)						
13.4.2 All oral vs single vaginal						
von Hertzen 2003	28	740	32	738	100.0%	0.87 [0.53 , 1.43]
Subtotal (95% CI)		**740**		**738**	**100.0%**	**0.87 [0.53 , 1.43]**
Total events:	28		32			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.54 (P = 0.59)						
13.4.3 Vaginal and continuous oral vs single vaginal						
von Hertzen 2003	35	741	32	738	100.0%	1.09 [0.68 , 1.74]
Subtotal (95% CI)		**741**		**738**	**100.0%**	**1.09 [0.68 , 1.74]**
Total events:	35		32			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.36 (P = 0.72)						

Test for subgroup differences: Chi² = 0.85, df = 2 (P = 0.65), I² = 0%



Analysis 13.5. Comparison 13: Combined regimen mifepristone/prostaglandin:single vs continuous prostaglandin, Outcome 5: Ongoing pregnancy at follow-up

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
13.5.1 All oral vs vaginal and continuous oral					
Tang 2002	0	50	2	50	0.20 [0.01 , 4.06]
von Hertzen 2003	9	740	1	741	9.01 [1.14 , 70.96]
13.5.2 All oral vs single vaginal					
Tang 2002	0	50	0	50	Not estimable
von Hertzen 2003	9	740	2	738	4.49 [0.97 , 20.70]
13.5.3 Vaginal and continuous oral vs single vaginal					
Tang 2002	0	50	2	50	0.20 [0.01 , 4.06]
von Hertzen 2003	1	741	2	738	0.50 [0.05 , 5.48]



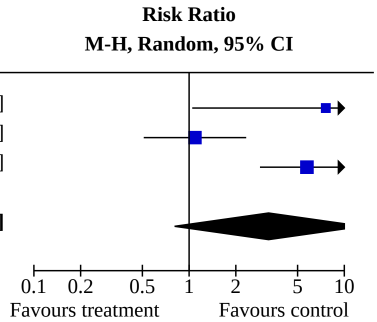
Comparison 14. Mifepristone alone vs combined regimen mifepristone/prostaglandin

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
14.1 Failure to achieve complete abortion	3	273	Risk Ratio (M-H, Random, 95% CI)	3.25 [0.81, 13.09]

Analysis 14.1. Comparison 14: Mifepristone alone vs combined regimen mifepristone/prostaglandin, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Cameron 1986	8	20	1	19	23.0%	7.60 [1.05 , 55.14]
Swahn 1989	6	14	11	28	38.1%	1.09 [0.51 , 2.33]
Zheng 1989	45	95	8	97	38.9%	5.74 [2.86 , 11.53]
Total (95% CI)		**129**		**144**	**100.0%**	**3.25 [0.81 , 13.09]**
Total events:	59		20			

Heterogeneity: Tau² = 1.17; Chi² = 12.09, df = 2 (P = 0.002); I² = 83%
Test for overall effect: Z = 1.66 (P = 0.10)
Test for subgroup differences: Not applicable



Comparison 15. Prostaglandin alone vs combined regimen (all)

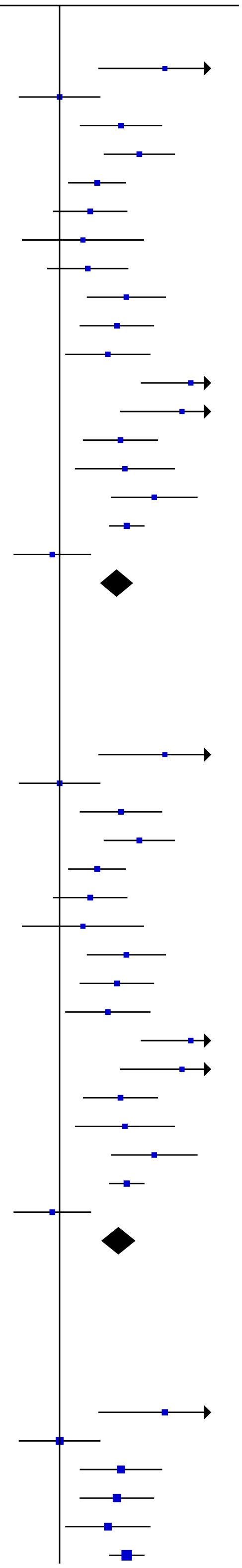
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
15.1 Failure to achieve complete abortion	18		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
15.1.1 All	18	3471	Risk Ratio (M-H, Random, 95% CI)	2.39 [1.89, 3.02]
15.1.2 All without Jain 1999	17	3321	Risk Ratio (M-H, Random, 95% CI)	2.45 [1.92, 3.14]
15.1.3 With letrozole combined	6	1008	Risk Ratio (M-H, Random, 95% CI)	2.29 [1.62, 3.24]
15.1.4 ≤ 49 days	4	785	Risk Ratio (M-H, Random, 95% CI)	4.65 [2.65, 8.16]
15.1.5 > 49 days	4	447	Risk Ratio (M-H, Random, 95% CI)	2.33 [1.48, 3.65]
15.1.6 With methotrexate combined regimen	3	333	Risk Ratio (M-H, Random, 95% CI)	1.97 [0.74, 5.26]
15.1.7 With estradiol valerate	1	100	Risk Ratio (M-H, Random, 95% CI)	4.25 [2.19, 8.25]
15.2 Time until passing of conceptus	6	988	Std. Mean Difference (IV, Random, 95% CI)	0.43 [0.05, 0.80]
15.3 Surgical evacuation	7	1307	Risk Ratio (M-H, Random, 95% CI)	2.90 [2.26, 3.73]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
15.4 Additional uterotonics used	3	592	Risk Ratio (M-H, Random, 95% CI)	2.05 [1.26, 3.34]
15.5 Ongoing pregnancy	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
15.5.1 All	5	1353	Risk Ratio (M-H, Random, 95% CI)	3.63 [1.18, 11.16]
15.5.2 ≤ 49 days	3	632	Risk Ratio (M-H, Random, 95% CI)	10.12 [3.07, 33.44]
15.5.3 > 49 days	3	358	Risk Ratio (M-H, Random, 95% CI)	3.33 [0.86, 12.89]
15.6 Side effects	13		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
15.6.1 Diarrhoea	11	2342	Risk Ratio (M-H, Random, 95% CI)	1.26 [1.11, 1.43]
15.6.2 Nausea	12	2722	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.74, 1.10]
15.6.3 Vomiting	9	2029	Risk Ratio (M-H, Random, 95% CI)	0.87 [0.66, 1.15]
15.6.4 Pelvic infection	1	300	Risk Ratio (M-H, Random, 95% CI)	0.99 [0.14, 6.91]
15.6.5 Abdominal pain	6	1827	Risk Ratio (M-H, Random, 95% CI)	1.11 [0.90, 1.37]
15.7 Women's dissatisfaction with the procedure	5	1485	Risk Ratio (M-H, Random, 95% CI)	1.65 [0.75, 3.64]
15.8 Blood transfusion	1	300	Risk Ratio (M-H, Random, 95% CI)	0.33 [0.03, 3.13]
15.9 Days of bleeding	2	420	Mean Difference (IV, Random, 95% CI)	-1.13 [-1.69, -0.57]
15.10 Haemoglobin level	1	128	Mean Difference (IV, Random, 95% CI)	-0.41 [-0.75, -0.07]

Analysis 15.1. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 1: Failure to achieve complete abortion

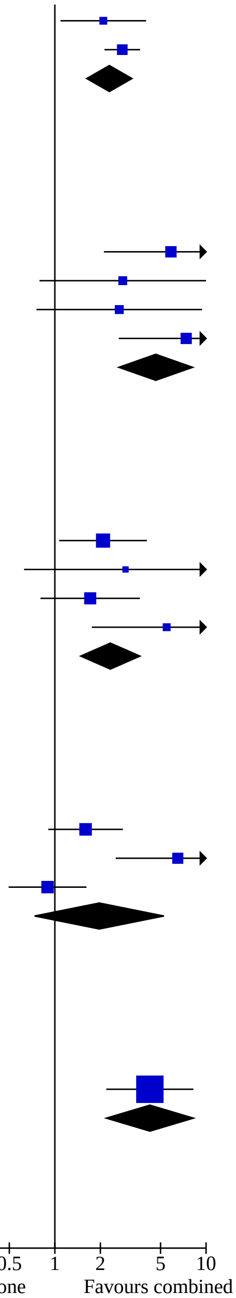
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
15.1.1 All						
Abbasalizadeh 2018	20	64	4	64	3.4%	5.00 [1.81 , 13.81]
Allameh 2020	15	63	15	63	5.7%	1.00 [0.54 , 1.87]
Behroozi Lak 2018	23	39	9	39	5.6%	2.56 [1.36 , 4.80]
Blum 2012	51	221	15	220	6.3%	3.38 [1.96 , 5.83]
Cheng 1994	36	76	20	75	7.1%	1.78 [1.14 , 2.77]
Creinin 1994	17	30	11	31	6.1%	1.60 [0.90 , 2.82]
Fekih 2010	10	126	7	126	3.8%	1.43 [0.56 , 3.63]
Jain 1999	20	75	13	75	5.7%	1.54 [0.83 , 2.86]
Jain 2002	35	125	12	119	5.8%	2.78 [1.52 , 5.09]
Javadi 2015	24	35	10	35	6.1%	2.40 [1.36 , 4.24]
Lee 2011	23	84	11	84	5.5%	2.09 [1.09 , 4.01]
Ngoc 2011	51	198	7	202	4.7%	7.43 [3.46 , 15.98]
Ozeren 1999	26	36	4	36	3.7%	6.50 [2.52 , 16.74]
Schreiber 2018	36	151	14	149	6.0%	2.54 [1.43 , 4.51]
Sinha 2018	19	46	7	46	4.7%	2.71 [1.26 , 5.83]
Teimoori 2019	34	50	8	50	5.4%	4.25 [2.19 , 8.25]
Torky 2018	134	219	48	219	8.4%	2.79 [2.13 , 3.66]
Vahid Roudsari 2010	17	100	19	100	5.9%	0.89 [0.49 , 1.62]
Subtotal (95% CI)		**1738**		**1733**	**100.0%**	**2.39 [1.89 , 3.02]**
Total events:	591		234			
Heterogeneity: Tau² = 0.15; Chi² = 46.65, df = 17 (P = 0.0001); I² = 64%						
Test for overall effect: Z = 7.24 (P < 0.00001)						
15.1.2 All without Jain 1999						
Abbasalizadeh 2018	20	64	4	64	3.7%	5.00 [1.81 , 13.81]
Allameh 2020	15	63	15	63	6.0%	1.00 [0.54 , 1.87]
Behroozi Lak 2018	23	39	9	39	6.0%	2.56 [1.36 , 4.80]
Blum 2012	51	221	15	220	6.6%	3.38 [1.96 , 5.83]
Cheng 1994	36	76	20	75	7.5%	1.78 [1.14 , 2.77]
Creinin 1994	17	30	11	31	6.5%	1.60 [0.90 , 2.82]
Fekih 2010	10	126	7	126	4.1%	1.43 [0.56 , 3.63]
Jain 2002	35	125	12	119	6.2%	2.78 [1.52 , 5.09]
Javadi 2015	24	35	10	35	6.4%	2.40 [1.36 , 4.24]
Lee 2011	23	84	11	84	5.8%	2.09 [1.09 , 4.01]
Ngoc 2011	51	198	7	202	5.0%	7.43 [3.46 , 15.98]
Ozeren 1999	26	36	4	36	4.0%	6.50 [2.52 , 16.74]
Schreiber 2018	36	151	14	149	6.4%	2.54 [1.43 , 4.51]
Sinha 2018	19	46	7	46	5.0%	2.71 [1.26 , 5.83]
Teimoori 2019	34	50	8	50	5.7%	4.25 [2.19 , 8.25]
Torky 2018	134	219	48	219	8.8%	2.79 [2.13 , 3.66]
Vahid Roudsari 2010	17	100	19	100	6.3%	0.89 [0.49 , 1.62]
Subtotal (95% CI)		**1663**		**1658**	**100.0%**	**2.45 [1.92 , 3.14]**
Total events:	571		221			
Heterogeneity: Tau² = 0.16; Chi² = 44.70, df = 16 (P = 0.0002); I² = 64%						
Test for overall effect: Z = 7.20 (P < 0.00001)						
15.1.3 With letrozole combined						
Abbasalizadeh 2018	20	64	4	64	8.5%	5.00 [1.81 , 13.81]
Allameh 2020	15	63	15	63	15.8%	1.00 [0.54 , 1.87]
Behroozi Lak 2018	23	39	9	39	15.7%	2.56 [1.36 , 4.80]
Javadi 2015	24	35	10	35	17.4%	2.40 [1.36 , 4.24]
Lee 2011	23	84	11	84	15.1%	2.09 [1.09 , 4.01]
Torky 2018	134	219	48	219	27.4%	2.79 [2.13 , 3.66]

Risk Ratio
M-H, Random, 95% CI



Analysis 15.1. (Continued)

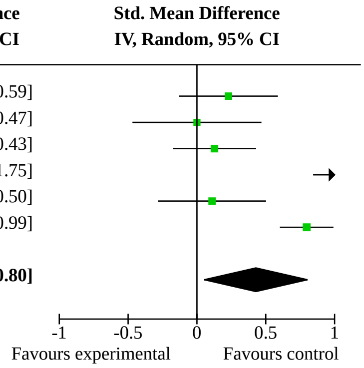
Lee 2011	23	84	11	84	15.1%	2.09 [1.09 , 4.01]
Torky 2018	134	219	48	219	27.4%	2.79 [2.13 , 3.66]
Subtotal (95% CI)		**504**		**504**	**100.0%**	**2.29 [1.62 , 3.24]**
Total events:	239		97			
Heterogeneity: Tau² = 0.09; Chi² = 10.95, df = 5 (P = 0.05); I² = 54%						
Test for overall effect: Z = 4.72 (P < 0.00001)						
15.1.4 ≤ 49 days						
Blum 2012	26	121	4	109	30.4%	5.86 [2.11 , 16.24]
Jain 2002	9	80	3	75	19.7%	2.81 [0.79 , 10.00]
Lee 2011	8	45	3	45	19.9%	2.67 [0.76 , 9.41]
Ngoc 2011	27	148	4	162	30.0%	7.39 [2.65 , 20.62]
Subtotal (95% CI)		**394**		**391**	**100.0%**	**4.65 [2.65 , 8.16]**
Total events:	70		14			
Heterogeneity: Tau² = 0.00; Chi² = 2.37, df = 3 (P = 0.50); I² = 0%						
Test for overall effect: Z = 5.35 (P < 0.00001)						
15.1.5 > 49 days						
Blum 2012	22	97	11	101	42.6%	2.08 [1.07 , 4.06]
Jain 2002	6	45	2	44	8.4%	2.93 [0.63 , 13.76]
Lee 2011	13	37	8	39	33.7%	1.71 [0.80 , 3.65]
Ngoc 2011	19	45	3	39	15.3%	5.49 [1.76 , 17.16]
Subtotal (95% CI)		**224**		**223**	**100.0%**	**2.33 [1.48 , 3.65]**
Total events:	60		24			
Heterogeneity: Tau² = 0.01; Chi² = 3.11, df = 3 (P = 0.38); I² = 3%						
Test for overall effect: Z = 3.67 (P = 0.0002)						
15.1.6 With methotrexate combined regimen						
Creinin 1994	17	30	11	31	35.5%	1.60 [0.90 , 2.82]
Ozeren 1999	26	36	4	36	29.4%	6.50 [2.52 , 16.74]
Vahid Roudsari 2010	17	100	19	100	35.1%	0.89 [0.49 , 1.62]
Subtotal (95% CI)		**166**		**167**	**100.0%**	**1.97 [0.74 , 5.26]**
Total events:	60		34			
Heterogeneity: Tau² = 0.63; Chi² = 12.52, df = 2 (P = 0.002); I² = 84%						
Test for overall effect: Z = 1.35 (P = 0.18)						
15.1.7 With estradiol valerate						
Teimoori 2019	34	50	8	50	100.0%	4.25 [2.19 , 8.25]
Subtotal (95% CI)		**50**		**50**	**100.0%**	**4.25 [2.19 , 8.25]**
Total events:	34		8			
Heterogeneity: Not applicable						
Test for overall effect: Z = 4.28 (P < 0.0001)						



Analysis 15.2. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 2: Time until passing of conceptus

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Std. Mean Difference IV, Random, 95% CI
Allameh 2020	8.2	3.84	60	7.35	3.54	60	16.8%	0.23 [-0.13 , 0.59]
Javadi 2015	9	1.7	35	9	2.4	35	15.1%	0.00 [-0.47 , 0.47]
Lee 2011	334.17	704.07	84	254.29	534.27	84	17.6%	0.13 [-0.18 , 0.43]
Sinha 2018	8.03	2.77	46	4.74	2.24	46	15.4%	1.30 [0.84 , 1.75]
Teimoori 2019	5	10	50	4	8	50	16.3%	0.11 [-0.28 , 0.50]
Torky 2018	3.05	1.55	219	2.09	0.7	219	18.8%	0.80 [0.60 , 0.99]
Total (95% CI)			**494**			**494**	**100.0%**	**0.43 [0.05 , 0.80]**

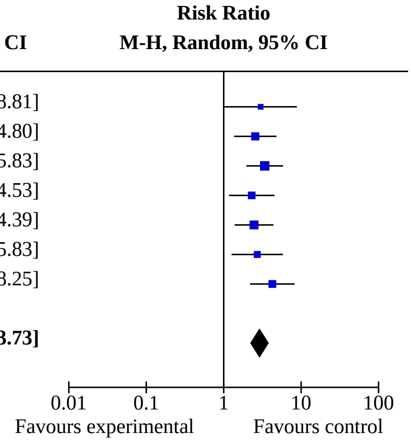
Heterogeneity: Tau² = 0.18; Chi² = 37.03, df = 5 (P < 0.00001); I² = 86%
Test for overall effect: Z = 2.24 (P = 0.03)
Test for subgroup differences: Not applicable



Analysis 15.3. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 3: Surgical evacuation

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Abbasalizadeh 2018	12	64	4	64	5.4%	3.00 [1.02 , 8.81]
Behroozi Lak 2018	23	39	9	39	15.8%	2.56 [1.36 , 4.80]
Blum 2012	51	221	15	220	21.2%	3.38 [1.96 , 5.83]
Lee 2011	23	84	10	84	13.7%	2.30 [1.17 , 4.53]
Schreiber 2018	35	151	14	149	18.9%	2.47 [1.39 , 4.39]
Sinha 2018	19	46	7	46	10.8%	2.71 [1.26 , 5.83]
Teimoori 2019	34	50	8	50	14.3%	4.25 [2.19 , 8.25]
Total (95% CI)		**655**		**652**	**100.0%**	**2.90 [2.26 , 3.73]**
Total events:	197		67			

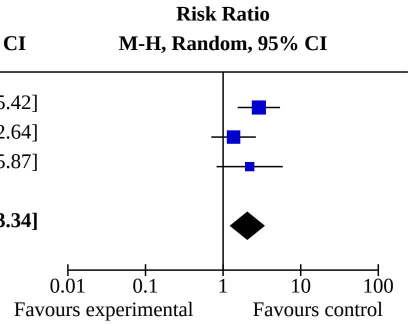
Heterogeneity: Tau² = 0.00; Chi² = 2.53, df = 6 (P = 0.87); I² = 0%
Test for overall effect: Z = 8.33 (P < 0.00001)
Test for subgroup differences: Not applicable



Analysis 15.4. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 4: Additional uterotonics used

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Ngoc 2011	34	198	12	202	41.1%	2.89 [1.54 , 5.42]
Sinha 2018	15	46	11	46	38.2%	1.36 [0.70 , 2.64]
Teimoori 2019	11	50	5	50	20.7%	2.20 [0.82 , 5.87]
Total (95% CI)		**294**		**298**	**100.0%**	**2.05 [1.26 , 3.34]**
Total events:	60		28			

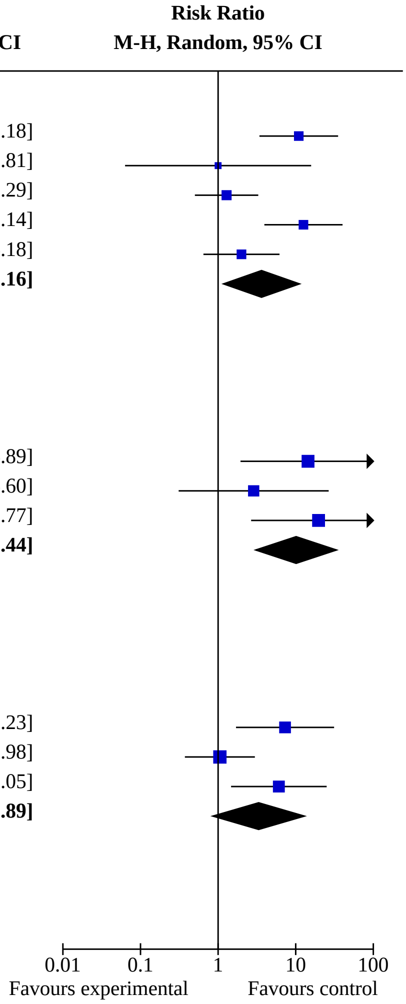
Heterogeneity: Tau² = 0.05; Chi² = 2.69, df = 2 (P = 0.26); I² = 26%
Test for overall effect: Z = 2.88 (P = 0.004)
Test for subgroup differences: Not applicable



Analysis 15.5. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 5: Ongoing pregnancy

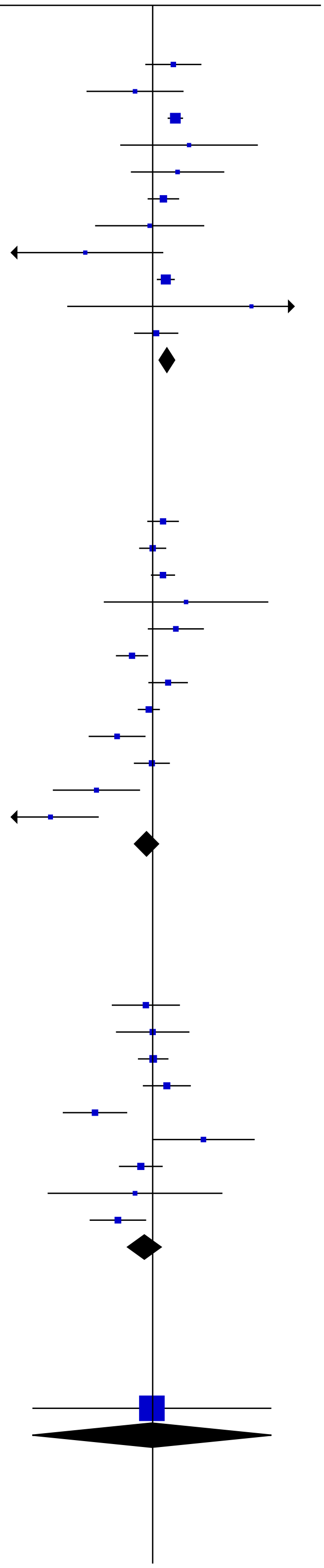
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
15.5.1 All						
Blum 2012	33	221	3	220	21.8%	10.95 [3.41 , 35.18]
Fekih 2010	1	126	1	126	10.5%	1.00 [0.06 , 15.81]
Lee 2011	9	84	7	84	23.7%	1.29 [0.50 , 3.29]
Ngoc 2011	37	198	3	202	21.9%	12.58 [3.94 , 40.14]
Sinha 2018	8	46	4	46	22.1%	2.00 [0.65 , 6.18]
Subtotal (95% CI)		**675**		**678**	**100.0%**	**3.63 [1.18 , 11.16]**
Total events:	88		18			
Heterogeneity: Tau² = 1.15; Chi² = 15.86, df = 4 (P = 0.003); I² = 75%						
Test for overall effect: Z = 2.25 (P = 0.02)						
15.5.2 ≤ 49 days						
Blum 2012	16	121	1	109	35.6%	14.41 [1.94 , 106.89]
Lee 2011	3	47	1	45	28.8%	2.87 [0.31 , 26.60]
Ngoc 2011	18	148	1	162	35.6%	19.70 [2.66 , 145.77]
Subtotal (95% CI)		**316**		**316**	**100.0%**	**10.12 [3.07 , 33.44]**
Total events:	37		3			
Heterogeneity: Tau² = 0.00; Chi² = 1.87, df = 2 (P = 0.39); I² = 0%						
Test for overall effect: Z = 3.80 (P = 0.0001)						
15.5.3 > 49 days						
Blum 2012	14	97	2	101	30.9%	7.29 [1.70 , 31.23]
Lee 2011	6	37	6	39	37.5%	1.05 [0.37 , 2.98]
Ngoc 2011	14	45	2	39	31.5%	6.07 [1.47 , 25.05]
Subtotal (95% CI)		**179**		**179**	**100.0%**	**3.33 [0.86 , 12.89]**
Total events:	34		10			
Heterogeneity: Tau² = 0.99; Chi² = 6.55, df = 2 (P = 0.04); I² = 69%						
Test for overall effect: Z = 1.74 (P = 0.08)						

Test for subgroup differences: Chi² = 2.00, df = 2 (P = 0.37), I² = 0%



Analysis 15.6. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 6: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
15.6.1 Diarrhoea						
Abbasalizadeh 2018	28	64	20	64	6.5%	1.40 [0.89 , 2.21]
Allameh 2020	9	63	12	63	2.4%	0.75 [0.34 , 1.65]
Blum 2012	186	221	128	220	31.0%	1.45 [1.28 , 1.64]
Creinin 1994	7	30	4	31	1.2%	1.81 [0.59 , 5.55]
Fekih 2010	15	126	10	126	2.6%	1.50 [0.70 , 3.21]
Jain 1999	50	75	42	75	15.7%	1.19 [0.92 , 1.54]
Jain 2002	9	125	9	119	1.9%	0.95 [0.39 , 2.32]
Lee 2011	3	84	9	84	1.0%	0.33 [0.09 , 1.19]
Ngoc 2011	142	198	117	202	27.9%	1.24 [1.07 , 1.43]
Ozeren 1999	2	36	0	36	0.2%	5.00 [0.25 , 100.63]
Schreiber 2018	44	151	41	149	9.6%	1.06 [0.74 , 1.52]
Subtotal (95% CI)		**1173**		**1169**	**100.0%**	**1.26 [1.11 , 1.43]**
Total events:	495		392			
Heterogeneity: Tau² = 0.01; Chi² = 13.47, df = 10 (P = 0.20); I² = 26%						
Test for overall effect: Z = 3.60 (P = 0.0003)						
15.6.2 Nausea						
Abbasalizadeh 2018	45	64	38	64	10.5%	1.18 [0.92 , 1.53]
Allameh 2020	45	63	45	63	11.0%	1.00 [0.80 , 1.25]
Blum 2012	114	221	96	220	11.3%	1.18 [0.97 , 1.44]
Creinin 1994	5	30	3	31	1.8%	1.72 [0.45 , 6.58]
Fekih 2010	35	126	24	126	7.5%	1.46 [0.92 , 2.30]
Jain 2002	51	125	68	119	10.4%	0.71 [0.55 , 0.93]
Lee 2011	45	84	35	84	9.5%	1.29 [0.93 , 1.77]
Ngoc 2011	104	198	113	202	11.6%	0.94 [0.78 , 1.12]
Ozeren 1999	14	36	25	36	7.4%	0.56 [0.35 , 0.89]
Schreiber 2018	56	151	56	149	9.9%	0.99 [0.74 , 1.32]
Sinha 2018	8	46	20	46	4.7%	0.40 [0.20 , 0.81]
Torky 2018	7	219	37	219	4.2%	0.19 [0.09 , 0.42]
Subtotal (95% CI)		**1363**		**1359**	**100.0%**	**0.90 [0.74 , 1.10]**
Total events:	529		560			
Heterogeneity: Tau² = 0.08; Chi² = 47.14, df = 11 (P < 0.00001); I² = 77%						
Test for overall effect: Z = 1.01 (P = 0.31)						
15.6.3 Vomiting						
Abbasalizadeh 2018	17	64	19	64	10.8%	0.89 [0.51 , 1.56]
Allameh 2020	16	63	16	63	10.1%	1.00 [0.55 , 1.82]
Blum 2012	80	221	79	220	16.8%	1.01 [0.79 , 1.29]
Jain 1999	34	75	27	75	13.9%	1.26 [0.85 , 1.86]
Jain 2002	16	125	39	119	11.3%	0.39 [0.23 , 0.66]
Lee 2011	16	84	7	84	6.9%	2.29 [0.99 , 5.27]
Ngoc 2011	42	198	52	202	14.6%	0.82 [0.58 , 1.18]
Ozeren 1999	3	36	4	36	3.1%	0.75 [0.18 , 3.11]
Schreiber 2018	23	151	40	149	12.5%	0.57 [0.36 , 0.90]
Subtotal (95% CI)		**1017**		**1012**	**100.0%**	**0.87 [0.66 , 1.15]**
Total events:	247		283			
Heterogeneity: Tau² = 0.10; Chi² = 22.54, df = 8 (P = 0.004); I² = 65%						
Test for overall effect: Z = 0.98 (P = 0.33)						
15.6.4 Pelvic infection						
Schreiber 2018	2	151	2	149	100.0%	0.99 [0.14 , 6.91]
Subtotal (95% CI)		**151**		**149**	**100.0%**	**0.99 [0.14 , 6.91]**
Total events:	2		2			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.01 (P = 0.99)						
15.6.5 Abdominal pain						



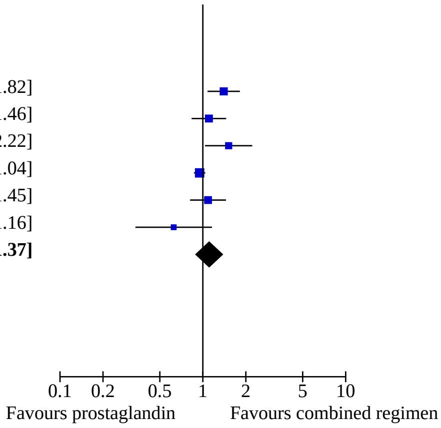
Analysis 15.6. (Continued)

Test for overall effect: Z = 0.01 (P = 0.99)

Study or Subgroup	Events	Total	Events	Total	Weight	Risk Ratio
15.6.5 Abdominal pain						
Abbasalizadeh 2018	49	64	35	64	18.4%	1.40 [1.08 , 1.82]
Blum 2012	72	221	65	220	17.7%	1.10 [0.83 , 1.46]
Fekih 2010	47	126	31	126	14.0%	1.52 [1.04 , 2.22]
Lee 2011	75	84	79	84	24.5%	0.95 [0.87 , 1.04]
Ngoc 2011	65	198	61	202	17.3%	1.09 [0.81 , 1.45]
Torky 2018	15	219	24	219	8.0%	0.63 [0.34 , 1.16]
Subtotal (95% CI)		**912**		**915**	**100.0%**	**1.11 [0.90 , 1.37]**
Total events:	323		295			

Heterogeneity: Tau² = 0.05; Chi² = 19.02, df = 5 (P = 0.002); I² = 74%

Test for overall effect: Z = 0.94 (P = 0.35)



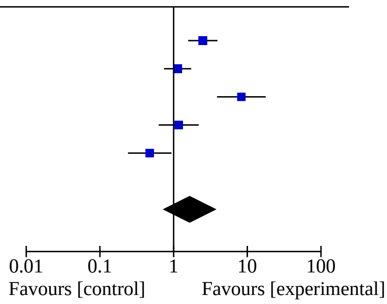
Analysis 15.7. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 7: Women's dissatisfaction with the procedure

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Blum 2012	55	221	22	220	21.0%	2.49 [1.57 , 3.93]
Fekih 2010	34	126	30	126	21.2%	1.13 [0.74 , 1.73]
Ngoc 2011	57	198	7	202	18.7%	8.31 [3.88 , 17.77]
Schreiber 2018	19	151	16	149	19.8%	1.17 [0.63 , 2.19]
Sinha 2018	9	46	19	46	19.3%	0.47 [0.24 , 0.93]
Total (95% CI)		**742**		**743**	**100.0%**	**1.65 [0.75 , 3.64]**
Total events:	174		94			

Heterogeneity: Tau² = 0.72; Chi² = 39.25, df = 4 (P < 0.00001); I² = 90%

Test for overall effect: Z = 1.24 (P = 0.21)

Test for subgroup differences: Not applicable



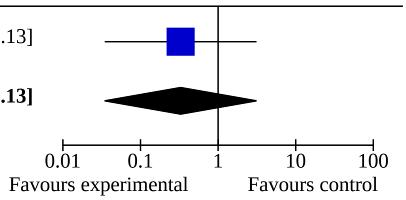
Analysis 15.8. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 8: Blood transfusion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Schreiber 2018	1	151	3	149	100.0%	0.33 [0.03 , 3.13]
Total (95% CI)		**151**		**149**	**100.0%**	**0.33 [0.03 , 3.13]**
Total events:	1		3			

Heterogeneity: Not applicable

Test for overall effect: Z = 0.97 (P = 0.33)

Test for subgroup differences: Not applicable

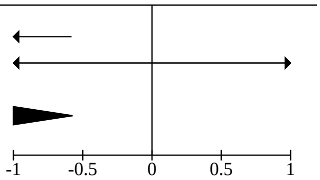


Analysis 15.9. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 9: Days of bleeding

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Fekih 2010	11.35	2.41	126	12.49	2.11	126	99.8%	-1.14 [-1.70 , -0.58]
Lee 2011	32.03	44.5	84	28.74	42.99	84	0.2%	3.29 [-9.94 , 16.52]
Total (95% CI)			**210**			**210**	**100.0%**	**-1.13 [-1.69 , -0.57]**

Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.43$, df = 1 (P = 0.51); $I^2 = 0\%$
Test for overall effect: Z = 3.97 (P < 0.0001)
Test for subgroup differences: Not applicable

Mean Difference IV, Random, 95% CI

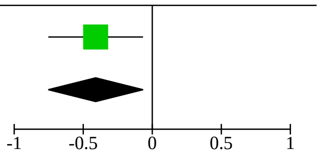


Analysis 15.10. Comparison 15: Prostaglandin alone vs combined regimen (all), Outcome 10: Haemoglobin level

Study or Subgroup	Treatment Mean	Treatment SD	Treatment Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Abbasalizadeh 2018	12.14	0.97	64	12.55	1.01	64	100.0%	-0.41 [-0.75 , -0.07]
Total (95% CI)			**64**			**64**	**100.0%**	**-0.41 [-0.75 , -0.07]**

Heterogeneity: Not applicable
Test for overall effect: Z = 2.34 (P = 0.02)
Test for subgroup differences: Not applicable

Mean Difference IV, Random, 95% CI



Comparison 16. Prostaglandin alone: route of administration

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
16.1 Failure to achieve complete abortion	10		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
16.1.1 Misoprostol oral vs vaginal	2	216	Risk Ratio (M-H, Random, 95% CI)	3.68 [1.56, 8.71]
16.1.2 Misoprostol sublingual vs vaginal	5	2705	Risk Ratio (M-H, Random, 95% CI)	0.69 [0.37, 1.28]
16.1.3 Misoprostol sublingual vs oral	1	109	Risk Ratio (M-H, Random, 95% CI)	0.24 [0.05, 1.06]
16.1.4 Misoprostol sublingual vs buccal	1	401	Risk Ratio (M-H, Random, 95% CI)	1.11 [0.71, 1.74]
16.1.5 Misoprostol repeat vs single	1	180	Risk Ratio (M-H, Random, 95% CI)	1.30 [0.78, 2.15]
16.1.6 Administration model of misoprostol	2	1164	Risk Ratio (M-H, Random, 95% CI)	2.09 [0.56, 7.81]
16.1.7 ≤ 49 days	1	248	Risk Ratio (M-H, Random, 95% CI)	1.00 [0.56, 1.76]
16.1.8 > 49 days	1	141	Risk Ratio (M-H, Random, 95% CI)	0.51 [0.18, 1.41]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
16.2 Blood transfusion	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.2.1 Misoprostol sublingual vs vaginal	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.3 Ongoing pregnancy	3		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
16.3.1 Misoprostol sublingual vs buccal	1	401	Risk Ratio (M-H, Random, 95% CI)	1.11 [0.50, 2.45]
16.3.2 Misoprostol sublingual vs vaginal	2	340	Risk Ratio (M-H, Random, 95% CI)	0.57 [0.21, 1.51]
16.3.3 > 49 days	1	141	Risk Ratio (M-H, Random, 95% CI)	0.25 [0.03, 2.21]
16.3.4 ≤ 49 days	1	248	Risk Ratio (M-H, Random, 95% CI)	0.16 [0.02, 1.26]
16.4 Additional uterotonics used	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.4.1 Misoprostol sublingual vs buccal	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.5 Surgical evacuation	8		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
16.5.1 Misoprostol sublingual vs buccal	1	401	Risk Ratio (M-H, Random, 95% CI)	1.49 [0.80, 2.79]
16.5.2 Misoprostol sublingual vs vaginal	3	540	Risk Ratio (M-H, Random, 95% CI)	0.47 [0.26, 0.85]
16.5.3 Misoprostol sublingual vs oral	2	173	Risk Ratio (M-H, Random, 95% CI)	2.21 [0.60, 8.10]
16.5.4 Misoprostol repeat vs single	1	180	Risk Ratio (M-H, Random, 95% CI)	1.30 [0.78, 2.15]
16.5.5 Misoprostol nurse vs doctor	1	1010	Risk Ratio (M-H, Random, 95% CI)	3.84 [2.16, 6.83]
16.6 Nausea	9		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
16.6.1 Misoprostol sublingual vs vaginal	4	2505	Risk Ratio (M-H, Random, 95% CI)	1.61 [0.78, 3.33]
16.6.2 Misoprostol sublingual vs buccal	1	401	Risk Ratio (M-H, Random, 95% CI)	1.15 [0.92, 1.44]

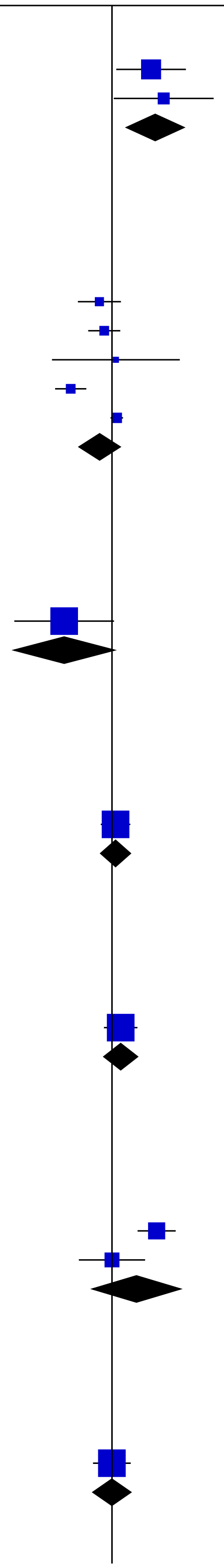
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
16.6.3 Misoprostol sublingual vs oral	1	109	Risk Ratio (M-H, Random, 95% CI)	1.89 [0.50, 7.19]
16.6.4 Misoprostol oral vs vaginal	2	216	Risk Ratio (M-H, Random, 95% CI)	1.21 [0.92, 1.61]
16.6.5 Misoprostol repeat vs single	1	180	Risk Ratio (M-H, Random, 95% CI)	1.46 [1.00, 2.14]
16.6.6 Administration model of misoprostol	2	1164	Risk Ratio (M-H, Random, 95% CI)	1.16 [1.02, 1.32]
16.7 Vomiting	7		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
16.7.1 Misoprostol sublingual vs vaginal	4	2605	Risk Ratio (M-H, Random, 95% CI)	1.83 [1.22, 2.73]
16.7.2 Misoprostol sublingual vs buccal	1	401	Risk Ratio (M-H, Random, 95% CI)	1.45 [1.00, 2.11]
16.7.3 Misoprostol sublingual vs oral	1	109	Risk Ratio (M-H, Random, 95% CI)	2.21 [0.60, 8.10]
16.7.4 Misoprostol repeat vs single	2	296	Risk Ratio (M-H, Random, 95% CI)	1.83 [0.31, 10.82]
16.7.5 Misoprostol nurse vs doctor	1	1010	Risk Ratio (M-H, Random, 95% CI)	1.35 [1.08, 1.68]
16.8 Diarrhoea	10		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
16.8.1 Misoprostol sublingual vs vaginal	5	2725	Risk Ratio (M-H, Random, 95% CI)	1.73 [1.41, 2.12]
16.8.2 Misoprostol sublingual vs buccal	1	401	Risk Ratio (M-H, Random, 95% CI)	1.13 [0.99, 1.29]
16.8.3 Misoprostol sublingual vs oral	1	109	Risk Ratio (M-H, Random, 95% CI)	0.71 [0.17, 3.02]
16.8.4 Misoprostol oral vs vaginal	2	216	Risk Ratio (M-H, Random, 95% CI)	1.42 [0.60, 3.37]
16.8.5 Misoprostol repeat vs single	1	180	Risk Ratio (M-H, Random, 95% CI)	1.12 [0.71, 1.76]
16.8.6 Administration model of misoprostol	2	1164	Risk Ratio (M-H, Random, 95% CI)	1.22 [1.00, 1.49]
16.9 Abdominal pain	6		Risk Ratio (M-H, Random, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
16.9.1 Misoprostol sublingual vs vaginal	3	459	Risk Ratio (M-H, Random, 95% CI)	1.38 [0.49, 3.86]
16.9.2 Misoprostol sublingual vs oral	1	109	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.96, 1.09]
16.9.3 Misoprostol oral vs vaginal	2	216	Risk Ratio (M-H, Random, 95% CI)	1.24 [0.48, 3.21]
16.9.4 Administration model of misoprostol	2	1164	Risk Ratio (M-H, Random, 95% CI)	1.28 [0.55, 2.97]
16.10 Women's dissatisfaction with the procedure	5		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.10.1 Misoprostol sublingual vs buccal	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.10.2 Misoprostol sublingual vs vaginal	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.10.3 Misoprostol sublingual vs oral	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.10.4 Misoprostol oral vs vaginal	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.10.5 Misoprostol nurse vs doctor	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
16.11 Days of bleeding	3		Std. Mean Difference (IV, Random, 95% CI)	Totals not selected
16.11.1 Misoprostol sublingual vs vaginal	1		Std. Mean Difference (IV, Random, 95% CI)	Totals not selected
16.11.2 Administration model of misoprostol	2		Std. Mean Difference (IV, Random, 95% CI)	Totals not selected
16.12 Time until passing of conceptus	4		Std. Mean Difference (IV, Random, 95% CI)	Subtotals only
16.12.1 Misoprostol sublingual vs vaginal	2	320	Std. Mean Difference (IV, Random, 95% CI)	-1.10 [-1.34, -0.87]
16.12.2 Misoprostol sublingual vs oral	1	64	Std. Mean Difference (IV, Random, 95% CI)	0.15 [-0.34, 0.64]
16.12.3 Misoprostol oral vs vaginal	1	100	Std. Mean Difference (IV, Random, 95% CI)	0.71 [0.31, 1.12]
16.13 Haemoglobin level	2		Std. Mean Difference (IV, Random, 95% CI)	Totals not selected

Analysis 16.1. Comparison 16: Prostaglandin alone: route of administration, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.1.1 Misoprostol oral vs vaginal						
Marwah 2016	13	50	4	50	67.3%	3.25 [1.14 , 9.29]
Souizi 2020	8	53	2	63	32.7%	4.75 [1.05 , 21.43]
Subtotal (95% CI)		**103**		**113**	**100.0%**	**3.68 [1.56 , 8.71]**
Total events:	21		6			
Heterogeneity: Tau² = 0.00; Chi² = 0.17, df = 1 (P = 0.68); I² = 0%						
Test for overall effect: Z = 2.97 (P = 0.003)						
16.1.2 Misoprostol sublingual vs vaginal						
Abdelshafy 2019	13	100	19	100	20.6%	0.68 [0.36 , 1.31]
Sonsanoh 2014	19	60	24	60	22.9%	0.79 [0.49 , 1.28]
Souizi 2020	2	56	2	63	7.4%	1.13 [0.16 , 7.72]
Tanha 2010	17	110	59	110	23.1%	0.29 [0.18 , 0.46]
von Hertzen 2007	191	1021	166	1025	26.0%	1.16 [0.96 , 1.40]
Subtotal (95% CI)		**1347**		**1358**	**100.0%**	**0.69 [0.37 , 1.28]**
Total events:	242		270			
Heterogeneity: Tau² = 0.37; Chi² = 30.23, df = 4 (P < 0.00001); I² = 87%						
Test for overall effect: Z = 1.19 (P = 0.24)						
16.1.3 Misoprostol sublingual vs oral						
Souizi 2020	2	56	8	53	100.0%	0.24 [0.05 , 1.06]
Subtotal (95% CI)		**56**		**53**	**100.0%**	**0.24 [0.05 , 1.06]**
Total events:	2		8			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.88 (P = 0.06)						
16.1.4 Misoprostol sublingual vs buccal						
Sheldon 2019	34	199	31	202	100.0%	1.11 [0.71 , 1.74]
Subtotal (95% CI)		**199**		**202**	**100.0%**	**1.11 [0.71 , 1.74]**
Total events:	34		31			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.47 (P = 0.64)						
16.1.5 Misoprostol repeat vs single						
Mizrachi 2017	26	90	20	90	100.0%	1.30 [0.78 , 2.15]
Subtotal (95% CI)		**90**		**90**	**100.0%**	**1.30 [0.78 , 2.15]**
Total events:	26		20			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.02 (P = 0.31)						
16.1.6 Administration model of misoprostol						
Klingberg Allvin 2015	54	506	14	504	54.8%	3.84 [2.16 , 6.83]
Ng 2015	7	77	7	77	45.2%	1.00 [0.37 , 2.72]
Subtotal (95% CI)		**583**		**581**	**100.0%**	**2.09 [0.56 , 7.81]**
Total events:	61		21			
Heterogeneity: Tau² = 0.74; Chi² = 5.28, df = 1 (P = 0.02); I² = 81%						
Test for overall effect: Z = 1.10 (P = 0.27)						
16.1.7 ≤ 49 days						
Sheldon 2019	19	118	21	130	100.0%	1.00 [0.56 , 1.76]
Subtotal (95% CI)		**118**		**130**	**100.0%**	**1.00 [0.56 , 1.76]**
Total events:	19		21			
Heterogeneity: Not applicable						

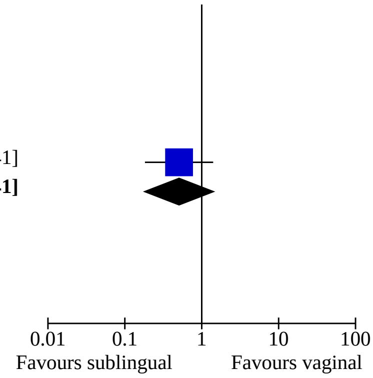
Risk Ratio
M-H, Random, 95% CI



Analysis 16.1. (Continued)

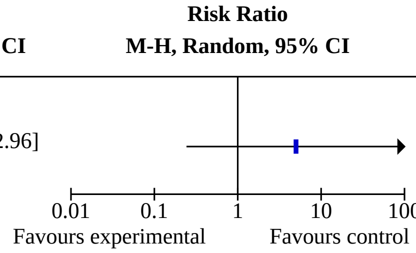
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Total events:	19		21			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.01 (P = 0.99)						
16.1.8 > 49 days						
Sheldon 2019	5	70	10	71	100.0%	0.51 [0.18 , 1.41]
Subtotal (95% CI)		**70**		**71**	**100.0%**	**0.51 [0.18 , 1.41]**
Total events:	5		10			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.30 (P = 0.19)						

Test for subgroup differences: Chi² = 17.43, df = 7 (P = 0.01), I² = 59.8%



Analysis 16.2. Comparison 16: Prostaglandin alone: route of administration, Outcome 2: Blood transfusion

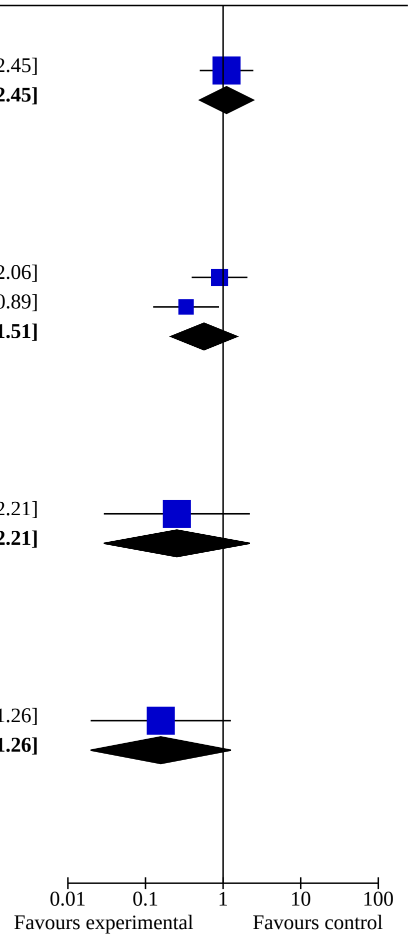
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
16.2.1 Misoprostol sublingual vs vaginal					
Tanha 2010	2	110	0	110	5.00 [0.24 , 102.96]



Analysis 16.3. Comparison 16: Prostaglandin alone: route of administration, Outcome 3: Ongoing pregnancy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.3.1 Misoprostol sublingual vs buccal						
Sheldon 2019	12	199	11	202	100.0%	1.11 [0.50 , 2.45]
Subtotal (95% CI)		**199**		**202**	**100.0%**	**1.11 [0.50 , 2.45]**
Total events:	12		11			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.25 (P = 0.80)						
16.3.2 Misoprostol sublingual vs vaginal						
Sonsanoh 2014	9	60	10	60	53.5%	0.90 [0.39 , 2.06]
Tanha 2010	5	110	15	110	46.5%	0.33 [0.13 , 0.89]
Subtotal (95% CI)		**170**		**170**	**100.0%**	**0.57 [0.21 , 1.51]**
Total events:	14		25			
Heterogeneity: Tau² = 0.29; Chi² = 2.35, df = 1 (P = 0.13); I² = 57%						
Test for overall effect: Z = 1.14 (P = 0.26)						
16.3.3 > 49 days						
Sheldon 2019	1	70	4	71	100.0%	0.25 [0.03 , 2.21]
Subtotal (95% CI)		**70**		**71**	**100.0%**	**0.25 [0.03 , 2.21]**
Total events:	1		4			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.24 (P = 0.21)						
16.3.4 ≤ 49 days						
Sheldon 2019	1	118	7	130	100.0%	0.16 [0.02 , 1.26]
Subtotal (95% CI)		**118**		**130**	**100.0%**	**0.16 [0.02 , 1.26]**
Total events:	1		7			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.74 (P = 0.08)						
Test for subgroup differences: Chi² = 4.28, df = 3 (P = 0.23), I² = 29.8%						

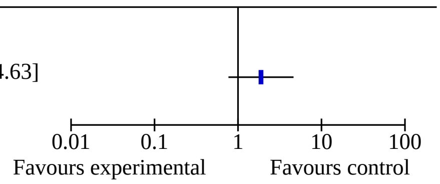
Risk Ratio
M-H, Random, 95% CI



Analysis 16.4. Comparison 16: Prostaglandin alone: route of administration, Outcome 4: Additional uterotonics used

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
16.4.1 Misoprostol sublingual vs buccal					
Sheldon 2019	13	199	7	202	1.89 [0.77 , 4.63]

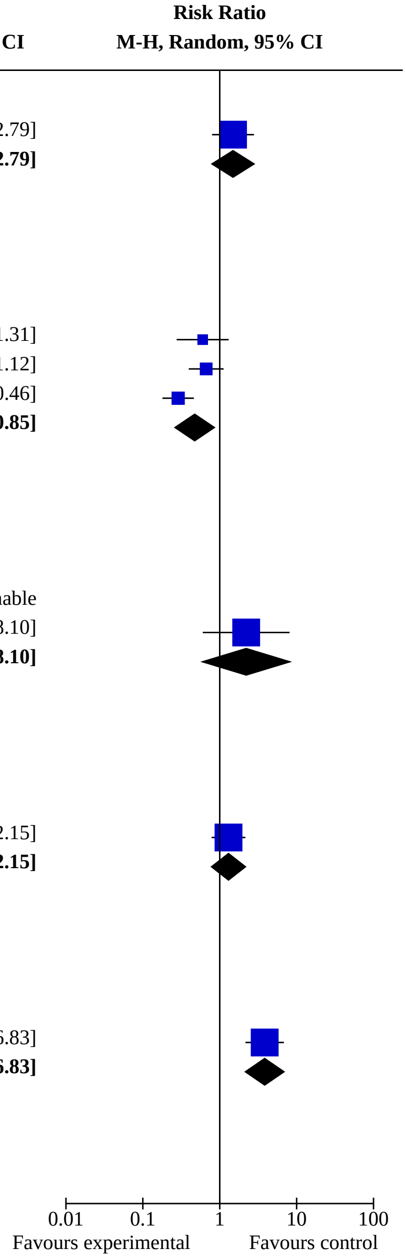
Risk Ratio
M-H, Random, 95% CI



Analysis 16.5. Comparison 16: Prostaglandin alone: route of administration, Outcome 5: Surgical evacuation

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.5.1 Misoprostol sublingual vs buccal						
Sheldon 2019	22	199	15	202	100.0%	1.49 [0.80 , 2.79]
Subtotal (95% CI)		**199**		**202**	**100.0%**	**1.49 [0.80 , 2.79]**
Total events:	22		15			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.25 (P = 0.21)						
16.5.2 Misoprostol sublingual vs vaginal						
Abdelshafy 2019	9	100	15	100	26.5%	0.60 [0.28 , 1.31]
Sonsanoh 2014	16	60	24	60	35.7%	0.67 [0.40 , 1.12]
Tanha 2010	17	110	59	110	37.7%	0.29 [0.18 , 0.46]
Subtotal (95% CI)		**270**		**270**	**100.0%**	**0.47 [0.26 , 0.85]**
Total events:	42		98			
Heterogeneity: $Tau^2 = 0.18$; $Chi^2 = 6.21$, df = 2 (P = 0.04); $I^2 = 68\%$						
Test for overall effect: Z = 2.51 (P = 0.01)						
16.5.3 Misoprostol sublingual vs oral						
Paritakul 2010	0	32	0	32		Not estimable
Souizi 2020	7	56	3	53	100.0%	2.21 [0.60 , 8.10]
Subtotal (95% CI)		**88**		**85**	**100.0%**	**2.21 [0.60 , 8.10]**
Total events:	7		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.20 (P = 0.23)						
16.5.4 Misoprostol repeat vs single						
Mizrachi 2017	26	90	20	90	100.0%	1.30 [0.78 , 2.15]
Subtotal (95% CI)		**90**		**90**	**100.0%**	**1.30 [0.78 , 2.15]**
Total events:	26		20			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.02 (P = 0.31)						
16.5.5 Misoprostol nurse vs doctor						
Klingberg Allvin 2015	54	506	14	504	100.0%	3.84 [2.16 , 6.83]
Subtotal (95% CI)		**506**		**504**	**100.0%**	**3.84 [2.16 , 6.83]**
Total events:	54		14			
Heterogeneity: Not applicable						
Test for overall effect: Z = 4.59 (P < 0.00001)						

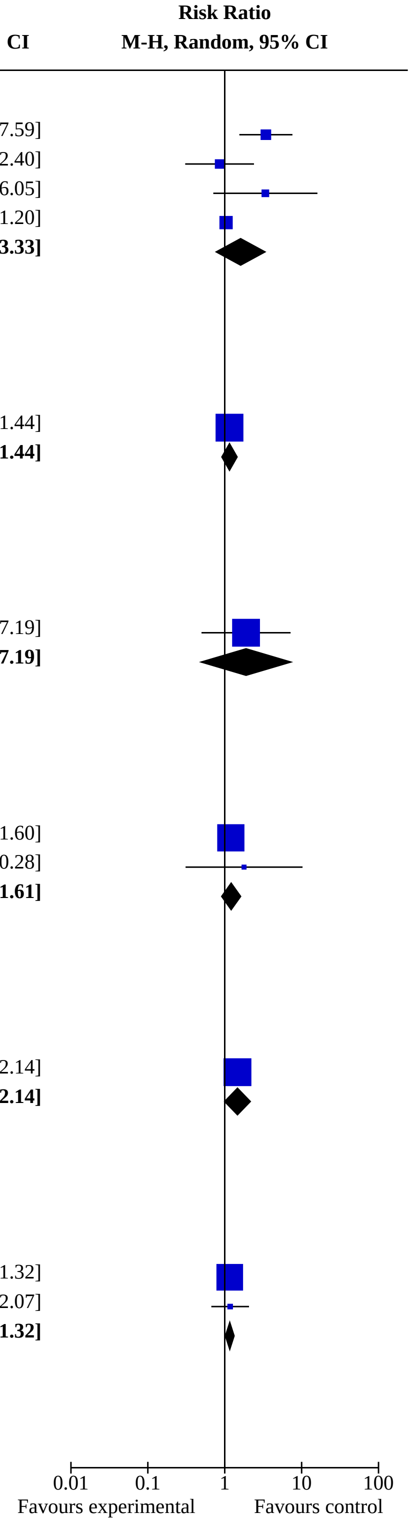
Test for subgroup differences: $Chi^2 = 25.66$, df = 4 (P < 0.0001), $I^2 = 84.4\%$



Analysis 16.6. Comparison 16: Prostaglandin alone: route of administration, Outcome 6: Nausea

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.6.1 Misoprostol sublingual vs vaginal						
Abdelshafy 2019	24	100	7	100	26.4%	3.43 [1.55 , 7.59]
Sonsanoh 2014	6	60	7	60	21.7%	0.86 [0.31 , 2.40]
Souizi 2020	6	56	2	63	13.9%	3.38 [0.71 , 16.05]
von Hertzen 2007	279	1033	268	1033	38.0%	1.04 [0.90 , 1.20]
Subtotal (95% CI)		**1249**		**1256**	**100.0%**	**1.61 [0.78 , 3.33]**
Total events:	315		284			
Heterogeneity: Tau² = 0.36; Chi² = 10.70, df = 3 (P = 0.01); I² = 72%						
Test for overall effect: Z = 1.28 (P = 0.20)						
16.6.2 Misoprostol sublingual vs buccal						
Sheldon 2019	92	199	81	202	100.0%	1.15 [0.92 , 1.44]
Subtotal (95% CI)		**199**		**202**	**100.0%**	**1.15 [0.92 , 1.44]**
Total events:	92		81			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.24 (P = 0.22)						
16.6.3 Misoprostol sublingual vs oral						
Souizi 2020	6	56	3	53	100.0%	1.89 [0.50 , 7.19]
Subtotal (95% CI)		**56**		**53**	**100.0%**	**1.89 [0.50 , 7.19]**
Total events:	6		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.94 (P = 0.35)						
16.6.4 Misoprostol oral vs vaginal						
Marwah 2016	36	50	30	50	97.4%	1.20 [0.90 , 1.60]
Souizi 2020	3	53	2	63	2.6%	1.78 [0.31 , 10.28]
Subtotal (95% CI)		**103**		**113**	**100.0%**	**1.21 [0.92 , 1.61]**
Total events:	39		32			
Heterogeneity: Tau² = 0.00; Chi² = 0.21, df = 1 (P = 0.65); I² = 0%						
Test for overall effect: Z = 1.34 (P = 0.18)						
16.6.5 Misoprostol repeat vs single						
Mizrachi 2017	41	90	28	90	100.0%	1.46 [1.00 , 2.14]
Subtotal (95% CI)		**90**		**90**	**100.0%**	**1.46 [1.00 , 2.14]**
Total events:	41		28			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.96 (P = 0.05)						
16.6.6 Administration model of misoprostol						
Klingberg Allvin 2015	259	506	222	504	95.0%	1.16 [1.02 , 1.32]
Ng 2015	20	77	17	77	5.0%	1.18 [0.67 , 2.07]
Subtotal (95% CI)		**583**		**581**	**100.0%**	**1.16 [1.02 , 1.32]**
Total events:	279		239			
Heterogeneity: Tau² = 0.00; Chi² = 0.00, df = 1 (P = 0.97); I² = 0%						
Test for overall effect: Z = 2.33 (P = 0.02)						

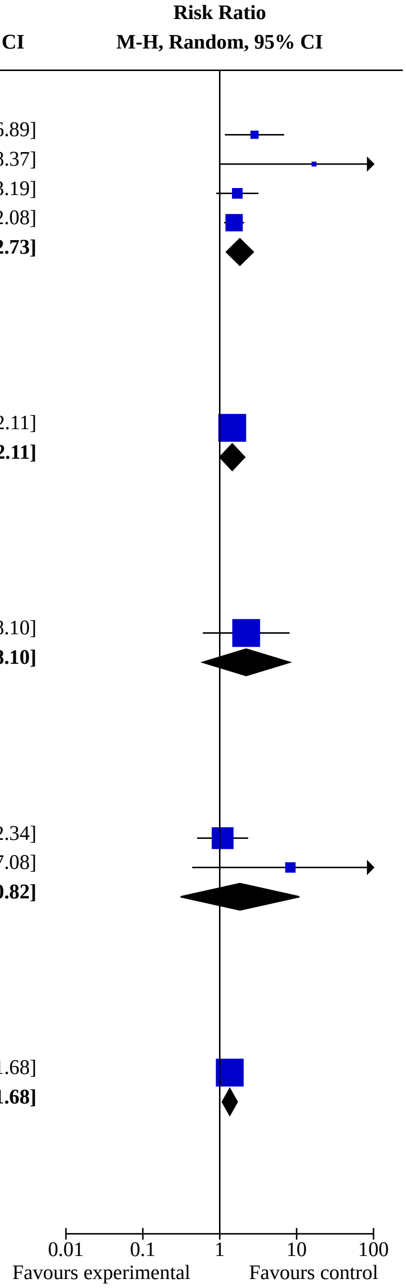
Test for subgroup differences: Chi² = 2.48, df = 5 (P = 0.78), I² = 0%



Analysis 16.7. Comparison 16: Prostaglandin alone: route of administration, Outcome 7: Vomiting

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.7.1 Misoprostol sublingual vs vaginal						
Abdelshafy 2019	17	100	6	100	16.2%	2.83 [1.17 , 6.89]
Souizi 2020	7	56	0	63	1.9%	16.84 [0.98 , 288.37]
Tanha 2010	22	110	13	110	26.7%	1.69 [0.90 , 3.19]
von Hertzen 2007	100	1033	65	1033	55.1%	1.54 [1.14 , 2.08]
Subtotal (95% CI)		**1299**		**1306**	**100.0%**	**1.83 [1.22 , 2.73]**
Total events:	146		84			
Heterogeneity: Tau² = 0.05; Chi² = 4.26, df = 3 (P = 0.23); I² = 30%						
Test for overall effect: Z = 2.94 (P = 0.003)						
16.7.2 Misoprostol sublingual vs buccal						
Sheldon 2019	53	199	37	202	100.0%	1.45 [1.00 , 2.11]
Subtotal (95% CI)		**199**		**202**	**100.0%**	**1.45 [1.00 , 2.11]**
Total events:	53		37			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.98 (P = 0.05)						
16.7.3 Misoprostol sublingual vs oral						
Souizi 2020	7	56	3	53	100.0%	2.21 [0.60 , 8.10]
Subtotal (95% CI)		**56**		**53**	**100.0%**	**2.21 [0.60 , 8.10]**
Total events:	7		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.20 (P = 0.23)						
16.7.4 Misoprostol repeat vs single						
Mizrachi 2017	12	90	11	90	74.4%	1.09 [0.51 , 2.34]
Souizi 2020	3	53	0	63	25.6%	8.30 [0.44 , 157.08]
Subtotal (95% CI)		**143**		**153**	**100.0%**	**1.83 [0.31 , 10.82]**
Total events:	15		11			
Heterogeneity: Tau² = 0.95; Chi² = 1.79, df = 1 (P = 0.18); I² = 44%						
Test for overall effect: Z = 0.67 (P = 0.50)						
16.7.5 Misoprostol nurse vs doctor						
Klingberg Allvin 2015	141	506	104	504	100.0%	1.35 [1.08 , 1.68]
Subtotal (95% CI)		**506**		**504**	**100.0%**	**1.35 [1.08 , 1.68]**
Total events:	141		104			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.66 (P = 0.008)						

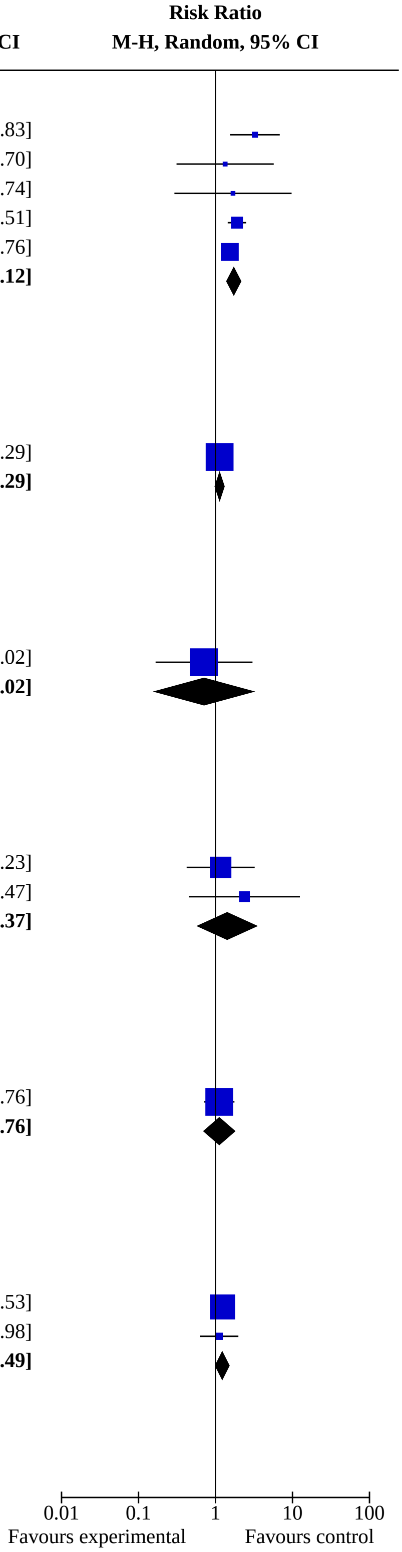
Test for subgroup differences: Chi² = 2.12, df = 4 (P = 0.71), I² = 0%



Analysis 16.8. Comparison 16: Prostaglandin alone: route of administration, Outcome 8: Diarrhoea

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.8.1 Misoprostol sublingual vs vaginal						
Abdelshafy 2019	26	100	8	100	6.9%	3.25 [1.55 , 6.83]
Sonsanoh 2014	4	60	3	60	1.9%	1.33 [0.31 , 5.70]
Souizi 2020	3	56	2	63	1.3%	1.69 [0.29 , 9.74]
Tanha 2010	76	110	40	110	32.2%	1.90 [1.44 , 2.51]
von Hertzen 2007	363	1033	237	1033	57.6%	1.53 [1.33 , 1.76]
Subtotal (95% CI)		**1359**		**1366**	**100.0%**	**1.73 [1.41 , 2.12]**
Total events:	472		290			
Heterogeneity: Tau² = 0.01; Chi² = 5.34, df = 4 (P = 0.25); I² = 25%						
Test for overall effect: Z = 5.23 (P < 0.00001)						
16.8.2 Misoprostol sublingual vs buccal						
Sheldon 2019	147	199	132	202	100.0%	1.13 [0.99 , 1.29]
Subtotal (95% CI)		**199**		**202**	**100.0%**	**1.13 [0.99 , 1.29]**
Total events:	147		132			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.85 (P = 0.06)						
16.8.3 Misoprostol sublingual vs oral						
Souizi 2020	3	56	4	53	100.0%	0.71 [0.17 , 3.02]
Subtotal (95% CI)		**56**		**53**	**100.0%**	**0.71 [0.17 , 3.02]**
Total events:	3		4			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.46 (P = 0.64)						
16.8.4 Misoprostol oral vs vaginal						
Marwah 2016	7	50	6	50	72.6%	1.17 [0.42 , 3.23]
Souizi 2020	4	53	2	63	27.4%	2.38 [0.45 , 12.47]
Subtotal (95% CI)		**103**		**113**	**100.0%**	**1.42 [0.60 , 3.37]**
Total events:	11		8			
Heterogeneity: Tau² = 0.00; Chi² = 0.52, df = 1 (P = 0.47); I² = 0%						
Test for overall effect: Z = 0.79 (P = 0.43)						
16.8.5 Misoprostol repeat vs single						
Mizrachi 2017	28	90	25	90	100.0%	1.12 [0.71 , 1.76]
Subtotal (95% CI)		**90**		**90**	**100.0%**	**1.12 [0.71 , 1.76]**
Total events:	28		25			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.49 (P = 0.62)						
16.8.6 Administration model of misoprostol						
Klingberg Allvin 2015	143	506	115	504	87.9%	1.24 [1.00 , 1.53]
Ng 2015	19	77	17	77	12.1%	1.12 [0.63 , 1.98]
Subtotal (95% CI)		**583**		**581**	**100.0%**	**1.22 [1.00 , 1.49]**
Total events:	162		132			
Heterogeneity: Tau² = 0.00; Chi² = 0.11, df = 1 (P = 0.74); I² = 0%						
Test for overall effect: Z = 1.98 (P = 0.05)						

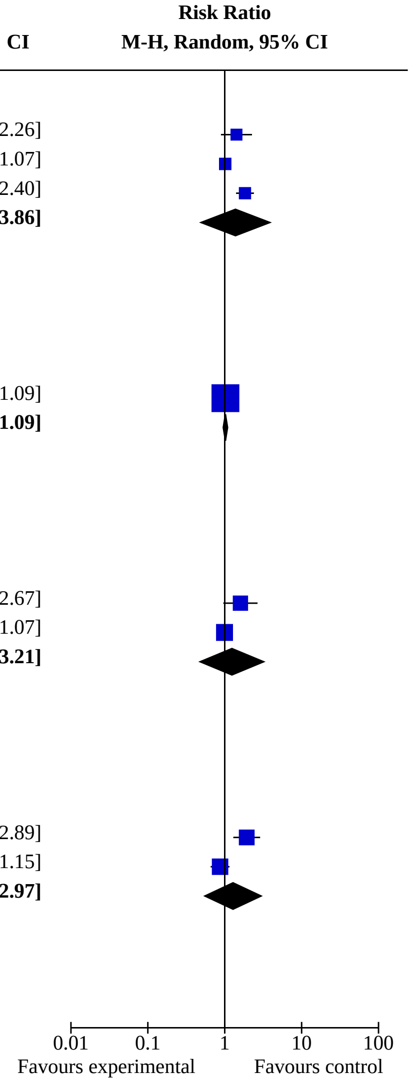
Test for subgroup differences: Chi² = 12.81, df = 5 (P = 0.03), I² = 61.0%



Analysis 16.9. Comparison 16: Prostaglandin alone: route of administration, Outcome 9: Abdominal pain

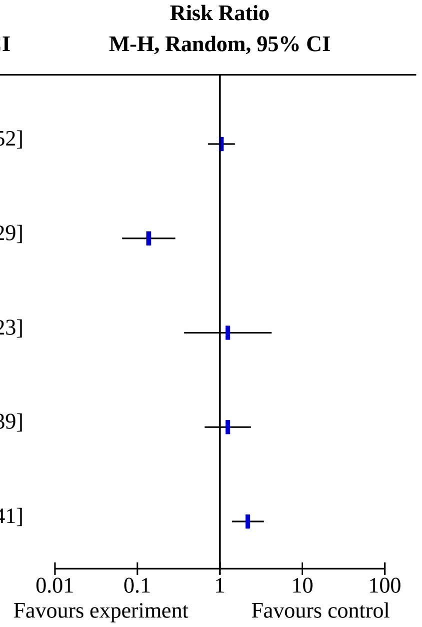
Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
16.9.1 Misoprostol sublingual vs vaginal						
Sonsanoh 2014	27	60	19	60	32.1%	1.42 [0.89 , 2.26]
Souizi 2020	55	56	61	63	34.3%	1.01 [0.96 , 1.07]
Tanha 2010	77	110	42	110	33.6%	1.83 [1.40 , 2.40]
Subtotal (95% CI)		**226**		**233**	**100.0%**	**1.38 [0.49 , 3.86]**
Total events:	159		122			
Heterogeneity: Tau² = 0.80; Chi² = 111.53, df = 2 (P < 0.00001); I² = 98%						
Test for overall effect: Z = 0.61 (P = 0.54)						
16.9.2 Misoprostol sublingual vs oral						
Souizi 2020	55	56	51	53	100.0%	1.02 [0.96 , 1.09]
Subtotal (95% CI)		**56**		**53**	**100.0%**	**1.02 [0.96 , 1.09]**
Total events:	55		51			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.63 (P = 0.53)						
16.9.3 Misoprostol oral vs vaginal						
Marwah 2016	24	50	15	50	46.5%	1.60 [0.96 , 2.67]
Souizi 2020	51	53	61	63	53.5%	0.99 [0.93 , 1.07]
Subtotal (95% CI)		**103**		**113**	**100.0%**	**1.24 [0.48 , 3.21]**
Total events:	75		76			
Heterogeneity: Tau² = 0.44; Chi² = 13.58, df = 1 (P = 0.0002); I² = 93%						
Test for overall effect: Z = 0.44 (P = 0.66)						
16.9.4 Administration model of misoprostol						
Klingberg Allvin 2015	64	506	33	504	48.6%	1.93 [1.29 , 2.89]
Ng 2015	40	77	46	77	51.4%	0.87 [0.66 , 1.15]
Subtotal (95% CI)		**583**		**581**	**100.0%**	**1.28 [0.55 , 2.97]**
Total events:	104		79			
Heterogeneity: Tau² = 0.34; Chi² = 11.74, df = 1 (P = 0.0006); I² = 91%						
Test for overall effect: Z = 0.58 (P = 0.56)						

Test for subgroup differences: Chi² = 0.76, df = 3 (P = 0.86), I² = 0%



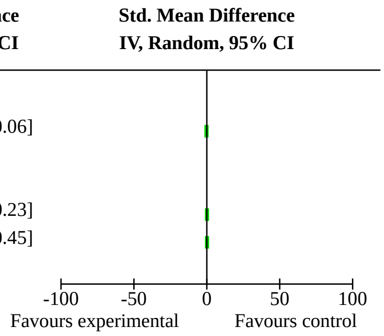
Analysis 16.10. Comparison 16: Prostaglandin alone: route of administration, Outcome 10: Women's dissatisfaction with the procedure

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
16.10.1 Misoprostol sublingual vs buccal					
Sheldon 2019	43	199	42	202	1.04 [0.71 , 1.52]
16.10.2 Misoprostol sublingual vs vaginal					
Tanha 2010	7	110	51	110	0.14 [0.07 , 0.29]
16.10.3 Misoprostol sublingual vs oral					
Paritakul 2010	5	32	4	32	1.25 [0.37 , 4.23]
16.10.4 Misoprostol oral vs vaginal					
Marwah 2016	15	50	12	50	1.25 [0.65 , 2.39]
16.10.5 Misoprostol nurse vs doctor					
Klingberg Allvin 2015	57	506	26	504	2.18 [1.40 , 3.41]



Analysis 16.11. Comparison 16: Prostaglandin alone: route of administration, Outcome 11: Days of bleeding

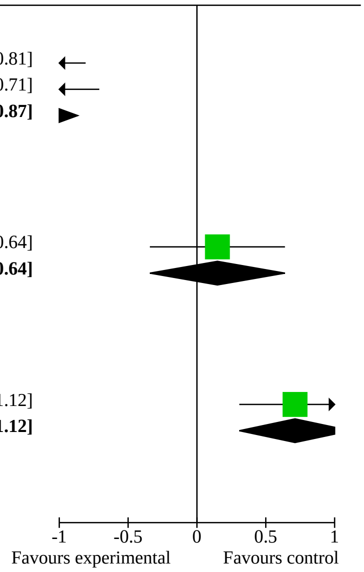
Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Std. Mean Difference IV, Random, 95% CI
16.11.1 Misoprostol sublingual vs vaginal							
Abdelshafy 2019	14.6	2.5	100	15.2	2.9	100	-0.22 [-0.50 , 0.06]
16.11.2 Administration model of misoprostol							
Klingberg Allvin 2015	5.3	2.9	506	5	2.7	504	0.11 [-0.02 , 0.23]
Ng 2015	6	1.51	77	5.82	1.13	77	0.13 [-0.18 , 0.45]



Analysis 16.12. Comparison 16: Prostaglandin alone: route of administration, Outcome 12: Time until passing of conceptus

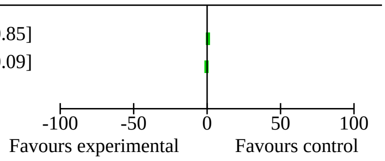
Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Weight	Std. Mean Difference IV, Random, 95% CI
16.12.1 Misoprostol sublingual vs vaginal								
Abdelshafy 2019	12.3	3.1	100	16.4	4.2	100	62.5%	-1.11 [-1.40 , -0.81]
Sonsanoh 2014	12.6	3.9	60	21.1	10.2	60	37.5%	-1.09 [-1.48 , -0.71]
Subtotal (95% CI)			**160**			**160**	**100.0%**	**-1.10 [-1.34 , -0.87]**
Heterogeneity: Tau² = 0.00; Chi² = 0.00, df = 1 (P = 0.96); I² = 0%								
Test for overall effect: Z = 9.17 (P < 0.00001)								
16.12.2 Misoprostol sublingual vs oral								
Paritakul 2010	21.1	17.1	32	18.6	16.1	32	100.0%	0.15 [-0.34 , 0.64]
Subtotal (95% CI)			**32**			**32**	**100.0%**	**0.15 [-0.34 , 0.64]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 0.59 (P = 0.55)								
16.12.3 Misoprostol oral vs vaginal								
Marwah 2016	13.24	3.1	50	10.87	3.49	50	100.0%	0.71 [0.31 , 1.12]
Subtotal (95% CI)			**50**			**50**	**100.0%**	**0.71 [0.31 , 1.12]**
Heterogeneity: Not applicable								
Test for overall effect: Z = 3.45 (P = 0.0006)								

Test for subgroup differences: Chi² = 66.18, df = 2 (P < 0.00001), I² = 97.0%



Analysis 16.13. Comparison 16: Prostaglandin alone: route of administration, Outcome 13: Haemoglobin level

Study or Subgroup	Experimental Mean	Experimental SD	Experimental Total	Control Mean	Control SD	Control Total	Std. Mean Difference IV, Random, 95% CI
Abdelshafy 2019	101	10	100	95	11	100	0.57 [0.29 , 0.85]
Ng 2015	0.47	0.45	77	0.71	0.68	77	-0.41 [-0.73 , -0.09]

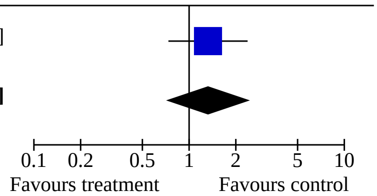


Comparison 17. Mifepristone alone: high dose vs low dose

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
17.1 Failure to achieve complete abortion	1	101	Risk Ratio (M-H, Random, 95% CI)	1.32 [0.74, 2.38]
17.2 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
17.2.1 Nausea	1	101	Risk Ratio (M-H, Random, 95% CI)	0.40 [0.19, 0.84]
17.2.2 Vomiting	1	101	Risk Ratio (M-H, Random, 95% CI)	0.36 [0.07, 1.78]

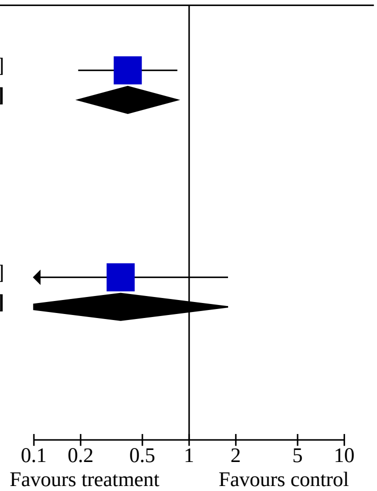
Analysis 17.1. Comparison 17: Mifepristone alone: high dose vs low dose, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Birgerson 1988	19	53	13	48	100.0%	1.32 [0.74 , 2.38]
Total (95% CI)		**53**		**48**	**100.0%**	**1.32 [0.74 , 2.38]**
Total events:	19		13			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.94 (P = 0.35)						
Test for subgroup differences: Not applicable						



Analysis 17.2. Comparison 17: Mifepristone alone: high dose vs low dose, Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
17.2.1 Nausea						
Birgerson 1988	8	53	18	48	100.0%	0.40 [0.19 , 0.84]
Subtotal (95% CI)		**53**		**48**	**100.0%**	**0.40 [0.19 , 0.84]**
Total events:	8		18			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.42 (P = 0.02)						
17.2.2 Vomiting						
Birgerson 1988	2	53	5	48	100.0%	0.36 [0.07 , 1.78]
Subtotal (95% CI)		**53**		**48**	**100.0%**	**0.36 [0.07 , 1.78]**
Total events:	2		5			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.25 (P = 0.21)						

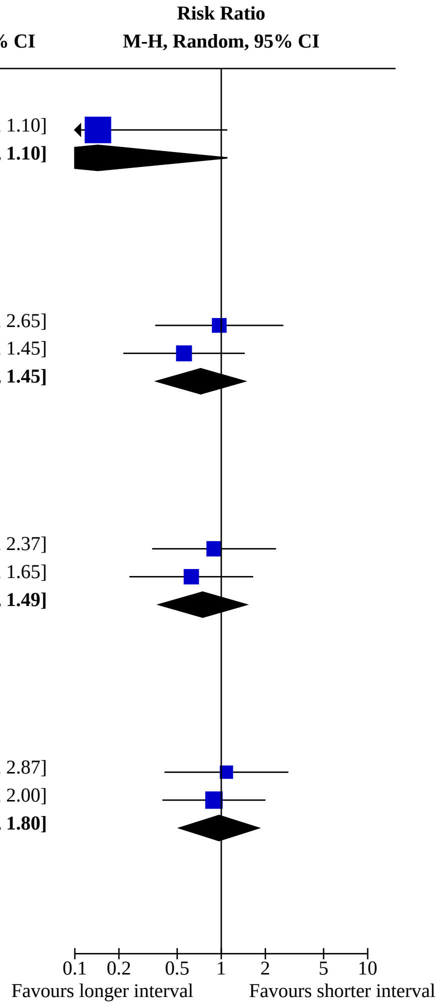


Comparison 18. Combined regimen methotrexate/prostaglandin: timing of prostaglandin

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
18.1 Failure to achieve complete abortion	3		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
18.1.1 Misoprostol day 7 vs day 3	1	86	Risk Ratio (M-H, Random, 95% CI)	0.14 [0.02, 1.10]
18.1.2 Misoprostol day 5 vs day 3	2	387	Risk Ratio (M-H, Random, 95% CI)	0.72 [0.36, 1.45]
18.1.3 Misoprostol day 5 vs day 4	2	394	Risk Ratio (M-H, Random, 95% CI)	0.75 [0.38, 1.49]
18.1.4 Misoprostol day 4 vs day 3	2	393	Risk Ratio (M-H, Random, 95% CI)	0.97 [0.52, 1.80]

Analysis 18.1. Comparison 18: Combined regimen methotrexate/prostaglandin: timing of prostaglandin, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
18.1.1 Misoprostol day 7 vs day 3						
Creinin 1995	1	40	8	46	100.0%	0.14 [0.02 , 1.10]
Subtotal (95% CI)		**40**		**46**	**100.0%**	**0.14 [0.02 , 1.10]**
Total events:	1		8			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.87 (P = 0.06)						
18.1.2 Misoprostol day 5 vs day 3						
Carbonell 1997b	7	96	7	93	47.3%	0.97 [0.35 , 2.65]
Carbonell 1998	6	98	11	100	52.7%	0.56 [0.21 , 1.45]
Subtotal (95% CI)		**194**		**193**	**100.0%**	**0.72 [0.36 , 1.45]**
Total events:	13		18			
Heterogeneity: Tau² = 0.00; Chi² = 0.61, df = 1 (P = 0.43); I² = 0%						
Test for overall effect: Z = 0.92 (P = 0.36)						
18.1.3 Misoprostol day 5 vs day 4						
Carbonell 1997b	7	96	8	98	49.9%	0.89 [0.34 , 2.37]
Carbonell 1998	6	98	10	102	50.1%	0.62 [0.24 , 1.65]
Subtotal (95% CI)		**194**		**200**	**100.0%**	**0.75 [0.38 , 1.49]**
Total events:	13		18			
Heterogeneity: Tau² = 0.00; Chi² = 0.26, df = 1 (P = 0.61); I² = 0%						
Test for overall effect: Z = 0.83 (P = 0.41)						
18.1.4 Misoprostol day 4 vs day 3						
Carbonell 1997b	8	98	7	93	40.9%	1.08 [0.41 , 2.87]
Carbonell 1998	10	102	11	100	59.1%	0.89 [0.40 , 2.00]
Subtotal (95% CI)		**200**		**193**	**100.0%**	**0.97 [0.52 , 1.80]**
Total events:	18		18			
Heterogeneity: Tau² = 0.00; Chi² = 0.09, df = 1 (P = 0.76); I² = 0%						
Test for overall effect: Z = 0.11 (P = 0.91)						

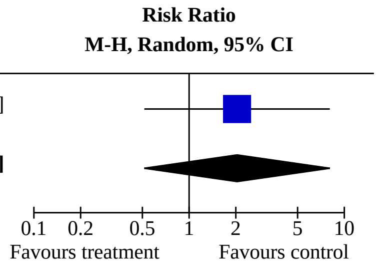


Comparison 19. Combined regimen methotrexate/prostaglandin: methotrexate intramuscular vs oral

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
19.1 Failure to achieve complete abortion	1	100	Risk Ratio (M-H, Random, 95% CI)	2.04 [0.51, 8.07]
19.2 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
19.2.1 Nausea	1	100	Risk Ratio (M-H, Random, 95% CI)	0.52 [0.22, 1.25]
19.2.2 Vomiting	1	100	Risk Ratio (M-H, Random, 95% CI)	4.89 [0.57, 42.21]
19.2.3 Diarrhoea	1	100	Risk Ratio (M-H, Random, 95% CI)	1.22 [0.18, 8.34]

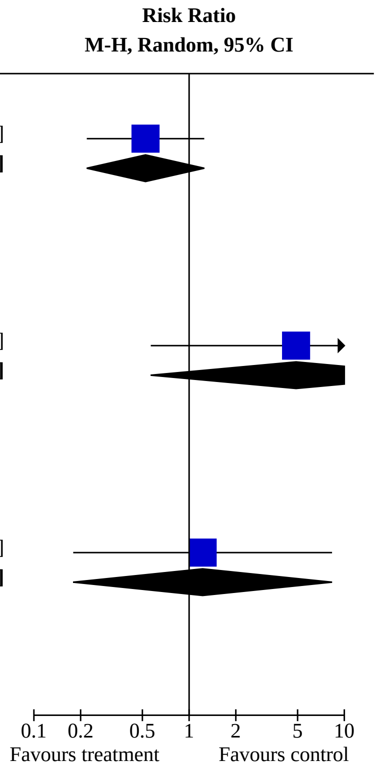
Analysis 19.1. Comparison 19: Combined regimen methotrexate/prostaglandin: methotrexate intramuscular vs oral, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Wiebe 1999b	5	45	3	55	100.0%	2.04 [0.51 , 8.07]
Total (95% CI)		**45**		**55**	**100.0%**	**2.04 [0.51 , 8.07]**
Total events:	5		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.01 (P = 0.31)						
Test for subgroup differences: Not applicable						



Analysis 19.2. Comparison 19: Combined regimen methotrexate/prostaglandin: methotrexate intramuscular vs oral, Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
19.2.1 Nausea						
Wiebe 1999b	6	45	14	55	100.0%	0.52 [0.22 , 1.25]
Subtotal (95% CI)		**45**		**55**	**100.0%**	**0.52 [0.22 , 1.25]**
Total events:	6		14			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.45 (P = 0.15)						
19.2.2 Vomiting						
Wiebe 1999b	4	45	1	55	100.0%	4.89 [0.57 , 42.21]
Subtotal (95% CI)		**45**		**55**	**100.0%**	**4.89 [0.57 , 42.21]**
Total events:	4		1			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.44 (P = 0.15)						
19.2.3 Diarrhoea						
Wiebe 1999b	2	45	2	55	100.0%	1.22 [0.18 , 8.34]
Subtotal (95% CI)		**45**		**55**	**100.0%**	**1.22 [0.18 , 8.34]**
Total events:	2		2			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.20 (P = 0.84)						



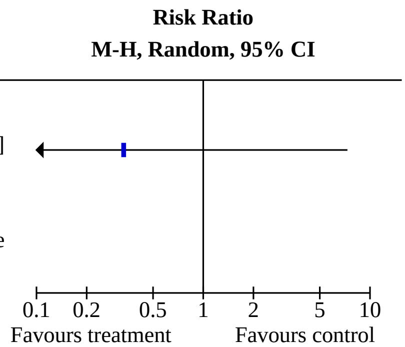
Comparison 20. Combined regimen methotrexate/prostaglandin: dose of methotrexate

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
20.1 Failure to achieve complete abortion	2		Risk Ratio (M-H, Random, 95% CI)	Totals not selected
20.1.1 methotrexate 60 mg vs 50 mg	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
20.1.2 methotrexate 50 mg vs 25 mg	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 20.1. Comparison 20: Combined regimen methotrexate/prostaglandin: dose of methotrexate, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
20.1.1 methotrexate 60 mg vs 50 mg					
Creinin 1997	0	10	1	10	0.33 [0.02 , 7.32]
20.1.2 methotrexate 50 mg vs 25 mg					
Creinin 1996a	0	10	0	10	Not estimable

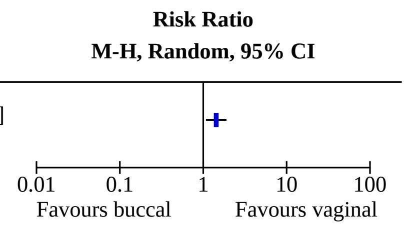


Comparison 21. Combined regimen methotrexate/prostaglandin: route of prostaglandin (misoprostol)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
21.1 Failure to achieve complete abortion	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
21.2 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
21.2.1 Nausea	1	309	Risk Ratio (M-H, Random, 95% CI)	1.09 [0.84, 1.42]
21.2.2 Vomiting	1	309	Risk Ratio (M-H, Random, 95% CI)	1.02 [0.67, 1.56]
21.2.3 Diarrhoea	1	309	Risk Ratio (M-H, Random, 95% CI)	1.40 [0.94, 2.07]

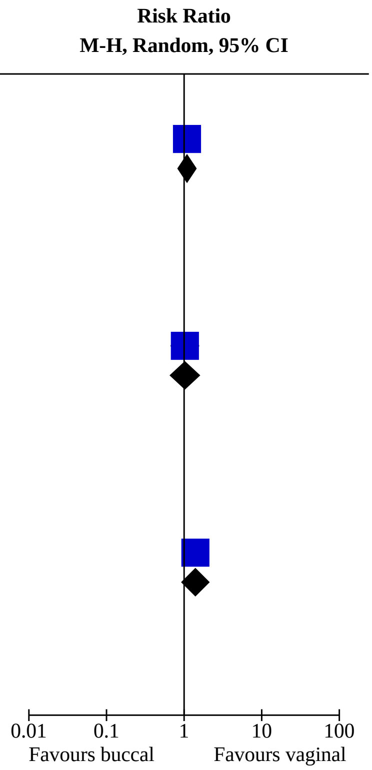
Analysis 21.1. Comparison 21: Combined regimen methotrexate/prostaglandin: route of prostaglandin (misoprostol), Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
Wiebe 2004	72	155	50	154	1.43 [1.08 , 1.90]



Analysis 21.2. Comparison 21: Combined regimen methotrexate/prostaglandin: route of prostaglandin (misoprostol), Outcome 2: Side effects

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
21.2.1 Nausea						
Wiebe 2004	68	155	62	154	100.0%	1.09 [0.84 , 1.42]
Subtotal (95% CI)		**155**		**154**	**100.0%**	**1.09 [0.84 , 1.42]**
Total events:	68		62			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.64 (P = 0.52)						
21.2.2 Vomiting						
Wiebe 2004	34	155	33	154	100.0%	1.02 [0.67 , 1.56]
Subtotal (95% CI)		**155**		**154**	**100.0%**	**1.02 [0.67 , 1.56]**
Total events:	34		33			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.11 (P = 0.91)						
21.2.3 Diarrhoea						
Wiebe 2004	45	155	32	154	100.0%	1.40 [0.94 , 2.07]
Subtotal (95% CI)		**155**		**154**	**100.0%**	**1.40 [0.94 , 2.07]**
Total events:	45		32			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.66 (P = 0.10)						

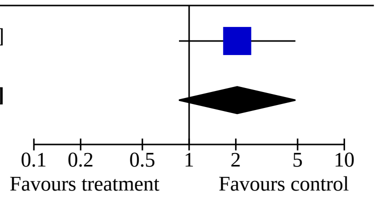


Comparison 22. Tamoxifen vs methotrexate (combined with prostaglandin): low-dose tamoxifen (40 mg)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
22.1 Failure to achieve complete abortion	1	198	Risk Ratio (M-H, Random, 95% CI)	2.04 [0.86, 4.84]
22.2 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
22.2.1 Nausea	1	198	Risk Ratio (M-H, Random, 95% CI)	0.56 [0.33, 0.97]
22.2.2 Vomiting	1	198	Risk Ratio (M-H, Random, 95% CI)	1.70 [0.42, 6.92]
22.2.3 Diarrhoea	1	198	Risk Ratio (M-H, Random, 95% CI)	1.53 [0.26, 8.96]

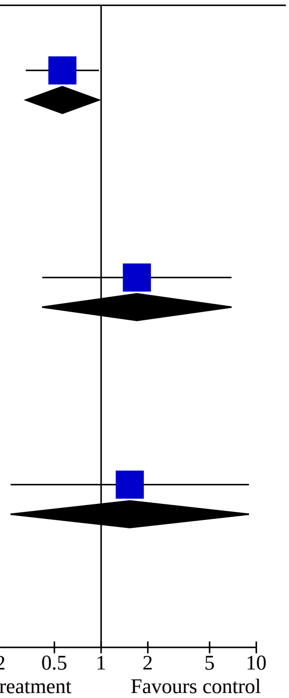
Analysis 22.1. Comparison 22: Tamoxifen vs methotrexate (combined with prostaglandin): low-dose tamoxifen (40 mg), Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Wiebe 1999a	14	98	7	100	100.0%	2.04 [0.86 , 4.84]
Total (95% CI)		**98**		**100**	**100.0%**	**2.04 [0.86 , 4.84]**
Total events:	14		7			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.62 (P = 0.11)						
Test for subgroup differences: Not applicable						



Analysis 22.2. Comparison 22: Tamoxifen vs methotrexate (combined with prostaglandin): low-dose tamoxifen (40 mg), Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
22.2.1 Nausea						
Wiebe 1999a	16	98	29	100	100.0%	0.56 [0.33 , 0.97]
Subtotal (95% CI)		**98**		**100**	**100.0%**	**0.56 [0.33 , 0.97]**
Total events:	16		29			
Heterogeneity: Not applicable						
Test for overall effect: Z = 2.07 (P = 0.04)						
22.2.2 Vomiting						
Wiebe 1999a	5	98	3	100	100.0%	1.70 [0.42 , 6.92]
Subtotal (95% CI)		**98**		**100**	**100.0%**	**1.70 [0.42 , 6.92]**
Total events:	5		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.74 (P = 0.46)						
22.2.3 Diarrhoea						
Wiebe 1999a	3	98	2	100	100.0%	1.53 [0.26 , 8.96]
Subtotal (95% CI)		**98**		**100**	**100.0%**	**1.53 [0.26 , 8.96]**
Total events:	3		2			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.47 (P = 0.64)						



Comparison 23. Tamoxifen vs methotrexate (combined with prostaglandin): high-dose tamoxifen (160 mg)

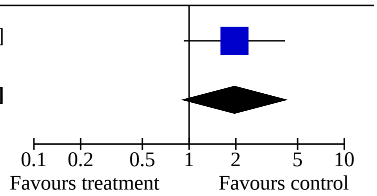
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
23.1 Failure to achieve complete abortion	1	200	Risk Ratio (M-H, Random, 95% CI)	1.96 [0.93, 4.15]
23.2 Side effects	1		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
23.2.1 Nausea	1	200	Risk Ratio (M-H, Random, 95% CI)	0.78 [0.54, 1.10]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
23.2.2 Vomiting	1	200	Risk Ratio (M-H, Random, 95% CI)	0.65 [0.28, 1.53]
23.2.3 Diarrhoea	1	200	Risk Ratio (M-H, Random, 95% CI)	1.23 [0.34, 4.43]

Analysis 23.1. Comparison 23: Tamoxifen vs methotrexate (combined with prostaglandin): high-dose tamoxifen (160 mg), Outcome 1: Failure to achieve complete abortion

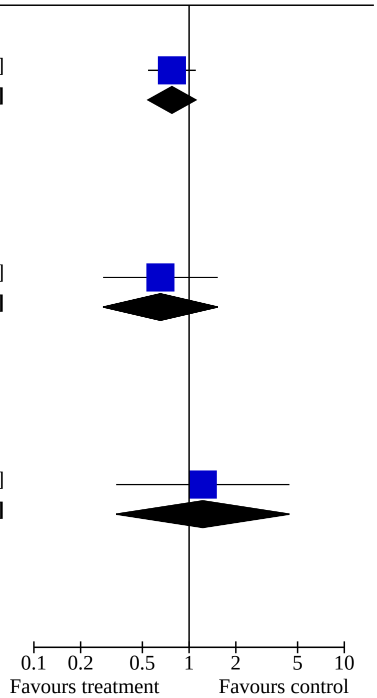
Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Wiebe 1999a	18	101	9	99	100.0%	1.96 [0.93 , 4.15]
Total (95% CI)		**101**		**99**	**100.0%**	**1.96 [0.93 , 4.15]**
Total events:	18		9			

Heterogeneity: Not applicable
Test for overall effect: Z = 1.76 (P = 0.08)
Test for subgroup differences: Not applicable



Analysis 23.2. Comparison 23: Tamoxifen vs methotrexate (combined with prostaglandin): high-dose tamoxifen (160 mg), Outcome 2: Side effects

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
23.2.1 Nausea						
Wiebe 1999a	34	101	43	99	100.0%	0.78 [0.54 , 1.10]
Subtotal (95% CI)		**101**		**99**	**100.0%**	**0.78 [0.54 , 1.10]**
Total events:	34		43			
Heterogeneity: Not applicable						
Test for overall effect: Z = 1.41 (P = 0.16)						
23.2.2 Vomiting						
Wiebe 1999a	8	101	12	99	100.0%	0.65 [0.28 , 1.53]
Subtotal (95% CI)		**101**		**99**	**100.0%**	**0.65 [0.28 , 1.53]**
Total events:	8		12			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.98 (P = 0.33)						
23.2.3 Diarrhoea						
Wiebe 1999a	5	101	4	99	100.0%	1.23 [0.34 , 4.43]
Subtotal (95% CI)		**101**		**99**	**100.0%**	**1.23 [0.34 , 4.43]**
Total events:	5		4			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.31 (P = 0.76)						

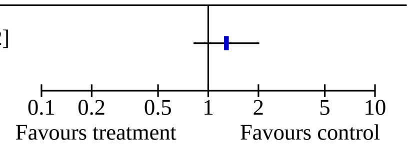


Comparison 24. Combined regimen mifepristone/prostaglandin vs mifepristone/prostaglandin and tamoxifen

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
24.1 Failure to achieve complete abortion	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Analysis 24.1. Comparison 24: Combined regimen mifepristone/prostaglandin vs mifepristone/prostaglandin and tamoxifen, Outcome 1: Failure to achieve complete abortion

Study or Subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Risk Ratio M-H, Random, 95% CI
Wu 1993	39	461	31	471	1.29 [0.82 , 2.02]



ADDITIONAL TABLES

Table 1. Other studies included in the review

Study	Intervention	Outcomes
Chen 2015a	Day 1-3: antibiotic and oestrogen. After gestational sac passed out: oxytocin, 20 U, IM Day 3: mifepristone 150 mg, oral Day 4: misoprostol 600 µg, vaginal Group 1: Shenghua decoction from day 1 to 2 weeks after gestational sac passed out Group 2: Shenghua decoction from day 4 to 2 weeks after gestational sac passed out Group 3: without Shenghua decoction	Bleeding duration Time until passing of conceptus
Li 2015	Day 1: oral mifepristone from 150-50 mg Day 2: 200 µg of oral misoprostol Group 1: 6 pills of mifepristone (150 mg) Group 2: 5 pills of mifepristone plus 1 pill of placebo (125 mg) Group 3: 4 pills of mifepristone plus 2 pills of placebo (100 mg) Group 4: 3 pills of mifepristone plus 3 pills of placebo (75 mg) Group 5: 2 pills of mifepristone plus 4 pills of placebo (50 mg)	Failure to achieve complete abortion Group 1: 24/500 Group 2: 27/500 Group 3: 25/500 Group 4: 22/500 Group 5: 28/500 Ongoing pregnancy Group 1: 18/500 Group 2: 20/500

Table 1. Other studies included in the review *(Continued)*

		Group 3: 18/500 Group 4: 19/500 Group 5: 25/500 Nausea Group 1: 211/500 Group 2: 203/500 Group 3: 197/500 Group 4: 109/500 Group 5: 84/500 Vomiting Group 1: 64/500 Group 2: 62/500 Group 3: 55/500 Group 4: 43/500 Group 5: 32/500 Diarrhoea Group 1: 50/500 Group 2: 50/500 Group 3: 37/500 Group 4: 29/500 Group 5: 24/500 Abdominal pain Group 1: 394/500 Group 2: 380/500 Group 3: 354/500 Group 4: 299/500 Group 5: 119/500
Liao 2004	Group 1: mifepristone given: 50 mg, then 12 h later 25 mg, then 12 h later 50 mg, and finally, 12 h later, 25 mg mifepristone. 24 h after, 600 µg misoprostol oral (total: 150 mg) Group 2: mifepristone given 30 mg, then 15 mg every 12 h for 3 doses. 24 h after last dose, 600 µg misoprostol given oral (total: 75 mg)	Failure to achieve complete abortion Group 1: 11/240 Group 2: 9/240 Ongoing pregnancy Group 1: 2/240 Group 2: 1/240
Wang 2000	Group 1:	Failure to achieve complete abortion Group 1: 18/1118

Table 1. Other studies included in the review *(Continued)*

	Day 1: mifepristone 50 mg/oral/12-hourly, 2 doses, day 2-7: mifepristone 25 mg oral/day Day 3: misoprostol 600 µg/oral, day 4-6: misoprostol 200 µg/day Group 2: Day 1: mifepristone 50 mg, followed by 25 mg/12-hourly/4 times Day 3: misoprostol 600 µg, oral	Group 2: 59/494 Ongoing pregnancy Group 1: 2/1118 Group 2: 6/494
WHO 1989	Group 1: mifepristone 25 mg/twice daily for 3 days (total 150 mg) and sulprostone 0.25 mg IM on morning of 3rd day Group 2: mifepristone 25 mg /twice daily for 4 days (total 200 mg) and sulprostone 0.25 mg IM on morning of fourth day	Failure to achieve complete abortion Group 1: 15/125 Group 2: 13/126 Ongoing pregnancy Group 1: 3/125 Group 2: 3/126
WHO 1991	Group 1: mifepristone 25 mg/12-hourly, 5 doses (total 125 mg) and gemeprost 1 mg vaginally 60 h after the start of the treatment Group 2: mifepristone 600 mg single dose and gemeprost 1 mg vaginally 60 h after the start of the treatment	Failure to achieve complete abortion Group 1: 12/181 Group 2: 15/187
Wiebe 2006	Group 1: methotrexate 50 mg/m^2 followed by >/72 h by 400 µg misoprostol vaginal Group 2: misoprostol 400 µg sublingual and 400 µg misoprostol vaginal	Failure to achieve complete abortion Group 1: 62/149 Group 2: 57/149 Nausea Group 1: 53/49 Group 2: 54/149 Vomiting Group 1: 17/149 Group 2: 21/149 Diarrhoea Group 1: 16/149 Group 2: 41/149 Surgical abortion Group 1: 9/149 Group 2: 18/149

IM: intramuscular

APPENDICES

Appendix 1. Update review search strategies

Cochrane Central Register of Controlled Trials (CENTRAL; Ovid EBM Reviews) February 2021

1 (abortion* or "menstrual regulation" or pre-abort* or preabort* or post-abort* or postabort* or post-terminat* or postterminat* or pre-terminat* or preterminat* or terminat* or ((gestation* or pregnan* or trimester) adj5 (interrup* or terminate* or termination*))).ti,ab. (13435)
2 (MTOP or abortifacient* or ((chemical* or drug or medical or medically or medicinal or medication or medicine or mifepristone or misoprostol or pharma*) adj3 (abortion* or induc* or interrupt* or termination*)) or ((dinoprost* or carboprost or epostane or ethacridine or gemeprost or isosorbide or lilopristone or meteneprost or methotrexate or mifepristone or misoprostol or onapristone or oxytocin or "potassium chloride" or prostaglandin* or saline or sulprostone) adj10 (abortion* or induc* or interrupt* or terminat*))).ti,ab. (11965)
3 (Abo-pill or Colestone or Cytotec or Elmif or GyMiso or Korlym or Medabon or Mefeprin or Mefipil or Mifebort or Mifegest or Mifegyne or Mifeprex or Miferiv or Mifty or Mtpill or Pitocin or RU-486 or RU486 or Syntocinon or T-Pill or Termipil).ti,ab. (470)
4 (((1st or first) adj trimester) or (early adj2 (abortion* or pregnan*)) or (("5" or five or fifth or "6" or six or sixth or "7" or seven or seventh or "8" or eight or eighth or "9" or nine or ninth or "10" or ten or tenth or "11" or eleven or eleventh or "12" or twelve or twelfth or "13" or thirteen or thirteenth or "14" or fourteen or fourteenth) adj (week or weeks))).ti,ab. or days.ti. (160519)
5 or/2-3 (12258)
6 and/1,4-5 (776)
7 limit 6 to yr="2010 -Current" (332)

MEDLINE ALL (Ovid) 1946 to February 28 2021

1 Abortion, Induced/ or Abortion, Eugenic/ or Abortion, Legal/ or Abortion, Therapeutic/ (38970)
2 (abortion* or "menstrual regulation" or pre-abort* or preabort* or post-abort* or postabort* or post-terminat* or postterminat* or pre-terminat* or preterminat* or terminat* or ((gestation* or pregnan* or trimester) adj5 (interrup* or terminate* or termination*))).ti,ab,kf. (172407)
3 or/1-2 (182824)
4 Abortifacient Agents/ or Abortifacient Agents, Nonsteroidal/ or Abortifacient Agents, Steroidal/ or Isosorbide Dinitrate/ or Nitric Oxide Donors/ or Potassium Chloride/ or exp Prostaglandins A/ or exp Prostaglandins A, Synthetic/ or exp Prostaglandins E/ or exp Prostaglandins E, Synthetic/ or exp Prostaglandins F/ or exp Prostaglandins F, Synthetic/ (93413)
5 (MTOP or abortifacient* or ((chemical* or drug or medical or medically or medicinal or medication or medicine or mifepristone or misoprostol or pharma*) adj3 (abortion* or induc* or interrupt* or termination*)) or ((dinoprost* or carboprost or epostane or ethacridine or gemeprost or isosorbide or lilopristone or meteneprost or methotrexate or mifepristone or misoprostol or onapristone or oxytocin or "potassium chloride" or prostaglandin* or saline or sulprostone) adj10 (abortion* or induc* or interrupt* or terminat*))).ti,ab,kf. or dt.fs. (2264873)
6 (Abo-pill or Colestone or Cytotec or Elmif or GyMiso or Korlym or Medabon or Mefeprin or Mefipil or Mifebort or Mifegest or Mifegyne or Mifeprex or Miferiv or Mifty or Mtpill or Pitocin or RU-486 or RU486 or Syntocinon or T-Pill or Termipil).ti,ab,kf. (4577)
7 or/4-6 (2335333)
8 Pregnancy Trimester, First/ (16245)
9 (((1st or first) adj trimester) or (early adj2 (abortion* or pregnan*)) or (("5" or five or fifth or "6" or six or sixth or "7" or seven or seventh or "8" or eight or eighth or "9" or nine or ninth or "10" or ten or tenth or "11" or eleven or eleventh or "12" or twelve or twelfth or "13" or thirteen or thirteenth or "14" or fourteen or fourteenth) adj (week or weeks))).ti,ab,kf. or days.ti. (438332)
10 or/8-9 (442254)
11 randomized controlled trial.pt. (497720)
12 controlled clinical trial.pt. (93488)
13 randomized.ab. (465143)
14 placebo.ab. (203911)
15 drug therapy.fs. (2169045)
16 randomly.ab. (324249)
17 trial.ab. (488824)
18 groups.ab. (1991209)
19 or/11-18 (4600230)
20 and/3,7,10,19 (2521)
21 20 not ((exp animals/ not humans/) or (animal model* or bovine or canine or capra or cat or cats or cattle or cow or cows or dog or dogs or equine or ewe or ewes or feline or goat or goats or horse or hamster* or horses or invertebrate or invertebrates or macaque or macaques or mare or mares or mice or monkey or monkeys or mouse or murine or nonhuman or non-human or mosquito* or ovine or pig or pigs or porcine or primate or primates or rabbit or rabbits or rat or rats or rattus or rhesus or rodent* or ruminant* or sheep or simian or sow or sows or vertebrate or vertebrates or zebrafish).ti.) (2280)
22 limit 21 to yr="2010 -Current" (753)

Embase.com

#1 'abortion'/de OR 'pregnancy termination'/de OR 'induced abortion'/de OR 'medical abortion'/de OR 'legal abortion'/de OR 'therapeutic abortion'/de (65,601)
#2 abortion*:ti,ab,kw OR 'menstrual regulation':ti,ab,kw OR 'pre abort*':ti,ab,kw OR preabort*:ti,ab,kw OR 'post abort*':ti,ab,kw OR postabort*:ti,ab,kw OR 'post terminat*':ti,ab,kw OR postterminat*:ti,ab,kw OR 'pre terminat*':ti,ab,kw OR preterminat*:ti,ab,kw OR terminat*:ti,ab,kw OR (((gestation* OR pregnan* OR trimester) NEAR/5 (interrup* OR terminate* OR termination*)):ti,ab,kw) (206,013)
#3 #1 OR #2 (226,628)
#4 'abortive agent'/de OR '9 deoxo 9 methyleneprostaglandin e2'/de OR 'aglepristone'/de OR 'anordrin'/de OR 'carboprost'/de OR 'carboprost methyl'/de OR 'carboprost trometamol'/de OR 'dilapan'/de OR 'epostane'/de OR 'fenprostalene'/de OR 'fluprostenol'/de OR 'gemeprost'/de OR 'hydrocortisone acetate plus urea'/de OR 'lamicel'/de OR 'lilopristone'/de OR 'meteneprost'/de OR 'mifepristone'/de OR 'mifepristone plus misoprostol'/de OR 'misoprostol'/de OR 'onapristone'/de OR 'prostaglandin e2'/de OR 'prostaglandin e2 trometamol'/de OR 'prostaglandin f2 alpha'/de OR 'prostaglandin f2 alpha trometamol'/de OR 'prostalene'/de OR 'sulprostone'/de OR 'urea'/de (167,689)
#5 mtop:ti,ab,kw OR abortifacient*:ti,ab,kw OR ((chemical*:ti,ab,kw OR drug:ti,ab,kw OR medical:ti,ab,kw OR medically:ti,ab,kw OR medicinal:ti,ab,kw OR medication:ti,ab,kw OR medicine:ti,ab,kw OR mifepristone:ti,ab,kw OR misoprostol:ti,ab,kw OR pharma*:ti,ab,kw) AND adj3:ti,ab,kw AND (abortion*:ti,ab,kw OR induc*:ti,ab,kw OR interrupt*:ti,ab,kw OR termination*:ti,ab,kw)) OR (((dinoprost* OR carboprost OR epostane OR ethacridine OR gemeprost OR isosorbide OR lilopristone OR meteneprost OR methotrexate OR mifepristone OR misoprostol OR onapristone OR oxytocin OR 'potassium chloride' OR prostaglandin* OR saline OR sulprostone) NEAR/10 (abortion* OR induc* OR interrupt* OR terminat*)):ti,ab,kw) (44,575)
#6 'abo pill':ti,ab,kw OR colestone:ti,ab,kw OR cytotec:ti,ab,kw OR elmif:ti,ab,kw OR gymiso:ti,ab,kw OR korlym:ti,ab,kw OR medabon:ti,ab,kw OR mefeprin:ti,ab,kw OR mefipil:ti,ab,kw OR mifebort:ti,ab,kw OR mifegest:ti,ab,kw OR mifegyne:ti,ab,kw OR mifeprex:ti,ab,kw OR miferiv:ti,ab,kw OR mifty:ti,ab,kw OR mtpill:ti,ab,kw OR pitocin:ti,ab,kw OR 'ru 486':ti,ab,kw OR ru486:ti,ab,kw OR syntocinon:ti,ab,kw OR 't pill':ti,ab,kw OR termipil:ti,ab,kw (5,388)
#7 #4 OR #5 OR #6 (197,576)
#8 'first trimester pregnancy'/exp (39,939)
#9 (((1st OR first) NEAR/1 trimester):ti,ab,kw) OR ((early NEAR/2 (abortion* OR pregnan*)):ti,ab,kw) OR ((('5' OR five OR fifth OR '6' OR six OR sixth OR '7' OR seven OR seventh OR '8' OR eight OR eighth OR '9' OR nine OR ninth OR '10' OR ten OR tenth OR '11' OR eleven OR eleventh OR '12' OR twelve OR twelfth OR '13' OR thirteen OR thirteenth OR '14' OR fourteen OR fourteenth) NEAR/1 (week OR weeks)):ti,ab,kw) OR days:ti (684,086)
#10 #8 OR #9 (691,975)
#11 'crossover procedure':de OR 'double-blind procedure':de OR 'randomized controlled trial':de OR 'single-blind procedure':de OR random*:de,ab,ti OR factorial*:de,ab,ti OR crossover*:de,ab,ti OR ((cross NEXT/1 over*):de,ab,ti) OR placebo*:de,ab,ti OR ((doubl* NEAR/1 blind*):de,ab,ti) OR ((singl* NEAR/1 blind*):de,ab,ti) OR assign*:de,ab,ti OR allocat*:de,ab,ti OR volunteer*:de,ab,ti (2,511,701)
#12 #3 AND #7 AND #10 AND #11 (668)
#13 'animal'/exp NOT 'human'/exp (5,374,483)
#14 'animal model*':ti OR bovine:ti OR canine:ti OR capra:ti OR cat:ti OR cats:ti OR cattle:ti OR cow:ti OR cows:ti OR dog:ti OR dogs:ti OR equine:ti OR ewe:ti OR ewes:ti OR feline:ti OR goat:ti OR goats:ti OR horse:ti OR hamster*:ti OR horses:ti OR invertebrate:ti OR invertebrates:ti OR macaque:ti OR macaques:ti OR mare:ti OR mares:ti OR mice:ti OR monkey:ti OR monkeys:ti OR mouse:ti OR murine:ti OR nonhuman:ti OR 'non human':ti OR mosquito*:ti OR ovine:ti OR pig:ti OR pigs:ti OR porcine:ti OR primate:ti OR primates:ti OR rabbit:ti OR rabbits:ti OR rat:ti OR rats:ti OR rattus:ti OR rhesus:ti OR rodent*:ti OR ruminant*:ti OR sheep:ti OR simian:ti OR sow:ti OR sows:ti OR vertebrate:ti OR vertebrates:ti OR zebrafish:ti (2,688,217)
#15 #13 OR #14 (5,811,088)
#16 #12 NOT #15 (647)
#17 #16 AND (2010:py OR 2011:py OR 2012:py OR 2013:py OR 2014:py OR 2015:py OR 2016:py OR 2017:py OR 2018:py OR 2019:py) (239)

Global Health (Ovid) 1973 to 2021

1 (abortion* or "menstrual regulation" or pre-abort* or preabort* or post-abort* or postabort* or post-terminat* or postterminat* or pre-terminat* or preterminat* or terminat* or ((gestation* or pregnan* or trimester) adj5 (interrup* or terminate* or termination*))).ti,ab. (17255)
2 (MTOP or abortifacient* or ((chemical* or drug or medical or medically or medicinal or medication or medicine or mifepristone or misoprostol or pharma*) adj3 (abortion* or induc* or interrupt* or termination*)) or ((dinoprost* or carboprost or epostane or ethacridine or gemeprost or isosorbide or lilopristone or meteneprost or methotrexate or mifepristone or misoprostol or onapristone or oxytocin or "potassium chloride" or prostaglandin* or saline or sulprostone) adj10 (abortion* or induc* or interrupt* or terminat*))).ti,ab. (10473)
3 (Abo-pill or Colestone or Cytotec or Elmif or GyMiso or Korlym or Medabon or Mefeprin or Mefipil or Mifebort or Mifegest or Mifegyne or Mifeprex or Miferiv or Mifty or Mtpill or Pitocin or RU-486 or RU486 or Syntocinon or T-Pill or Termipil).ti,ab. (154)
4 (((1st or first) adj trimester) or (early adj2 (abortion* or pregnan*)) or (("5" or five or fifth or "6" or six or sixth or "7" or seven or seventh or "8" or eight or eighth or "9" or nine or ninth or "10" or ten or tenth or "11" or eleven or eleventh or "12" or twelve or twelfth or "13" or thirteen or thirteenth or "14" or fourteen or fourteenth) adj (week or weeks))).ti,ab. or days.ti. (90577)
5 or/2-3 (10601)
6 and/1,4-5 (199)
7 (groups or random* or placebo or trial).ti,ab. (544497)
8 and/6-7 (76)

9 limit 8 to yr="2010 -Current" (48)

LILACS

WORDS (abortion OR abortions OR "interruption of pregnancy" OR "termination of pregnancy" OR "pregnancy termination" OR "pregnancy terminations" OR feticide OR feticidal OR foeticide OR foeticidal OR abortifacient OR abortifacients OR LTOP OR MTOP OR Abo-pill OR Colestone OR Cytotec OR Elmif OR GyMiso OR Korlym OR Medabon OR Mefeprin OR Mefipil OR Mifebort OR Mifegest OR Mifegyne OR Mifeprex OR Miferiv OR Mifty OR Mtpill OR Pitocin OR RU-486 OR RU486 OR Syntocinon OR T-Pill OR Termipil)

AND

WORDS (((1st OR first) AND trimester) OR "early abortion*" OR "early pregnancy" OR "early pregnancies" OR ((5 OR five OR fifth OR 6 OR six OR sixth OR 7 OR seven OR seventh OR 8 OR eight OR eighth OR 9 OR nine OR ninth OR 1o OR ten OR tenth OR 11 OR eleven OR eleventh OR 12 OR twelve OR twelfth OR 13 OR thirteen OR thirteenth OR 14 OR fourteen OR fourteenth) AND (week OR weeks))

(284)

ClinicalTrials.gov [EXPERT SEARCH]

EXPAND[Concept] ("1st trimester" OR "first trimester" OR "I trimester" OR "early abortion" OR "early pregnancy" OR "early pregnancies" OR "days" OR ("5" OR "five" OR "fifth" OR "6" OR "six" OR "sixth" OR "7" OR "seven" OR "seventh" OR "8" OR "eight" OR "eighth" OR "9" OR "nine" OR "ninth" OR "10" OR "ten" OR "tenth" OR "11" OR "eleven" OR "eleventh" OR "12" OR "twelve" OR "twelfth" OR "13" OR "thirteen" OR "thirteenth" OR "14" OR "fourteen" OR "fourteenth") AND ("week" OR "weeks"))

AND AREA[OverallStatus] EXPAND[Term] COVER[FullMatch] ("Recruiting" OR "Active, not recruiting" OR "Completed" OR "Enrolling by invitation" OR "Suspended" OR "Terminated")

AND AREA[ConditionSearch] (abortion OR interruption of pregnancy OR termination of pregnancy)

AND AREA[InterventionSearch] (abortifacient OR LTOP OR MTOP OR (chemical* OR drug OR medical OR medically OR medicinal OR medication OR medicine OR mifepristone OR misoprostol OR pharma*) AND (abortion* OR interruption OR termination) OR (dinoprost* OR carboprost OR epostane OR ethacridine OR gemeprost OR isosorbide OR lilopristone OR meteneprost OR mifepristone OR methotrexate OR misoprostol OR onapristone OR prostaglandin* OR sulprostone) AND (abortion* OR EXPAND[Concept] "interruption of pregnancy" OR EXPAND[Concept] "termination of pregnancy") OR Abo-pill OR Colestone OR Cytotec OR Elmif OR GyMiso OR Korlym OR Medabon OR Mefeprin OR Mefipil OR Mifebort OR Mifegest OR Mifegyne OR Mifeprex OR Miferiv OR Mifty OR Mtpill OR Pitocin OR RU-486 OR RU486 OR Syntocinon OR T-Pill OR Termipil)

AND AREA[StudyFirstPostDate] EXPAND[Term] RANGE[01/01/2010, 28/02/2021]

(85)

WHO ICTRP [ADVANCED SEARCH]

CONDITION (without synonyms checked) = abortion* OR "interruption of pregnancy" OR "termination of pregnancy" OR "pregnancy termination" OR MTOP

INTERVENTION (without synonyms checked) = dinoprost* OR carboprost OR epostane OR ethacridine OR gemeprost OR isosorbide OR lilopristone OR meteneprost OR mifepristone OR methotrexate OR misoprostol OR onapristone OR prostaglandin* OR sulprostone

RECRUITMENT STATUS = ALL

DATE OF REGISTRATION = 01/01/2010 and 28/02/2021

(130)

Appendix 2. 2011 Systematic review search strategies

Search methods for identification of studies section

The Cochrane Controlled Trials Register, MEDLINE and POPLINE were systematically searched. Reference lists of retrieved papers were also searched. Electronic literature search of MEDLINE(with the Cochrane 3-stage search strategy)(1966-2003) and POPLINE (1970-2003) databases with the following key words: (abortion OR pregnancy termination OR termination of pregnancy) AND(first trimester OR early) AND (mifepristone OR misoprostol OR methotrexate OR dinoprost* OR carboprost OR sulprostone OR gemeprost OR meteneprost OR lilopristone OR onapristone OR epostane OR oxytocin OR RU 486 OR mifegyne). There were no language preferences in the application of the search.

Index terms

Medical Subject Headings (MeSH)

Abortifacient Agents [administration & dosage]; Abortion, Incomplete [chemically induced]; Abortion, Induced [adverse effects; *methods]; Drug Therapy, Combination; Methotrexate [administration & dosage]; Mifepristone [administration & dosage]; Misoprostol [administration & dosage]; Pregnancy Trimester, First; Prostaglandins [administration & dosage]; Randomized Controlled Trials as Topic; Tamoxifen [administration & dosage]

MeSH check words

Female; Humans; Pregnancy

WHAT'S NEW

Date	Event	Description
28 February 2021	New citation required and conclusions have changed	We added 41 new studies and 2 new comparison groups (administration strategy of medical abortion; misoprostol sublingual versus buccal) in the updated review.

HISTORY

Protocol first published: Issue 4, 2000
Review first published: Issue 1, 2004

Date	Event	Description
3 October 2011	New citation required but conclusions have not changed	New author Nathalie Kapp helped updating this review and 19 new studies were added
15 April 2008	Amended	Converted to new review format.
17 October 2003	New citation required and conclusions have changed	Substantive amendment

CONTRIBUTIONS OF AUTHORS

ZJ and ZK did the data extraction and quality evaluation of the updated review.

ZJ, ZK, SD and LX reviewed and contributed substantially in all aspects of the updated review.

DECLARATIONS OF INTEREST

JZ: nothing to declare

KZ: nothing to declare

DS: nothing to declare

XL: nothing to declare

SOURCES OF SUPPORT

Internal sources

- Chinese Evidence Based Medicine Center, China

  Methodological support

- West China Second University Hospital, Sichuan University, China

  Technical support
- Science and Technology Program Project of Sichuan Province,China(2021YFS0127; 2019YFS0422), China

  Investigator support
- Scientific Research Program of the Health Commission of Sichuan Province (20PJ075), China

  Investigator support

External sources

- Effective Care Research Unit, University of the Witwatersrand, South Africa, South Africa

  Support for previous review

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

We used the random-effects model for statistical analysis in the updated review for significant clinical heterogeneity and inconsistence among the included studies.

We used risk ratios instead of odds ratios.

We added new comparison groups (administration strategy of medical abortion; misoprostol sublingual versus buccal) in the updated review.

For expulsion time of conceptus, we included both dichotomous data, and continuous data.

NOTES

none

INDEX TERMS

Medical Subject Headings (MeSH)

*Abortifacient Agents, Nonsteroidal [adverse effects]; *Abortion, Spontaneous [chemically induced]; Diarrhea [chemically induced]; Drug Therapy, Combination; Estradiol; Letrozole; Methotrexate; Mifepristone [adverse effects]; *Misoprostol; *Oxytocics; Pregnancy Trimester, First; Prostaglandins; Tamoxifen

MeSH check words

Female; Humans; Pregnancy